

Introduction to AI argumentation

Liuwen Yu and Leendert van der Torre

25 September 2026

Contents

Preface	v
Introduction: making reasons accountable	vii
Chapter 0. Making a reasoning claim checkable	1
Part I. Reasons, agents and formal objects	8
1 Chapter 1. Choosing and representing a reasoning method	10
2 Chapter 2. Reasons across agents	20
3 Chapter 3. Reconstructing arguments	27
4 Chapter 4. Choosing the question	35
5 Chapter 5. Formalising argumentation	43
Part II. The attack–defense paradigm shift and its immediate generalizations	52
6 Chapter 6. Creating argumentation frameworks and semantics	53
7 Chapter 7. Representing nonmonotonic logics and solution concepts	62
8 Chapter 8. Rationality postulates for defining new logics	75
9 Chapter 9. Dialogue: public procedure and developing reasons	89
10 Chapter 10. Evaluating issues and comparing options	100
Part III. From argumentative methods to computational use	111
11 Chapter 11. Comparing argumentative methods	112
12 Chapter 12. Computing a specified answer	127
13 Chapter 13. Explaining an answer in context	145

14 Chapter 14. Argumentation as system architecture	159
15 Chapter 15. Argumentation in an acting and revising system	170
Reading routes and supplied studies	183
Optional study guide for Part II	184
Part III study guide: comparison, computation and explanation	187
Part III study guide: systems and the bounded agentic outlook	191
Directory of supplied studies	192
Chapter 2 study: coalition witnesses	193
Chapter 2 study: grounded majority comparison	196
Supplement S3.1. Evaluating a supplied extraction	198
Chapter 7 mathematical supplement: Stable models of normal programs	199
Supplement S9.1. Comparative disclosure and withdrawal	203
Supplement S9.2. From a dialogue theory to an application	207
Supplement S10.1. Scores and rankings	208
Supplement S10.2. Values and audiences	210
Supplement S10.3. Uncertainty about graphs and beliefs	211
Supplement S10.4. Where do weights get their authority?	213
Study S11. Mechanisms and change	214
Study S12. Queries, projections and certificates	217
Study S13. Evidence for a classifier explanation	219

Supplement S14.1. When support changes a score	221
Supplement S14.2. Selecting the content of a report	223
Supplement S15.1. A reference for the finite revision and action handoff	225
Solutions	227
Reader reference	297
References	306

Preface

This book grew from the intellectual programme of *Thirteen Challenges in Formal and Computational Argumentation* [1], from teaching in the field, and from the research brought together in the three volumes of the *Handbook of Formal Argumentation* [2, 3, 4]. That programme asks how argumentation can connect the analysis of reasons with their formal representation, evaluation and use. Its questions invite a journey across traditions: from individual and collective reasoning to dialogue, practical judgment and computational systems. They supply an intellectual starting point, rather than a syllabus whose entries can simply be expanded into independent lectures.

Our aim is to develop those questions as a connected introduction. A reader should be able to work with a model, understand why it answers one question, and recognize what must be added when the question changes. This requires more than a survey of results. We develop small cases far enough to expose their assumptions, prove selected results whose structure matters for later reasoning, and return to earlier examples with a new task. The logical preparation and comparison of source methods in Chapters 0–1 make that journey accessible before the specialist machinery is introduced.

The historical questions receive the following development in this book. The descriptions state their intellectual roles here; they do not prescribe equal space or a uniform mathematical treatment.

Challenge	Development in this book	Chapter
1	Agency and individual/collective reasoning	2
2	Reconstructing and analysing arguments	3
3	Choosing the question and conceptualization	4
4	Formalizing content-bearing reasons	5
5	Semantic evaluation by attack and defense	6
6	Representing reasoning methods and solution concepts	7
7	Rationality requirements and limits of construction	8
8	Public dialogue and the development of reasons	9
9	Issue acceptance, burdens and practical balancing	10
10	Principle-based comparison of methods	11
11	Computing a specified answer	12
12	Explaining an answer in context	13
13	Integrating argumentation into systems	14

The resulting textbook substantially extends this starting programme. Representation becomes a recurring comparison between a source method and an argumentative construction, including the possibility that their complete families of answers differ. Dialogue and practical evaluation are developed as ways of reasoning with their

own objects and outputs. They return when we compare principles, compute a next response or a permitted set, explain a changed decision, and design a service. Chapter 15 extends the final movement to an acting and revising system: information can arrive without changing a belief, a previously checked response can become stale, and an authorized action can achieve something useful while leaving part of the original goal unresolved. These developments use finite cases to make the new questions precise.

The Handbooks remain companions for deeper study. This book offers its own cumulative route through the field they represent, with annotated readings that explain where a specialist treatment begins. It is written for readers who want both mathematical command of selected methods and a considered view of what argumentative methods contribute to reasoning.

Introduction: making reasons accountable

A reason does more than contribute to a final answer. It gives us something to examine: a premise that might be disputed, a step whose use needs justification, or a consideration that counts differently when another option is available. When several reasons conflict, keeping their structure visible helps us ask what survives, what would change the answer, and what someone must explain. Formal argumentation studies how to make these relationships precise enough to reason about and compute with.

This becomes useful in different ways. A reasoning system may need to draw conclusions from information that cannot all be retained. Participants in a discussion may need to know which reply is due and what has actually been conceded. A decision maker may have grounds for several options and need to distinguish what is permitted from what a further policy selects. In each case, representing reasons makes some part of the reasoning inspectable. The useful representation depends on the task.

Content, acts and choices

Consider the difference between offering a reason and asking for its grounds. The two acts may concern the same proposition, but they do different things in a discussion. An opposing reason adds another consideration; a request for justification calls on a participant to develop what has already been said. To reconstruct an argument, we therefore need to distinguish its content from the act performed with that content, and what a source supplies from what an analyst adds. A graph can record selected relationships only after those choices have been made.

Practical reasoning introduces a further comparison. A consideration in favour of one dinner option may defeat an objection when that option is compared with one alternative and fail against another. Computing the permitted options can explain which comparisons succeed. Selecting one of them requires an additional rule, and carrying it out requires an occasion and authority to act. These distinctions give us more to explain than a single accepted proposition. The book develops them through worked cases, then uses them again when it studies computation, explanation and systems.

The common question is what a method makes accountable. We ask what it receives, what it retains of the reasons and their setting, which answer it returns, and what that answer establishes. Making these choices explicit allows methods to be compared without pretending that they all return the same kind of object.

What abstraction makes possible

Dung’s attack–defense abstraction is the central mathematical pivot of the book [5]. It starts with arguments and a directed attack relation. At that level, very different sources of reasons can give rise to common questions: which sets avoid internal conflict, which defend their members, and which satisfy a chosen notion of acceptability? A shared representation makes semantic definitions, finite algorithms and structural comparisons available across many settings. It also permits a precise representation question: does an argumentative construction preserve the answers of an independently specified reasoning method?

The gain comes from selecting what the abstraction retains. The internal content of an argument, the responsibility incurred by an utterance, and the grounds for a comparative weight need further representation when a task depends on them. We can then study the connection between levels. An extraction maps selected arguments to conclusions; a public protocol relates moves to records and pending replies; an authorized revision identifies which stored weight may change. Such connections have assumptions and obligations of their own. Establishing one does not supply a universal translation between inference, dialogue and decision.

Three connected movements

Part I moves from reasoning and argumentative activity to formal objects. Chapter 1 asks which source method is to be represented and works through repair, preference and base-argument comparisons. Agency and collective reasoning then make the source of reasons matter. Reconstruction and the choice of an inferential, public or practical question prepare structured argumentation, where premises, rules, attacks and preferences have distinct roles. These are substantive questions before they become prerequisites for later mathematics.

Part II develops the attack–defense abstraction and tests its reach. Semantic choices lead to representation, and representation leads to requirements on both outputs and constructions. A successful encoding of a graph can still fail to preserve the intended source method. The public and practical returns then recover questions that extension membership does not settle: which response is due, how reasons are developed, whether an issue meets its standard, and which options survive comparison. Earlier examples return because there is new work for them to do.

Part III compares methods, computes answers, explains them and puts them to use. Properties of a semantic function differ from properties of a public transition or a practical comparison. Checking a supplied result differs from finding one; an explanation must retain the context needed by its target. The systems cases connect these procedures to specified purposes, and the final agentic case asks what receiving information, revising a model and acting add to the reasoning problem. Mathematical guarantees, computational performance and evidence of

human understanding answer different questions throughout this movement.

Preparation and independent study

The intended reader is a master's student, beginning doctoral researcher, or comparably prepared independent reader who is comfortable with elementary discrete mathematics and logical reasoning. Chapter 0 provides a working route through the sets, relations, propositional logic, quantifiers and proof habits used here. It is worth working its examples even when the notation is familiar: distinctions such as maximal versus maximum, or an empty answer versus no answer, become consequential later. Argumentation expertise, a previous complexity course and programming are not assumed.

The main argument and ordinary exercises use material supplied in the book. A task may rely on an earlier definition or on one of the studies collected at the back; the studies supply their own additional preparation. Solving an exercise can require discovering a proof or a construction, but an external paper or an answer is not a missing premise. The visibly labelled literature/research exercises have a different purpose: they ask the reader to consult specified external sources and compare what those sources do. An optional or difficult ordinary exercise is still internally prepared.

The chapter-end bank supports several routes rather than one weekly assignment. Complete public answers, including substantial proof work and the nonprogramming capstone, are collected by chapter. Each *[solution]* link opens the matching answer; its exercise number or *[prompt]* link returns to the task. Attempt the question and identify its assumptions before opening the answer; then compare the reasoning as well as the outcome. Paper algorithms, marked sets, tables and traces provide complete nonprogramming routes. Code is an optional way to implement the same specified calculation.

Book study follows the connected argument and its selected formal depth. A semester course can ask for independent reconstruction of a chosen range of proofs and transfers. An intensive course can use guided reconstruction and fewer independent tasks while retaining the point of the comparisons. The reading routes beginning on p. 183 and the chapter-level guidance distinguish these expectations more precisely. They do not make every book-core proof an examination requirement.

The supplied-study directory is on p. 192, complete answers begin on p. 227, and the definition spine, glossary and indices begin on p. 297. These lookup pages point back to explanations with their assumptions; they are not a vocabulary list to memorize before beginning Chapter 1.

Continuing into the literature

The chapter-end annotations connect a question developed here to foundations, alternatives and later work. They are the most useful starting point for deeper reading. Volume 1 of the Handbook offers broad foundational treatments of the main formalisms, semantic and computational questions [2]. Volume 2 develops extensions, dynamics and further dialogue and comparative perspectives [3]. Volume 3 opens applications, connections to other fields and directions for further investigation [4]. The three volumes overlap productively; the chapter routes identify the passage that serves a particular next question. The [Chapter 6 semantic route](#) and [Chapter 12 computational route](#) lead into Volume 1; the [Chapter 9 dialogue route](#) and [Chapter 11 comparisons](#) open extensions and investigations in Volume 2. The [Chapter 13 explanation route](#) and [Chapter 14 application routes](#) connect with Volume 3.

When following a route, keep the source's object and assumptions beside its result. A construction with a restricted domain, a pedagogical convention, a theorem and evidence from an implementation or a human study have different warrants. The book names those boundaries so that further reading can extend the argument without silently changing what has already been established.

Chapter 0. Making a reasoning claim checkable

Suppose a report contains p and $p \rightarrow q$. Does it support q ? Suppose a diagram contains the arrows $a \rightarrow b$ and $b \rightarrow c$. Is there an arrow from a to c ? The first answer is yes under classical consequence; the second is no for the displayed relation, although there is a path. Neither answer follows from recognizing familiar symbols alone. We must know which objects are given and which operation the question permits.

Formal argumentation repeatedly makes such distinctions consequential. A reason may be available without being accepted; a set may contain mutually incompatible reasons; different acceptable sets may yield a common conclusion. Before choosing a theory of argumentation, we need to be able to check claims of these kinds. This chapter develops the elementary logic, set reasoning and proof habits used for that purpose. It assumes familiarity with ordinary mathematical reasoning, but makes the conventions needed by the opening chapters explicit.

Objects, relations and families

Let $A = \{a, b, c\}$ and $R = \{(a, b), (b, c)\}$. The statement $a \in A$ concerns an object; $\{a, c\} \subseteq A$ concerns a set of objects. Inclusion means that every member of the first set is a member of the second. We write $X \subsetneq Y$ for inclusion with $X \neq Y$, and 2^A for the family of all subsets of A . Thus 2^A has eight members, including $\emptyset$ and A .

An ordered pair retains order: $(a, b) \neq (b, a)$. The Cartesian product $A \times A$ contains every ordered pair from A ; a binary relation is any subset of that product. Our R contains two pairs, neither their reverses nor (a, c) . A function $f : X \rightarrow Y$ assigns exactly one value in Y to each member of X . A relation need not do so: one object may be related to several others, or to none. For $U \subseteq X$, its image is $f[U] = \{f(u) : u \in U\}$.

The directed graph (A, R) has arguments as nodes only when we give the nodes that interpretation. Mathematically it has nodes and directed edges. A path $v_0, \dots, v_n$ follows an edge $(v_i, v_{i+1}) \in R$ at every step. A cycle is a path of positive length returning to its start, and a self-loop is an edge (v, v) . Our graph has the path a, b, c , but no cycle. Adding (c, a) creates a cycle without creating a self-loop. Whether these arrows mean attack, succession or something else remains a modelling choice; reachability does not automatically inherit the interpretation of a single edge.

We will often compare a *family* of possible sets. For example,

$$\mathcal{X} = \{\{a\}, \{b, c\}\}$$

has two members, of different sizes. Its union is $\{a, b, c\}$: an object belongs to the union when it belongs to at least one member. Its intersection is empty: no object belongs to every member. These operations can lose information about the alternatives. The union does not say that $\{a, b, c\}$ was itself an available member.

In particular, $\emptyset$ and $\{\emptyset\}$ are different families. The first has no member; the second has one member, itself empty. Both have empty union. Later a reasoning method may return no acceptable set, or may return the empty set as its sole acceptable set. A readback that records only their union would conceal this difference.

Truth, consequence and countermodels

A propositional language L consists of formulas built from atoms such as p, q , using negation $\neg$, conjunction $\wedge$, disjunction $\vee$ and implication $\rightarrow$. A valuation assigns each atom either 1 (true) or 0 (false). Negation reverses the value; a conjunction is true when both conjuncts are true; a disjunction is true when at least one disjunct is true. The material implication $p \rightarrow q$ is false exactly when p is true and q is false. It does not assert a causal connection.

Write $v \models \varphi$ when valuation v satisfies formula φ , and $v \models \Gamma$ when it satisfies every formula in the premise set Γ . In the classical propositional calculations in this book,

$$\Gamma \vdash \varphi \iff \forall v (v \models \Gamma \Rightarrow v \models \varphi).$$

Here $\vdash$ denotes classical consequence; elsewhere a different consequence relation is declared explicitly. A set of formulas, often called a knowledge base, is consistent if some valuation satisfies all of it. Equivalently in this setting, it does not entail the always-false formula $\perp$.

The four valuations of p, q show what the opening question requires:

p	q	$p \rightarrow q$	satisfies $\{p, p \rightarrow q\}$?
0	0	1	no
0	1	1	no
1	0	0	no
1	1	1	yes

Every model of $\{p, p \rightarrow q\}$ makes q true, so that set entails q . But $\{p \rightarrow q\}$ alone does not: the valuation $p = q = 0$ is a *countermodel*, satisfying the premise and falsifying the conclusion. The valuation $p = q = 1$ satisfies both premise and conclusion, but that successful case cannot prove a universal entailment. One countermodel suffices to refute it.

The set $\{p, \neg p\}$ has no model. Consequently it classically entails every formula: there is no model of the premises that could falsify a query. This is explosion. A method that repairs inconsistent information must change which premises are retained or which consequence operation is used; it cannot obtain that behaviour merely by drawing the original premises as a graph.

Write

$$\begin{aligned}\text{Mod}(\Gamma) &= \{v : v \models \Gamma\}, \\ \text{Th}(W) &= \{\varphi : \forall v \in W, v \models \varphi\}, \\ \text{Cn}(\Gamma) &= \{\varphi : \Gamma \vdash \varphi\}.\end{aligned}$$

Thus $\text{Cn}(\Gamma) = \text{Th}(\text{Mod}(\Gamma))$. The closure $\text{Cn}(\Gamma)$ contains consequences beyond the formulas explicitly written in Γ . For instance, q belongs to $\text{Cn}(\{p, p \rightarrow q\})$ although it is not a displayed premise. The closure of the empty premise set consists of the tautologies, whereas $\text{Th}(\emptyset) = L$: no valuations and no premises are different inputs.

Who may choose a witness?

Quantifiers say how much a witness must accomplish. Let $B = \{b_1, b_2\}$ be two tasks and $D = \{d_1, d_2\}$ two possible answers. Suppose the relation $Q \subseteq D \times B$ is $\{(d_1, b_1), (d_2, b_2)\}$, where dQb means that d answers b . Then

$$\forall b \in B \exists d \in D dQb$$

is true: choose d_1 for b_1 , and d_2 for b_2 . The choice of answer may depend on the task. In contrast,

$$\exists d \in D \forall b \in B dQb$$

is false. Neither answer handles both tasks. Chapter 2 will use this distinction twice: a different counterargument may answer each attack, and a different agent may supply each counterargument.

To refute the first statement we need one unanswered task, not a claim that every task is unanswered. Its negation is

$$\exists b \in B \forall d \in D \neg(dQb).$$

If only d_1 is available, b_2 is that failure certificate. The universal remainder matters: naming b_2 and an answer that fails against it would not exclude another available answer that succeeds.

Empty domains deserve the same literal reading. If $B = \emptyset$, the assertion that every task has an answer is true even for $D = \emptyset$. The assertion that *one answer* handles every task still demands a member of D . Likewise $\forall E \in \emptyset a \in E$ is logically true, but it supplies no acceptable set containing a . Where the book asks whether a is skeptically accepted in an output family, it additionally requires that family to be nonempty. Such a query convention must be stated; it does not change classical truth or the value of $\text{Th}(\emptyset)$.

Maximal, minimal and least

A selection policy needs an order as well as a family of candidates. Inclusion compares sets by containment. In $\mathcal{X} = \{\{a\}, \{b, c\}\}$, neither member contains the

other. Both are inclusion-maximal: neither has a proper superset in $\mathcal{X}$. Both are also inclusion-minimal: neither has a proper subset in $\mathcal{X}$. Yet the second has larger cardinality.

More generally, for $E \in \mathcal{X}$,

$$\begin{array}{ll} E \text{ is minimal} & \text{iff no } T \in \mathcal{X} \text{ has } T \subsetneq E, \\ E \text{ is least} & \text{iff } E \subseteq T \text{ for every } T \in \mathcal{X}, \\ E \text{ is maximal} & \text{iff no } T \in \mathcal{X} \text{ has } E \subsetneq T, \\ E \text{ is greatest} & \text{iff } T \subseteq E \text{ for every } T \in \mathcal{X}. \end{array}$$

Our two-member family has no least or greatest member. A least member, when it exists, is unique: if E and F are both least, then $E \subseteq F$ and $F \subseteq E$, hence $E = F$. This small two-inclusion proof explains why leastness says more than minimality. A maximum-cardinality member, by contrast, maximizes a number and need not contain any other candidate.

For $K = \{p, \neg p, q\}$, consider all consistent subsets of K . The set $\{p\}$ is consistent but not maximal: q can be added. Both $\{p, q\}$ and $\{\neg p, q\}$ are maximal. Neither can absorb the omitted literal. Chapter 1 will make this a repair method. The important test is whether a consistent extension exists, not whether a set looks large or plausible.

Other policies use other orders. A preorder is a reflexive, transitive relation; it may tie distinct candidates. Relative to a preference preorder, a candidate is maximal when no candidate is strictly preferred to it. Numerical ranks also require a declared direction. In Chapter 1's valuation example a lower rank is better; this convention is not imposed on later rule priorities.

What a short proof establishes

Classical consequence is monotonic: if $\Gamma \vdash \varphi$ and $\Gamma \subseteq \Delta$, then $\Delta \vdash \varphi$. For the proof, take an arbitrary model of Δ . It satisfies every formula of Γ , hence satisfies φ . This proves the claim for every model of Δ , including the case in which there is none. Equivalently, $\text{Cn}(\Gamma) \subseteq \text{Cn}(\Delta)$. A method is nonmonotonic when some added premises can withdraw a conclusion; Chapter 1 supplies a complete finite instance of such withdrawal.

A set equality usually needs two inclusions. For example,

$$\text{Mod}(\Gamma \cup \Delta) = \text{Mod}(\Gamma) \cap \text{Mod}(\Delta).$$

A model of the union satisfies each component, giving one inclusion. Conversely, a model of both components satisfies every formula in their union, giving the other. A calculation on one chosen valuation would not establish this identity for arbitrary premise sets.

The same discipline applies to algorithms. For the graph $a \rightarrow b \rightarrow c$, start with $X_0 = \{a\}$, then repeatedly add every successor of a node already present:

$$X_{n+1} = X_n \cup \{y : \exists x \in X_n, (x, y) \in R\}.$$

The successive sets are $\{a\}, \{a, b\}, \{a, b, c\}, \{a, b, c\}$. An *invariant* is a property preserved at each step: here every listed node is reachable from a . It holds initially by the zero-length path, and a new node has a path obtained by adding one edge to a previous path. Since no node is removed and the graph is finite, strict enlargement eventually stops.

Termination and the invariant do not by themselves prove that this is the least successor-closed set containing a . For that, let Y be any set containing a and containing every successor of each of its members. Then $X_0 \subseteq Y$; if $X_n \subseteq Y$, the update also lies in Y . By induction, the final set is contained in every such Y . At termination it is itself successor-closed. These are the two parts of its leastness claim.

For a map $F : 2^A \rightarrow 2^A$, a fixed point satisfies $F(X) = X$. *Monotonicity* of the map means $X \subseteq Y \Rightarrow F(X) \subseteq F(Y)$; *inflationarity* means $X \subseteq F(X)$. They are different properties. The constant map $F(X) = \emptyset$ is monotonic but not inflationary on a nonempty A . Our reachability update is both. Later defense operators will need their own proofs: this elementary example prepares the reasoning, not a permission to assume every iteration has the same properties.

Calculating, checking and explaining

A decision question asks whether a property holds. Search asks for one object satisfying it; enumeration asks for all such objects. Verification starts with a proposed object and checks it. For a graph (A, R) , call $E \subseteq A$ conflict-free when no $x, y \in E$ satisfy $(x, y) \in R$. Given E , we can inspect its internal pairs. An internal edge is a short certificate of failure; the absence of one requires checking all relevant pairs. Finding a conflict-free set and listing every such set are different tasks, even though the same definition governs them.

A certificate explains why a particular answer passes a test. A proof of an algorithm explains why its answers always satisfy a specification and why required answers are not missed. Neither establishes that the chosen graph captured the intended reasons. The book will keep these questions connected: what was represented, what follows under the stated method, and what the resulting answer warrants.

Exercises

All ordinary exercises below use only this chapter. Complete solutions are collected at the back under Chapter 0. Programming is not required.

- 0.1 *[solution]* For $A = \{a, b, c\}$ and $R = \{(a, b), (c, b)\}$, list the nodes related to b , the nodes to which b is related, and the image of A under the relation $R[A] = \{y : \exists x \in A, (x, y) \in R\}$.
- 0.2 *[solution]* For the graph on $\{a, b, c\}$ with edges $(a, b), (b, c), (c, a)$, give a path from a to c , a cycle, and the complete set of self-loops.
- 0.3 *[solution]* Decide whether $\{p, p \rightarrow q, \neg q\}$ is consistent. Separately decide whether $\{p \rightarrow q\} \vdash q$. Give a proof or a complete countermodel as appropriate.
- 0.4 *[solution]* Explain why $\{p, p \rightarrow q, r\} \vdash q$. Prove the corresponding monotonicity statement for arbitrary classical premise sets, and explain why adding a formula that makes the set inconsistent does not refute it.
- 0.5 *[solution]* For $K = \{p, p \rightarrow q, \neg q, s\}$, find one inclusion-maximal consistent subset containing s . Give a satisfying valuation of all three atoms and show why every omitted formula cannot be added consistently. You need not enumerate all repairs.
- 0.6 *[solution]* What certificate refutes conflict-freeness of E in (A, R) ? Give one for $E = \{a, b\}$ when $A = \{a, b, c\}$ and $R = \{(a, b), (b, c)\}$. Explain why one tested nonedge would not certify conflict-freeness.
- 0.7 *[solution]* Compare $\mathcal{E}_0 = \emptyset$ and $\mathcal{E}_1 = \{\emptyset\}$. Give their numbers of members and unions. For an object a , evaluate whether some member contains a , and whether every member contains a , for each family. Then apply the additional nonempty-family guard for skeptical acceptance.
- 0.8 *[solution]* Let $B = \{b_1, b_2\}$, $D = \{d_1, d_2\}$, and $Q = \{(d_1, b_1), (d_2, b_2)\}$. Compare $\forall b \exists d dQb$ and $\exists d \forall b dQb$, with variables ranging over the indicated sets. Negate both statements. After deleting d_2 from the available answer set, give a complete failure certificate for the first statement. Finally evaluate both statements for $B = D = \emptyset$.
- 0.9 *[solution]* In $\mathcal{X} = \{\{a\}, \{b, c\}\}$, list the minimal, maximal, least, greatest and maximum-cardinality members. Repeat after adding $\emptyset$ as a third member.
- 0.10 *[solution]* On $A = \{a, b, c\}$, take $R = \{(a, b), (b, a)\}$. Run the successor-addition update from $X_0 = \{a\}$ until equality. Prove that the result is the least successor-closed set containing a , stating both an invariant and the comparison with an arbitrary closed set. Is the constant map $H(X) = \{a\}$ monotonic? Is it inflationary on 2^A ?
- 0.11 *[solution]* Prove $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$ by two inclusions. State what an arbitrary member of each side must satisfy before passing to the other side.

Reading routes

For another use of these mathematical habits, read Section 2 and the beginning of Section 3 of the Handbook chapter on abstract frameworks [6, Vol. 1, Ch. 4, pp. 162 onward]. Identify the set, relation and quantified test before studying the different semantics; Chapter 6 will develop those in the book. For the difference between verifying, finding and enumerating, use the computational-problems chapter [7, Vol. 1, Ch. 13, Sec. 3.2, pp. 647–648]. These passages apply the present preparation; they are not additional prerequisites.

A reader needing more practice with elementary consequence can consult Enderton or Ebbinghaus, Flum and Thomas [8, 9], selecting the needed topic rather than undertaking a new logic course. To proceed with the book’s question, Makinson’s short account of assumptions, valuations and rules is the useful next reading [10, Sec. 1.4, p. 18]. Chapter 1 supplies the comparison internally.

Literature exercise 0.L1 (optional). *[solution]* This is deliberately an external reading task. Read Makinson, Section 1.4, p. 18, and the two specified Handbook passages above. Write a short comparison identifying each passage’s question, formal object, and contribution to one worked example in this chapter. Distinguish introducing a reasoning method from defining its mathematical objects and from asking a computational question. The collected solutions give a model comparison and criteria; the ordinary exercises require none of these readings.

For book study, the worked examples and the distinctions they establish belong to the opening route, including maximality on first reading. A semester course can select countermodel, quantifier, order and proof transfers from the bank. An intensive course can work the displayed cases together and choose fewer independent reconstructions. Neither route requires programming. The reference pages at the back (p. 297) are for later lookup; their specialist terms need not be learned before Chapter 1.

Part I. Reasons, agents and formal objects

An inconsistent report, an unanswered objection and a disputed practical choice can all give rise to argumentation. They do not pose the same question. In the first case we may want conclusions that survive a repair of the information. In the second we may want to know who has grounds for an answer, or who has undertaken to give one. In the third we may need to compare reasons for incompatible options. A formal account earns its place by making one of these questions precise while allowing us to see what its answer establishes.

This part develops that choice of question together with the objects needed to answer it. Chapter 1 begins with reasoning methods whose outputs can be specified before drawing an argument graph. Keeping assumptions, repairing them and selecting preferred models have different consequences. The chapter then constructs arguments that represent one of those methods exactly. Representation is useful because the conflicts and surviving positions become inspectable; the choice of source method remains substantive.

Chapter 2 adds agents. A collection of reasons may answer every objection although no single participant supplies all the answers. That is a question about distributed defense. Forming a collective view from several participants' graphs or judgments is another question: a property of each input can fail in the aggregate. Knowing which agent is associated with an argument supplies useful information, but does not by itself establish what that agent has publicly asserted, is entitled to assert, or ought to believe.

That gap leads to Chapter 3. Before evaluating an actual exchange we must reconstruct the contribution made by its words. Offering a reason, asking for a reason, conceding a consideration and withdrawing a claim are different acts, even when they concern the same content. A reconstruction may add a warrant or normalize several expressions into one claim. Making those additions visible permits criticism of the formal input itself, as well as criticism within the resulting model.

Chapter 4 then holds a custody case in view while changing the question. Inferential support for a proposition differs from the responsibilities created by a public exchange and from reasons permitting or selecting an option. These perspectives can inform each other. They do not combine automatically into a sequence in which an accepted proposition becomes an authorized and executed action. Chapter 5 develops one inferential route in depth: reasons are constructed from premises and rules, their internal parts determine where they can be attacked, and preferences can determine which attacks succeed as defeats.

The emerging commonality makes Dung's attack-defense abstraction powerful. Once a construction has supplied arguments and attacks, we can ask which sets survive the specified conflicts without repeating every detail of their origin. Chapters 1 and 2 use a limited selection test where it already serves a comparison; Chapter 6 develops the general semantic choices. What was abstracted away remains relevant

when we interpret an output, compare it with a source method, reconstruct a dialogue, or evaluate practical reasons.

The chapters therefore introduce ideas for use now and return to them when a further question calls for more structure. Chapter 1's repairs return in Chapter 7 as a problem about representation and extraction. Chapter 2's source association prepares, but does not replace, the public sponsorship of Chapters 3–4 and the exact dialogue rules of Chapter 9. Chapter 4's distinction between support, permission and selection returns in Chapter 10's practical evaluations. The methods become cumulative because each return retains the earlier question while making an additional choice explicit.

Read the five chapters as an argument about what should be made accountable: the source policy, the distribution of reasons, the reconstruction of an act, the purpose of an evaluation, and the construction of its arguments. The worked cases establish those distinctions before the full abstraction. The exercises let the reader change an input or policy and discover which part of an answer depended on it.

1 Chapter 1. Choosing and representing a reasoning method

An agent receives information that cannot all be true. Which conclusions should it retain? Classical consequence gives a precise answer to the premises as supplied, but an inconsistent premise set entails everything. Repairing the input offers another answer, and choosing a preferred repair offers a third. The difference is already present before we construct any arguments.

The source-method question will govern this chapter: what operation on information do we want an argumentative construction to represent? We first make several operations inspectable on small inputs. Then we construct arguments whose accepted sets correspond exactly to the repairs of one specified method. The construction exposes conflicts and surviving positions; it does not decide, without further justification, which repair policy the agent ought to adopt.

1.1 What may be revised?

Let K be a finite set of revisable background formulas and P a finite set of current hard premises in a classical propositional language L . Hardness is a role in the present reasoning problem. It means that a proposed repair must retain P , not that every premise in P is infallible. A different inquiry might make some of them revisable.

We require $\text{Cons}(P)$, meaning that P has a classical model. If the hard premises already contradict one another, deleting background formulas cannot repair them. We reject that input for the repair methods below rather than interpreting the absence of repairs as support for every query.

Keep the roles disjoint: $K \cap P = \emptyset$. If the same formula is used in both roles, give its occurrences different labels and let $\pi(\ell, \varphi) = \varphi$ recover the underlying formula. All logical tests on a labelled set are tests on its image under π ; labels do not become additional propositional atoms. Source labels will also matter when we read a repair back from its argumentative representation.

The small case

$$K_0 = \{p, \neg p, q\}, \quad P_0 = \emptyset$$

will let us compare methods without changing the input. The conflict is between p and $\neg p$; the separate formula q can survive either choice. Chapter 0 supplied the consistency and maximality tests. We now ask what policy should use them.

1.2 Route 1: from a fixed pivot to default-assumption consequence

One possible policy keeps every formula in K . For a fixed background set K , define

$$P \vdash_K \varphi \iff K \cup P \vdash \varphi.$$

This is pivotal-assumption consequence [10, Def. 2.1, p. 24]. The pivot makes background assumptions explicit but does not repair them. For K_0 , every query follows. The problem is not an error in classical logic; we have chosen an operation that retains incompatible premises.

Instead retain as much of the background as consistency with P permits. A *maximal P -consistent subset* is a set $S \subseteq K$ such that

$$\text{Cons}(S \cup P) \quad \text{and} \quad \nexists T (S \subsetneq T \subseteq K \text{ and } \text{Cons}(T \cup P)).$$

Write $\text{MCS}(K, P)$ for the family of all such S . Maximality uses inclusion, not cardinality. On our finite domain this family is nonempty: start with the empty background subset, which is consistent with P , and enlarge it while possible.

For K_0 , both choices must retain q , since adding q to either consistent literal choice does no harm. They cannot retain both opposing literals. Hence

$$\text{MCS}(K_0, P_0) = \{\{p, q\}, \{\neg p, q\}\}.$$

A satisfying valuation $p = q = 1$ witnesses the first repair; $p = 0, q = 1$ witnesses the second. Adding the omitted opposing literal makes each inconsistent. The two repairs are alternatives, not two halves of a larger consistent premise set.

A cautious conclusion follows in every repair; a credulous conclusion follows in at least one. With an explicit guard, the definitions are

$$\begin{aligned} P \sim_K^\forall \varphi &\iff \text{Cons}(P), \text{MCS}(K, P) \neq \emptyset, \forall S \in \text{MCS}(K, P), S \cup P \vdash \varphi, \\ P \sim_K^\exists \varphi &\iff \text{Cons}(P), \text{MCS}(K, P) \neq \emptyset, \exists S \in \text{MCS}(K, P), S \cup P \vdash \varphi. \end{aligned}$$

The cautious operation is the finite propositional specialization of Makinson's default-assumption consequence [10, Def. 2.4, p. 31], with the book's nonvacuous query convention displayed. It differs from the fixed-pivot operation above. Here q is cautious; p and $\neg p$ are each credulous but not cautious.

Notice where consequence is taken. The cautious output is

$$\bigcap_{S \in \text{MCS}(K, P)} \text{Cn}(S \cup P).$$

It consists of conclusions common to the separate closures. Simply intersecting the written premise sets is a different operation. For the present literal example the distinction is easy to miss; Chapter 7 returns to it with formulas whose common consequences are not written as common premises.

1.3 Assumptions, valuations and rules

Repairs are one way to make consequence responsive to an input. Makinson's broader comparison distinguishes three possible pivots: assumptions, valuations

and rules [10, Sec. 1.4, p. 18]. These are types of object, not three names for the same algorithm. An assumption pivot adjoins formulas. A valuation pivot restricts or selects the truth assignments against which a query is checked. A rule pivot supplies inference pairs, closing the premises under classical consequence and the permitted inferences [10, Sec. 4.1, Def. 4.1, pp. 85–86].

For example, the inference pair (p, q) permits adding q when p is available. It is not automatically the premise $p \rightarrow q$. With that material implication and $\neg q$, classical reasoning gives $\neg p$; merely having the inference pair gives no such backward application. Chapters 5 and 7 will specify particular rule-based constructions. Naming the rule pivot here does not identify those constructions with every rule method in the literature.

The common question is what remains fixed and what is selected again when premises change. Fixed assumptions remain monotonic under classical consequence. So does restriction to a fixed set of valuations. To see the latter, let V contain all Boolean valuations of the finite vocabulary, fix $W_0 \subseteq V$, and put

$$C_{W_0}^{\text{fix}}(P) = \text{Th}(W_0 \cap \text{Mod}(P)).$$

Adding premises can only reduce the models being tested, so it cannot falsify a formula that was true at every previous model [10, Def. 3.1, p. 60].

If $W_0 = \text{Mod}(K)$, this valuation description agrees exactly with the fixed-assumption description:

$$C_{\text{Mod}(K)}^{\text{fix}}(P) = \text{Cn}(K \cup P).$$

Indeed, a valuation belongs to the set on which the left-hand consequence test operates exactly when it satisfies K and P , equivalently when it satisfies their union. Both operations quantify over the same valuations. This proves the identity, including the case with no satisfying valuation; it is the restricted form of Makinson’s Lemma 3.2 [10, p. 61]. The identity does not contain a preference relation.

Now make valuation selection depend on the premises. Fix an irreflexive, transitive relation $<$ on V , with $u < v$ meaning that u is preferred to v , and select

$$\begin{aligned} \text{Best}_{<}(P) &= \{v \in \text{Mod}(P) : \nexists u \in \text{Mod}(P) (u < v)\}, \\ C_{<}(P) &= \text{Th}(\text{Best}_{<}(P)). \end{aligned}$$

This is the minimal-valuation idea of preferential consequence [10, Def. 3.8, pp. 67–68]. Our finite example below has consistent premises and numerical ranks, so a preferred valuation always exists. The relation stays fixed; the set over which we select does not.

1.4 One changed input, different answers

Let f abbreviate the whole material formula $p \rightarrow q$. Keep $K = \{f\}$, first take $P = \{p\}$, and then add $\neg q$, giving $P' = \{p, \neg q\}$. Both hard inputs are consistent.

For fixed valuation restriction take $W_0 = \text{Mod}(K)$; for preference use all four valuations V , with the following pedagogical ranks. Lower ranks are better and ties are incomparable.

valuation	p	q	belongs to W_0 ?	rank in V
v_{00}	0	0	yes	0
v_{01}	0	1	yes	0
v_{10}	1	0	no	1
v_{11}	1	1	yes	0

Before the change, p permits v_{10} and v_{11} . Keeping f removes v_{10} ; preferring the lower rank also selects v_{11} . Both operations therefore support q . After the change, only v_{10} satisfies the hard premises. Keeping f excludes this remaining valuation, whereas preference selects it as the best available valuation. The difference is now visible.

Source operation	output for $P = \{p\}$	output for $P' = \{p, \neg q\}$
Hard premises alone	$\text{Cn}(\{p\})$	$\text{Cn}(\{p, \neg q\})$
Fixed K , equivalently fixed W_0	$\text{Cn}(\{p, q\})$	L
Cautious MCS	$\text{Cn}(\{p, q\})$	$\text{Cn}(\{p, \neg q\})$
Minimal valuations in V	$\text{Cn}(\{p, q\})$	$\text{Cn}(\{p, \neg q\})$

For the MCS row, $\{f\}$ is initially the sole maximal compatible subset. After the change it is incompatible with P' , so $\text{MCS}(K, P') = \{\emptyset\}$. There is one repair, with no background formula in it, and it still includes the hard premises when consequences are taken. The output family is not empty. For the fixed-valuation row, in contrast, $W_0 \cap \text{Mod}(P') = \emptyset$, so its classical universal consequence is L . The nonempty-repair guard does not alter that distinct definition.

Thus MCS selection and minimal-valuation selection withdraw q in this case; the fixed pivot does not. The hard premises alone never supported q in the first place. A ranking can remain unchanged while its selected models change. This is how the preference example permits withdrawal without deleting any input premise. Its agreement with the MCS outputs is a finite comparison, not a theorem representing arbitrary valuation preferences by argumentation. Exercise 1.9 changes only the ranking to test that distinction.

1.5 Route 2: make the preference over repairs explicit

Return to K_0 . We may have a reason to choose between its two repairs, rather than retaining both. Let $\succeq$ be a preorder over $\text{MCS}(K, P)$, and retain its maximal

members:

$$\text{Pref}_{\succeq}(K, P) = \max_{\succeq} \text{MCS}(K, P).$$

A strict improvement means $T \succeq S$ but not $S \succeq T$. Since the repair family is finite and nonempty, at least one maximal member exists. Cautious or credulous consequence then quantifies over the selected repairs. A flat preorder retains all repairs.

For an explicit teaching policy, give the formulas a strict priority list. Compare two repairs at the highest-priority formula on which they differ; prefer the repair containing it. With

$$p \succ \neg p \succ q,$$

the preferred repair is $\{p, q\}$. The common q does not decide the comparison; p does. We now obtain cautious support for p , which the unprioritized repair family did not give. Changing the priority can change that answer while leaving the consistent repair candidates unchanged.

This lexicographic policy is a pedagogical construction. Makinson's selection among maximal consistent subsets supplies its methodological context [10, Sec. 2.3, pp. 44–47]; it is not the valuation preference defined earlier. A repair is a set of source formulas, whereas a valuation assigns truth values to every atom. Nor is this policy yet a preference between the structured arguments of Chapter 5. We must identify the objects being compared before asking whether another construction preserves the comparison.

1.6 Route 3: exact representation by base argumentation

We can now state the representation problem. Starting from a finite base B , construct arguments and attacks, select sets by a declared test, and compare those sets with the maximal consistent source bases. The question is whether the selected argumentative positions preserve the repair method already defined.

The arguments must be rich enough to expose collective inconsistency. For instance, take $K = \{p, q, \neg(p \wedge q)\}$ and hard premise t , where t is a separate atom. Each pair of background formulas is consistent with t , but all three are not. The maximal repairs contain exactly two of them. A node for each formula with attacks only between inconsistent pairs would have no attack and would miss the conflict. We therefore use sets of premises as arguments.

Base arguments and attacks

For finite B , a *base argument* is a consistent $\Gamma \subseteq B$ such that no $\Gamma' \subsetneq \Gamma$ has $\text{Cn}(\Gamma') = \text{Cn}(\Gamma)$. Write $\rho(B)$ for all such arguments. Define

$$\Gamma \rightarrow_B \Delta \iff \Gamma \vdash \neg\psi \text{ for some } \psi \in \Delta, \quad F_B = (\rho(B), \rightarrow_B).$$

For role-labelled premises, closures and entailments evaluate their underlying formulas through π , while the argument retains the labels.

The requirement is irredundancy for the *whole closure*, not minimal support for a chosen conclusion. The empty set is a base argument: it is consistent and has no proper subset. These details matter when Chapter 7 compares different argument objects.

To evaluate this finite attack graph, use the following local test. A set E of its arguments is *stable* when it is conflict-free and attacks every argument outside it:

$$\nexists \Gamma, \Delta \in E (\Gamma \rightarrow_B \Delta), \quad \forall \Delta \in \rho(B) \setminus E \exists \Gamma \in E (\Gamma \rightarrow_B \Delta).$$

On any finite argument graph the same test means no internal attack and a counterattack from the set to every outsider. Dung gives the conflict-free and stable definitions [5, Defs. 5 and 13, pp. 326, 328]. Chapter 6 develops the wider semantic choices; this test is sufficient for the present representation.

The base construction above is that of Rienstra, van der Torre and Yu [11, supplied manuscript, Sec. 3.2, Defs. 3–5, pp. 10–11]. Its correspondence result states that, for $\text{MC}(B) = \text{MCS}(B, \emptyset)$,

$$\begin{aligned} S \in \text{MC}(B) &\implies \rho(S) \text{ is stable in } F_B, \\ E \text{ stable in } F_B &\implies E = \rho(S) \text{ for some } S \in \text{MC}(B). \end{aligned}$$

This is the cited Proposition 1 [11, supplied manuscript, Sec. 3.2, Prop. 1, p. 11]. It relates the whole families on the stated finite classical domain, not just a selected query in one example.

For K_0 , the six base arguments are

$$\rho(K_0) = \{\emptyset, \{p\}, \{\neg p\}, \{q\}, \{p, q\}, \{\neg p, q\}\}.$$

Put $\mathcal{P} = \{\{p\}, \{p, q\}\}$ and $\mathcal{N} = \{\{\neg p\}, \{\neg p, q\}\}$. The entire attack relation is $(\mathcal{P} \times \mathcal{N}) \cup (\mathcal{N} \times \mathcal{P})$: eight directed attacks. Neither the empty base nor $\{q\}$ attacks anything. The stable sets are

$$\begin{aligned} E_+ &= \{\emptyset, \{p\}, \{q\}, \{p, q\}\} = \rho(\{p, q\}), \\ E_- &= \{\emptyset, \{\neg p\}, \{q\}, \{\neg p, q\}\} = \rho(\{\neg p, q\}). \end{aligned}$$

Each contains no opposing pair and attacks both outsiders. For completeness, the two unattacked arguments must belong to any stable set; an omitted unattacked

argument could not be attacked. A stable set must also choose a side of the $p/\neg p$ conflict. Once it chooses one side, omitting another argument on that side would leave an outsider unattacked. Hence there are exactly these two sets. Their source bases are exactly the two MCS repairs.

The graph has made the incompatible positions visible without changing which repairs the method returns. Selecting only E_+ would require the additional preference policy, not merely the representation theorem.

1.7 Keeping hard premises through the representation

For nonempty P , a maximal consistent subset of $K \cup P$ might discard a hard premise. We must express the hard-premise policy in the readback. Let $B = K \cup P$ and retain source-labelled pairs

$$\text{Pair}_P(B) = \{(S, \rho(S)) : S \in \text{MC}(B), P \subseteq S\}.$$

Then, for finite role-disjoint K, P with consistent P ,

$$\{S \cap K : (S, \rho(S)) \in \text{Pair}_P(B)\} = \text{MCS}(K, P).$$

This is a restricted textbook corollary of the base correspondence, with a policy filter on source-labelled outputs.

Both directions explain the filter. If $S \in \text{MC}(B)$ contains P , its projection $T = S \cap K$ is consistent with P . A strictly larger compatible subset of K would, after adding P , consistently enlarge S , contradicting maximality. Conversely, if $T \in \text{MCS}(K, P)$, then $S = T \cup P$ is consistent. Any consistent proper superset within B would add a background premise and contradict the maximality of T . Thus $S \in \text{MC}(B)$, and the cited correspondence supplies $\rho(S)$. Disjoint roles make projection back to K exact.

The source set in the pair is deliberate. In the literal example one can recover S by unioning the visible base arguments. That is not a general extraction convention: redundant or tautological source premises need not be recoverable that way. For example, a tautological premise belongs to a maximal source base but contributes nothing to an irredundant base argument. Keep the source labels, filter by $P \subseteq S$, and take consequences from $S = (S \cap K) \cup P$.

Apply this to the changed two-atom input. With $B = \{f, p, \neg q\}$, the maximal consistent source bases are $\{f, p\}, \{f, \neg q\}, \{p, \neg q\}$. Only the last contains both hard premises. Its retained pair is

$$(\{p, \neg q\}, \{\emptyset, \{p\}, \{\neg q\}, \{p, \neg q\}\}).$$

Projection onto $K = \{f\}$ gives the empty repair, and taking consequences with P' gives $\text{Cn}(\{p, \neg q\})$. Thus the argumentative route represents the MCS row of the earlier table. Nothing in this reasoning proves a representation for every preferential model.

1.8 What the representation has explained

The comparison has four distinct answers. A fixed inconsistent pivot gives every classical formula. MCS retains both maximal repairs of K_0 . The pedagogical priority selects one of those repairs. Base argumentation returns stable sets corresponding to the unprioritized repairs. Its output is a set of arguments rather than a set of premises, so even exact agreement needs a declared readback.

A reasoning-alignment diagram records these routes and the equality one intends to justify. It is a specification of a comparison, not evidence that the comparison succeeds. The source method, construction, selection and extraction must all be stated. This use of alignment follows the supplied manuscript's architecture [11, Sec. 2, pp. 3–5]. A tested small case can reveal a defect; the correspondence theorem is what licenses the family claim on its full stated domain.

The positive contribution is therefore more than tolerating inconsistency. We can expose which information supports competing positions, make selection policies explicit, and ask whether a different computational representation preserves the intended answers. Chapter 7 will deepen that question for other source methods and extraction choices. First, Chapter 2 asks what changes when the reasons and judgments are distributed across agents.

1.9 Exercises

Use the chapter's finite classical setting and explicit source-labelled readback. All tasks have a paper route and complete collected solutions.

- 1.1 *[solution]* For $K_0 = \{p, \neg p, q\}$ and $P_0 = \emptyset$, enumerate every maximal consistent repair. For each give a consistency witness and a maximality certificate. Explain why your enumeration is complete.
- 1.2 *[solution]* Compute fixed-pivot, cautious-MCS and credulous-MCS status for $p, \neg p, q$. Give repair witnesses or countermodels for the failures of cautious support.
- 1.3 *[solution]* Apply $p \succ \neg p \succ q$. Identify the decisive comparison, preferred repair and its cautious conclusions among $p, \neg p, q$.
- 1.4 *[solution]* Enumerate $\rho(K_0)$, every attack, and all stable sets. Check both clauses of stability and justify completeness. Read each back to its underlying source base.
- 1.5 *[solution]* Give a three-row table comparing the default-assumption, pedagogical lexicographic and base-argumentation routes: selected objects, outputs and cautious status of p . State which two align exactly under the specified readback and which refines the repair family.
- 1.6 *[solution]* For $P = \{p, \neg p\}$ and $K = \emptyset$, compute the compatible-background

family and explain why the repair query is rejected. Contrast the result with classical consequence and its empty-model convention.

- 1.7 *[solution]* Let $P = \{p\}$ and $K = \{p \rightarrow q, q \rightarrow r, \neg r, s\}$. Enumerate every MCS, with consistency and maximality certificates, and determine cautious status of q, r, s . Give a countermodel for each negative answer. Explain why a graph with one node per background formula and attacks only for pairwise inconsistency cannot represent this repair family, even when consistency of each pair is tested with P .
- 1.8 *[solution]* Change only the priority of the common K_0, P_0 case to $\neg p \succ p \succ q$. By paper calculation or pseudocode, recompute the preferred repair and explain why the MCS family and unprioritized stable outputs stay invariant. Optional programming may implement the same tests; executing a notebook is not required.
- 1.9 *[solution]* In the four-valuation case of [One changed input, different answers](#), make every valuation rank zero while keeping V, K, P, P' and both repair constructions fixed. (a) Compute $\text{Best}(P)$, $\text{Best}(P')$ and their consequence sets. (b) Explain whether q is withdrawn and compare with the unchanged MCS outputs. (c) Prove the fixed-assumption/fixed-valuation identity and explain why it does not require the changed preferential outputs to coincide with MCS. (d) Explain why the MCS family for P' is $\{\emptyset\}$, not $\emptyset$, and give the source-labelled stable pair retained for P' , including its base arguments.

1.10 Reading routes

To develop the source-method comparison, follow Makinson’s Definitions 2.1 and 2.4, pp. 24 and 31, then the selection methods in Section 2.3, pp. 44–47 [10]. Compare these with fixed valuations, Definition 3.1 and Lemma 3.2, pp. 60–61, and preferential selection, Definition 3.8, pp. 67–68. Section 4.1, pp. 85–86, introduces the rule pivot. The purpose is to identify the selected object and the place where dependence on premises enters, not to treat every construction as the same kind of repair. KLM’s preferential and cumulative approach offers a further relation-level comparison [12]; its general metatheory is beyond the finite calculation supplied here.

The Handbook’s deductive-argument chapter [13, Vol. 1, Ch. 9, Secs. 3–4] compares support–conclusion objects with our closure-irredundant bases. Keep that difference in view when Chapter 7 asks about extraction. For that return, read the logic-based approaches chapter of Volume 2, Section 2.3.1, especially Definition 44 and Example 45, p. 749 [3]: selected bases, common written premises and common consequences need not be the same object.

Dung’s Definitions 5 and 13, pp. 326 and 328, anchor the local conflict-free and stable tests [5]; Chapter 6 supplies the broader semantic comparison. For the

particular representation used here, read the *supplied manuscript* of Rienstra, van der Torre and Yu, Section 3.2, Definitions 3–5 and Proposition 1, pp. 10–11, together with the alignment architecture in Section 2, pp. 3–5 [11]. The published article [14] is a separate continuation into the broader alignment programme; its publication pagination is not the internal pagination just cited. The textbook’s hard-premise filter remains a restricted corollary, not a new attribution to an unqualified general theorem.

For book study, reconstruct both the repair and valuation comparisons and understand what the correspondence preserves. A semester course can combine a repair calculation, the changed-ranking transfer and one short proof. An intensive course can guide the same distinctions through the displayed cases, requiring a narrower independent reconstruction. The extended source readings and all-proof mastery are different commitments; neither is needed to solve the ordinary exercises.

2 Chapter 2. Reasons across agents

A set of reasons can contain an answer to every objection even though no one participant supplies all the answers. Conversely, a participant may have a useful counterargument without being associated with the position it would defend. These differences disappear if we retain only an anonymous graph. Adding agents lets us ask how a defense is distributed, while preserving the question about which arguments actually do the defending.

There is also a different collective problem. Several participants may supply different graphs, or may reach different judgments from those graphs. How should the group combine their inputs or their answers? We will see that these two orders of combination can give incompatible results. Neither problem is solved merely by counting how many agents are associated with a claim.

2.1 Arguments and their sources

An *agent argumentation framework* has the form

$$F = (A, R, S, \text{src}),$$

where A is a finite set of arguments, $R \subseteq A \times A$ is attack, S is a set of agents or sources, and $\text{src} \subseteq A \times S$ associates arguments with sources. Write

$$\begin{aligned} A_i &= \{x \in A : \text{src}(x, i)\}, \\ S_x &= \{i \in S : \text{src}(x, i)\}, \\ \text{Att}(x) &= \{b \in A : (b, x) \in R\}. \end{aligned}$$

An argument can have no source, one source or several. This is the supplied proof's Definition 5 [15, internal p. 4], with src used for its association relation.

The association is deliberately neutral. It can record that an argument belongs to, is held by, or is associated with a participant. It does not, without additional conventions, certify that participant's expertise, entitlement to assert it, or acceptance of its conclusion. It also does not record an actual public act. Chapters 3 and 4 will distinguish what a speaker has offered, questioned or undertaken to defend. Here we first examine what the stated association allows a source-sensitive test to count.

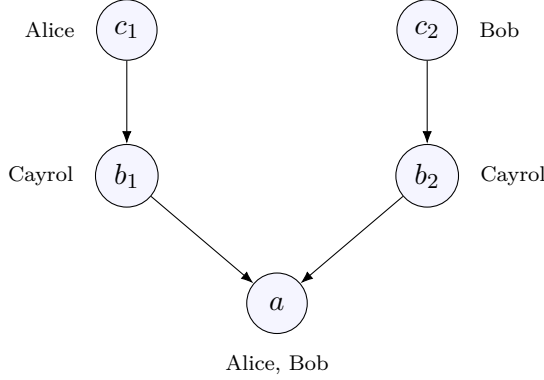
Consider five arguments and three agents:

$$\begin{aligned} A &= \{a, b_1, b_2, c_1, c_2\}, \\ R &= \{(b_1, a), (b_2, a), (c_1, b_1), (c_2, b_2)\}, \\ S &= \{\text{Alice}, \text{Bob}, \text{Cayrol}\}, \quad E = \{c_1, c_2\}. \end{aligned}$$

Let the complete source relation be

$$\begin{aligned} \text{src} &= \{(a, \text{Alice}), (a, \text{Bob}), (b_1, \text{Cayrol}), (b_2, \text{Cayrol}), \\ &\quad (c_1, \text{Alice}), (c_2, \text{Bob})\}. \end{aligned}$$

All unlisted attacks and associations are absent. This is the source’s collective-defense example with named agents [15, Example 5, supplied-proof internal p. 6, Fig. 3, p. 7].



The target a has two attackers. The explicit set E supplies c_1 against b_1 and c_2 against b_2 . Alice is associated with a, c_1 , while Bob is associated with a, c_2 . Together they cover the two attacks. Neither alone supplies both answers. Cayrol is associated with the attackers, but is not a source of the target or its displayed defenders. We can now state exactly which of these facts each defense test uses.

2.2 One defending set, different source conditions

Ordinary defense asks whether an explicit set $E \subseteq A$ counterattacks every attacker of x :

$$\text{OrdDef}_F(E, x) \iff \forall b \in \text{Att}(x) \exists d \in E (d, b) \in R.$$

This is Dung’s acceptability with respect to a set [5, Def. 6(1), p. 326], written here as a defense predicate. It does not require E to be conflict-free or assert that x has been selected by an acceptance semantics. For our $E = \{c_1, c_2\}$, ordinary defense of a succeeds.

To inspect a particular agent, define

$$\text{IndDef}_F(i, E, x) \iff i \in S_x \wedge \forall b \in \text{Att}(x) \exists d \in E \cap A_i (d, b) \in R.$$

This fixed-agent notation requires the agent to be associated with the target and to supply, within E , an answer to every attack. The canonical individual predicate asks whether such an agent exists:

$$\text{IndDef}_F^\exists(E, x) \iff \exists i \in S_x \text{IndDef}_F(i, E, x).$$

By contrast, collective agent defense permits a different target-associated agent for each attacker:

$$\text{CollDef}_F(E, x) \iff \forall b \in \text{Att}(x) \exists i \in S_x \exists d \in E \cap A_i (d, b) \in R.$$

These are the individual and collective tests of Definition 6 [15, supplied-proof internal p. 5]. The source’s collective predicate has no coalition parameter.

The quantifier order captures the difference seen in the graph.

attacker of a	answer in E	target-associated source	missing for
b_1	c_1	Alice	Bob
b_2	c_2	Bob	Alice

Ordinary and collective defense hold, while the existential individual predicate fails. The relevant agent may vary with the attacker under the collective test; the individual test requires one agent to handle the whole task. The set E is a set of arguments, not a set of agents.

The restriction to E matters independently of source association. Replacing E by $\{c_1\}$ leaves b_2 unanswered, even though c_2 still exists in the framework. The source restriction matters independently of ordinary defense: if only Alice is associated with a target and only Bob with its available counterargument, ordinary defense can succeed while collective agent defense fails. These tests make different information relevant; none is a disguised vote count.

2.3 Derived coalition certificates

When collective defense succeeds, we may want to report which agents suffice. A *coalition witness* is a subset $C \subseteq S_x$ that covers every attacker through the available arguments of its members:

$$\text{CWit}_F(C, E, x) \iff C \subseteq S_x \wedge \forall b \in \text{Att}(x) \exists i \in C \exists d \in E \cap A_i (d, b) \in R.$$

It is inclusion-minimal if it is a witness and no proper subset is one:

$$\text{MinCWit}_F(C, E, x) \iff \text{CWit}_F(C, E, x) \wedge \nexists D \subsetneq C \text{ CWit}_F(D, E, x).$$

These are derived textbook certificates for the canonical collective predicate, not additional clauses of the source’s Definition 6.

In the running case the pair {Alice, Bob} is the unique minimal witness. It covers both attackers; deleting Alice loses the answer to b_1 , and deleting Bob loses the answer to b_2 . Adding an irrelevant target-associated agent would preserve coverage without making the larger coalition minimal. A minimal coalition need not have the smallest cardinality among all witnesses: a singleton and an incomparable pair can both be minimal. The supplied [coalition study](#) develops enumeration and supplies complete inputs for the transfer exercises.

Empty cases reveal what these certificates mean. An attacked target with no associated source cannot be collectively defended: one of its attackers would require a member of the empty source set. For an *unattacked* unsourced target, however,

collective defense is true because there is no attacker to answer. The empty coalition is then the unique minimal witness. Individual defense remains false, since it asks for an agent to exist. Requiring a nonempty responsible group would be a different convention. The empty certificate here records an empty coverage task; it does not identify a person responsible for a public claim.

2.4 Forming a collective graph or a collective answer

So far there has been one graph with a source relation. Now suppose three participants report different attack relations on the common arguments $\{a, b, c\}$:

$$\begin{array}{l|l} F_1 & a \rightarrow b, b \rightarrow c \\ F_2 & b \rightarrow c, c \rightarrow a \\ F_3 & c \rightarrow a, a \rightarrow b \end{array}$$

There are no other attacks. This profile is the Handbook’s graph-majority example [16, Vol. 2, Ch. 4, Sec. 3, Def. 3.8 and example, pp. 261–262]. Each reported graph is a directed path and hence acyclic. Include an edge in the collective graph exactly when at least two of the three participants report it. Every displayed edge has two supporters, so the collective graph is

$$F_{\text{maj}} : \quad a \rightarrow b, b \rightarrow c, c \rightarrow a.$$

Acyclicity has been lost. The counterexample concerns this rule and this property; it does not establish an impossibility result for every collective method.

We can compare the order of aggregation and evaluation without introducing a new selection rule. Use Chapter 1’s stable test: a set must be conflict-free and attack every outsider. In F_1 , a has no attacker, so a stable set must include it; otherwise the outsider a could not be attacked. Conflict-freeness excludes b . The set must then include c , because its only possible attacker b is excluded. The resulting $\{a, c\}$ is conflict-free and attacks its sole outsider b . Rotating the names gives

$$E_1 = \{a, c\}, \quad E_2 = \{a, b\}, \quad E_3 = \{b, c\}$$

as the unique stable sets of the three inputs.

The majority cycle has no stable set. The empty set attacks no outsider. A singleton attacks only one of the two outsiders, and every two-element set contains an attack. The full set also contains attacks. These cases exhaust all subsets. Thus its stable-extension family is $\emptyset$.

Instead evaluate each input first and accept an argument into the collective candidate if it belongs to at least two of E_1, E_2, E_3 . Each argument occurs twice, so the candidate is $\{a, b, c\}$. It is not conflict-free in the majority graph. The first order produced a graph with no stable candidate; the second produced a particular candidate that fails the same selection test. The output types are explicit, and there is no common stable answer to read back.

This comparison puts a substantive question to any group method: what is combined, and which property should the combination preserve? Here the inputs were attack relations and the alternative inputs were already evaluated sets. Neither operation was source-indexed defense in a fixed graph. The supplied [grounded study](#) works the same four graphs under a different, fully defined evaluation procedure. Chapter 6 later explains its general semantic basis.

2.5 Opinions, acceptance and public acts

Participants can also report status ballots. Let $\beta_i(x) \in \{\text{in}, \text{out}, \text{undec}\}$ be agent i 's reported status for argument x , without assuming that the reports satisfy any argumentation semantics. For a finite voter set define only

$$\text{MajIn}(x) \iff |\{i : \beta_i(x) = \text{in}\}| > |S|/2.$$

This Boolean indicator says whether a strict majority marked x in. Failure does not decide between out and undecided. Exercise [2.9](#) retains a small unconstrained profile to make that limitation calculable. Legal individual labellings and rules for aggregating them require additional conditions; merely using the three familiar status words does not impose those conditions.

We now have three positive uses of agent information. Source-sensitive defense identifies who can contribute to answering each attack under the stated association. Coalition certificates explain which sources suffice. Aggregation compares ways to form collective inputs or answers from individual ones. Each use supplies a different kind of accountability. None tells us, by itself, what a participant actually asserted or what obligation a challenge created. Chapter 3 turns to that prior work of reconstructing an argumentative act; Chapter 9 will return to its public consequences with exact interaction rules.

2.6 Exercises

Unless changed in a prompt, use the five-node case in [Arguments and their sources](#) with $E = \{c_1, c_2\}$. All tasks below are ordinary internal exercises. The supplied studies and collected solutions require no programming.

- 2.1 [\[solution\]](#) List $A, R, S, \text{src}, \text{Att}(a)$ and S_a , keeping argument and agent sets distinct.
- 2.2 [\[solution\]](#) Certify ordinary defense of a by giving one counterattack from E per attacker.
- 2.3 [\[solution\]](#) Audit fixed-agent individual defense for Alice and Bob. Give a failed attacker for each and evaluate the canonical existential individual predicate.
- 2.4 [\[solution\]](#) Certify collective agent defense, identifying the target-associated agent and the argument used against each attacker.

- 2.5 *[solution]* Prove that $\{\text{Alice}, \text{Bob}\}$ is the unique inclusion-minimal witness. Add Dana as an agent and add only (a, Dana) to the source relation, leaving all attacks and other pairs unchanged. Show that the triple is a witness but not minimal, and determine whether any other minimal witness is created.
- 2.6 *[solution]* Replace E by $\{c_1\}$. Evaluate ordinary, individual and collective defense of a and explain why the existing $c_2 \in A \setminus E$ cannot be used.
- 2.7 *[solution]* Compare two fully specified cases. In the first, $A = \{x, b, d\}$, $R = \{(b, x), (d, b)\}$, $S = \{\text{Alice}\}$, $\text{src} = \{(d, \text{Alice})\}$ and $E = \{d\}$. In the second, $A = \{x\}$, $R = \emptyset$, $S = \emptyset$, $\text{src} = \emptyset$ and $E = \emptyset$. All unlisted pairs are absent. For target x , calculate individual defense, collective defense and every inclusion-minimal witness in each case.
- 2.8 *[solution]* Use the complete Transfer and Co-owned-defender inputs in the [coalition study's complete inputs](#). In *each* case compute every inclusion-minimal witness. Supply attacker coverage and a proper-subset failure certificate for each, and explain why no other minimal witnesses exist. Why is minimum cardinality a different request?
- 2.9 *[solution]* For this unconstrained ballot profile, compute MajIn for each argument:

argument	Alice	Bob	Cayrol
a	in	in	out
b_1	out	undec	in
b_2	undec	out	in
c_1	in	undec	out
c_2	undec	in	out

Explain why the indicator does not by itself specify a three-valued collective labelling or establish semantic legality.

- 2.10 *[solution]* Let $A = \{x, b, d\}$, $R = \{(b, x), (d, b)\}$, $S = \{\text{Alice}, \text{Bob}\}$, $\text{src} = \{(x, \text{Alice}), (d, \text{Bob})\}$, $E = \{d\}$, and target x , with every unlisted pair absent. Compute ordinary, individual and collective defense and all minimal witnesses. Add only (x, Bob) to src and recompute. Use paper calculation or the study's pseudocode; optional programming is an alternative, not a prerequisite.
- 2.11 *[solution]* For the three graphs in [Forming a collective graph or a collective answer](#), (a) count support for every possible directed edge, including loops, and obtain the majority graph; (b) verify the three unique stable outputs and rule out all eight candidate subsets for stability in the majority graph; (c) compute membership majority over those outputs and identify an internal attack; (d) state exactly which preservation failure has been demonstrated and why it says nothing by itself about which agents collectively defend a target in an AAF.

2.7 Reading routes

Start the source comparison with Dung’s defense by an explicit set [5, Def. 6(1), p. 326], then read the supplied proof’s agent framework and individual/collective tests [15, Defs. 5–6, internal pp. 4–5, Ex. 5, p. 6]. Ask which witness can depend on the attacker and why the target-source restriction matters. Full agent semantics and principle analysis extend this defense calculation; they are not hidden prerequisites to it. These locators identify the supplied 52-page proof. The published 2026 article is a separate reading [17]; its pagination is not interchangeable with the internal proof pages.

For collective choice, the Handbook’s *Collective Acceptability in Abstract Argumentation* [16, Vol. 2, Ch. 4, Secs. 2–3, especially pp. 261–262] is a route through the choice of aggregation object and the conditions a group rule might preserve. Caminada and Pigozzi’s judgment-aggregation approach [18] continues the question for constrained judgments. Our unconstrained ballots do not implement that theory. The introduction to *Social Argumentation Systems* [19, Vol. 3, Ch. 1, starting p. 15] shows why social input can matter to a system while requiring an explicit interpretation. Source association should not silently become a trust score, an entitlement or a social strength.

The book route retains both the agency question and the aggregation comparison, with the supplied studies available for complete independent reconstruction. A semester course can require one coverage calculation, a minimality transfer and the graph-majority argument. An intensive route can guide the same distinctions through the worked cases while selecting less enumeration. Later report acceptance and trust in Chapter 14 concern an additional question; no such calculus is presupposed here.

3 Chapter 3. Reconstructing arguments

Before evaluating an argument, we must decide what argument has been made. A paragraph may state a reason, concede a consideration, imply a warrant and invite a reply without marking these functions separately. Even an accurate transcription leaves work for the analyst: what is the issue, which conclusion is proposed, and what is each consideration meant to do? These choices determine the questions a formal representation can answer.

The diversity of reasoning methods encountered so far therefore raises a prior question. What should we preserve when we turn an exchange of reasons into an object on which a method can operate? Reconstruction makes that choice accountable. It records what someone is doing with a claim, which reasons are offered and what they address. A formal object may retain only some of this information. Reconstruction need not discover a unique structure hidden in the words; it must make its additions and alternatives available for criticism.

3.1 What the text says, and what the analyst supplies

Consider this fictional passage:

Governments should ban indoor public smoking because smoking harms bystanders. Although adults have liberty interests, liberty does not protect activities that impose serious risks on others.

The proposal, its reason, the concession and the limiting consideration have different roles. We normalize their contents as a, b, c, d , respectively. For a larger debate, add three further statements e, f, g :

Claim	Content	Provenance
a	Governments should ban indoor public smoking.	Paragraph
b	Smoking causes serious harm to bystanders.	Paragraph
c	Adults have a liberty interest in choosing whether to smoke.	Paragraph
d	Liberty is limited when an activity harms others.	Paragraph
e	Public smoking bans reduce hospital admissions.	Added debate material
f	Some small businesses may lose revenue.	Added debate material
g	Revenue loss can be mitigated by transition support.	Added debate material

These are stipulated claims in a modelling example. In particular, the paragraph contains neither e nor expert testimony supporting it. A system that attributes such testimony to the speaker has changed the evidence, not merely expressed it more formally.

A *text occurrence* is a particular use of words; a *claim* is the content assigned to it. Different occurrences can express the same claim. An *argument unit* (P, q) records premises P offered for a conclusion q . Here $(\{b\}, a)$ records one offered reason, but does not explain why harm supports a ban. That inferential connection is a warrant which reconstruction must make explicit if the task requires it. The claim c , meanwhile, is conceded rather than simply denied. Reading d as a response to the *use* of liberty against the proposal differs from reading it as evidence that liberty interests do not exist.

To reconstruct an exchange, we must also identify the acts performed with these contents. Suppose, as a constructed continuation of the paragraph, that its speaker asserts a and an interlocutor asks, “Why should there be a ban?” The assertion presents a as a position the speaker sponsors; the question requests grounds without asserting $\neg a$. Answering “because b ” offers a justification. The interlocutor can instead take an opposing position: “The ban restricts the liberty described in c .” Now there is an offered consideration against the proposal. The original reply d can answer that use of c , while continuing to concede the liberty interest. A record of contents alone would miss the difference between asking for a reason and providing one against the claim.

Withdrawal is useful for the same reason. If the speaker adds e but cannot substantiate it when asked, she may withdraw her endorsement of e . This corrects what she publicly sponsors; it does not assert $\neg e$, erase the earlier utterance or withdraw the distinct reason b . Whether a particular procedure permits withdrawal at that moment is a further question. For reconstruction, the immediate task is to record the speaker, act and target accurately, keeping this invented continuation separate from the original paragraph.

An annotated graph can retain these relations without yet giving a full logic of arguments. In this chapter $G = (V, E^+, E^-)$ has claim-labelled vertices, support edges E^+ and attack edges E^- . More detailed annotations may distinguish replies, undercuts or relevance links. A reply relation alone does not determine any of those argumentative relations. Nor is a claim-labelled vertex already the recursively constructed argument introduced in Chapter 5. Here the graph records an interpretation of the exchange; later we will ask how premises and rules generate arguments.

3.2 Schemes expose the warrant

Argumentation schemes identify recurring defeasible patterns and the questions on which their application depends. They help us see what a reconstruction has supplied, rather than replacing that judgment with a label. Compare two patterns relevant to the smoking debate.

An expert-opinion reconstruction of e would require a source E , a domain D ,

expertise in that domain, a claim within it and testimony:

$$\frac{\text{Expert}(E, D) \quad \text{InDomain}(e, D) \quad \text{Says}(E, e)}{\text{Plausible}(e)}.$$

The pattern supports a defeasible assessment of a proposition. A practical reconstruction instead supplies an actor’s goal H , reducing serious harm to bystanders, and an action B , introducing the ban. Its basic pattern is

$$\text{Goal}(\text{actor}, H), \text{Means}(B, H) \Rightarrow \text{PracticallyOught}(\text{actor}, B).$$

The second output is represented by the recommendation a . The arrow expresses a defeasible scheme, not material implication. The expert and practical patterns ask different questions: whether to credit a statement, and what there is reason to do. Both require premises absent from the short paragraph. The expert premises are *linked*: testimony, relevant expertise and domain fit work together in this instance. Removing one breaks that route to its conclusion. Independent arguments can instead *converge* on a conclusion: once their warrants are supplied, harm to bystanders and a reduction in admissions can offer separate routes to the ban. A list of supporting claims does not distinguish these structures.

In the practical scheme, the goal is a state the actor seeks to bring about, reduced harm; the means claim says that introducing the ban promotes it. Public health is a value invoked to explain why that goal matters. Liberty supplies a competing value, and f identifies a possible adverse consequence for another interest. A goal does not establish that the proposed means works, and a valued consequence does not establish that this is the only feasible way to obtain it. An interlocutor might propose designated smoking areas as an alternative, then ask whether they would protect bystanders. That proposal supplies neither feasibility nor an answer about effectiveness. These questions examine different parts of the practical warrant: ends, means, alternatives and consequences [20, Sec. 4.3, p. 35] [21, Sec. 6, Table 6 and critical questions, pp. 547–548].

Critical questions make this scrutiny systematic, but they have different functions. Asking what E actually asserted requests support for a premise. Under our local convention, an unanswered request leaves the testimony premise unsubstantiated and the request open. It supplies neither $\neg e$ nor an attack edge by itself. A protocol may make the unmet burden consequential, but that is an additional rule about the use of an argument. Asking about adverse consequences of B , by contrast, seeks a competing reason. The question alone supplies no evidence of cost. The answer f provides such a consideration, and g provides a response to it. Whether that response suffices is another question. Different critical questions therefore need not have the same burden or defeat effect [21, p. 558].

Neither scheme settles the whole debate. Crediting e does not itself recommend the ban; instantiating the practical pattern does not compare every alternative or establish that the means premise is true. Conversely, failure to substantiate a

premise does not establish its contrary. The countermodels in Exercises 3.9 and 3.10 separate these possibilities without turning a request into an inference.

3.3 Changing the target changes the answer

Fix one reconstruction, R_0 , of the larger debate:

$$E_0^+ = \{(b, a), (e, a), (g, a)\}, \quad E_0^- = \{(c, a), (f, a), (d, c), (g, f)\}.$$

The edge (d, c) represents a response to the liberty objection. This coarse attack does not assert that adults lack liberty interests. A second reconstruction, R_1 , removes that attack and adds (d, a) as support: it uses the limiting norm directly for the proposal, leaving the liberty objection without its former recorded response. All other edges and all claim contents stay fixed. Neither choice is uniquely forced by the text.



Figure 1: One changed argumentative target. Solid arrows are support; dashed arrows are attack. Only the affected part of each reconstruction is shown. The words expressing d have not changed.

To make the consequence precise, consider three deliberately modest queries. Here $\text{Supporters}_2(x)$ returns vertices from which one or two consecutive support edges lead to x . Define also

$$\text{Attackers}(x) = \{y : (y, x) \in E^-\},$$

$$\text{Defenders}(x) = \{z : \exists y ((z, y) \in E^- \wedge (y, x) \in E^-)\}.$$

The last name denotes a local attack–attack query. It is not the set-based defense or acceptance test of Chapter 6.

Query	R_0	R_1
$\text{Supporters}_2(a)$	$\{b, e, g\}$	$\{b, d, e, g\}$
$\text{Attackers}(a)$	$\{c, f\}$	$\{c, f\}$
$\text{Defenders}(a)$	$\{d, g\}$	$\{g\}$

Every support walk ending at a here has length one. In R_0 , the attack–attack witnesses are $d \rightarrow c \rightarrow a$ and $g \rightarrow f \rightarrow a$; in R_1 , only the second remains. The positive-reason query gains d while the response-to-objection query loses it. A different representation of the same consideration has produced a different answer. No accepted conclusion or final recommendation follows from these query results.

3.4 Correct calculation and the wrong requirement

Suppose a policy analyst specifies the following requirement independently of the query implementation: use the agreed R_0 reconstruction, return its positive reasons and objections, and expose both recorded responses to those objections. Now supply a predicted graph P which is exactly R_0 , except that (d, c) is labelled support instead of attack. The prediction is fictional; no trained system is being evaluated.

A correct implementation on P returns $\text{Defenders}_P(a) = \{g\}$. It cannot return d as a defender, because the required attack edge is absent. Its answer is correct for its input, yet fails the independently specified requirement: the response to the liberty objection has disappeared. The support query still returns $\{b, e, g\}$; the mixed support–attack route through c is not a support walk. The fault is not repaired by checking the query code again. One must revisit the representation supplied to it.

A moderator may have a different requirement: distinguish what participants said from added context and supplied warrants, and keep unanswered requests separate from contrary assertions. Attributing $\text{Says}(E, e)$ to the paragraph fails that requirement even if every graph computation is exact. These two examples separate verification against a formal specification from assessment against stakeholder requirements [22, pp. 149–152]. Our requirements are stipulated for the examples; we have not established that real users would find the resulting interface useful.

This distinction also limits what we can conclude about generated arguments. A transparency convention may require each factual claim to be marked as grounded in supplied evidence, referenced or unsupported. An explicit unsupported label meets that requirement. Thus transparency about support, the quality of an argument and the truth of its claims are three different judgments. A reference can be irrelevant; a coherent argument can start from a false premise; an honest admission of missing evidence can make a reconstruction transparent without making its conclusion well supported. Exercise 3.7 formalizes the transparency condition; Exercise 3.6 explores a local quality rubric; Supplement S3.1 explores how an extraction score can conceal a consequential query failure.

Reconstruction has made our choices inspectable, not completed the work of evaluation. We can now ask what information and output a proposed use of argumentation requires. Chapter 4 compares three such questions before the book develops inference in detail.

3.5 Exercises

Complete solutions are collected at the end of the book. Exercises marked with an asterisk use the optional study or extend the main development.

- 3.1.** *[solution]* [Reconstruction] Segment the smoking paragraph into its proposal, reason, concession and response. Give one argument unit and distinguish it from its premise’s text occurrence. Which of e, f, g , the expert testimony and the

practical goal can be attributed to this paragraph? Identify each supplied addition. In the constructed exchange, how do assertion, question, opposition and answer differ when they concern a ? What does withdrawing e change?

- 3.2. *[solution]* [Comparison] Describe the kind of reasoning in each pattern in your own words and state the distinctive warrant it needs; standard scheme names are optional: (a) an expert in the relevant domain reports p ; (b) a symptom regularly associated with a condition is observed; (c) an action has a harmful consequence; (d) a new case resembles a prior case in relevant respects; (e) an action is a means to an agent's goal. For each, give a possible challenge to the warrant. Does a scheme name itself answer that challenge?
- 3.3. *[solution]* [Diagnosis] Give three critical questions for the expert instance and three for the practical instance. Classify each as a request for a premise or warrant, or a search for a competing reason. Explain what an unanswered question adds to the record and what it does not prove.
- 3.4. *[solution]* [Modelling] Construct the complete R_0 graph. Defend every edge's type and direction. Then construct R_1 . Explain to a moderator why two defensible reconstructions can serve different queries, and to an implementer what version information must accompany the graph. Why is a surface reply order insufficient?
- 3.5. *[solution]* [Calculation and computation] Recompute all three queries for R_0, R_1, P , giving witnesses. Specify a procedure for the following general contract: for a nonnegative integer k , $\text{Supporters}_k(x)$ returns vertices with an all-support walk of length 1 to k ending at x . Walks may repeat vertices; length zero is excluded, so $\text{Supporters}_0(x) = \emptyset$. Return witness walks and handle $k = 0$. As a boundary case, take support edges $x \rightarrow y, y \rightarrow x$ and no attacks: compute $\text{Supporters}_2(x)$. Explain why excluding repeated vertices would change the contract.
- 3.6. *[solution]* [Evaluation convention] Use the explicitly local rubric $Q(u) = (\text{Rel} + \text{Evid} + CQ + Coh + Dial)/10$, each component in $\{0, 1, 2\}$: relevance, evidential support, coverage of salient critical questions, coherence and engagement with counterarguments. A fictional generated answer supports the ban, acknowledges liberty, provides no evidence and ignores cost and alternatives. For this task stipulate component scores $(2, 0, 1, 2, 0)$. Compute Q , then change only evidence to 2. Next defend or contest at least two component assignments by citing specific features of the supplied answer; state a revised vector and its score if you contest one. Give a paper algorithm or pseudocode for the sum; a program is optional. Explain why these numbers and the resulting score are neither a validated quality instrument nor a truth probability. What justification would scoring a new answer require?
- 3.7. *[solution]* [Generation] Let $\text{Claims}(u)$ be the factual claims of generated text u .

Use the transparency condition

$$\forall q \in \text{Claims}(u), \quad \text{Grounded}(q) \vee \text{Referenced}(q) \vee \text{UnsupportedLabel}(q).$$

A fictional generated answer says, “The ban reduced admissions by forty percent, so it is beyond reasonable objection.” No source or data are supplied. Identify the unsupported factual content and rewrite the answer with an explicit support-status marker. Does your revision meet the transparency condition? Does it thereby establish the number or answer all objections?

- 3.8. *[solution]* [Computation] Specify `open_critical_questions(unit)` by a paper algorithm or pseudocode; a program is optional. The input is a finite dictionary `unit["critical_questions"]` mapping identifiers to "open" or "answered". Return exactly the open identifiers. Check empty and mixed dictionaries. State why this is a query over recorded status, not a test of answer adequacy.
- 3.9. *[solution]* [Countermodel] Let $u = \text{Says}(E, e)$. A moderator requests its source and receives no answer. Describe the record. A changed fictional theory supplies r and the strict rule $r \rightarrow \neg u$. What contrary now follows? Give a model showing that $\neg e$ still need not follow.
- 3.10. *[solution]* [Proof and diagnosis] Compare an unanswered side-effects question; f with a defeasible scheme $f \Rightarrow \neg a$; and a changed strict theory containing f and $f \rightarrow \neg a$. Which supplies a request, a counterargument, or a strict proof? Does the second settle the debate when g is present? Give a countermodel to the claim that f alone strictly entails $\neg a$.
- 3.11. *[solution]* ^{*}[Extraction study] Using Supplement S3.1, calculate relation error, accuracy and recall of the defender answer. Explain why unlisted pairs cannot be added to the denominator and why R_1 's defensibility does not make P correct against the declared R_0 reference.
- 3.12. *[solution]* [Literature investigation] Choose a foundational scheme source, a Handbook discussion and one mining, quality or question-generation source from the reading routes. Compare their objects, their evaluation targets and the evidence needed for their claims. Use the smoking example to explain one question which none of three graph queries answers. Separate a mathematical contract from an annotation judgment and an empirical claim.

3.6 Further reading

From ordinary reasoning to reconstructed objects. Begin with the distinctions among logical, dialectical and rhetorical purposes in [20, Sec. 2.2, pp. 10–11, Sec. 4.3, pp. 34–36], then read the scheme taxonomy and critical-question discussion in [21, Secs. 1–2, pp. 519–521, Sec. 6, pp. 547–548, pp. 558–560]. Toulmin's claim, data, warrant and backing distinguish what an inference uses from what authorizes

it [23]; Walton, Reed and Macagno provide a much wider repertoire of defeasible schemes [24]. Perelman and Olbrechts-Tyteca’s audience-oriented account [25] and pragma-dialectical critical discussion, introduced in the first Handbook route, change what is being evaluated. They should be read as alternative accounts of argumentative practice, rather than as incomplete versions of an attack graph.

Extraction, querying and quality. The Handbook’s treatment of natural-language argumentation separates annotation choices from computational tasks [26, pp. 577–578, 584–585]. Lawrence and Reed’s survey organizes the interacting problems of argument mining [27, Secs. 3–5]; the earlier survey [28] supplies another entry. The Argument Interchange Format [29] and the AIF/ArgQL material in [30, pp. 70–91] ask how represented arguments can be exchanged and searched. Our three queries are a small local interface, not an implementation of those languages. For quality, Wachsmuth and colleagues distinguish cogency, effectiveness and reasonableness and study their annotation [31, Secs. 3–4, pp. 181–184]. Compare their targets with Exercise 3.6’s toy sum: what is measured, for whom, and against which evidence?

Generated questions and evidence about use. The 2025 critical-questions shared task gives a concrete route into question generation and task-specific evaluation [32, Secs. 2–3 and 5, pp. 243–247]. A useful-question judgment or a match to a reference question establishes neither the truth of an answer nor fulfillment of a dialogue obligation. The Handbook’s cognitive-study comparison asks a different question: were participants reconstructing relations, or judging an already supplied graph [3, Ch. 13, pp. 860–865]? Chapter 11 takes up that distinction. The argumentation-and-machine-learning account [33, pp. 597, 601–608] leads to Chapter 13’s optional AXPLR study. Mathematical prediction correspondence, fidelity on a finite test set and human usefulness require different evidence. They are not interchangeable endorsements of a reconstructed argument.

4 Chapter 4. Choosing the question

A request to “analyse the arguments” leaves much undecided. We might be asked which conclusions the available reasons support, how participants may challenge one another, or which actions the reasons permit. Chapter 3 showed that the same content can perform different acts and address different targets. We now use one case to compare three questions that such a reconstruction makes possible.

Inference examines consequences and conflicts within supplied information. Dialogue examines what participants publicly undertake and what responses are called for. Practical evaluation examines reasons for choosing among alternatives. A question asked in dialogue may expose a missing inferential premise; comparing options may reveal a consequence that needs investigation. These connections are useful precisely because the questions and their answers remain distinguishable.

4.1 The custody case and its inferential conflict

Take a fictional custody dispute, adapted from an illustrative source case [34, Sec. 2.1, Examples 1–2]. The following facts and the civil-code premise are stipulations, not the law of any jurisdiction:

w_1	The child wants to live with the mother.
w_2	The child understands what she wants.
w_3	The mother is less wealthy than the father.
w_4	The father is often away for work.
w_5	The civil code says that the child’s opinion must be considered.

Let m be the recommendation that the child should live with the mother. Distinguish this proposition from the option M , making that arrangement, and the alternative option F , living with the father. We may ask whether reasons support m , whether M is permissible, or which arrangement is to be selected and carried out. A later best-interest inquiry can inform these questions without being identical to any of them.

An inferential reconstruction supplies schematic defeasible rules:

$$\begin{aligned} r_1 : w_1, w_2 &\Rightarrow m, & r_2 : w_3 &\Rightarrow \neg m, \\ r_3 : w_4 &\Rightarrow m, & r_4 : w_5, w_1 &\Rightarrow \text{legallyRelevantPreference}. \end{aligned}$$

A defeasible arrow records an inference that can be challenged; it does not make the conditional exceptionless. Let A_1 use r_1 , and A_2 use r_2 . Their contrary conclusions generate attacks in both directions. For this pair stipulate the strict priority $A_1 \succ A_2$, and let an attack from A to B succeed as a defeat exactly when $B \not\succ A$. Then A_1 defeats A_2 , while the reverse attack fails.

The priority resolves this local conflict. Evaluating acceptance across the resulting arguments requires a further rule, developed in Chapter 6 after Chapter 5 makes construction and attack targets precise. Even here, r_4 has a different conclusion from r_1 : establishing that the child’s preference is relevant does not establish that it decides the recommendation. The supplied rules make that gap visible.

4.2 Public sponsorship and responsibility

Now ask who has undertaken to defend what. For the following fragment, use limited local conventions. Asserting a claim publicly sponsors it and makes its speaker responsible for grounds. A question requests grounds without taking the opposing position; a challenge also undertakes opposition, whose grounds can in turn be requested. A requested speaker answers the same target before the exchange proceeds: support for her assertion, or opposing reasons for her challenge. Supplying that answer closes the particular request and leaves responsibility for the offered support. It does not certify that support’s truth or adequacy. These conventions make a small exchange intelligible; Chapter 9 will specify exact moves, targets and state changes.

Time	Public act	Resulting responsibility
t_0	The mother asserts m .	She sponsors m and is responsible for grounds.
t_1	The father challenges m .	She must answer; he undertakes opposition.
t_2	The mother offers w_1, w_2 in support of m .	The request is answered; she is responsible for the offered pair.
t_3	The mother asks for the grounds of his challenge.	He must give his opposing reason.
t_4	The father offers w_3 against m .	That request is answered; he is responsible for his offered ground.

The child’s wish and understanding form one offered unit, just as the expert premises were linked in Chapter 3. The father has first challenged and then supplied a consideration; the mother’s intervening question supplied no additional evidence for m . At this point both requests have been answered, although the disagreement remains. Repeating the inferential priority calculation would not tell us these facts about the exchange.

An alternative continuation after t_4 illustrates withdrawal. With no reply waiting, let the mother withdraw her assertion of m . Under the local convention, she ceases to sponsor that assertion but retains her separately offered w_1, w_2 . The father’s reason and the fact that he challenged remain recorded. Withdrawing the recommendation need not mean that the child has changed her wish or lost understanding. Public

commitment records what a participant has undertaken in the exchange; it does not establish a private belief.

The distinction between requests also matters. Returning to the record at t_4 , suppose the judge asks at t_5 why the child's wish is legally relevant. The supplied w_5 and r_4 provide the material for an answer. The fragment above does not yet specify a right to demand the relevance of an individual premise, or what would satisfy that demand. The question is intelligible before a complete protocol is available. To certify this further move, the model must add its target and response conditions.

A public responsibility and an evidential burden need not be discharged together. Suppose the mother bears the burden of substantiating the child's understanding if it is disputed, and the father bears the burden of substantiating the claimed financial disadvantage. This *allocation* says who must do the work. A *standard* says what showing suffices. Producing some relevant supporting material can satisfy a demand to bring an issue into consideration, while establishing that issue may require answering specified objections. Thus a mother who supplies evidence for w_2 may have answered a request without yet establishing understanding to the deciding body's standard. Chapter 10 will make premise status, allocated burden and issue evaluation precise in this same case.

Purpose affects which open question is worth pursuing. In an inquiry into understanding, asking for evidence about w_2 advances the shared aim of finding out. In persuasion, a participant seeks to defend a position before an interlocutor and may ask for the grounds of the opposing case. In deliberation, the participants seek what to do; comparing future care under M and F may be more useful than repeating an undisputed fact. The same answer can contribute to more than one purpose. A permitted reply is therefore not automatically the best next move for a particular aim [35, Secs. 3–4].

4.3 Reasons for an arrangement

For practical comparison, stipulate that M and F are the feasible alternatives under consideration and that the institution's criterion is the child's welfare. A goal such as providing stable daily care describes a desired outcome. A value, such as respect for the child's developing autonomy, explains why an outcome or way of deciding matters. The reported wish w_1 is neither that institutional criterion nor a complete policy for choosing: w_5 requires considering it, not treating it as decisive. An individual participant's objective, such as securing her proposed arrangement, can differ from the institution's purpose.

The same facts now enter comparative reasons. The child's preference favours M over F ; the father's greater resources can favour F over M ; his absence raises a concern about care under F . Each use needs a connection to the purpose. Greater wealth matters through the benefits it could provide, and absence matters through

its effects on care, not simply because one fact has a favourable and the other an unfavourable word in it. As in the smoking case, an empirical claim about consequences and an evaluation of those consequences do different work. Their relevance and comparative force can be challenged separately.

A reason can justify an option without requiring it in preference to its rival. Suppose the father's extra resources would provide worthwhile benefits, while resources with the mother would still be adequate. The advantage can make F reasonable to choose without making M unreasonable. By contrast, an unmet essential care need would press strongly against choosing the arrangement that leaves it unmet: choosing that option would require sufficient countervailing justification. We distinguish the force that supports permission for an option from the force that calls for rejecting its rival unless the rival has enough justification. They need not increase together. This is the elementary distinction later represented by justifying and requiring weights.

For a first qualitative assessment, stipulate that the child's preference and available care together justify M despite the resource advantage of F , while that advantage alone does not justify F against the other considerations. This permits M alone among the two alternatives. If the resource advantage is instead judged sufficient to justify F , without increasing its demand against M , both can remain permissible. The changed judgment concerns one reason's practical force; no fact in w_1 – w_5 has changed. Exercise 4.3 supplies a numerical realization of both assessments.

Choosing between permitted arrangements requires a further policy and the information it uses. A policy preferring the least disruption could select one, while a different stated priority could select the other. Permission alone supplies neither policy. Chapter 10 develops this distinction with three alternatives: an option may pass comparison with one rival yet fail against another. The practical question must therefore retain its alternatives as well as its grounds.

4.4 What must the model preserve?

The three views have specified different problems within the same case:

Question	Needed information	What the example establishes
Which attacks survive the priority?	Rules, contrary conclusions and a priority.	The displayed defeat pair; acceptance requires its own rule.
Who is responsible for a reply?	Participants, acts, targets and response conventions.	The two requests have answers; responsibility for their offered grounds remains.
Which arrangements are permitted?	Alternatives, comparative grounds and an assessment of their force.	The stated assessment permits M ; its variant permits both, leaving selection open.

A hybrid explanation should state what each component establishes and which assumptions connect its output to another component. The mother’s answer could supply an inferential premise, but its public availability does not establish its truth. A reason acceptable under an issue standard could enter practical evaluation, but its comparative force still needs justification. Agreement on one example is not an equivalence between methods.

Chapter 3 showed that an exact query can miss a requirement. Here the same problem arises before computation. Stakeholder purpose and an acceptance criterion should constrain model choice, rather than being inferred from whatever the implementation happens to return [22, pp. 149–152]. An interface for contesting the relevance of w_1 , for example, needs more than an option score. An interface asked only for the local priority result need not simulate an entire conversation.

4.5 When the right answer identifies a missing input

Suppose an assistant receives w_1 – w_5 but none of the comparative assessments, and is asked which arrangements are permitted. It cannot infer the grounds’ justifying or requiring force from the facts alone. The useful answer identifies the missing assessments and who is authorized to supply them. In the numerical version, that means missing weights, not zero weights. An explicitly empty reason list is different: under a policy permitting either option when neither faces an opposing demand, both pass. The exercise calculation preserves this distinction.

Likewise, the judge’s further relevance request needs target and response conditions before its protocol legality can be certified. Our small public fragment establishes responsibilities for its specified exchanges; it does not settle every possible continuation. And when both M and F are permitted, a request for a unique arrangement needs a selection policy. The permission assessment is complete; repeating it cannot supply that rule.

Such refusals identify the dependency between a question and the information that would make its answer warranted. They conclude a positive comparison: we have seen what the supplied rules, public conventions and practical assessments each allow us to say, and can now locate what a further request needs.

The next four chapters pursue the inferential perspective in depth. They construct arguments and conflicts, compare acceptance standards, ask when a method represents independently specified reasoning, and evaluate it by principles. Chapters 9 and 10 then formalize and deepen the interaction and evaluation concepts used here. This order develops our understanding of the field; it is not a universal algorithm through which every use of argumentation must pass.

4.6 Exercises

- 4.1. [\[solution\]](#) [Construction] Add the fictional fact w_6 , “the mother can provide a stable care schedule”, and a defeasible rule using it. Identify a resulting conflict with an existing conclusion. Which additional comparison would be needed to decide that attack as defeat? Why does adding the reason not yet prove an accepted conclusion?
- 4.2. [\[solution\]](#) [Dialogue diagnosis] Explain the responsibilities after t_2 and t_4 , and what withdrawing the mother’s assertion at that point would leave. Give two plausible continuations after the judge’s t_5 relevance request, including an answer. State exactly what is missing for certifying this further exchange’s legality and discharge. Distinguish responsibility to provide material from the standard for establishing an issue. Do not assume Chapter 9’s rules apply to the judge’s request.
- 4.3. [\[solution\]](#) [Calculation] Use these stipulated comparative records. They assign justifying weight JW and requiring weight RW , not measurements or legal authority:

Ground	Favoured option	Compared with	JW	RW
Child’s preference	M	F	8	4
Father’s wealth	F	M	3	1
Father’s absence	M	F	5	2

Let $J(o, f)$ and $R(o, f)$ sum the respective weights favouring o against f , with empty sums zero. Permit o against f exactly when $J(o, f) \geq R(f, o)$. Compute both tests and the original permitted set. Then change only the father’s wealth JW from 3 to 7, keeping its $RW = 1$, and recompute. Which practical force has changed? Does the new permission set select an arrangement? Give two different stipulated policies that could select from it.

- 4.4. [\[solution\]](#) [Model comparison] For each example name information preserved and information absent. Write a procedure with the contract: input a case, a question and the required output; return an encoding, its scoped result and a list of needed inputs not supplied. Apply it to the custody question, then explain to a nontechnical requester why an omitted feature matters. How would the answer change if the required output were a public obligation rather than an option?
- 4.5. [\[solution\]](#) [Transfer] Consider a fictional triage explanation interface, with all clinical rules and priorities supplied by an authorized domain team. Its tasks are to report the result of those rules, let a user contest an input and compare two permitted scheduling options under a separate resource policy. Assign each task a suitable perspective, state its missing inputs, and identify which outputs must remain distinct. This is a modelling task, not a request for a medical recommendation.

- 4.6. *[solution]* [Explanation] Design a short hybrid explanation using the custody case and the original weights in Exercise 4.3. State the local defeat and permission results separately, show their assumptions, and identify the public responsibilities that the fragment establishes. Leave acceptance and the judge’s further protocol legality unresolved where appropriate. Explain why neither calculation proves the other’s result. Specify the extra interface needed if a dialogue answer changes a comparative weight.
- 4.7. *[solution]* [Computation] Give a paper algorithm or pseudocode for Exercise 4.3’s permission tests; a program is optional. The input consists of two distinct options and a finite list of records (*ground*, *favoured*, *contrasted*, *JW*, *RW*). Weights are nonnegative integers or exact fractions, and each record concerns those two distinct options. Count each record once. Test the original case, the changed weight, an empty list and a nonzero double tie. Give a case with equality in one test in which the other option still fails.
- 4.8. *[solution]* [Requirements] Distinguish three requests: calculate permissions when weights are missing; calculate them when the reason list is explicitly empty; select one option when both are permitted and no policy is supplied. For each, state the available result and the exact remaining input, if any.
- 4.9. *[solution]* [Literature investigation] Compare Gordon’s requirements account, A-BDI and one practical or dialogue alternative from the reading routes. Identify a shared purpose and a nonshared output. After Chapter 10, use the modular legal-reasoning reading to describe one interface between two methods without claiming that they are equivalent or that the framework is a deployed system.

4.7 Further reading

Requirements before machinery. Gordon’s account of systems engineering makes stakeholder roles, goals and acceptance criteria central [22, pp. 145–153, especially pp. 149–152]. Read it with the Handbook’s introduction to dialogue [20, Sec. 4.4, pp. 36–38] and the separation of protocol, effects, outcomes and strategy in Prakken [35, Secs. 3–4]. A-BDI offers a useful later synthesis of inference, dialogue and balancing, explicitly as a discussion proposal [34, Secs. 1–2]. Its questions can help compare model contracts; it does not establish an exhaustive taxonomy or translations among every method bearing those names.

Different practical and public objects. Walton and Krabbe distinguish dialogue types through their purposes and commitments [36]; the Handbook develops the formal choices [37, pp. 517–523] and their application settings [38, pp. 230–248]. For practical reasoning, begin with [39, pp. 300–315] and [40, pp. 380–389]. Their routes to Atkinson and Bench-Capon’s practical-reasoning account [41] and Amgoud and Prade’s decision framework [42] invite comparison of actions, promoted values, reasons and preference assumptions. Cardinal *JW/RW* is one selected approach,

not the definition of every practical question. Keeney’s multi-criteria tradition [43] asks how objectives and trade-offs structure a choice; no equivalence with the present preview is assumed.

Combining methods without erasing their contracts. Return after Chapter 10 to Prakken and Sartor’s modular framework [44, Sec. 4]. It combines legal reasoning methods while allowing their applicability to be contested. The source supplies a semantic framework, not evidence of a deployed application. Compare what a module exports with what another requires. Chapters 7 and 13 revisit representation and explanation; Chapters 14 and 15 develop interfaces, normative multi-agent systems and reasoning alignment. The common question is what each component may legitimately supply to another.

5 Chapter 5. Formalising argumentation

What should a formal model of argumentation explain? An inference model asks which conclusions follow from information; a dialogue protocol asks what participants may say and what their moves commit them to; a model of practical evaluation compares reasons for action. These are different questions, even when the same dispute raises all three. Our task is to choose and connect methods that answer them. Here we develop one inference route through structured arguments; Chapters 9 and 10 return to the interaction and practical evaluation introduced in Chapter 4, making their selected rules precise.

Within inference, a chain of reasoning can be challenged at more than one place. We may dispute a premise, oppose a defeasible conclusion, or question whether a rule applies. Identifying a conflict does not yet settle how a priority should resolve it: the strongest final step may rest on the weakest earlier step. An argument must therefore retain more than its conclusion.

The fitness-loving Scot makes this choice concrete. A route through Scottishness supports liking whisky, while the fitness consideration supports the contrary conclusion. We use the source’s compressed input, with a shared unchallenged premise $\top$, rather than rebuilding the full story [45, p. 2, Ex. 1, pp. 5–6]:

$$d_1 : \top \Rightarrow a \quad (1), \quad d_2 : \top \Rightarrow \neg b \quad (2), \quad d_3 : a \Rightarrow b \quad (3).$$

Larger numbers give stronger priorities. One derivation reaches b through a ; another reaches $\neg b$ directly. Comparing final rules favours the first, whereas comparing every defeasible step favours the second. We first construct both arguments and locate their conflict; choosing a comparison comes afterwards.

Figure 2 places this task in the longer argument of the book. The lower route first constructs arguments and their conflicts, then evaluates a graph and extracts an answer. A source method may already supply answers independently. Chapter 7 will compare the two routes. For now, we stop at the graph: justifying its arrows and selecting its acceptable sets are separate tasks.

5.1 Premises, rules and derivations

ASPIC+ provides an abstract rule-based account in which these choices remain visible. An argumentation system is a tuple $AS = (\mathcal{L}, \bar{\cdot}, \mathcal{R}, n)$, with $\mathcal{R} = \mathcal{R}_s \dot{\cup} \mathcal{R}_d$. Here $\mathcal{L}$ is the language, $\bar{\cdot}$ maps a formula to a set of incompatible formulas, $\mathcal{R}_s$ and $\mathcal{R}_d$ are respectively strict and defeasible inference rules, and $n : \mathcal{R}_d \rightarrow \mathcal{L}$ names defeasible rules. A knowledge base $\mathcal{K} \subseteq \mathcal{L}$ is a disjoint union $\mathcal{K} = \mathcal{K}_n \dot{\cup} \mathcal{K}_p$. The axioms $\mathcal{K}_n$ are not open to premise attack; the ordinary premises $\mathcal{K}_p$ are. These are roles in the model, not guarantees that an axiom is true in the world [46, Defs. 2 and 4, pp. 365–366].

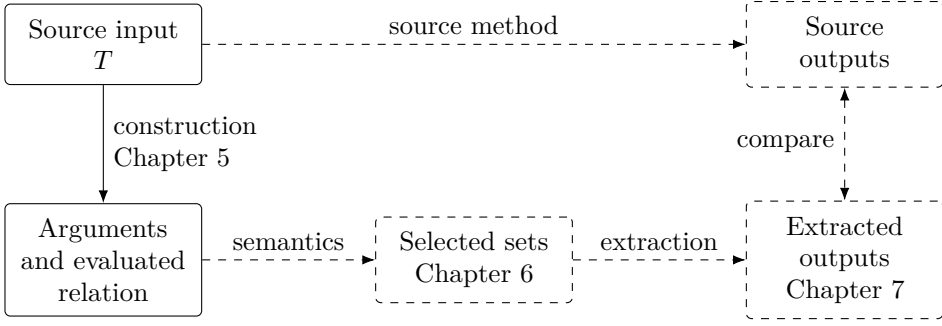


Figure 2: Two routes from an input to answers. Solid elements mark the construction studied here; dashed elements mark later operations, not different kinds of attack. In this chapter construction separates raw attacks from preference-resolved defeats.

The contrary map need not be ordinary negation. More exactly, ψ is an *asymmetric contrary* of φ when $\psi \in \overline{\varphi}$ but $\varphi \notin \overline{\psi}$; they are *contradictories* when both memberships hold. The canonical signature requires every formula to have at least one contradictory [46, Def. 2, p. 365]. Our finite examples display only the contrary entries relevant to their available conclusions and attack targets. These sparse tables are computational fragments of the ambient signature, not complete contrary maps. Unprinted ambient entries contribute no additional attacks among the displayed arguments. In these examples a defeasible rule r has name ok_r ; any relevant rule-name contrary is stated explicitly.

A strict rule $\varphi_1, \dots, \varphi_m \rightarrow \psi$ licenses an inference without a defeasible step; a defeasible rule $\varphi_1, \dots, \varphi_m \Rightarrow \psi$ licenses an inference that can be challenged in the ways defined below. Rules are inference permissions, not automatically formulas of $\mathcal{L}$. In particular, a rule $p \rightarrow q$ does not silently add a material implication to $\mathcal{K}$, or authorize all classical inferences [47, Sec. 2, pp. 293–296].

Arguments are constructed recursively. A premise $\varphi \in \mathcal{K}$ gives an argument $\langle \varphi \rangle$. If arguments $A_1, \dots, A_m$ conclude the antecedents of a rule r , applying it gives

$$[A_1, \dots, A_m \rightarrow \psi]_r \quad \text{or} \quad [A_1, \dots, A_m \Rightarrow \psi]_r,$$

according to the rule's type. Only arguments obtained by this recursive construction are included. Two derivations remain distinct arguments even when their conclusions coincide [46, Def. 5, pp. 366–367].

For the Scot input, take $\mathcal{K}_n = \{\top\}$, $\mathcal{K}_p = \mathcal{R}_s = \emptyset$, and the three displayed defeasible rules. They generate exactly

$$P = \langle \top \rangle, \quad A = [P \Rightarrow a]_{d_1}, \quad B = [A \Rightarrow b]_{d_3}, \quad C = [P \Rightarrow \neg b]_{d_2}.$$

Thus B contains A , which contains P . The rule d_3 is not an independent reason for b : its use depends on an argument for a .

Several functions recover information from a derivation. $\text{Conc}(X)$ is its final conclusion; $\text{Prem}(X)$ is the set of knowledge-base premises used; and $\text{Sub}(X)$ contains its subarguments, including X itself. For a composite argument, the premise set is the union of its children's premise sets, and its subarguments are itself together with all its children's subarguments. For a premise argument these sets are, respectively, $\{\varphi\}$ and $\{\langle\varphi\rangle\}$. $\text{TopRule}(X)$ is its final inference rule, and is undefined for a premise argument.

The two rule-set functions needed for preferences differ: $\text{DefRules}(X)$ contains every defeasible rule used anywhere in X , while $\text{LastDefRules}(X)$ is defined recursively by

$$\text{LastDefRules}(X) = \begin{cases} \emptyset, & X \text{ is a premise argument,} \\ \{r\}, & \text{TopRule}(X) = r \in \mathcal{R}_d, \\ \bigcup_i \text{LastDefRules}(X_i), & X = [X_1, \dots, X_m \rightarrow \psi]_r. \end{cases}$$

These are sets of rule identities, not counts of occurrences [46, Def. 5, p. 367]. On the running input:

Argument	Conc	DefRules	LastDefRules
P	$\top$	$\emptyset$	$\emptyset$
A	a	$\{d_1\}$	$\{d_1\}$
B	b	$\{d_1, d_3\}$	$\{d_3\}$
C	$\neg b$	$\{d_2\}$	$\{d_2\}$

An argument is *strict* when it uses no defeasible rule anywhere, and *firm* when all its premises are axioms. A strict final step does not make an argument strict: the whole derivation must be free of defeasible rules. Exercise 5.3 tests that distinction alongside the different vulnerabilities of premises and inference steps.

5.2 Where a challenge lands

An attack has a source argument, a target argument, and a particular place in the target at which the objection applies. Let $Y' \in \text{Sub}(Y)$. The following three clauses use only the attacker's *final* conclusion [46, Def. 8, p. 368]:

$$\begin{aligned} X \text{ undermines } Y \text{ on } Y' &\iff Y' = \langle\varphi\rangle, \quad \varphi \in \mathcal{K}_p, \quad \text{Conc}(X) \in \overline{\varphi}; \\ X \text{ rebuts } Y \text{ on } Y' &\iff Y' = [Y_1, \dots, Y_m \Rightarrow \varphi]_r, \quad \text{Conc}(X) \in \overline{\varphi}; \\ X \text{ undercuts } Y \text{ on } Y' &\iff \text{TopRule}(Y') = r \in \mathcal{R}_d, \quad \text{Conc}(X) \in \overline{n(r)}. \end{aligned}$$

Undermining disputes an ordinary premise. Rebutting opposes a defeasibly reached conclusion. Undercutting challenges a named rule's applicability without having to conclude the opposite of its head. An axiom cannot be undermined, and a strict

top step cannot be rebutted at that step. This is *restricted rebut*; a defeasible subargument below that strict step may still be a target.

The direction of dependence matters. An incoming attack reaches a larger argument through its attacked subargument. An outgoing attack does not automatically pass to a larger argument with a different final conclusion. Figure 3 shows both points on a small variant. Take axioms p, q , no strict rules, and

$$r : p \Rightarrow a \text{ (1)}, \quad s : a \Rightarrow b \text{ (5)}, \quad t : q \Rightarrow \neg a \text{ (2)},$$

with $a, \neg a$ contradictory and no other relevant conflicts. Write $A = [\langle p \rangle \Rightarrow a]_r$, $T = [A \Rightarrow b]_s$, and $U = [\langle q \rangle \Rightarrow \neg a]_t$. Then U rebuts T on A , not on its final conclusion b . Conversely A rebuts U , but T , whose conclusion is b , does not inherit that attack.

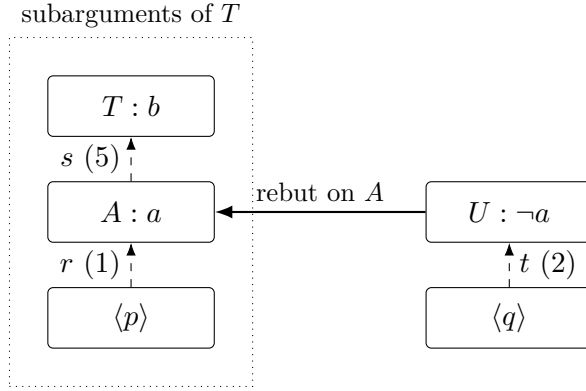


Figure 3: Dashed upward arrows are defeasible inferences, labelled by rule and rank; the solid horizontal arrow is an attack. Its target inside T is A . The relevant contrary information is stated in the text.

5.3 From attack to defeat

An attack identifies a vulnerability. A *defeat* is an attack that succeeds under the declared preference policy. Write $X \prec Y$ for “ X is strictly less preferred than Y .” Undercuts are preference-independent. So are undermining and rebutting attacks whose initiating formula is an asymmetric contrary, rather than a symmetric contradictory, of the target formula. All other undermining and rebutting attacks are preference-dependent. An argument X defeats Y precisely when some attack by X on Y , at a subargument Y' , is preference-independent, or is preference-dependent and $X \not\prec Y'$. The comparison is with the attacked subargument Y' , not automatically with the whole target Y [46, Def. 9, p. 368]. One successful witness suffices; an argument may have more than one vulnerability.

An undercut challenges the permission to use a rule, rather than the truth of its head. It can therefore succeed even from an argument with a lower rule rank.

Exercise 5.9 asks the reader to locate that vulnerability and test why a universal “stronger argument wins” policy would change the formalism.

5.4 Corrected weakest and last link

To calculate the preference-dependent cases we must say how priorities on ingredients become priorities on arguments. The corrected strict elitist comparison of rule sets is

$$\Gamma \triangleleft_{\text{Eli}} \Delta \iff \exists r \in \Gamma \forall s \in \Delta : r < s.$$

It is defined directly from the strict rule order. It must not be reconstructed as the strict part of the older non-strict lifting; the 2018 corrigendum changes precisely that construction [48, Eq. (3), p. 109, Sec. 4, p. 110].

We use a restricted instance: all premises are axioms, only defeasible-rule priorities are compared, and the distinct rule ranks form a strict total order. On the finite nonempty rule sets used here, the displayed comparison amounts to comparing minimum ranks. The quantified definition also handles empty selected sets; its boundary cases are derived in Exercise 5.4. Define

$$\begin{aligned} X \prec_{WL} Y &\iff \text{DefRules}(X) \triangleleft_{\text{Eli}} \text{DefRules}(Y), \\ X \prec_{LL} Y &\iff \text{LastDefRules}(X) \triangleleft_{\text{Eli}} \text{LastDefRules}(Y). \end{aligned}$$

This firm, rule-only fragment does not supply ASPIC+’s full treatment of preferences on ordinary premises [46, Sec. 5.1, pp. 374–376].

Return first to Figure 3. Under last link, the attack by U compares rank 2 with the attacked A ’s rank 1. It succeeds against both A and T . Comparing with T ’s rank 5 would instead reject it. A strong subsequent rule does not change the operand of the local test. Exercise 5.2 gives the converse error, in which a weak final rule would make whole-target comparison admit an attack that should fail.

For Scot, the only relevant contradictory pair is $b, \neg b$. The raw attacks are precisely B rebutting C on C and C rebutting B on B . The attacked subargument happens to be the entire target in both directions. Figure 4 shows why the rule selection reverses which rebut survives.

Nothing about semantic acceptance has yet been inferred. The construction has exposed a choice: count the early default as a weakness of the whole derivation, or let the last defeasible step determine this comparison.

The numbers are ordinal priorities. Their differences are not amounts of evidence, and rank 3 is not three times as credible as rank 1. An audience’s ordering of values, a numerical strength calculation, and a probability distribution require different interpretations and additional inputs. Chapters 10 and 11 return to those distinctions. Even abstract preference policies need not resolve the same raw attack in the same way; Exercise 6.9 compares two policies after semantics has been introduced.

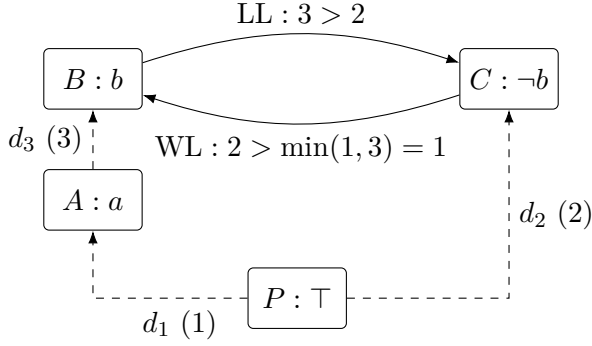


Figure 4: Scot’s structure and conflict. Dashed links are inferences, labelled by rule and rank. The solid arrows are the two raw rebuts, each targeting the other’s defeasible top; their labels identify the surviving defeat under each policy. Only solid arrows enter the defeat comparison: P , A are isolated in both evaluated graphs.

5.5 Other ways to retain reasons

What an argument retains determines where it can be challenged. ASPIC+ retains a derivation tree. Flat assumption-based argumentation locates defeasibility in designated assumptions; classical deductive arguments can instead retain a minimal consistent support and its conclusion; DeLP retains a minimal defeasible-rule support relative to a strict part. These are different objects, with different evaluation requirements. Exercises 5.7, 5.8 and 6.7 reconstruct small cases; the annotated reading below identifies the fuller constructions. Finding a derivation or a counterargument does not by itself supply any of their complete evaluation procedures.

The construction hands Chapter 6 the evaluated framework

$$F_T = (\text{Arg}(T), \text{Defeat}(T)).$$

The theory T includes the system, knowledge base and preference policy; $\mathcal{K}$ denotes only its premise set. Structural attack and surviving defeat remain distinct even when a later source calls its already preference-resolved relation an “attack” relation. That evaluated relation is passed to semantics once, with no second preference filter. Chapter 7 then asks a further question: whether the resulting conclusions reproduce those of an independently specified default procedure. A reversal between weakest and last link does not by itself answer that question. Our running inputs generate finite graphs. A finite rule base need not always do so: the repeated-rule construction in Exercise 5.4 can be continued indefinitely.

5.6 Further reading

Modgil and Prakken’s *Abstract Rule-Based Argumentation* is the Handbook entry point: its introduction and Section 2 explain why inference permissions, premise types and subarguments are useful abstractions [47, pp. 287–303]. Its historical discussion places this approach alongside Pollock, Vreeswijk and Prakken–Sartor; structured argumentation has a history of its own, not merely a role filling Dung’s nodes. For the exact definitions used here, read the 2013 article’s Definitions 2, 4–5 and 8–9 [46, pp. 365–368], together with the 2018 corrigendum [48, pp. 109–110]. The full premise-and-rule orderings and their rationality conditions are a further step; the numerical fragment is not a substitute for them.

To compare what counts as a reason, use Handbook I.7’s ABA definitions [49, pp. 368–371], I.9’s distinction between general deductive pairs and the minimal-support classical variant [13, pp. 443–446], and I.8’s progression from DeLP argument structures to warrant [50, pp. 413–422]. The starred exercises work through small cases from these routes. Notice especially where defeasibility is located and what additional evaluation an answer needs.

For a different use of preferences, Handbook II.3 compares abstract reductions and discusses sources of preference such as specificity and values [51, pp. 211–217, 238–241]. Read it alongside the local target test, not as a replacement for that test. The wider comparison with dialogue and practical evaluation returns in Chapters 9 and 10, where the formal question itself changes. Pardo and colleagues’ priority examples lead onward to Chapter 7’s source comparison and Chapter 8’s construction principles [45, Secs. 3–6]. The next chapter takes the more immediate step: given an evaluated graph, what should it mean to accept an argument?

A useful comparison across these readings is to hold the conclusion fixed and ask what makes two reasons different: their derivation trees, their assumptions, or their minimal supports. Then locate the object against which an attack is tested. This connects the alternative-formalism exercises to the chapter’s central distinction between a vulnerability and the procedure that evaluates it.

5.7 Exercises

Complete public solutions are collected in the end-of-book Solutions section, under [Chapter 5](#). Exercise roles indicate the kind of work, not a requirement that every course assign every task.

The starred exercises develop optional alternative constructions. The remaining exercises use the definitions above; their sparse contrary data follow the ambient-signature convention in Section 5.1. Unless stated otherwise, there are no additional premises or rules and no further relevant attack entries.

5.1. [\[solution\]](#) *Calculation and transfer.* For the Scot theory, calculate Prem, Sub

and **TopRule** for each of P, A, B, C , and explain why these are all the arguments. Name every raw attack and its target subargument. Change only the rank of d_1 from 1 to 4. Which parts of the construction remain unchanged, and what are the two defeat relations?

- 5.2.** *[solution] Conceptual diagnosis.* In Figure 3, replace r 's rank by 3 and s 's rank by 1, leaving t 's rank at 2. Compare the correct last-link test of U against T with a whole-target test, and explain why the original and changed cases refute that shortcut in opposite directions. In each case, does T attack U merely because A does?
- 5.3.** *[solution] Boundary diagnosis.* Locate the vulnerability, or explain its absence, in each independent case. Do not infer an attack from conflicting formula labels alone.
- (a) Let $\mathcal{K}_n = \{p, z\}$, $\mathcal{K}_p = \{q\}$, and $d : z \Rightarrow \neg p$. Declare $\bar{p} = \bar{q} = \{\neg p\}$ and $\neg\bar{p} = \{p, q\}$ on the relevant fragment. Which premise arguments can $N = [\langle z \rangle \Rightarrow \neg p]_d$ undermine? Is the information sufficient to decide every preference-dependent defeat?
 - (b) Let $\mathcal{K}_n = \{p, q\}$, $d : p \Rightarrow a$ (1), $n : q \Rightarrow \neg b$ (2), and $s : a \rightarrow b$, with only $b, \neg b$ contradictory in the relevant fragment. Does $N = [\langle q \rangle \Rightarrow \neg b]_n$ attack $T = [[\langle p \rangle \Rightarrow a]_d \rightarrow b]_s$? Is T a strict argument? Calculate its last defeasible rules.
 - (c) Let $\mathcal{K}_n = \{z\}$, $\mathcal{K}_p = \{\eta\}$, $r : z \Rightarrow p$ (1), with $p \in \bar{\eta}$ and $\eta \notin \bar{p}$. Does the argument for p defeat $\langle \eta \rangle$? Here η is a formula label, not a newly defined negation.
 - (d) Take axioms $\neg b, z, p$, strict rule $z \rightarrow \neg b$, and defeasible rule $d : p \Rightarrow b$ (9), with $b, \neg b$ contradictory. Compare the premise argument for $\neg b$ and its strict-rule derivation with the argument for b , under each rule-only link policy.
- 5.4.** *[solution] Proof and boundary calculation.* First derive the minimum-rank test from the quantified strict elitist definition for finite nonempty sets in the ranked fragment. Then compare $\emptyset$ with $\{r\}$ in both directions, and $\{r\}$ with itself. Explain why assigning an empty selected set minimum rank ∞ gives the same comparisons in this fragment. Why do equal selected strengths not block either direction of a symmetric rebut? If $\mathcal{K}_n = \{p\}$ and $r : p \Rightarrow p$, compare **DefRules** and **LastDefRules** for $[\langle p \rangle \Rightarrow p]_r$ and for the argument obtained by applying r once more. Are those arguments identical?
- 5.5.** *[solution] Modelling transfer.* Take axioms *snores*, *professor*, no ordinary premises or strict rules, and

$$r_1 : \textit{snores} \Rightarrow \textit{misbehaves} \quad (1),$$

$$r_2 : \textit{misbehaves} \Rightarrow \textit{accessDenied} \quad (3),$$

$$r_3 : \textit{professor} \Rightarrow \neg \textit{accessDenied} \quad (2).$$

The only relevant contradictory pair is *accessDenied*, $\neg$ *accessDenied*. Construct S_0, S_1, S_2 along the first chain and P_0, P_1 along the second, and find both defeat relations. What is preserved by comparison with Scot, and why are the argument universes not literally identical? Can the instruction “choose the best policy” be answered from these ranks alone?

- 5.6. *[solution]* *Cross-chapter comparison. Later use, after Chapter 7.* Revisit the original Scot input and the changed rank in Exercise 5.1, using Chapter 7’s source-first Scot calculation and solved Exercise 7.1. Which comparisons of full source and argumentation outputs agree? Explain why neither a defeat direction nor agreement on these two inputs establishes a general PDL representation. The source procedure and its complete calculations are supplied there, not assumed as Chapter 5 prerequisites.
- 5.7. *[solution]* * *Alternative formalism.* Fix classical propositional consequence and $K = \{p, p \rightarrow q, \neg q\}$. For this exercise an argument (Γ, φ) requires $\Gamma \subseteq K$, consistency of Γ , entailment of φ , and no proper subset of Γ entailing φ [13, Def. 4.8, p. 446]. Check $(\{p, p \rightarrow q\}, q)$, $(\{\neg q\}, \neg q)$, and (K, q) . Use countermodels where needed. Why would replacing the formula $p \rightarrow q$ by an uninterpreted rule name invalidate the first entailment check? What remains unspecified about evaluation?
- 5.8. *[solution]* * *Alternative formalism.* In the following DeLP construction task let $\Pi = \{bird(t)\}$ and

$$\Delta = \{flies(t) - \prec bird(t), \quad \sim flies(t) - \prec bird(t)\}.$$

The left side of each defeasible rule is its head, and $\sim$ is the displayed explicit negation. An argument support is an inclusion-minimal subset of Δ that, with Π , derives its conclusion without deriving contradictory literals [50, Defs. 4.5–4.6, pp. 416–417]. For this input, opposite conclusions give a counterargument at the opposing argument’s conclusion. Find the supports for the two opposing conclusions and explain their counterargument relation. Here the question of *warrant* is whether a conclusion is ultimately accepted after evaluating counterarguments, rather than merely derived from a support. Does either derivation alone settle that question? State what a further evaluation policy would have to decide; the technical DeLP evaluation procedure is not assumed.

- 5.9. *[solution]* *Conceptual diagnosis.* Take axioms p, z , rules $t : p \Rightarrow q$ of rank 9 and $u : z \Rightarrow \neg ok_t$ of rank 1, and name $n(t) = ok_t$. Let $\neg ok_t \in \overline{ok_t}$ be the only relevant contrary entry. Construct the two rule-based arguments and identify the attack kind and exact target. Is the attack a defeat under either rule-only link policy? Explain what a policy that always rejects a lower-ranked attacker would misunderstand. Does the attacker have to conclude $\neg q$?

Part II. The attack–defense paradigm shift and its immediate generalizations

Chapter 5 made the internal structure of a reason visible: which premises it uses, where an inference is defeasible, and where an objection can reach it. The next step deliberately forgets much of that structure. Once the arguments and their evaluated conflicts are fixed, can acceptance be understood through attack and defence alone? That abstraction is the pivot of this part. It separates the question of why an arrow is present from the question of what to do with a graph containing it.

The separation makes comparison possible. A position may answer every objection to its own members without resolving the entire dispute. Another may also attack every excluded alternative. A cautious position may accept only what can be defended without choosing a side in a cycle. Chapter 6 develops these standards on one linked graph and shows why they can select different sets. Acceptance is therefore already a choice of evaluative requirements, even before we return to the meanings of the arguments.

Restoring those meanings creates a further obligation. A default logic may already specify its answers through a rule-application procedure; a matching problem seeks an assignment of partners. If we represent either through argumentation, the selected sets must recover the right kind of output. Chapter 7 compares the direct source route with the route through construction, semantics and extraction. The recurring Scot example now tests fidelity to a priority policy, rather than merely illustrating how that policy changes an arrow. An exact encoding of a graph can preserve its answers without establishing that the graph was the right representation of the source.

Chapter 8 then turns from fidelity to design. Closure and consistency ask what an accepted output must retain. Redundancy and independence ask which changes to an input should leave old conclusions intact. Principles governing attack assignment ask whether the same reasons receive consistent treatment across contexts. These requirements act at different points in the reasoning process, and attractive requirements need not be jointly attainable. The PDL incompatibility proof makes that tension explicit: local constraints on attacks can conflict with a source procedure that inspects the whole current theory.

Chapters 9 and 10 return to the participants, commitments, purposes and burdens encountered in Chapters 3–4. Dialogue makes public responsibilities exact: a static acceptable set does not say who may speak next or what a challenge calls for. Issue evaluation and practical comparison then formalize how supplied reasons meet a standard or bear on competing options.

6 Chapter 6. Creating argumentation frameworks and semantics

Can attack and defence provide a common language for different forms of reasoning? Instead of beginning with one language of premises and rules, Dung's abstraction begins with arguments and a relation between them. The promise is that the same calculus can evaluate disputes whose reasons were constructed in different ways. Its adequacy still has to be shown: what should count as an argument or an attack, and which distinctions would be lost by retaining only that relation?

This chapter holds the graph fixed to develop the calculus itself. A graph of conflicts does not yet say which arguments to accept. One position may defend everything it contains while leaving a cycle of objections unresolved; another may answer every argument it excludes; a cautious position may commit to neither alternative. These are different standards of acceptance. Their comparison explains what the abstraction can do once its nodes and arrows have been justified.

Chapter 5 supplied one construction route. We now define the principal extension semantics and follow their differences on one linked graph. Chapter 7 will return to the larger representation question: do these selected sets recover the intended source answers? We work with finite frameworks; Dung's original theory is not restricted to finite argument sets [5, Sec. 2].

6.1 A graph with two kinds of cycle

An *abstract argumentation framework* (AF) is a pair $F = (A, R)$, where A is a set of arguments and $R \subseteq A \times A$ is an attack relation [5, Def. 2, p. 326]. Write xRy for $(x, y) \in R$. Self-attacks are permitted. When the framework comes from Chapter 5, R is its already preference-resolved *defeat* relation. The abstract terminology calls these arrows attacks, but does not apply the preferences a second time.

Our principal example is the linked-cycle framework

$$\begin{aligned} A_{\text{link}} &= \{a, b, c, d, e\}, \\ R_{\text{link}} &= \{(a, b), (b, a), (b, c), (c, d), (d, e), (e, c)\}. \end{aligned}$$

It links a mutual pair to a directed three-cycle [6, Fig. 9, p. 185] [1, Ex. 3.2, pp. 881–883]. The choice made at a, b can therefore affect the remaining dispute.

For an argument x and a set $S \subseteq A$, put

$$x^- = \{y \in A : yRx\}, \quad S^+ = \{y \in A : \exists x \in S (xRy)\}.$$

Thus x^- lists the attackers of x , while S^+ lists the arguments attacked by S . For example, $c^- = \{b, e\}$ in Figure 5, and $\{b\}^+ = \{a, c\}$. The arrow $b \rightarrow c$ is not a support edge to d ; its role in defending d will follow from the attack–defence definition.

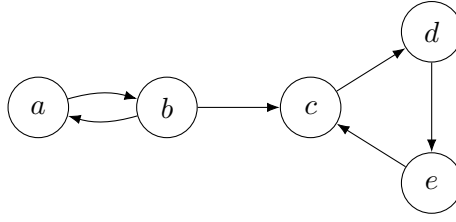


Figure 5: The linked-cycle framework F_{link} . All arrows are attacks in the evaluated AF. The distinct five-node framework reused for computation and explanation is supplied in Exercise 6.2.

6.2 Defending a position

A set S is *conflict-free* when it contains no attack between its members, that is, $R \cap (S \times S) = \emptyset$. This excludes a self-attacking argument as well as a pair of distinct arguments joined by an attack. A set S *defends* x when it attacks every attacker of x :

$$\forall y \in A \quad (yRx \implies \exists z \in S : zRy).$$

The defender z may depend on the attacker y ; it need not be x , and x need not belong to S . An unattacked argument is defended even by the empty set. Finally, S is *admissible* if it is conflict-free and defends each of its members [5, Defs. 5–6, p. 326].

In the linked graph, $\{a\}$ defends itself by attacking its only attacker b . The set $\{b\}$ likewise defends b . It also defends d , because it attacks d 's sole attacker c . Consequently $\{b, d\}$ is admissible: b answers the attacks on both its members. Defence is thus relative to a set, not a status permanently attached to a node.

The *characteristic function* collects everything a set defends:

$$\mathcal{F}_F(S) = \{x \in A : S \text{ defends } x\}.$$

Admissibility says that S is conflict-free and $S \subseteq \mathcal{F}_F(S)$ [5, Def. 16, pp. 328–329]. For the linked graph, $\mathcal{F}_F(\{b\}) = \{b, d\}$. This raises a natural question: should a position omit an argument it already defends?

6.3 Complete, preferred, stable and grounded

A *complete extension* is an admissible set containing every argument it defends. Equivalently, it is a conflict-free fixed point of $\mathcal{F}_F$. A *preferred extension* is an inclusion-maximal admissible set. A *stable extension* is a conflict-free set attacking every outsider, so $S \cup S^+ = A$. The *grounded extension*, denoted $G_F = \mathbf{Ground}(F)$, is the least fixed point of $\mathcal{F}_F$; it is also the least complete extension. The finite construction below explains its existence and leastness [5, Def. 7, p. 327, Def. 13, p. 328, Defs. 20 and 23, p. 329, Lem. 24 and Thm. 25, p. 330].

The admissible sets of the linked graph are exactly $\emptyset, \{a\}, \{b\}, \{b, d\}$. To see the decisive restrictions, an admissible set containing c would have to attack e , hence contain d , contrary to conflict-freeness. One containing e would have to contain c to attack d , again a conflict. With e excluded, defending d requires b . And a, b cannot coexist. The four remaining possibilities do pass the admissibility test.

The set $\{b\}$ is not complete because it omits defended d . The other three admissible sets are fixed points. Thus the semantic families are

Semantics	Extension family for F_{link}
Complete	$\{\emptyset, \{a\}, \{b, d\}\}$
Preferred	$\{\{a\}, \{b, d\}\}$
Stable	$\{\{b, d\}\}$
Grounded	$\{\emptyset\}$

Here the grounded *set* is $G_F = \emptyset$, whereas grounded semantics returns the singleton *family* $\{G_F\} = \{\emptyset\}$. We use $\text{Ext}_\sigma(F)$ for a semantic family, and the abbreviations $\text{Pref}(F)$ and $\text{Stable}(F)$ for the preferred and stable families.

Two distinctions are visible without changing the graph. First, $\{a\}$ is preferred although $\{b, d\}$ has more members. There is no admissible proper *superset* of $\{a\}$; the larger alternative does not contain it. Preferredness is not maximum cardinality. Second, $\{a\}$ is not stable: it attacks b but leaves the three-cycle without an answer. By contrast, $\{b, d\}^+ = \{a, c, e\}$, covering every outsider. Maximal defensibility and decisive outsider coverage are different requirements.

Theorem 6.1. In every finite AF, each admissible set is contained in a preferred extension. In particular, at least one preferred extension exists. Every stable extension is preferred, and every preferred extension is complete:

$$\text{Stable}(F) \subseteq \text{Pref}(F) \subseteq \text{Ext}_{\text{co}}(F).$$

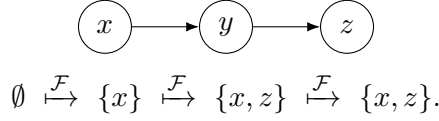
Here co denotes complete semantics [5, Thm. 11(2) and Cor. 12, p. 327, Lem. 15, p. 328, Thm. 25(1), p. 330].

The inclusions connect three demands: defending the selected position, including everything it already defends, and answering every outsider. Their proofs require more than the names of the semantics. Exercises 6.4–6.5 build the argument from defence witnesses and Dung’s Fundamental Lemma. The existence result will also justify Chapter 12’s use of an admissible witness for preferred credulous acceptance.

6.4 Constructing the grounded extension

Grounded semantics begins with no commitments and repeatedly accepts what the current set defends. Let F_{chain} have exactly the three nodes and two attacks

displayed below; its successive grounded iterates appear beneath it:



The unattacked x enters first. Its attack on y then defends z , illustrating reinstatement through a counterattack. The grounded extension is $\{x, z\}$. In the linked-cycle graph every argument is attacked, so the iteration is already fixed at $\emptyset$.

Theorem 6.2. Let $F = (A, R)$ be finite and define $S_0 = \emptyset$, $S_{i+1} = \mathcal{F}_F(S_i)$. Every S_i is admissible and $S_i \subseteq S_{i+1}$. The iteration reaches a fixed point after finitely many updates; that set is the unique least fixed point and the least complete extension [5, Lems. 18–19 and Def. 20, p. 329, Lem. 24 and Thm. 25(2), p. 330].

The essential argument is that every stage stays inside every fixed point: it starts inside one because it is empty, and monotonicity preserves that containment. Admissibility is a separate invariant, ensuring that the final fixed point is complete. Finite growth gives termination, but does not by itself give either of these properties. The full proof and two failed shortcuts are in Exercise 6.6, which also derives a bound on the number of updates. A bound on updates is not yet a bound on the cost of carrying out each one.

It follows that complete extensions always exist in a finite AF, and that G_F is contained in every complete, preferred and existing stable extension. Stable existence is different: a finite AF may have no stable extension. We must therefore distinguish the family $\emptyset$, which reports no output, from $\{\emptyset\}$, which reports one empty output. Exercise 6.3 constructs minimal witnesses of this distinction and tests a further grounded chain.

6.5 Labels and acceptance

An extension states which arguments are selected. A labelling also records why the others are not selected. A total function $\lambda : A \rightarrow \{\text{in}, \text{out}, \text{undec}\}$ is a *complete labelling* when every argument has a legal label: *in* requires all attackers to be *out*; *out* requires some attacker to be *in*; *undec* requires no attacker to be *in* and at least one attacker not to be *out*. A *stable labelling* is complete and has no *undec* arguments [6, Defs. 3.1, 3.19–3.20 and 3.36, pp. 170, 178, 184].

For a complete extension E , the corresponding complete labelling gives

$$\text{in} = E, \quad \text{out} = E^+, \quad \text{undec} = A \setminus (E \cup E^+).$$

Conversely, the *in*-set of a complete labelling is a complete extension [6, Def. 3.6, p. 173, Prop. 3.23, p. 179]. Conflict-freeness alone makes the displayed sets a partition; it does not ensure legal labels. For the linked graph all three complete labellings are:

in	out	undec
$\emptyset$	$\emptyset$	$\{a, b, c, d, e\}$
$\{a\}$	$\{b\}$	$\{c, d, e\}$
$\{b, d\}$	$\{a, c, e\}$	$\emptyset$

The middle row exposes what preferredness leaves unresolved. The last row is stable. This later labelling presentation is not attributed to Dung’s 1995 definitions; the Handbook develops its relationship to them.

Acceptance asks a different question: how often is a specified argument selected across a semantic family? For $x \in A$, the book uses

$$\begin{aligned}
 x \text{ is credulously accepted} &\iff \exists E \in \text{Ext}_\sigma(F) : x \in E, \\
 x \text{ is skeptically accepted} &\iff \text{Ext}_\sigma(F) \neq \emptyset \text{ and } \forall E \in \text{Ext}_\sigma(F) : x \in E.
 \end{aligned}$$

The Handbook formulates these questions on a nonempty-output domain [6, Defs. 4.1 and 4.3, p. 204]. We explicitly extend them to Boolean queries on every finite input by returning false for both statuses when the family is empty. The nonempty guard prevents vacuous skeptical acceptance. A separate *no-output* diagnostic should still report that no extension exists; a false query in that situation does not show rejection by an existing extension.

For the linked graph, preferred credulous acceptance holds for a, b, d , but preferred skeptical acceptance holds for none of them. Under stable semantics both statuses hold exactly for b, d . Under grounded semantics no argument is accepted. Thus changing the semantics can change an answer without changing any argument or attack.

When arguments have conclusions, one further operation is needed:

$$\text{Conc}(E) = \{\text{Conc}(x) : x \in E\}, \quad \{\text{Conc}(E) : E \in \text{Ext}_\sigma(F)\}.$$

The first expression is one conclusion set; the second is a family of such sets. Neither is an argument extension. Different arguments can support the same conclusion, so skeptical support for a formula need not amount to skeptical acceptance of one fixed argument. Chapter 7 develops that quantifier distinction and fixes the extraction appropriate to a representation claim.

We can now finish the semantic step left open by Scot in Section 5.4. In the weakest-link graph, the unattacked P, A, C must all belong to any stable extension; they exclude and attack the sole outsider B . The unique stable extension is therefore $\{P, A, C\}$. Last link similarly gives $\{P, A, B\}$. The calculus now settles both graphs. The broader ambition still requires a justified representation: why should either selected set answer the source problem? Chapter 7 makes that obligation precise by comparing the argumentative route with an independently specified reasoning policy.

6.6 Further reading

The principal Handbook route is Baroni, Caminada and Giacomin’s *Abstract Argumentation Frameworks and Their Semantics*. Sections 2–3.6 organize the extension and labelling accounts; Section 4.1 discusses acceptance across alternative outputs [6, pp. 159–185, 204–207]. Read these with Dung’s original definitions and results on pages 326–330 [5]. The finite proof exercises explain the relationships needed by this book without taking on the full order-theoretic treatment of arbitrary frameworks.

Two different questions lead beyond the quartet. If the issue is how to refine a choice among ordinary AF extensions, the Handbook’s Section 3.7 introduces semi-stable semantics and its treatment of undecided arguments [6, pp. 186–187]. If the input needs conjunction, support or other acceptance conditions, Handbook I.5 on ADFs changes the representation itself [2, Ch. 5, pp. 247–248 and following sections]. Exercise 6.8 is the two-valued-model entry point; the further question is how to evaluate partially decided information.

For the labelling lineage, the Handbook points to Caminada and Gabbay’s *A Logical Account of Formal Argumentation* (2009) [6, Sec. 3.1, pp. 173–174]. Caminada’s separate Handbook chapter interprets semantics through formal discussion [52, Defs. 17–18, pp. 512–513]. The relevant object there is a winning strategy, not one successful play. Chapter 9 develops the dialogue connection; Chapters 12 and 13 return to computation and explanation.

Later work also asks whether ordinary admissibility treats every attacker in the right way. *Thirteen Challenges*, Section 4.1, offers a route into weak-admissibility and related questions [1, pp. 917–919]. Liao and van der Torre’s *Attack-Defense Semantics of Argumentation* makes defender–attacker–target relations explicit and studies reasons for acceptance [53, Secs. 2, 4–5]. This is a further representational and explanatory programme, not the agent-indexed defence of Chapter 2 or an additional calculus assumed here. Chapter 11 compares semantic principles and separately examines empirical cognitive evidence. Correctness of an extension calculation alone establishes neither human plausibility nor the suitability of the graph for its purpose.

For a proof-based continuation, compare the role of finiteness in Exercises 6.5 and 6.6 with Dung’s general treatment. The finite cardinality arguments establish existence and termination here; they should not be carried over to an infinite framework merely by keeping the same notation. This gives a focused question for the original paper while preserving the finite calculus as a complete first account.

6.7 Exercises

Complete public solutions are collected in the end-of-book Solutions section, under Chapter 6. Exercise roles indicate the kind of work, not a requirement that every course assign every task.

Exercises 6.4–6.6 form a connected finite proof route. The starred exercises change the representation or preference policy; they are optional calculations, not additional definitions of ordinary Dung semantics.

- 6.1.** *[solution] Calculation and interpretation.* Verify the complete admissible family of F_{link} . Calculate the characteristic function on each member, then recover all complete, preferred and stable extensions. Justify the legal labels in the table and give witnesses for the preferred and stable acceptance claims. Why would retaining only largest-cardinality admissible sets lose a preferred extension?
- 6.2.** *[solution] Computation and transfer.* The following *computational-reuse framework* will return in Chapters 12 and 13:

$$F_{\text{reuse}} = (\{a, b, c, d, e\}, \{(a, b), (b, a), (b, c), (c, d), (d, c), (d, e)\}).$$

It has $a \leftrightarrow b$, $b \rightarrow c$, $c \leftrightarrow d$, and $d \rightarrow e$; it is not the linked-cycle framework.

- (a) List every conflict-free and admissible set, then every complete, preferred and stable extension, and the grounded set.
 - (b) For $S = \{a, c\}$, give a defender against every attacker of c . Must the same member answer both attackers? Negate the general defence formula, and use it to explain why $\{a, e\}$ does not defend e .
 - (c) Produce the complete labelling for $\{a\}$ and stable labelling for $\{b, d\}$. Classify every argument under stable credulous and skeptical acceptance.
 - (d) Add only (e, a) . Find the preferred and stable families, and explain what happens to the former extension $\{a, c, e\}$.
- 6.3.** *[solution] Boundary calculation.* Compare the preferred and stable families of the empty AF, the single self-attacker $F_{\circ} = (\{u\}, \{(u, u)\})$, and the directed cycle

$$F_3 = (\{p, q, r\}, \{(p, q), (q, r), (r, p)\}).$$

Distinguish an empty extension from an empty family. What does the guarded acceptance convention report when no stable extension exists? As a separate transfer, compute every grounded iterate for

$$F_4 = (\{w, x, y, z\}, \{(w, x), (x, y), (y, z)\}).$$

- 6.4.** *[solution] Proof.* Let $F = (A, R)$ be finite. Prove that $\mathcal{F}_F$ is monotone and that it sends conflict-free sets to conflict-free sets. Deduce that, for admissible S , $S \subseteq \mathcal{F}_F(S)$ and $\mathcal{F}_F(S)$ is admissible. Then prove Dung’s *Fundamental Lemma*: if S is admissible and $x, y \in \mathcal{F}_F(S)$, then $S \cup \{x\}$ is admissible and still defends y . Account explicitly for possible attacks from S to x , from x to S , and from x to itself [5, Lem. 10, p. 327, Lems. 18–19, p. 329].

- 6.5.** *[solution] Proof and counterexamples.* Using Exercise 6.4, prove every assertion of Theorem 6.1. In the existence argument, explain why choosing a largest set from a suitable finite family does not redefine preferredness as maximum cardinality. Refute the converse of each semantic inclusion using the mutual pair

$$F_2 = (\{u, v\}, \{(u, v), (v, u)\})$$

and the self-attacker. For a query $q \in A$, deduce that preferred credulous acceptance is equivalent to the existence of an admissible set containing q . Finally, why is repeatedly adding currently defended outsiders not a general preferred-extension algorithm?

- 6.6.** *[solution] Proof repair.* Repair the following proof: monotonicity makes the iterates from any starting set grow; finite growth stops at a fixed point; every fixed point is complete and grounded. Use F_{chain} with starting set $\{y\}$, and $F_{\circ}$ with starting set $\{u\}$, to locate the errors. Then prove Theorem 6.2 from the empty starting set, separating admissibility, termination and leastness. For $n = |A|$, prove that an implementation testing equality after each update detects termination within $n + 1$ evaluations of $\mathcal{F}_F$, including the final equality test. Include the empty-AF case, and distinguish this count from the cost of an update. Does the Fundamental Lemma remain valid if its hypothesis that S is admissible is weakened to conflict-freeness?

- 6.7.** *[solution] * Alternative formalism.* Consider a flat ABA framework with language $\{p, q, x, y\}$, assumptions $\mathcal{A} = \{p, q\}$, rules

$$x \leftarrow q, \quad y \leftarrow p, \quad \bar{p} = x, \quad \bar{q} = y.$$

These arrows license deductions, and the contrary map assigns a formula to each assumption. A set $S \subseteq \mathcal{A}$ attacks $T \subseteq \mathcal{A}$ when a deduction from S derives the contrary of some assumption in T . All assumption subsets are closed here, since no rule derives an assumption. An admissible assumption set does not attack itself and counterattacks every assumption set attacking it; a preferred one is inclusion-maximal admissible. A stable one does not attack itself and attacks every omitted assumption [49, Defs. 2.1–2.10, pp. 368–371]. Evaluate all four assumption subsets. Give every admissible, preferred and stable assumption set and the consequences of each stable set. Delete only $x \leftarrow q$ and recompute stability. Does this finite calculation prove a general ABA-to-AF correspondence?

- 6.8.** *[solution] * Alternative formalism.* An abstract dialectical framework (ADF) can specify an acceptance condition for each statement, rather than interpreting every parent link as an independent attack. Use statements a, b, c, d and

$$\varphi_a = \top, \quad \varphi_b = b, \quad \varphi_c = a \wedge b, \quad \varphi_d = \neg b.$$

The parent links are $(b, b), (a, c), (b, c), (b, d)$. A *two-valued model* assigns each statement the truth value of its acceptance formula [2, Ch. 5, Def. 3.1, Ex. 3.2

and Prop. 3.4, pp. 247–248]. Find every model, represented by its true-statement set. Then replace only φ_c by $a \vee b$ and repeat. What distinction would be lost by retaining just two untyped arrows into c ? The task does not ask for ADF stable or grounded semantics.

- 6.9.** *[solution]*^{*} *Policy comparison.* Take arguments a, b , the sole raw attack $a \rightarrow b$, and strict preference $b \succ a$. Compare two abstract policies. Reduction 1 keeps a raw attack only when its target is not strictly preferred to its source. Reduction 2 also reverses a critical asymmetric attack: on this input it replaces the blocked $a \rightarrow b$ by $b \rightarrow a$ [51, Defs. 2.4–2.7 and Exs. 2.6/2.8, pp. 216–217]. Find the two defeat graphs and their grounded and stable outputs. Is each output conflict-free relative to the raw attacks? What must be specified before claiming that the preferred argument wins? Explain why neither reduction may silently replace Chapter 5’s local ASPIC+ defeat clause.

7 Chapter 7. Representing nonmonotonic logics and solution concepts

What is gained by replacing a default theory with a graph of arguments? The graph can make its conflicts visible, but visibility is not yet representation. The theory already has a way of determining its conclusions. Unless the argumentative route recovers those conclusions, we may have designed a different reasoning method rather than represented the one with which we began.

This distinction gives Dung’s abstraction a role beyond the evaluation of individual disputes. Different nonmonotonic logics manage exceptions, inconsistency and alternative conclusions in different ways. Their representations through arguments and attacks can expose a common structure without making their differences disappear. The programme also reaches beyond logical inference: a selected object may be a matching or a set of payoff distributions, not a body of beliefs [5, Secs. 3–4]. We need to turn this ambition into representations that preserve the independently specified answers: which constructions work, over which inputs, and with what extraction?

Chapters 5 and 6 separated argument construction from semantic evaluation. We now ask whether that combined route answers an independently specified question. Priorities provide a particularly revealing comparison: the same rules can yield different outputs depending on what a priority is intended to do. After examining that choice, a compact exact encoding will show both the force and the limits of a correspondence theorem.

7.1 Two routes to an answer

The source method must first be chosen. Makinson’s bridges distinguish background assumptions, valuations and inference rules as starting points [10, Sec. 1.4, p. 18]. One may select compatible assumptions, prefer some premise-satisfying valuations, or regulate the application of rules. These are different choices about what licenses a conclusion; they need not agree merely because each is called nonmonotonic.

Recall Chapter 1, “Route 1: from a fixed pivot to default-assumption consequence.” With background $K = \{p \rightarrow q\}$ and hard premise p , both retaining K and selecting its maximal compatible subsets give q . Add the hard premise $\neg q$. Retaining all of K now produces classical inconsistency, not withdrawal of q ; selecting compatible subsets instead drops $p \rightarrow q$ and gives $\text{Cn}(\{p, \neg q\})$. The fixed background remains the same in both descriptions. What changes is whether the effective assumptions may depend on the input [10, Defs. 2.1 and 2.4, pp. 24, 30–31]. Likewise, restricting attention to a fixed set of valuations remains monotonic; selecting the best currently available valuations is a different operation [10, Defs. 3.1 and 3.8, pp. 60–61, 67–68]. An attack graph cannot choose between these source policies for us.

We can now complete the comparison in Figure 2. The upper route applies the source method directly. The lower route constructs a graph, selects extensions and extracts an output from each. The question at the right of the figure is whether the resulting families agree, not merely whether both routes produce plausible answers. Let $\text{Sol}(T)$ denote the output family already specified by a source method on input T . An argumentative route constructs F_T , evaluates it under a semantics σ , and applies an extraction Out_T to each extension. An exact representation over a domain $\mathcal{D}$ requires

$$\text{Sol}(T) = \{\text{Out}_T(E) : E \in \text{Ext}_\sigma(F_T)\} \quad \text{for every } T \in \mathcal{D}. \quad (7.1)$$

The construction, semantics, extraction and domain all belong to the claim. Agreement on one input verifies only an instance. Agreement on selected queries can be weaker still: it need not identify every possible output. A bijection with an explicit inverse says more than family equality under a possibly many-to-one extraction: it also tells us how to recover each represented selection.

Extraction is a substantive part of the route. Chapter 1 represents a family of repairs and then takes consequences from each repair. That is not the same as first retaining only their common premises. For example, the incompatible packed premises $p \wedge q$ and $\neg p \wedge q$ give two singleton repairs. Their common premise set is empty, yet both repairs imply q . This is the Handbook’s packed-premise example [3, Ch. 12, Def. 44 and Ex. 45, p. 749]. Exercise 7.2 reconstructs its base arguments and proves the two different consequence results. The representation of the repairs can be exact while the subsequent extraction policies differ.

As in Section 6.5, for arguments with a designated conclusion write $\text{Conc}(E) = \{\text{Conc}(a) : a \in E\}$. The distinction between a common argument and a common conclusion is equally important. With $\mathcal{E} = \text{Ext}_\sigma(F_T)$, the book’s consequence conventions are

$$\begin{aligned} T \vdash_\sigma^\exists \varphi & \text{ iff } \exists E \in \mathcal{E} \exists a \in E : \text{Conc}(a) = \varphi, \\ T \vdash_\sigma^\forall \varphi & \text{ iff } \mathcal{E} \neq \emptyset \text{ and } \forall E \in \mathcal{E} \exists a \in E : \text{Conc}(a) = \varphi. \end{aligned}$$

The nonempty-family guard extends the Boolean convention of Section 6.5: both queries are false when no extension exists, and that absence is reported separately as *no output*. The supporting argument for a skeptical conclusion may vary between extensions; it need not be one argument accepted in all of them [6, Sec. 4.1, pp. 204, 209]. We shall compare whole output families before summarizing them by such queries.

7.2 What should a priority preserve?

Return to Chapter 5’s three-default Scot case. Its compressed rule pattern is

$$d_1 : \top \Rightarrow a \quad (1), \quad d_2 : \top \Rightarrow \neg b \quad (2), \quad d_3 : a \Rightarrow b \quad (3).$$

Larger ranks mean stronger rules. The unchallenged fact is $\top$; there are no ordinary premises, strict rules or undercuts. The conflicting domain literals are $b, \neg b$. These are the precise inputs of the source example, not priorities inferred from the informal story [45, Ex. 1, pp. 5–6].

Before constructing an AF, determine the answer supplied by the source procedure. We use the cited article’s normal-default fragment of *prioritized default logic* (PDL). A rule is active when its prerequisite is present and neither its head nor its contrary is present. Starting from the facts, choose a rule that is highest-ranked among the active rules and whose addition preserves consistency, add its head, and repeat until no permitted choice remains. In a theory with strict rules, strict closure is also taken at each stage; the present Scot input has none. Where priorities tie, all permitted choice sequences contribute to the PDL output family [45, Rem. 1 and Def. 15, pp. 12–13]. This is a bounded procedure for the fragment under discussion, not a definition of every prioritized default logic.

Here d_1, d_2 are initially active. Rule d_3 , despite having the highest rank, lacks its prerequisite a . Consequently the only permitted sequence is

$$\{\top\} \xrightarrow{d_2} \{\top, \neg b\} \xrightarrow{d_1} \{\top, \neg b, a\}.$$

Once a is obtained, the already present $\neg b$ prevents d_3 from applying. Thus the complete source family is $\{\{a, \neg b\}\}$. In these comparisons we report the derived domain literals, with the common $\top$ suppressed on both routes; we are not silently taking classical closure.

On the argumentative route, retain the four arguments P, A, B, C of Section 5.1, with conclusions $\top, a, b, \neg b$, respectively. The only structural rebuts are $B \leftrightarrow C$, each aimed at the other’s defeasible top. Sections 5.3–5.4 already determine which survives: weakest link compares B ’s minimum rank 1 with C ’s rank 2, giving $C \rightarrow B$; last link compares their last ranks, 3 and 2, giving $B \rightarrow C$. These are the corrected strict comparisons, applied to the attacked subargument.

The source article calls these surviving, preference-resolved relations *attacks*. We retain Chapter 5’s distinction between structural attack and defeat; the relation passed to Dung semantics is the surviving one, with no further preference filter.

Section 6.5 calculated the unique stable extension of each graph. What we can now compare is their extracted output families with the independently obtained PDL family:

Route	Stable-extension family	Domain-output family
PDL	not an AF calculation	$\{\{a, \neg b\}\}$
Weakest link	$\{\{P, A, C\}\}$	$\{\{a, \neg b\}\}$
Last link	$\{\{P, A, B\}\}$	$\{\{a, b\}\}$

The weakest-link route meets the PDL target on this input.

Does this make last link the wrong comparison? It does if the stated task is to reproduce PDL. A different task can favor its answer. Suppose the requirement is to choose, between these two complete outcomes, the one using d_1, d_3 rather than d_1, d_2 : their common d_1 does not distinguish them, and applying d_3 is stipulated to be more desirable than applying d_2 . The target is then $\{a, b\}$. Last link meets that requirement here, while weakest link does not. This is a supplied comparison of these outcomes, not a general ordering over sets of defaults.

The example separates priority as *order of application* from priority as *contribution to a desirable outcome*. These are the prescriptive and descriptive readings distinguished in the source [1, Sec. 3.2, Ex. 3.5, pp. 891–892]. “Descriptive” here does not mean that either answer has been validated as a description of human behavior. Nor can a label such as “legal” or “epistemic” replace the missing requirement. Chapter 5’s Scot and snoring-professor interpretations make the choice salient; they do not determine it by themselves.

Exercise 7.1 changes the rank of d_1 . The change affects which rule becomes available when, even though the two directly conflicting rules retain their ranks. This is a change in the source input, not a post hoc adjustment of an attack graph to obtain a preferred answer.

7.3 The significance of a shared rule

The agreement just obtained is fragile in an instructive way. Keep d_1, d_3 , but replace d_2 by $d'_2 : a \Rightarrow \neg b$ of rank 2. This is a second theory, with the same fact and no strict rules. Both competing conclusions now depend on a . On the source side, only d_1 is initially active; it enables d_3 and d'_2 , of which d_3 is stronger. PDL consequently gives the unique sequence (d_1, d_3) and the output family $\{\{a, b\}\}$ [45, Ex. 2, p. 7].

On the argumentation side, P, A, B remain, while C is replaced by $\hat{C} = [A \Rightarrow \neg b]_{d'_2}$. The only structural rebuts are now $B \leftrightarrow \hat{C}$. Both arguments use the same rank-1 rule d_1 , so simple weakest link gives them both strength 1. Both defeats survive; call the resulting framework F_{sh} . The unattacked P, A must be in every stable extension, and exactly one of $B, \hat{C}$ must accompany them. Selecting both violates conflict-freeness; selecting neither leaves both outsiders unattacked. Hence

$$\text{Stable}(F_{\text{sh}}) = \{\{P, A, B\}, \{P, A, \hat{C}\}\}.$$

Extracting domain conclusions gives the family $\{\{a, b\}, \{a, \neg b\}\}$. Simple weakest link retains an extra output. Its error as a proposed PDL representation is not that it loses the desired answer, but that it also permits an answer the source rejects.

Disjoint weakest link asks whether the shared rule should count against either side of this comparison. Write $D(X) = \text{DefRules}(X)$ for the set of defeasible rule identities used by an argument X . For a rebut by X at a defeasible-top subargument Y' of Y , compare

$$D(X) \setminus D(Y') \quad \text{with} \quad D(Y') \setminus D(X).$$

In the ranked fragment, the rebut survives when the minimum rank on the first side is at least the minimum on the second; an empty set has strength ∞ . All conflicting differences in our two examples are nonempty. This is the disjoint comparison of the source’s Definitions 12–14 [45, pp. 10–11].

For B and $\hat{C}$, subtraction removes the common *rule* d_1 , leaving $\{d_3\}$ against $\{d'_2\}$. The ranks are now $3 > 2$, so only $B \rightarrow \hat{C}$ survives. Its unique stable extension is $\{P, A, B\}$, whose domain conclusions match the PDL output. The cancellation concerns rule identities, not equal numerical ranks; two different rules of rank 1 would not thereby cancel. Moreover, disjoint strength is relational: which rules are ignored depends on the argument with which an argument is being compared.

The two examples together distinguish three constructions. In the Scot case, the opposing arguments share no default, so disjoint weakest link agrees with simple weakest link, not last link. In the shared-rule case, disjoint weakest link agrees with last link, not simple weakest link. Agreement on one input therefore cannot identify the policies. They use different information about an argument: all its defeasible rules, its last defeasible rules, or the rules not shared with the particular opponent.

These finite agreements do not establish a general PDL correspondence. Disjoint weakest link applied to the original ranks can fail even when all ranks are distinct; Exercise 7.7 gives a small example with a directional strict rule. The classical/SP correspondence reported by Young, Modgil and Rodrigues uses a different construction.

Start with a consistent set W of closed first-order formulas, finitely many closed normal defaults $\theta : \varphi/\varphi$, and a strict total priority. Translate each default to $\theta \Rightarrow \varphi$, take W as axioms, and use all classically valid strict inferences, with classical syntactic negation as contrary. Construct all finite arguments from these premises and rules; finitely many defaults need not yield a finite argument universe. All arguments are firm, and there are no undercuts [54, Sec. 3, p. 629]. Assume also that every translated default occurs in some freely constructed argument, as the later account explicitly requires [55, Sec. 3.2, author manuscript p. 6].

The supplied rule priority is first transformed into a *structure-preference* (SP) order, which accounts for when a rule’s prerequisite becomes constructible. This uses freely constructed arguments, without a consistency test; it need not follow the defaults actually accepted by PDL. Exercise 7.8 supplies the enumeration and applies it to both running theories. The disjoint comparison and local rebut-defeat test use this transformed order [54, Eqs. (4.13)–(4.16), p. 630, Ex. 5, p. 631]. The cited sources report that the resulting defeat graph has a unique stable extension S with

$$\text{Conc}(S) = E_{\text{PDL}},$$

where E_{PDL} is the unique classically closed PDL extension [54, Thms. 5.2–5.3, pp. 632–633]. This is a claim about full conclusion sets. Under the recorded free classical/SP construction, however, a five-default counterexample shows that, on one legal input, no stable extension has the specified PDL full conclusion set. The

reported conjunction is therefore refuted under this construction and is not used here as an established general correspondence. This does not separately disprove stable existence or uniqueness. The preceding finite calculations remain instance comparisons, not proofs of the general claim. For the developed witness and proof, see the [mathematical companion](#), §7.3, [Counterexample 7.5d](#) (approved edition of 28 September 2026, pp. 32–33).

A default blocked by PDL can nevertheless be selected by free SP enumeration. If its head conflicts with an axiom, the inconsistent prefix admits further prerequisites by classical inference and can change the order of the remaining rules. Rejecting arguments using that default during later semantic evaluation does not undo the ordering change already made.

Both classical strict inference and the SP transformation belong to the reported construction: the source itself shows that disjoint comparison of the original priorities can fail even in its classical setting [54, Ex. 3, p. 630].

Allowing tied ranks enlarges the comparison domain from total orders to total preorders. In that domain, Pardo et al. give mismatches between disjoint weakest link and PDL in both directions [45, Exs. 3, 7–8, pp. 7, 13–14]. The next chapter will use such a domain to ask a stronger question than whether one familiar link policy succeeds. Before taking that step, we can isolate exactly what a successful encoding does guarantee.

7.4 An exact encoding of the resulting graph

Keep the simple-weakest-link graph F_{sh} , including its extra non-PDL output. Can a rule program represent its stable extensions exactly? This question runs in the other direction: it starts with an AF already chosen, rather than trying to construct one from a source logic. The answer will preserve the graph’s output, including any mismatch with the intended source method.

Construction 7.1. Let $F = (A, R)$ be any finite AF, including empty frameworks and self-attacks. Choose injective atom names $\iota : A \rightarrow U$, with atom universe exactly $U = \iota[A]$. Writing $\text{Att}(a) = \{b \in A : (b, a) \in R\}$, form

$$P_F = \left\{ \begin{array}{l} \iota(a) \leftarrow \text{not } \iota(b_1), \dots, \text{not } \iota(b_t) \\ : a \in A, \{b_1, \dots, b_t\} = \text{Att}(a) \end{array} \right\}.$$

There is one rule per argument. All its attackers occur together in the conjunctive body; an unattacked argument gives a fact, with empty body. The construction is AA2LP, from abstract argumentation to logic programming [56, Sec. 3.2, Def. 11, p. 438].

Here $\text{not } x$ tests absence from a candidate set of atoms; it is not classical negation. The required program semantics is especially simple. Given $M \subseteq U$, delete every

rule having some body atom $x \in M$ under not, then erase the remaining default-negated conditions. This produces the *reduct* P_F^M . Every surviving rule is now a fact. The least model of a fact-only program is exactly the set of its fact heads, denoted $\text{LM}(P_F^M)$. The candidate is a *stable model* precisely when $M = \text{LM}(P_F^M)$. This is the usual two-valued reduct test specialized to these programs [5, Sec. 4.3.1, p. 344, paragraph before Thm. 49]. The public mathematical supplement, *Stable models of normal programs*, develops the general positive-body case, least-model construction and Nixon example. None of that additional machinery is needed to evaluate a fact-only reduct here.

Theorem 7.2. For every F , ι , U and P_F as in Construction 7.1, the maps

$$E \mapsto \iota[E], \quad M \mapsto \text{ExtOut}(M) = \{a \in A : \iota(a) \in M\}$$

are mutually inverse bijections between $\text{Stable}(F)$ and the stable-model family $\text{SM}(P_F)$.

The Handbook supplies the construction and semantic correspondence [56, Sec. 3.2, pp. 438–440, especially Cor. 2]. The key to its finite, two-valued proof is that

$$\text{LM}(P_F^{\iota[E]}) = \{\iota(a) : \text{Att}(a) \cap E = \emptyset\}. \quad (7.2)$$

Equality with $\iota[E]$ says that selected arguments have no selected attackers, while every omitted argument has a selected attacker. These are precisely conflict-freeness and stable outsider coverage [5, Lem. 14, p. 328]. Thus the translation both retains every stable extension and excludes spurious stable models. Exercise 7.3 gives the full proof in both directions, including the inverse maps.

For the already evaluated shared-rule graph the program is

$$x_P \leftarrow, \quad x_A \leftarrow, \quad x_B \leftarrow \text{not } x_{\widehat{C}}, \quad x_{\widehat{C}} \leftarrow \text{not } x_B.$$

Its stable models are exactly $\{x_P, x_A, x_B\}$ and $\{x_P, x_A, x_{\widehat{C}}\}$; each leaves precisely its own atoms as facts in the reduct. Inverse extraction recovers the two stable extensions of F_{sh} . Figure 6 follows them back to the source question: exact encoding preserves the unwanted output too.

Two details of the theorem prevent common overreadings. Atom names identify *arguments*, so different arguments with the same conclusion still receive distinct atoms. Also, the result translates finite AFs into a particular class of programs: one purely default-negative rule per argument. Its two proof directions are not a translation of arbitrary normal programs into AFs. Positive bodies and alternative rules for one head require further machinery [56, Sec. 3.2, p. 439, n. 6, Sec. 3.3].

7.5 Solutions, not only consequences

Representation is not confined to selecting beliefs. The following source-output distinctions are already present in Dung’s programme:

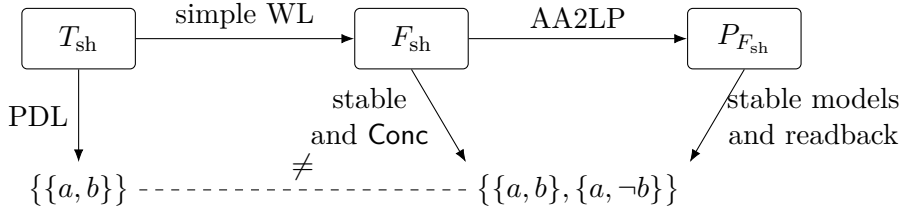


Figure 6: The shared-rule theory T_{sh} along three routes. AA2LP preserves the chosen AF’s stable extensions exactly; reading their conclusions still yields a larger family than PDL. The arrows denote constructions and output operations, not attacks.

Source	Selected object	Argumentation semantics
Default theory	A default extension	Stable, with the source construction
Normal logic program	Stable or well-founded model	Stable or grounded, respectively
Cooperative game	A stable set of payoff distributions	Stable in a domination framework
Matching problem	A complete pairing without a blocking pair	Stable in a preference-based framework

The corresponding constructions and results occur in Dung’s Sections 3–4 [5, Thms. 37, 39, 43, 49–50, pp. 336–344]. The matching construction assumes equally sized sides and complete strict preference lists. Each row has its own construction and output type.

Matching makes the change of output concrete. A complete matching assigns each participant one partner; a pair *blocks* it when both participants prefer one another to their assigned partners. Exercise 7.6 compares direct selection of unblocked matchings with a small AF whose arguments are complete candidate matchings. The construction, preferences and extraction are supplied there, so the comparison can be worked from both ends. This is a bounded whole-matching abstraction; Dung’s general construction instead uses individual possible pairs as nodes [5, Sec. 3.2, Thm. 39, pp. 337–338]. The recovered object is a partner assignment. Propositions about that assignment could be extracted subsequently, but they are not the solution concept itself.

7.6 From correspondence to principles

The comparisons locate three different kinds of revision. We can change the *source input*, as when a rule’s prerequisite or rank changes. We can change the *argumentative construction*, as when shared defaults cease to determine a rebut’s

strength. Or we can change the *extraction*, as when we intersect premises instead of their consequence sets. A correct semantic calculation does not make these changes interchangeable. AA2LP is exact after an AF has been specified; it cannot decide which of those revisions the application warrants.

The next question therefore concerns a construction method across inputs. For prioritized normal-default theories allowing tied ranks, can one attack-assignment function on the prescribed induced arguments recover exactly the entire PDL output family under stable semantics throughout a declared domain of related theories?

Two proposed constraints give the question its force. *Context independence* (P2) requires the same arguments to retain their attack status across inputs when the priority relation on their combined defaults is unchanged. *Attack closure* (P6) permits an assigned attack only when the attacker undercuts or contradicts a subargument of the target, and requires undercuts to count as attacks [45, Principle 2, p. 16, Principle 6, p. 21]. In our undercut-free fragment only the restriction to subargument contradictions remains. P6 does *not* require every such contradiction to become an attack.

P2 seeks consistent treatment of the same reasons across contexts; P6 rules out arbitrary attacks inserted merely to engineer an answer. PDL, however, chooses according to what becomes active in the whole theory. Can exact representation and both constraints be maintained by one assignment? Chapter 8 states the witness-containing domain and develops the bounded incompatibility proof [45, Sec. 6, Thm. 6.1, pp. 25–27]. Neither a failed weakest-link example nor agreement on individual skeptical queries settles that question. Its subject is the joint commitment to a construction, stable semantics, complete output-family equality and principles across inputs, not the impossibility of representing PDL by any conceivable argumentative means.

Representation and design thus return us to different uses of the same machinery. We may keep a source method and seek a faithful construction, or choose requirements and design a new method. A correspondence theorem makes the relation between routes exact. It does not choose the destination.

7.7 Further reading

The source-choice question has several research lineages. Makinson’s assumption, valuation and rule bridges [10, Secs. 1.4, 2–4] deepen the source-choice question; they should not be read as three interchangeable translations. To follow the valuation route, turn to Heyninck et al.’s Handbook chapter on conditionals [57, pp. 514–515, Secs. 3, 5]. It asks which premise-satisfying worlds a conditional should select, then examines connections and mismatches with argumentation. Its preferential-model route leads to Kraus, Lehmann and Magidor [12]; selecting plausible worlds and evaluating an attack graph are not equivalent merely because both yield defeasible conclusions.

For the source side of the priority examples, Reiter’s *A Logic for Default Reasoning* [58] explains the role of a default’s justification; Brewka’s *Reasoning about Priorities in Default Logic* [59, pp. 940–945] makes priority-sensitive application the subject of study. A normal default is neither a material implication nor a program’s default-negated condition. These are foundational readings, not interchangeable definitions of the particular PDL procedure used here.

For the reported classical/SP correspondence and its source argument, read Young, Modgil and Rodrigues [54, Secs. 3–5, pp. 629–633]. The full-conclusion claim is refuted under the recorded construction, as explained in Section 7.3. Their later account makes the rule-availability condition explicit and develops the interaction between logical structure and preference [55, Secs. 3.2–4.1, author manuscript pp. 6–10]. Keep full classical strict inference, transformed priority, all finite arguments and full conclusion extraction together when reading this claim. The correspondence reported as Theorem 4.2 by Pardo et al. [45, p. 13] does not transfer to the directional-rule instance in Exercise 7.7.

Pardo et al. provide the exact simple/disjoint definitions, the normal-default PDL procedure and examples separating their outputs [45, Secs. 3–4]. Their principle analysis and incompatibility result prepare Chapter 8 [45, Secs. 5–6]. Readers pursuing further variants should distinguish the earlier lookahead proposal from the later paper’s PDL-based attack; the revision and the lookahead subargument-closure problem are expressly noted there [45, p. 7, n. 9, pp. 23–24].

For logic programming, the accompanying public mathematical supplement *Stable models of normal programs* supplies the general reduct calculus and solved calculations. Gelfond and Lifschitz’s original stable-model paper [60] gives the semantic lineage of the candidate-dependent reduct, not the authorship of the AA2LP construction. Sá, Dvořák and Caminada’s Handbook chapter develops that translation and its more involved reverse direction [56, Secs. 3.2–3.3, pp. 438–448]. The broader three-valued correspondences distinguish stable from well-founded reasoning; our theorem covers only the stated two-valued case. The Handbook also explains why changing the semantics can break an expected correspondence: optimizing at argument level and optimizing after conclusion extraction need not coincide [56, Thm. 5 and discussion, pp. 446–447]. Follow its reference to Caminada et al.’s 2015 study [61] for this separate problem. Chapter 12 returns to representations as computational tools.

Two Handbook routes broaden the logical programme. Bochman’s *Argumentation, Nonmonotonic Reasoning and Logic* [2, Ch. 19, pp. 981–984] reconstructs argumentation through collective attack and a four-valued logical basis. Arieli, Borg, Heyninck and Straßer distinguish a logic used to construct arguments from a logic used to describe and reason about argumentation [3, Ch. 12, pp. 719–722]; their extraction discussion is the source of Exercise 7.2. For concrete alternatives, return to Handbook I.7’s assumption-based route and I.9’s deductive support–conclusion pairs [49, 13]. The former leads to Bondarenko et al.’s original argumentation-

theoretic approach to default reasoning [62]. These change what a reason contains and where it can be challenged. Similarly, DeLP’s argument construction and dialectical evaluation [50] should not be identified with encoding an already given AF as a program.

Dung’s original game and matching constructions [5, Secs. 3.1–3.2, pp. 335–338] show why the common framework is not merely another notation for logical consequence. Read them with the output type fixed: a stable set of payoff distributions and a stable matching are different solution concepts, even when both admit an argumentative representation.

For a short continuation of the source-choice question, Baumann and van der Torre vary what is held fixed and what counts as input to an AF-based inference operator [63, Secs. 3–4]. For a systems-level sequel, the supplied *Reasoning Alignment for Agentic AI* manuscript connects exact readback to belief revision and dialogue [11, Secs. 2–3, pp. 3–11]. Keep that restricted representation’s domain and the supplied version distinct from the published article [14] when following its theorem statements.

The SP exercise provides a useful way into that literature: mark the point at which freely constructing a prerequisite differs from accepting a consistent default conclusion. Then compare the two directions claimed for the classical/SP representation, whose full-conclusion requirement is obstructed, with the independently proved AA2LP theorem. In one case the construction itself must answer a source-policy question; in the other, the graph has already fixed the answers to be preserved.

7.8 Exercises

Complete public solutions are collected in the end-of-book Solutions section, under [Chapter 7](#). Exercise roles indicate the kind of work, not a requirement that every course assign every task.

- 7.1. [\[solution\]](#) *Transfer and modelling diagnosis.* In the original Scot theory, change only the rank of d_1 from 1 to 4. Compute the complete PDL output family and the stable extensions under simple weakest link, last link and disjoint weakest link. Explain why changing a rule not itself concluding b or $\neg b$ can affect the answer. Does agreement on both this and the original input prove a general representation result? An application supplies the original rules and ranks but asks only for “the best priority policy.” What information is missing?
- 7.2. [\[solution\]](#) *Proof and representation comparison.* For $K = \{u, v\}$, where $u = p \wedge q$, $v = \neg p \wedge q$, and there are no hard premises, reconstruct the repairs and the complete base-argument AF of Chapter 1, “Route 3: exact representation by base argumentation.” Determine every stable extension and recover its repair.

For $\mathcal{M} = \text{MCS}(K, \emptyset)$, prove

$$\text{Cn}\left(\bigcap_{S \in \mathcal{M}} S\right) = \text{Cn}(\emptyset), \quad \bigcap_{S \in \mathcal{M}} \text{Cn}(S) = \text{Cn}(\{q\}).$$

Why is the second consequence set strictly larger? More generally, prove

$$\text{Cn}\left(P \cup \bigcap_{S \in \mathcal{M}} S\right) \subseteq \bigcap_{S \in \mathcal{M}} \text{Cn}(P \cup S)$$

for finite K , consistent hard premises P , and $\mathcal{M} = \text{MCS}(K, P)$. Finally add a fresh atom r to the original K , keeping $P = \emptyset$. Compute the two extraction results and explain why their difference does not contradict the repair representation. Compare two further questions about $\Gamma = \{u, r\}$. Is Γ a Chapter 1 base argument? Is (Γ, q) a classical argument, where the support must be consistent, entail the specified conclusion, and have no proper subset entailing that conclusion [2, Ch. 9, Def. 4.8, p. 446]?

7.3. *[solution] Proof.* Prove Theorem 7.2 on its entire stated domain. First establish equation (7.2) for every $E \subseteq A$, without assuming stability. Then prove that every stable extension gives a stable model and that every stable model extracts to a stable extension. Establish that the maps are inverses, identifying where injective naming and the exact image universe are used. Does any step require a nonempty AF or the absence of self-attacks?

7.4. *[solution] Computation and diagnosis.* Let

$$F = (\{a, b, c, d\}, \{(a, b), (b, c), (c, b), (c, d)\}).$$

Using argument letters as distinct atom names, construct P_F and find all its stable models. A translator accidentally omits not a from the body for b . Find the additional stable model and explain its failure in the original AF. Is the erroneous program a correct AA2LP translation after deleting the edge (a, b) from the source? Explain why c then loses skeptical acceptance although no attack on c was added.

For a separate change, return to F and add (d, a) . Construct the new program, determine its complete stable-model family, and check the extracted extensions directly. Which step of the original uniqueness argument is no longer available?

7.5. *[solution] Boundary diagnosis.* Let $H = (\{u, v\}, \{(u, v), (v, u)\})$, with $\text{Conc}(u) = \text{Conc}(v) = p$. Give its stable-extension family, AA2LP stable-model family using names x_u, x_v , and family of conclusion sets. Explain the witnesses for skeptical conclusion p , and whether a particular argument is skeptical. Compare an empty AF and a singleton self-attacker concluding p . Finally, does adding a self-attack to an AF necessarily eliminate every stable extension? Test the AF with attacks $s \rightarrow t$ and $t \rightarrow t$.

- 7.6. *[solution]* *Transfer to a solution concept.* With two participants on each side, the two complete matchings are

$$M_1 = \{(m_1, w_1), (m_2, w_2)\}, \quad M_2 = \{(m_1, w_2), (m_2, w_1)\}.$$

Use complete strict preferences

$$\begin{array}{c|c} m_1 & w_2 \succ w_1 \\ \hline m_2 & w_1 \succ w_2 \end{array} \quad \begin{array}{c|c} w_1 & m_2 \succ m_1 \\ \hline w_2 & m_1 \succ m_2. \end{array}$$

Check every possible blocking pair before computing an AF. Then use nodes A_{M_1}, A_{M_2} , with an attack from A_{M_i} to A_{M_j} exactly when $i \neq j$ and M_i contains a pair blocking M_j . Determine the entire relation and every stable extension. Compare the source family with the outputs of $\text{Out}_M(E) = \bigcup_{A_{M_i} \in E} M_i$. What kind of object is extracted? What would still be needed for a general matching representation theorem?

- 7.7. *[solution]* *Counterexample and scope.* For this exercise, strict closure uses only the displayed directional rule:

$$\mathcal{K}_n = \{\top\}, \quad \mathcal{R}_s = \{s : a \rightarrow \neg b\}, \quad d_1 : \top \Rightarrow a \text{ (1)}, \quad d_2 : \top \Rightarrow b \text{ (2)}.$$

There are no ordinary premises or undercuts. Construct every argument, find every structural rebut, and apply disjoint weakest link to the original ranks. Compare the complete stable-output family with the PDL domain-output family. Does totality of the ranks suffice for agreement? Explain why this calculation alone does not test the full classical/SP construction.

- 7.8. *[solution]* * *Construction and conceptual diagnosis.* For the classical/SP construction in Section 7.3, form the SP order by repeatedly choosing the highest-priority unused default whose prerequisite is the conclusion of an argument constructible from W and the already chosen defeasible rules, using all strict inferences. Earlier choices receive higher SP priority [54, Eqs. (4.13)–(4.16), p. 630]. The construction assumes every translated default eventually becomes available to this free enumeration.

Take $W = \{\top\}$ and first the original Scot defaults, then the shared-rule variant replacing d_2 by d'_2 . List the entire enumeration and resulting SP priority in each case. Compare each list with the accepted PDL choice sequence already calculated in the main text. Why is enumeration allowed to continue after the accepted PDL sequence has stopped? Does replacing original ranks by these orders alone turn the chapter's four-argument examples into the full classical representation construction?

8 Chapter 8. Rationality postulates for defining new logics

A faithful representation preserves the answers of its source. Reasoning with defeasible or inconsistent information raises a further question: what should those answers satisfy? A selection may omit a subargument on which one of its members depends; its conclusions may conflict or omit a consequence of the strict rules. Correctly evaluating a graph does not by itself prevent these failures. We therefore formulate requirements on a reasoning method and determine which constructions meet them.

Chapter 7 separated fidelity to a source procedure from the choice of a desirable outcome. We now examine those requirements in their own right. If accepted conclusions satisfy the premises of a strict rule, may its conclusion be omitted? If the evidence stays fixed but a second route to one of its facts is added, should an old conclusion disappear? These questions move from the quality of one output to the behaviour of a construction across inputs.

Their location matters. In Figure 2, closure and consistency constrain selected arguments or extracted conclusions; construction principles constrain the route that generates the evaluated relation. Semantic principles concern the selection rule on graphs, the broader topic of Chapter 11. Keeping these levels distinct lets us identify which choice a failed test calls into question. It also exposes a deeper difficulty: individually attractive requirements may conflict. The final part proves a bounded incompatibility between exact PDL representation and two principles governing attack assignment.

8.1 What an accepted output must retain

Let F_T be the evaluated graph of a structured theory T . For an extension E , write $C_E = \text{Conc}(E)$. An *output postulate* constrains E or C_E . To say that a method satisfies it over a domain is to quantify over every input in that domain and every relevant extension, not just the output preferred by the modeller. A single violating extension refutes that universal claim. A successful example does not prove it.

Let $\text{Cl}_s(C)$ be the least set containing C and closed under the theory's strict rules. Only those rules are used: a directional rule is not silently supplemented by classical inference. A set C is *directly consistent* when there are no $\varphi, \psi \in C$ such that $\psi \in \bar{\varphi}$. Thus consistency uses the declared contrary relation, not only visible negation

symbols [46, Def. 3, p. 366]. The four requirements are

- subargument closure: $A \in E \implies \text{Sub}(A) \subseteq E$,
- strict closure: $\text{Cl}_s(C_E) = C_E$,
- direct consistency: C_E is directly consistent,
- indirect consistency: $\text{Cl}_s(C_E)$ is directly consistent.

We apply these tests to individual complete extensions, hence also to grounded, preferred and any existing stable extensions. Subargument closure concerns argument identities. The other three concern formulas. If $B = [A \rightarrow q]_r$, accepting B without A violates the first condition even when a different argument supports $\text{Conc}(A)$. Accepting p but omitting q , when $p \rightarrow q$ is strict, instead violates strict closure.

Consistency must likewise be tested on one output at a time. The alternatives $\{p\}$ and $\{\neg p\}$ may each be consistent although their union is not. And a universal statement that every extension passes a test supplies no existence guarantee: when there are no extensions, there is no output to examine. The guarded query convention of Section 6.5 and its separate no-output diagnosis remain in force.

Consider the proposed conclusion set $C = \{p, r\}$ and sole strict rule $p, r \rightarrow \perp$. Declare $\perp$ self-contrary, with no contrary pair inside $\{p, r\}$. As in Chapter 5, this is the contrary information relevant to the calculation, not a claim to print the entire ambient signature. The initial set is directly consistent. Both rule premises are present, however, so $\text{Cl}_s(C) = \{p, r, \perp\}$. The self-contrary $\perp$ makes this closure inconsistent. Direct consistency alone has missed an incompatibility exposed by the strict rules. No explosion rule is needed. This is a diagnostic proposed output, not an assertion that an ASPIC+ construction satisfying its rationality hypotheses produces it.

The relation between the tests is exact: indirect consistency implies direct consistency, and strict closure together with direct consistency implies indirect consistency. Under strict closure, the two consistency requirements are therefore equivalent. The displayed witness shows why the closure hypothesis matters. Exercise 8.1 proves these implications, constructs the other boundary cases, and develops the finite closure procedure with its termination and leastness proof.

Positive rationality results depend on how arguments are constructed. The ASPIC+ results for complete extensions use well-defined systems and reasonable argument orderings: consistent strict closure of the axioms, suitable transposition or contraposition of strict inference, well-formed contraries, and the additional classicality condition for the consistent-premise variant are among the source's requirements [46, Defs. 12 and 18, pp. 369–372, Thms. 12–15 and Cor. 17, p. 373]. These hypotheses are not conferred by drawing a defeat graph. The corrected preference definitions in Section 5.4 must also be retained [48, pp. 109–110]. The full conditional proofs are an onward reading, not a guarantee attached to every small graph considered here.

8.2 When the same postulates permit different answers

Return to the snoring professor in Exercise 5.5. The arguments S_0, S_1, S_2 proceed from snoring through misbehavior to denial of access; P_0, P_1 proceed from professor status to the contrary conclusion. The respective rule ranks are 1, 3 on the first chain and 2 on the second. Chapter 5 found exactly one defeat under each policy: $P_1 \rightarrow S_2$ for weakest link and $S_2 \rightarrow P_1$ for last link.

Applying the same stable test as in Section 6.5 gives the unique extensions

$$E_{WL} = \{S_0, S_1, P_0, P_1\}, \quad E_{LL} = \{S_0, S_1, S_2, P_0\}.$$

Each contains its selected arguments' subarguments. There are no strict rules, so each conclusion set is strictly closed. Neither contains both *accessDenied* and $\neg$ *accessDenied*, the only relevant contradictory pair. Both direct and indirect consistency hold. Thus all four postulates pass, although the two policies disagree about access.

This is not a defect in the postulates: they exclude certain unacceptable outputs without ordering all remaining ones. Weakest link treats the early defeasible classification as a vulnerability of the whole chain; last link lets the final rule settle this conflict. To choose between them we still need the application's purpose. Chapter 7's distinction between a priority as an application order and as a contribution to an outcome is relevant precisely because consistency does not settle it. Exercise 8.2 completes the output check without reconstructing the arguments.

8.3 Adding a second route to an existing fact

A different requirement concerns change. In the Sherlock example of Dung's axiomatic analysis, three suspects are P_1, P_2 and a child S . Abbreviate their innocence statements by i_1, i_2, i_s , respectively, taking each literal and its explicit negation as a contradictory pair. The evidence already contains i_s , and the strict rules are

$$r_1 : i_1, i_s \rightarrow \neg i_2, \quad r_2 : i_2, i_s \rightarrow \neg i_1, \quad r_3 : i_1, i_2 \rightarrow \neg i_s.$$

Presumptions $d_1 :\Rightarrow i_1$ and $d_2 :\Rightarrow i_2$, with $d_1 \prec d_2$, favor the second suspect's innocence. Now add only $d :\Rightarrow i_s$, retaining the original evidence [64, Ex. 1.2, pp. 108–109]. The addition supplies a defeasible route to an undisputed fact. It does not replace that fact in the theory.

One can nevertheless replace an occurrence of the fact inside a particular derivation by the new defeasible argument for i_s . The resulting argument is weaker in the source's structural sense: it depends on an additional defeasible step. That a weaker argument becomes available is different from the evidence becoming weaker. The intended redundancy requirement asks whether this alternative route should change the method's belief-set outputs.

The question is about outputs, not the number of available arguments. If the weaker route introduces an alternative stable output that lacks $\neg i_1$, a formerly skeptical conclusion about the first suspect can cease to be skeptical even though the evidence is unchanged. Exercise 8.3 makes this possibility precise through fully specified finite diagnostic graphs and compares it with an addition that preserves the old output. Their arguments are selected from the story and their arrows are stipulated; they are not claimed to be the complete graphs generated by corrected ASPIC+. The exercise separates calculating the effect of an assigned relation from justifying that assignment.

Dung’s *irrelevance of redundant defaults* demands equality of both the stable and the complete belief-set families before and after adding $d_f \Rightarrow f$ for an already given fact f , within the source’s stipulated domain [64, Def. 4.6, p. 117]. A changed skeptical query would refute that stronger equality if the graphs came from an assignment in that domain. Agreement on just this query could not prove it.

There is also a construction-level question. *Attack monotonicity* requires a strengthening of an attacker to preserve its outgoing attacks, and a weakening of a target to retain attacks already directed at it. These are conditions on the source’s recursive strengthening relation, not monotonicity of nonmonotonic consequence [64, Defs. 4.4–4.5, p. 116] [65, Defs. 11–12, p. 9]. Exercise 8.4 distinguishes this local test from the full-output calculation. A failure of an edge condition and a change in skeptical belief are different findings, even when one story motivates both.

For a sensible source domain also closed under these redundant-default additions, Dung proves that attack monotonicity and context independence imply redundant-default irrelevance [64, Def. 4.7 and Thm. 4.1, pp. 117–118]. Context independence fixes attacks between shared arguments when the priorities relevant to that pair are unchanged. We will state it precisely below. This is a conditional guarantee about one assignment across inputs, not a theorem proved by the finite Sherlock calculation. The historical comparisons in those papers must also be distinguished from the corrected preference definitions of Chapter 5.

8.4 Independent additions and the existence of outputs

If a new body of information is unrelated to the old one, a natural expectation is that it should not change conclusions in the old vocabulary. At graph level, genuine disconnection makes this expectation precise, but stable semantics introduces an existence condition.

Proposition 8.1. Let $F = (A, R)$ and $H = (B, S)$ be finite AFs with disjoint argument sets, and let $F \sqcup H = (A \cup B, R \cup S)$. Then

$$\text{Stable}(F \sqcup H) = \{E_F \cup E_H : E_F \in \text{Stable}(F), E_H \in \text{Stable}(H)\}.$$

Equivalently, E is stable in the union exactly when $E \cap A$ and $E \cap B$ are stable in the respective components.

The reason is the locality of both parts of stability. An internal conflict belongs to one component, and every attack on an outsider must come from the same component because there are no cross edges. Exercise 8.5 gives the complete proof and the projection argument.

If the added component has a stable extension, projection of the combined argument family onto A recovers $\text{Stable}(F)$. With a fixed conclusion map and disjoint conclusion vocabularies, the corresponding output-family projection also agrees. If the added component has no stable extension, however, the product family is empty: the old graph is unchanged, but no combined output remains from which to read its conclusions. Both guarded queries then return false and the separate *no-output* diagnosis explains their failure.

Exercise 8.5 proves the projection statement and compares two independent rule blocks with an added self-attacker. Its calculation exposes why semantic existence belongs to an independence claim, even when the absence of cross edges is certain.

Disjoint vocabularies at source level do not automatically produce disconnected graphs. Classical strict inference can, for example, use a contradiction in one block to derive a formula about another through explosion. The resulting constructed arguments need not respect the supposed separation [66, Sec. 4, pp. 786–789]. This construction problem and the graph-level loss of stable existence are different. The small product result proves neither the general noninterference nor the crash-resistance postulates; their hypotheses and alternative treatments belong to the reading route below.

8.5 When representation and principles conflict

Failure of a particular policy leaves open the possibility of a better one. We now ask whether *any* attack assignment can recover the PDL outputs while treating the same reasons consistently across inputs. Two modest-looking requirements turn this question into an incompatibility theorem. One limits where an attack may be placed; the other limits what information may determine its presence.

Continue with the normal-default, domain-literal fragment of Section 7.2. In the shared notation an input is

$$T = (\mathcal{R}_s, \mathcal{R}_d, \preceq, \mathcal{K}_n).$$

The default set $\mathcal{R}_d$ is finite, $\preceq$ is a total preorder on it, and the evidence $\mathcal{K}_n$ has consistent strict closure. There are no ordinary premises. Ties are allowed; larger ranks mean stronger defaults. Write $\text{PDL}(T)$ for the entire family of outputs of the source procedure: one highest-ranked active default is chosen at a time, subject to consistency after strict closure [45, Defs. 2–3, p. 8, Rem. 1 and Def. 15, pp. 12–13].

Let $\text{Arg}(T)$ contain exactly the finite arguments induced by the premises and rules, using Chapter 5’s construction. An *attack assignment* is one function att on a

domain $\mathcal{D}$, assigning to every input a relation $\text{att}(T) \subseteq \text{Arg}(T) \times \text{Arg}(T)$ that never targets a strict argument. Here strict means $\text{DefRules}(A) = \emptyset$, not merely that its last inference is strict. The assigned relation is evaluated directly; no further preference filter is applied [45, Defs. 4–8, p. 9]. For $F_T = (\text{Arg}(T), \text{att}(T))$, exact representation requires

$$\{\text{Conc}(E) : E \in \text{Stable}(F_T)\} = \text{PDL}(T) \quad \text{for every } T \in \mathcal{D}. \quad (8.1)$$

Every source output must be recovered and every stable output must be a source output. The comparison is between families of conclusion sets, not between PDL outputs and sets of arguments [45, Defs. 9–10, p. 10].

The source calls $\mathcal{D}$ *sensible* when it is a nonempty class of consistent inputs and, whenever it contains an input, also contains every consistent change of its evidence with the rules and priorities fixed [45, Def. 7, p. 9]. This is closure under evidence changes, not under rule deletion. Our theorem will explicitly require all three witness inputs below to belong to the domain.

Context independence (P2). For any $T, T' \in \mathcal{D}$ and shared arguments $A, B \in \text{Arg}(T) \cap \text{Arg}(T')$, put $U = D(A) \cup D(B)$, where $D(X) = \text{DefRules}(X)$. If the restrictions of $\preceq_T$ and $\preceq_{T'}$ to U coincide, then

$$(A, B) \in \text{att}(T) \iff (A, B) \in \text{att}(T').$$

The same arguments, with the same relative priorities among their defaults, retain their attack status [45, Principle 2, p. 16]. P2 permits dependence on the pair’s own reasons and priorities, but excludes dependence on rules outside that pair.

Attack closure (P6). For every $T \in \mathcal{D}$, an assigned attack $(A, B) \in \text{att}(T)$ requires that A undercut B or contradict some subargument of B . Conversely, every undercut must be an assigned attack [45, Principle 6, p. 21]. Here contradiction means opposite domain literals at the indicated conclusions; the subargument may be B itself. Undercutting concerns a defeasible rule’s applicability [45, Def. 11, p. 10].

P6 does not require every contradiction to become an attack. Nor is its eligibility condition simply ASPIC+’s restricted-rebut clause. In the present PDL fragment there are no abnormality atoms giving undercuts, so only the restriction to subargument contradictions is operative. This limits the assigned relation; it is not closure of an accepted argument set or closure under strict inference.

Take evidence $\mathcal{K}_n = \{\top\}$, no strict rules, and four defaults:

$$\begin{aligned} d_1 : \top \Rightarrow a \quad (1), \quad d_2 : \top \Rightarrow b \quad (1), \\ d_3 : a \Rightarrow \neg b \quad (2), \quad d_4 : b \Rightarrow \neg a \quad (2). \end{aligned}$$

Let T contain all four; let T_1 delete only d_4 and T_2 delete only d_3 . Each fragment inherits the priority relation on its remaining defaults [45, Exs. 11–13, pp. 25–26].

First determine the source outputs. In T , only d_1, d_2 are initially active. Choosing d_1 supplies a and activates rank-2 d_3 , which must precede rank-1 d_2 . Its conclusion $\neg b$ then blocks d_2 . Beginning with d_2 symmetrically gives $b, \neg a$. PDL cannot first collect both rank-1 conclusions: it must reconsider active rules after each choice.

In T_1 , beginning with d_2 no longer enables d_4 . Rule d_1 can therefore supply a ; the already present b blocks d_3 . The branch beginning with d_1 is unchanged. The deletion in T_2 has the symmetric effect. Put

$$S_1 = \{\top, a, \neg b\}, \quad S_2 = \{\top, b, \neg a\}, \quad S_3 = \{\top, a, b\}.$$

Unlike the abbreviated Chapter-7 comparisons, we retain the common $\top$ in every output here. The entire calculation is

Input	All permitted rule-choice sequences	PDL family
T	$(d_1, d_3), (d_2, d_4)$	$\{S_1, S_2\}$
T_1	$(d_1, d_3), (d_2, d_1)$	$\{S_1, S_3\}$
T_2	$(d_2, d_4), (d_1, d_2)$	$\{S_2, S_3\}$

The two initial choices exhaust the possibilities; each continuation is forced. Thus S_3 is permitted in each fragment but excluded from the full theory, before any attack has been assigned.

The induced arguments are exactly

$$\begin{aligned} P &= \langle \top \rangle, \\ A_1 &= [P \Rightarrow a]_{d_1}, \quad A_2 = [P \Rightarrow b]_{d_2}, \\ A_3 &= [A_1 \Rightarrow \neg b]_{d_3}, \quad A_4 = [A_2 \Rightarrow \neg a]_{d_4}. \end{aligned}$$

There are no further premises or rules to generate more arguments. Consequently

$$\begin{aligned} \text{Arg}(T) &= \{P, A_1, A_2, A_3, A_4\}, \\ \text{Arg}(T_1) &= \{P, A_1, A_2, A_3\}, \\ \text{Arg}(T_2) &= \{P, A_1, A_2, A_4\}. \end{aligned}$$

These are the source's derivations with the premise argument explicit [45, Ex. 6, p. 9]. The strict P cannot be targeted, and under P6 its conclusion cannot attack another argument. It is isolated and belongs to every stable extension.

Each displayed conclusion has exactly one argument. Its output therefore fixes a unique argument preimage:

$$E_1 = \{P, A_1, A_3\}, \quad E_2 = \{P, A_2, A_4\}, \quad E_3 = \{P, A_1, A_2\}.$$

Exact capture would require

$$\begin{aligned} \text{Stable}(F_T) &= \{E_1, E_2\}, \\ \text{Stable}(F_{T_1}) &= \{E_1, E_3\}, \\ \text{Stable}(F_{T_2}) &= \{E_2, E_3\}. \end{aligned}$$

This inference uses the unique conclusion witnesses on these inputs; it is not valid for arbitrary constructions. The floating conclusion in Exercise 7.5 supplies the earlier warning.

P6 permits only the following attack positions in T :

$$(A_1, A_4), (A_2, A_3), (A_3, A_2), (A_3, A_4), (A_4, A_1), (A_4, A_3).$$

For example, A_3 's conclusion $\neg b$ contradicts the conclusion of A_4 's subargument A_2 , although the final conclusions of A_3, A_4 are not contradictory. In T_1 , only $A_2 \rightarrow A_3$ and $A_3 \rightarrow A_2$ are permitted; in T_2 , only $A_1 \rightarrow A_4$ and $A_4 \rightarrow A_1$. This is a list of possibilities, not an instruction to include them all.

Theorem 8.2. For every sensible domain $\mathcal{D}$ containing T, T_1, T_2 above, no attack assignment on the prescribed induced arguments can jointly satisfy exact PDL capture under stable semantics, context independence (P2), and attack closure (P6) [45, Thm. 6.1 and proof, pp. 26–27].

Proof. Suppose one assignment satisfies all three requirements. Exact capture on T_1 requires E_3 to be stable there: the source output S_3 must be recovered and E_3 is its only possible argument preimage. Its sole outsider is A_3 . Stable outsider coverage requires an attack from E_3 on A_3 , and P6 permits only A_2 to supply it. Hence

$$(A_2, A_3) \in \text{att}(T_1).$$

Similarly, exact capture on T_2 makes E_3 stable with sole outsider A_4 . P6 permits only A_1 to attack that outsider, so

$$(A_1, A_4) \in \text{att}(T_2).$$

Both pairs also exist in T . Their sets of defaults are exactly

$$D(A_2) \cup D(A_3) = \{d_1, d_2, d_3\}, \quad D(A_1) \cup D(A_4) = \{d_1, d_2, d_4\}.$$

The relative priorities on the first set are unchanged between T_1, T , and on the second between T_2, T . P2 therefore transports both attacks to the full input:

$$(A_2, A_3), (A_1, A_4) \in \text{att}(T).$$

No member of $E_3 = \{P, A_1, A_2\}$ contradicts a subargument of another member, and there are no undercuts. P6 therefore makes E_3 conflict-free in F_T . Its only outsiders are A_3, A_4 , both covered by the transported attacks. Thus E_3 is stable and gives $\text{Conc}(E_3) = S_3$. But $S_3 \notin \text{PDL}(T)$, contradicting equation (8.1). $\square$

Figure 7 isolates the proof's decisive move. Exactness first requires the fragments to recover S_3 , forcing two attacks; it then forbids the full theory from returning S_3 . P2 transports the attacks, while P6 prevents an internal attack from destroying

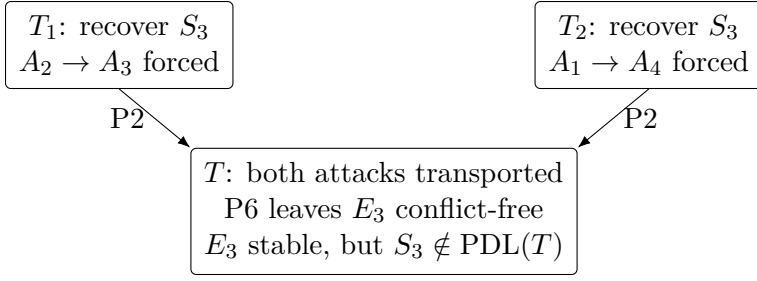


Figure 7: Fragment obligations become an unwanted full-theory output. The displayed argument attacks are forced by exact stable capture and P6; the connecting arrows denote their transport by P2. This is a proof schematic, not a drawing of the complete attack graphs.

E_3 . No particular link policy was assumed: the witness excludes every assignment meeting the stated contract.

The source procedure explains the tension. A stronger continuation interrupts the collection of a, b in T , but its deletion allows that output in a fragment. The compared argument pair does not use the deleted rule, so P2 cannot let its attack status respond to that change. Fidelity to a procedure inspecting all active rules conflicts here with the local invariance of attacks.

8.6 What follows for the design of a logic?

The theorem is not an impossibility of representing PDL by any argumentative means. It fixes the induced arguments, the domain, stable semantics, full-family extraction and two principles. P2 and P6 alone are compatible: on the undercut-free fragment, the empty attack assignment satisfies both. It is their conjunction with exact PDL capture on the stated domain that fails.

If PDL's application order is the specification, a context-dependent construction may be appropriate. If independence of the same reasons from their surroundings has priority, the target logic may instead change. The source leaves that choice open [45, p. 27, discussion after Thm. 6.1]. Changing the domain, arguments, semantics or preservation demand also changes the question. A restriction must exclude the obstruction; an encoding with additional arguments must explain their role and provide an extraction. None of these changes supplies a positive theorem merely by escaping the negative one. Exercise 8.7 verifies context-dependent capture on the three displayed inputs and identifies exactly why that is still short of a domain-wide representation.

The earlier examples impose complementary obligations. The professor requires a purpose beyond consistency. Sherlock requires a reason for sensitivity to an added route. Independent additions require both a construction that respects separation

and attention to whether the chosen semantics produces any output. A useful design therefore states what its requirements exclude, what they leave open, and which guarantees have actually been proved.

Chapter 9 returns from static inference to the interaction of Chapters 3–4: an extension postulate is not a rule governing a participant’s next move. Chapter 10 formalizes the earlier burdens and practical comparisons, while Chapter 11 compares semantic principles. The common discipline is to identify the object being constrained before asking whether the constraint is reasonable.

8.7 Further reading

Begin with Caminada’s *Rationality Postulates: Applying Argumentation Theory for Non-Monotonic Reasoning* in Handbook I.15 [66, Secs. 3–4]. It explains why selecting acceptable arguments must yield meaningful logical conclusions, and then extends the question from consistency to noninterference and contamination. Its account leads back to Caminada and Amgoud’s foundational formulation [67]. Read the tandem example as another way to test respect for a strict constraint, and compare the requirement it reveals with this chapter’s distinction between output conditions and principles governing a construction across inputs.

For conditional positive results, Modgil and Prakken’s *Abstract Rule-Based Argumentation* gives the broader construction setting [47, Sec. 3]; their original general account locates the hypotheses behind subargument closure, strict closure and consistency [46, Defs. 12 and 18, Secs. 4.2–4.3]. Use the 2018 corrigendum when following preference-based guarantees [48]. An older presentation of lifting in a survey does not supersede that correction. This route asks where the hypotheses do work in the proof, not simply which postulates appear in a framework’s description.

Dung’s 2016 axiomatic analysis develops the strengthening, redundancy and context conditions behind the Sherlock question [64, Defs. 4.4–4.7 and Thm. 4.1, pp. 116–118]. Dung and Thang’s subsequent *Fundamental Properties of Attack Relations in Structured Argumentation with Priorities* moves toward minimal attack removal and canonical assignments [65, pp. 1–3, Secs. 5–7]. The additional domain, regularity and existence assumptions matter: the programme is not an unrestricted claim that one canonical relation is available for every proposed theory. Keep its historical preference comparisons distinct from a new calculation with the corrected Chapter-5 lifting.

A substantial alternative starts from deductive arguments rather than defeasible rule trees. Besnard and Hunter’s Handbook I.9 survey [13] and Arieli et al.’s *Logic-Based Approaches to Formal Argumentation* [3, Ch. 12, Sec. 2.3] lead to Gorogiannis and Hunter’s study of postulates for classical-logic arguments [68]. Use Exercise 5.7 to recall the minimal consistent support–conclusion object, then compare how alternative attack relations affect the logical content of extensions. The rationality question survives the change of construction, but its hypotheses are not those of

the rule-only priority examples.

The positive-design literature also revisits the sufficient conditions themselves. Through Handbook I.6’s discussion [47, Sec. 3.3.2], Dung and Thang’s *Closure and Consistency in Logic-Associated Argumentation* [69] offers an alternative analysis without preferences and connections to assumption-based argumentation. For the restricted-versus-unrestricted-rebut question, follow Caminada’s afterword [66, pp. 793–794] to Heyninck and Straßer [70]. That route changes how an argument can challenge another, rather than treating a different extension semantics as the only design choice.

For noninterference, distinguish two responses described in Handbook I.15 [66, pp. 790–791]. Wu and Podlaszewski remove inconsistent arguments before evaluation [71]; the result discussed there assumes an empty rule-strength ordering. Grooters and Prakken instead investigate restrictions on construction and relevance [72], with different, weaker properties established. These are not interchangeable guarantees. In particular, neither route should be summarized merely as deleting a disconnected self-attacker: the Handbook’s source-level problem involves the reach of strict inference across apparently unrelated information.

Baumann’s *On the Nature of Argumentation Semantics* [73, pp. 839–841, Sec. 2] gives the existence boundary a broader setting. It distinguishes existence and uniqueness from which families a framework can express and how it can be replaced in a surrounding context. Van der Torre and Vesic’s *The Principle-Based Approach to Abstract Argumentation Semantics* [74] leads onward to Chapter 11’s comparison of selection functions, a different level from this chapter’s output and assignment requirements.

Finally, read Pardo et al.’s principles and three-input proof with the exact domain and extraction fixed [45, Secs. 5–6, especially pp. 25–27]. Their later discussion explores parallel default application, auxiliary arguments and weaker context independence [45, Secs. 7, 9, pp. 27–28, 30–31]. These change different commitments. The bounded obstruction motivates the alternatives but does not certify any of them as an exact replacement representation.

The three reading routes answer different questions. The output-postulate literature asks what accepted reasons must preserve; the strengthening and redundancy literature asks which changes should matter; the PDL obstruction asks whether a preservation contract and construction principles can coexist. Comparing the assumptions of those results is a substantive task of logical design. It is not resolved by collecting their postulate names into one list.

8.8 Exercises

Complete public solutions are collected in the end-of-book Solutions section, under [Chapter 8](#). Exercise roles indicate the kind of work, not a requirement that every course assign every task.

- 8.1. *[solution] Calculation and proof.* Use the relevant contrary pairs $p, \neg p$ and $\perp, \perp$, with no other contrary pair in the displayed sets. For each row, calculate the strict closure and test strict closure, direct consistency and indirect consistency:

C	strict rules
$\{p\}$	$p \rightarrow q, q \rightarrow t$
$\{p, r\}$	$p, r \rightarrow \perp$
$\{p, \neg p\}$	none.

Prove that indirect consistency implies direct consistency and that strict closure together with direct consistency implies indirect consistency. Use the rows to explain why direct consistency alone, strict closure alone, and indirect consistency alone do not imply all the other conditions.

Now give a finite procedure for $\text{Cl}_s(C)$, where $C \subseteq U$ and all rule bodies and heads lie in the fixed finite universe U . Empty rule bodies are allowed. Prove termination, closure of the result and leastness among closed supersets. Bound the number of strict increases. Finally, explain why none of these conclusion-set tests establishes subargument closure. Use S_2, S_1, S_0 from the professor case to make the missing obligation explicit.

- 8.2. *[solution] Interpretation and modelling.* Use the five arguments and two defeat relations already calculated in Exercise 5.5. Prove that E_{WL} and E_{LL} in Section 8.2 are the unique stable extensions of their respective graphs. Give their complete conclusion sets and check all four output postulates. Does a rule to select only outputs passing those four tests decide the access question? State what additional modelling choice remains.
- 8.3. *[solution] Controlled change and computation.* Use the following selected-argument diagnostics for the Sherlock story in Section 8.3. They omit the explicit fact argument and other composites. The arrows are stipulated, already evaluated relations; apply no further preference filter. They are not asserted to be the graphs generated by corrected ASPIC+. The conclusion map is

X	A_1	A_2	N_1	N_2	N'_1
$\text{Conc}(X)$	i_1	i_2	$\neg i_2$	$\neg i_1$	$\neg i_2$.

Here N_1 uses A_1 and fact i_s , while N_2 uses A_2 and that fact. Argument N'_1 replaces the fact occurrence in N_1 by its new defeasible route; the original evidence remains present in the story.

Before the addition, take

$$V_0 = \{A_1, A_2, N_1, N_2\}, \quad R_0 = \{(N_2, A_1), (N_2, N_1)\}.$$

For the enlarged universe $V_1 = V_0 \cup \{N'_1\}$, compare

$$R_1^{\text{bad}} = R_0 \cup \{(N_2, N'_1), (N'_1, N_2)\}$$

with

$$R_1^{good} = R_0 \cup \{(N_2, N'_1)\}.$$

Write $F_0 = (V_0, R_0)$ and $F_1^{good/bad} = (V_1, R_1^{good/bad})$. Find every stable extension of $F_0, F_1^{good}, F_1^{bad}$, proving exhaustiveness. Extract every conclusion set, then classify the fixed query $\neg i_1$ as credulous and skeptical before and after each addition. Identify a directly inconsistent output. Why does a changed query diagnose a failure of family invariance, whereas an unchanged query would not prove that invariance? Does this calculation establish that corrected ASPIC+ generates the bad graph?

- 8.4. *[solution] Conceptual diagnosis.* For N_1 a strengthening of N'_1 , the two attack-monotonicity clauses require, for every target or attacker B ,

$$N'_1 R B \implies N_1 R B, \quad B R N_1 \implies B R N'_1.$$

In a separate construction diagnostic, N'_1 weakens N_1 , $N'_1 \rightarrow A_2$ is present and $N_1 \rightarrow A_2$ is absent. Identify the precise attack-monotonicity clause that fails. What does the other clause require if $N_2 \rightarrow N_1$? Explain why these facts alone do not calculate the changed skeptical query in Exercise 8.3. Distinguish weakening an argument by replacing a fact occurrence from removing that fact from the evidence.

- 8.5. *[solution] Proof and boundary calculation.* Prove Proposition 8.1 in both directions. Show that if $\text{Stable}(H) \neq \emptyset$, projection of the combined family onto A is exactly $\text{Stable}(F)$. Does this conclusion require $\text{Stable}(F) \neq \emptyset$? When is a comparison of skeptical support based on existing outputs available?

For a concrete application, fact p and strict rule $p \rightarrow q$ give attack-free arguments P, Q . An independent fact r with $r \rightarrow s$ gives attack-free R, S . No rules, contraries or priorities connect the blocks. Combine them, then replace the added block by a singleton self-attacker z . Give the families, the Boolean status of query q , and the separate no-output diagnosis. Explain why disjoint source vocabularies alone are insufficient to invoke the proposition.

- 8.6. *[solution] Proof reconstruction.* Reconstruct the proof of Theorem 8.2 by answering the following questions.

- Why can the full input not obtain both a, b , although both corresponding defaults are initially active? Which deletion allows that output in each fragment?
- Which direction of exact family equality forces each fragment attack? Identify the unique preimage and the P6 restriction used.
- Give the precise default set for each P2 comparison. How does P6 establish conflict-freeness of the unwanted full-theory extension, and which direction of exactness does that extension violate?

- (d) Does sensibility alone guarantee the two fragments are in the domain? Does P6 demand all six permitted attacks? Does a proof for one selected link policy establish this theorem?

8.7. *[solution] Transfer and counterexample.* An implementation retains S_1, S_2 for T but also returns S_3 . Would checking credulous and skeptical answers for each individual domain literal $a, b, \neg a, \neg b$ detect this error? Use the entire families to justify your answer.

Now allow context dependence on the three witness inputs. Assign

$$\begin{aligned} Q_0 &= \{(A_3, A_2), (A_3, A_4), (A_4, A_1), (A_4, A_3)\} && \text{to } T, \\ Q_1 &= \{(A_2, A_3), (A_3, A_2)\} && \text{to } T_1, \\ Q_2 &= \{(A_1, A_4), (A_4, A_1)\} && \text{to } T_2. \end{aligned}$$

Keep the full argument universes, including isolated P . Verify P6 and determine every stable extension. Identify the P2 comparisons that fail. Does this construction establish exact PDL capture on a sensible domain containing the three inputs?

8.8. *[solution] Design and research extension.* For each proposal, identify what changes in the representation contract and what further evidence is needed:

- (a) Retain P2 and P6, but compare only individual literal queries rather than entire output families.
- (b) Allow context dependence and use the graphs from Exercise 8.7.
- (c) Restrict the source domain to strict total priorities because the witness contains ties.
- (d) Keep the domain and source outputs, but change the induced arguments or replace stable semantics.

Why do the professor and Sherlock examples remain relevant even after this incompatibility has been resolved?

9 Chapter 9. Dialogue: public procedure and developing reasons

Chapter 3 distinguished asking for a reason from offering one in opposition. Chapter 4 followed the resulting public responsibilities through the custody exchange. We now make those responsibilities exact, returning from the inferential methods of Chapters 5–8 to participants, targets and replies. An attack graph can retain conflict while losing which request awaits an answer.

The early distinction between a permitted reply and a useful one also returns. An exact protocol will support a finite choice between continuations; a second ledger will show how reasons remain unfinished; and a semantic game will reuse Chapter 6’s defence to characterize grounded acceptance. Public legality, purposeful continuation, a request’s completion and a winning strategy answer different questions about interaction.

9.1 What may be said?

A dialogue protocol specifies moves and state changes. Write

$$\Pi = (Ag, M, S, s_0, \text{Legal}, \delta, W) :$$

agents, instantiated moves, states, initial state, legal-next-move function, transition function and, when supplied, an outcome condition. Legality tells us which transition is available. An outcome condition says when the resulting record answers a further question. The former need not supply the latter.

Our public protocol is a restricted construction based on Fatio’s five locutions and six combination rules [75, Sec. 3, PDF pp. 4–5]. Let Ag and C be finite sets of agents and opaque claim labels. A support unit is a nonempty finite set $\Phi \in J = \mathcal{P}(C) \setminus \{\emptyset\}$. Use the tagged disjoint union $U = C \dot{\cup} J$ for targets: the claim P and the unit $\{P\}$ are different objects. A supplied support unit is not automatically a verified inference.

For agents a, b , claim c , target u , support Φ and polarity $\sigma \in \{+, -\}$, the moves are

$$\begin{aligned} \text{assert}(a, c), \quad & \text{question}(b, a, u), \quad & \text{challenge}(b, a, u), \\ \text{justify}(a, \Phi \vdash^\sigma u), \quad & & \text{retract}(a, u). \end{aligned}$$

The state retains active assertions, active justifications and recorded challenges, together with a pending immediate-reply field. Public claim commitments Com_i are the union of agent i ’s active assertion labels and the premises in its active justifications. The response ledger $\text{DOS} \subseteq Ag \times U \times \{+, -\}$ records support or opposition obligations. Neither Com nor DOS alone is a complete executable state: active witnesses and the pending field affect legality.

The following exact local rules govern the next move. When a reply is pending, it is the only permitted form; otherwise use the applicable rule below. Questions and challenges require distinct participants.

Rule	Local condition
CR1	Assert a claim when no immediate reply is pending.
CR2	A question or challenge targets the other speaker's still-active assertion or whole support unit. It requires a positive reply under CR3.
CR3	The targeted speaker immediately justifies the same target positively, with some nonempty support unit.
CR4	A question or challenge can instead target a prior challenge by the other speaker. It requires that challenger's opposing reply under CR5.
CR5	That challenger immediately justifies the same target negatively, with some nonempty support unit.
CR6	With no reply pending, a speaker may retract its still-active assertion or whole support unit, of either justification polarity.

If both CR2 and CR4 witnesses exist, CR2 has priority. No oldest-obligation or strategic selection rule is otherwise imposed. In particular, a pending CR3/CR5 reply cannot be evaded by a fresh assertion or retraction.

The public effects are equally explicit. Assertion adds its active record and $(a, c, +)$ to DOS. A question opens the appropriate pending reply but has no direct DOS effect. A challenge does the same and adds the challenger's opposition obligation $(b, u, -)$. Justification discharges the matching requested obligation (a, u, σ) , adds (a, Φ, σ) , records the active justification and clears the pending reply. Its premises now contribute to Com_a . Retraction withdraws the matching active assertion or all matching support units and removes the obligation for that target of either sign. Recompute commitments from the remaining active records. A premise still supplied elsewhere remains a commitment; retracting an assertion does not retract a recorded challenge.

Typed whole units, active targets, CR2 priority and symmetric unit retraction are local conventions. The response discharge also changes Fatio: the source's positive justification retains the claim obligation and adds an argument obligation, whereas our matching requested obligation is discharged first [75, Sec. 4.1, pp. 6–7]. This public construction is therefore more than an omission of private beliefs. It checks neither full belief–desire–intention preconditions nor the truth or validity of supplied support. Exercise 9.2 compares the source and local choices.

9.2 A six-turn public record

Let the agents be A, B, C . Write χ for the claim that the Brigade Restaurant is excellent. P says that A had a great meal, Q that the vegetarian food is awful, and

S that this is a reason against excellence. All remain opaque to the protocol. For a compact ledger write $D_A = (A, \{P\}, +)$ and $D_B = (B, \{Q, S\}, -)$ for the support-unit obligations, and $K_A = (A, \chi, +)$, $K_B = (B, \chi, -)$ for the claim obligations. Agent C 's commitment store stays empty. The pending field, when nonempty, names the required speaker and polarity; the target throughout is χ .

Turn	Move and rule	Com _A ; Com _B	DOS	Pending
0	Start	$\emptyset; \emptyset$	$\emptyset$	none
1	<i>assert</i> (A, χ) [CR1]	$\{\chi\}; \emptyset$	$\{K_A\}$	none
2	<i>challenge</i> (B, A, χ) [CR2]	$\{\chi\}; \emptyset$	$\{K_A, K_B\}$	$A, +$
3	<i>justify</i> ($A, \{P\} \vdash^+ \chi$) [CR3]	$\{\chi, P\}; \emptyset$	$\{D_A, K_B\}$	none
4	<i>question</i> (C, B, χ) [CR4]	$\{\chi, P\}; \emptyset$	$\{D_A, K_B\}$	$B, -$
5	<i>justify</i> ($B, \{Q, S\} \vdash^- \chi$) [CR5]	$\{\chi, P\}; \{Q, S\}$	$\{D_A, D_B\}$	none
6	<i>retract</i> (A, χ) [CR6]	$\{P\}; \{Q, S\}$	$\{D_A, D_B\}$	none

The certificate is $(CR1, CR2, CR3, CR4, CR5, CR6)$. The responsibilities are familiar from custody; the ledger now distinguishes the standing obligation from the forced immediate reply. At turn 3, D_A replaces K_A . Turn 4 targets B 's earlier challenge, so the question forces a negative reply while leaving DOS unchanged. At turn 6, P remains in an active justification after χ 's assertion is withdrawn. Whole-unit and premise-level retraction are different moves.

A projection retaining the two justifications for χ as mutually attacking arguments records disagreement. It discards speakers' burdens, the question/challenge distinction and forced immediacy. Semantic evaluation cannot recover those fields; a richer encoding could retain them.

Checking a proposed formalization differs from reconstructing the utterances. A trace checker diagnoses malformed records, then returns the first failed transition or a certificate and final state. A valid prefix can still have a pending reply. Neither its completion nor the adequacy of the original speech-act reconstruction follows from that certificate (Exercise 9.6).

9.3 What should be said next?

We can now give Chapter 4's question about purpose an exact finite instance. Return to the state after turn 3, with $\text{Com}_A = \{\chi, P\}$, $\text{DOS} = \{D_A, (B, \chi, -)\}$ and no immediate reply pending. Suppose A is offered two continuations: withdraw its assertion, or ask B about the challenge. These do not exhaust all legal moves. For this finite experiment, the public response policy says that B , if asked, will answer with $\{Q, S\}$. CR5 forces the speaker, target and polarity, but the response policy supplies this particular support content.

Give the experiment a budget of two further public moves. Let *retained*(*s*) say that *A*'s assertion of χ is active, and *exposed*(*s*) that *B* has an active negative justification of χ . Stop at the first state after at least one additional move where no immediate reply is pending and either the assertion has been withdrawn or the countergrounds exposed. Record these two Booleans, the public stores and the reason for stopping. Standing support obligations may remain. This is an observation boundary for the experiment, not termination of the protocol.

Withdrawal uses *retract*(*A*, χ) [CR6] and stops after one move. *A* retains *P* from its justification; DOS remains $\{D_A, (B, \chi, -)\}$. Its observation is (*retained*, *exposed*) = (0, 0), with reason “withdrawn”. Inquiry uses *question*(*A*, *B*, χ) [CR4], followed immediately by *justify*(*B*, $\{Q, S\} \vdash^- \chi$) [CR5]. It stops after two moves with $\text{Com}_A = \{\chi, P\}$, $\text{Com}_B = \{Q, S\}$, $\text{DOS} = \{D_A, D_B\}$ and observation (1, 1), with reason “countergrounds disclosed”. Stopping just after the question would be premature because its reply is still pending.

Now stipulate *A*'s objective: first expose the countergrounds, then retain public sponsorship of the claim, then use fewer moves. Formally, maximize (*exposed*, *retained*, $-n$) lexicographically among offered continuations finishing within budget, where *n* counts added moves. Withdrawal scores (0, 0, -1); inquiry scores (1, 1, -2). Inquiry is preferred for that stated purpose. Nothing here establishes an equilibrium, an optimal strategy over every legal history, or the restaurant's excellence. With a one-move budget, the question remains legal but its offered continuation cannot finish in time. Legality, completion and purpose have separate roles.

9.4 What reasons still need development?

Interaction can develop the material that evaluation will later use. A reason's content, its occurrence and its target then need separate identities. Consider a dinner context with options *R*, restaurant; *C*, café; and *T*, takeaway eaten separately. These options are distinct from the agents above. The general claim q_R , that the restaurant suits this dinner, is also distinct from χ , excellence. Neither claim by itself settles an option's permission.

Let the company ground say that *R* offers the fullest shared meal, *C* a shorter shared visit, and *T* none. An occurrence o_g uses this content for the general target $\text{General}(q_R)$. An organiser's invitation supplies a stipulated documentary answer to a request for its grounds. A later occurrence o_{rc} uses the same company content for the different target $\text{Compare}(R, C)$. This opens a request d_3 for the comparison-specific warrant. The invitation does not answer that request: evidence about attendance and a normative comparison have different targets.

Use a separate pedagogical ledger. Each primary reason-use occurrence has an identifier, content, primary target, speaker, time, context, source and source version. Documentary answers target demands about those occurrences. The open frontier

$\text{Front}(L)$ is the set of demand identifiers still open. An answer closes its matching demand only when target, context and required record type agree. These are explicit rules for this example, not additional Fatio moves.

Let L_5 be the record after the two primary occurrences and their requests above. In L_6 , a matching comparison-policy record answers d_3 : for this dinner, the extra shared time counts in favour of R over C , with its force limited by the shared visit already available at C . The policy supplies the requested assessment of that comparison, not new evidence that people will attend. No numerical scale is needed to inspect which demand it answers:

	L_5	L_6
Primary occurrences	$o_g : \text{General}(q_R), o_{rc} : \text{Compare}(R, C)$	The same two occurrences
Added documentary answer	No answer to the comparative demand	Matching comparison policy supplied for d_3
Open frontier	$\{d_3\}$	$\emptyset$

Define π_{occ} to retain primary occurrences as opaque AF nodes and only a separately supplied attack relation, dropping targets, provenance, documentary answers and demands. The supplied attack relation here is empty. Then

$$\pi_{occ}(L_5) = \pi_{occ}(L_6) = (\{o_g, o_{rc}\}, \emptyset), \quad \text{Front}(L_5) \neq \text{Front}(L_6).$$

The added documentary answer is not a primary reason-use node, which explains the equality. An even coarser projection $\pi_{content}$ identifies nodes with the same content; both records become $(\{g_{company}\}, \emptyset)$. That projection additionally loses the general/comparative distinction. The first adapter has retained occurrence identities and still lost the frontier.

This repeats Chapter 3’s target distinction as a Chapter 7 preservation test: equal projected graphs conceal different answers to the specified frontier query. Calling every candidate node accepted in an edgeless graph cannot recreate a discarded documentary demand or validate a source. A richer node encoding or metadata table could retain the missing information; no universal representation impossibility follows. Nor does one completed comparative record give a complete practical model. Exercise 9.10 supplies the full event ledger, numerical policy records and a changed-foil transfer. Supplying a reason can close one request while opening another, and repeated content need not answer the same question.

9.5 What does a winning strategy establish?

There is another use of dialogue in which the input is already fixed: a game can test semantic acceptance. It neither elicits missing restaurant facts nor allocates the public obligations above. The standard grounded game (SGG) is played on a finite

AF by a proponent P and opponent O [52, Sec. 7, Defs. 17–18, pp. 512–513]. P starts with the queried argument; O and P alternate, each attacking the immediately previous argument. P may not repeat an argument already used at a P-position on that play. O has no corresponding repetition restriction. Play terminates when the next player has no legal move; the last player wins. A prefix stopped while a legal next move exists is not terminated.

Winning a play is weaker than possessing a winning strategy. A strategy for query a is a rooted argument-occurrence tree with root a at level 1. Every odd-level P node has exactly one child for each attacker of its label and no other children; every even-level O node has exactly one child, a chosen counterattacker. Every root-to-leaf path must be a terminated P-winning play respecting the repetition rule. Thus every O choice is covered, while P supplies a response. Repetition is path-local: an argument can appear in separate branches.

The quantifiers recall Chapter 6: defence requires answering every attacker. A strategy additionally organizes those answers into terminated plays obeying the game's repetition rule.

For this game, an argument is in the grounded extension exactly when P has a winning strategy [52, p. 513, correspondence following Def. 18]. The general proof is a reading route; a complete finite witness shows why the strategy quantifiers matter. Use arguments a, b, c, d, e, f , with attacks

$$\{(a, b), (b, c), (c, d), (b, e), (d, e), (a, f), (b, f)\}.$$

Figure 8 displays the graph and a full strategy for e .

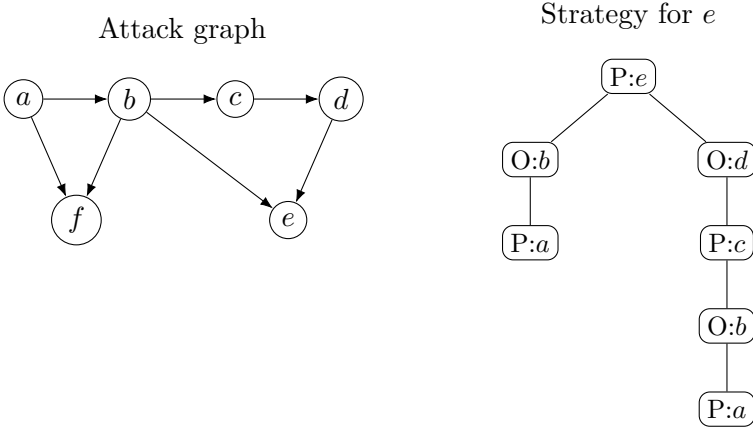


Figure 8: Attacks on the left, game-tree edges on the right. The two occurrences of a are in different plays. At each P node every attack must be represented by an O child; at each O node P chooses one reply.

At e , O can choose b or d . P answers b by a . Against d , P uses c ; O's only reply is then b , answered by a . Both paths, $[e, b, a]$ and $[e, d, c, b, a]$, terminate with P because

a is unattacked. Every O branch is covered, and no path repeats a P argument. Reusing Chapter 6’s grounded construction gives $\emptyset \mapsto \{a\} \mapsto \{a, c\} \mapsto \{a, c, e\}$, then fixed.

By contrast, $[f, b, a]$ is a legal P-winning play but not a strategy for f . O may instead attack f with a . The play $[f, a]$ then terminates with O, since no argument attacks a . A strategy tree for f must include that O child and cannot supply its required P reply. Thus no winning strategy exists. Equivalently, no set can defend f against its unattacked attacker a . Successful selected paths cannot replace universal opponent coverage.

The four questions now have distinct answers: a protocol certificate; a choice under an objective and information policy; a documentary frontier; and a semantic winning strategy. None is a collective acceptance decision, which additionally requires the aggregation choices encountered in Chapter 2. Chapter 10 returns to fixed-input evaluation with issue standards and comparative options. An interaction can supply or contest those inputs, but what a contribution establishes must be specified separately from its permission to be uttered.

9.6 Exercises

An asterisk marks an optional study; the Libra exercises are read after Chapter 10. All solutions are collected at the end of the book.

- 9.1. [\[solution\]](#) [Public-state calculation] Reconstruct the six-turn restaurant certificate and every public-store entry, including the pending reply. Explain why C ’s question has no direct DOS effect and why P remains after A retracts χ .
- 9.2. [\[solution\]](#) [Model comparison] Use the following supplied account of the cited Fatio report; consulting the report is not required for this exercise. In its axiomatic account, $B_i\psi$ means that agent i believes ψ , and $D_i\psi$ that i desires ψ . One precondition of i ’s assertion of φ is $\forall j \neq i, D_iB_jB_i\varphi$: the speaker wants every other agent to believe that the speaker believes φ . Its corresponding postcondition is $\forall k \neq i \forall j \neq i, B_kD_iB_jB_i\varphi$: every other agent believes that the speaker has that desire. The account also specifies public obligation effects. Positive justification retains the existing obligation to support the claim and adds an obligation to support the offered argument. The printed CR6 permits retraction of a claim at any time following its assertion by that speaker, and retraction of Φ at any time following that speaker’s positive justification $\Phi \vdash^+ \varphi$. No negative-justification branch is printed in CR6. The subsequent semantic discussion nevertheless treats negative justification and argument retraction by analogy with the positive case [75, Sec. 3, p. 5, Sec. 4.1, pp. 6–7].

Separate local combination-rule legality, the public-store projection and the source’s axiomatic reading. Give information represented by each and explain the different positive-justification discharge convention. Compare the source’s

printed CR6 justification branch with the local retraction rule, distinguishing the printed branch from the accompanying semantic discussion. Explain the scope of the six-turn certificate to a reader who knows only the natural-language exchange.

- 9.3. *[solution]* [Forced response] After $\text{assert}(A, \chi); \text{question}(B, A, \chi)$, state the only legal next move form, its speaker, target and polarity. Is its support uniquely determined? Could A instead assert P ?
- 9.4. *[solution]* [Obligation diagnosis] Replace that question by $\text{challenge}(B, A, \chi)$. State A 's forced response and B 's new obligation. Which part differs from the question case?
- 9.5. *[solution]* [Counterexample traces] Diagnose the first failure in each trace:
 - (a) $\text{assert}(A, r); \text{question}(B, A, r); \text{justify}(A, \{s, f\} \vdash r); \text{challenge}(B, A, s)$.
 - (b) $\text{assert}(A, s); \text{challenge}(B, A, s); \text{retract}(A, s)$.

Why are target typing and immediate response separate issues?

- 9.6. *[solution]* [Formalization and computation] Formalize “A calls the restaurant excellent; B asks for a reason; A cites a good meal”, and a variant in which the third utterance changes topic. Give a rule certificate or first illegal move. Specify a checker that distinguishes malformed records from illegal transitions and reports pending replies after a valid prefix. For the supplied JSON convention, a string in `claim` denotes an atom and a nonempty list one whole unit; `support` supplies justification premises. Explain why “P”, “[P]” and “Q&S” are not interchangeable.
- 9.7. *[solution]* [Boundary calculation] After turn 3, retract A 's whole unit $\{P\}$. Compute commitments and DOS. Why would retracting atom P require a different active record? Repeat with $\text{retract}(B, \{Q, S\})$ after turn 5, and explain which negative-retraction convention is local. What if another active record still supplies one of the premises?
- 9.8. *[solution]* [Challenge a challenge] After turn 3 let C challenge B 's earlier challenge of χ . Give the forced next speaker and polarity, and C 's new obligation. What priority applies if a CR2 witness also exists for the same target?
- 9.9. *[solution]* [Purpose and feasibility] Certify withdrawal and inquiry from the supplied prefix, give their complete terminal records and objective tuples, and explain why stopping after the question fails. Repeat the choice with a one-move budget. Does the budget change CR4 legality?
- 9.10. *[solution]* [Documentary reconstruction] Use context $D1$, agent A , company source `company@v1`, invitation source `invitation@v1` and policy source `policy-RC@S7-base`. Reconstruct these events with distinct occurrence identifiers, times, targets, sources and demand links: (0) empty record; (1) assert $t_g = \text{General}(q_R)$, opening d_0 ; (2) supply o_g from the company

source, answering d_0 and opening $d_1 = \text{GroundsFor}(o_g)$; (3) supply invitation occurrence o_h , answering d_1 ; (4) assert $t_{rc} = \text{Compare}(R, C)$, opening d_2 ; (5) supply o_{rc} with the same company content, answering d_2 and opening $d_3 = \text{ComparisonWarrantFor}(o_{rc}, R, C)$; (6) supply o_w from the policy source, answering d_3 with (4, 2). All demands belong to A . By stipulation, documentary answers close only a matching target/context/source-type demand and add no further demand. The pairs are the exercise's supplied policy data; their two coordinates receive an evaluative interpretation in Chapter 10, which is not needed to check these matches. Give every frontier and both projections. Can o_h close d_3 ? If the comparative target changes to (R, T) , can the RC row close it? The new stipulated policy row for that foil is (7, 5).

- 9.11. *[solution]* [Strategy proof] Give the complete occurrence tree for e , audit every branch, and give both the winning play and the refuting O choice for f . Why does showing only $[e, b, a]$ not prove that it is a winning strategy? Why are two occurrences of a permitted in the complete tree?
- 9.12. *[solution]* [Transfer and cycles] Delete only attack (a, f) from the six-node graph. Recompute grounded iterates and decide whether $[f, b, a]$ now supplies a complete winning strategy. Separately, take exactly the two attacks $g \rightarrow h, h \rightarrow g$. Starting at g , explain why $[g, h]$ terminates and who wins.
- 9.13. *[solution]* [Output types] Answer and identify the result type: (a) Was inquiry's first question legal and complete for the experiment? (b) Why choose inquiry with budget two? (c) Does a winning play for f establish grounded membership? (d) Does an edgeless projection close L_5 's frontier? (e) Does majority of individual grounded memberships equal grounded evaluation after majority aggregation of graphs? For (e), use the following supplied witness: individual grounded sets $\{a, c\}, \{a, b\}, \{b, c\}$, aggregate attacks $a \rightarrow b, b \rightarrow c, c \rightarrow a$, aggregate grounded set $\emptyset$, and no stable extension. Do not derive a new collective protocol.
- 9.14. *[solution]* * [Source protocol] From Supplement S9.1's initial data and moves, reconstruct all six states. Identify the first true permission variable, its first disclosed conjunctive explanation, a relevant but nonlive option, and what survives café withdrawal. Do empty targets and high/low labels supply a collective choice rule?
- 9.15. *[solution]* * [Disclosure transfer] Append to that state $question(B, A, j_1)$, $justify(A, \{g\})$, $retract(A, g)$. Check legality and compute equations, ownership, outstanding targets and $(p, q, z, j_1, j_2, r_{1w}, r_{2w}, g)$. Does withdrawing the ground withdraw the option, erase the café comparison or reopen its burdens? Distinguish this from changing a numerical weight.
- 9.16. *[solution]* * [Application evidence] Using Supplement S9.2, assess: "An acceptable argument answers every follow-up question"; "a sound and complete EQR theory proves patients understand the implementation"; and "downstream admissibility

repairs ungrounded sensor interpretation”. State the evidence or interface each claim would actually require.

- 9.17. *[solution]* [Literature investigation] Compare a foundational public protocol, the semantic game and a strategic or application route. State their different success conditions. In the abduction route, what extra input distinguishes searching for a hypothesis from testing acceptance in one fixed AF? In the mechanization route, why must an omitted precondition component be reported?

9.7 Further reading

Public procedures and their purposes. Start with the Handbook distinctions among dialogue purposes, locutions, commitments, protocols and strategy [37, Secs. 2.1–2.3, pp. 517–523]. Walton and Krabbe provide a foundational account of commitment and dialogue types [36]. For the selected Fatio construction, use the original ULCS report’s syntax and axiomatic effects [75, Sec. 3, pp. 4–5, Sec. 4.1, pp. 6–7]; the related chapter has different pagination [76]. Compare its standing obligations with our local response ledger. Prakken’s alternative discussion of dialectical obligations makes relevance and postponement policy consequential [35, Sec. 7.1]; immediate response is one design choice. FIPA communicative acts [77] and normative multi-agent systems [78] broaden the assumptions about agents and institutions. None licenses identifying a public commitment with sincerity or private belief.

Games, strategies and information. The exact SGG definitions and grounded correspondence are in [52, Sec. 7, Defs. 17–18 and following correspondence, pp. 512–513]; the earlier Theorem 6 there concerns a different game. Read the full proof route after understanding why every opponent branch must be covered. The strategic Handbook account [3, Ch. 10, pp. 578–583, 616–617] then adds aims, disclosure choices and information assumptions. Hypothetical dialogue and abduction add yet another input: a specified family of candidate framework changes from which a hypothesis is sought [79, Secs. 3–5, arXiv v1 PDF pp. 2–4]. They do not allow arbitrary independent edits or automatically return a minimal repair. Compare supporting an observation under a hypothesis with proving acceptance in a fixed model and with Chapter 7’s distinct task of representing an independently specified reasoning method while preserving its output family. Chapter 13 returns to explanatory targets.

Mechanization, comparative disclosure and applications. Pasetto and Benzmüller implement a Fatio encoding in LogiKey [80, Secs. 2–3, PDF pp. 2–8]; their omission of the **pre-con** component of **Done** matters when comparing formalizations. The local certificate here is not already that full encoding. Libra supplies a distinct comparative protocol with public relevance and Boolean explanatory dependencies [81, Defs. 3.1–4.13, pp. 4–9, Sec. 5, pp. 9–10]. Its worked account is supplied

in Supplement [S9.1](#); keep its abstract *Bal* separate from additive permission. The application Handbook route [[38](#), pp. 230–252] asks how formal accounts become implemented interfaces. The CONSULT/EQRbot study preserves the difference between reported theory, supported locutions and evidence about human benefit, preparing the further application and explanation questions of Chapters 13–15.

10 Chapter 10. Evaluating issues and comparing options

Chapter 4 separated responsibility for producing reasons from a standard for establishing an issue, then compared reasons for choosing an arrangement. Chapter 9 showed how interaction can develop and contest such inputs. We now hold those inputs fixed and formalize the two evaluative questions: does a proposition meet its assigned standard, and which feasible options do the comparative reasons permit? The first returns to custody; the second develops the dinner comparison through all its alternatives.

The early output distinction continues to govern both calculations:

$$\begin{aligned} \text{acceptable proposition} &\neq \text{permitted option} \\ &\neq \text{selected option} \neq \text{executed action.} \end{aligned}$$

These are different kinds of output, not automatically successive stages of one universal procedure. An application must specify which it needs and how, if at all, its methods connect them.

10.1 Premise roles and issue standards

In the 2007 Carneades model, an argument graph is finite, directed, acyclic and bipartite: statement and argument nodes alternate. Arguments have premises and pro or con orientation toward a conclusion. Evaluation also uses a context assigning statement statuses, issue-specific proof standards and a strict partial priority order on arguments [82, Defs. 4–5, pp. 882, 884–885]. The graph shows which considerations depend on which; the context determines how they may be used.

A statement’s status is *stated*, *questioned*, *accepted* or *rejected*. A particular occurrence as a premise has a role: *ordinary*, *assumption* or *exception*. The same statement can have different roles in different arguments. Write $A(q)$ for the Boolean result that statement q is acceptable under its assigned standard. The complete holding conditions are:

Status of q	Ordinary premise	Assumption	Exception
stated	$A(q)$	true	$\neg A(q)$
questioned	$A(q)$	$A(q)$	$\neg A(q)$
accepted	true	true	false
rejected	false	false	true

An argument is *defensible* exactly when all its premises hold [82, Defs. 9–10, p. 886]. A stated ordinary premise still needs an acceptable statement; a stated assumption

can be used until it is questioned. An exception names a blocking circumstance. In the stated or questioned status, its premise holds when that circumstance is not acceptable; the accepted and rejected statuses use the overrides in the table. Holding of an exception premise thus permits use of the argument; it does not establish the exceptional circumstance.

Accepted *status* is an input decision in the context. Acceptability is an evaluative output. The table does not identify the two. Dependencies also matter: if a premise needs an acceptability check, the arguments for that subsidiary statement must be evaluated first. Acyclicity supports this bottom-up reading. We use the source's holding conditions but isolate a smaller kernel whose currently defensible pro and con sets are supplied. It does not purport to implement the entire recursive evaluator or a speech-act protocol.

Reuse Chapter 4's fictional facts w_1 – w_5 , but change the question. There m expressed the recommendation that the child should live with the mother; here we reuse the letter for the issue that living with the mother is in the child's best interest. Establishing this issue could support the earlier recommendation under an appropriate connecting rule. It neither supplies that rule nor selects or executes option M . Take arguments

$$a_1 : w_1, w_2, w_5 \Rightarrow m, \quad a_2 : w_3 \Rightarrow \neg m, \quad a_3 : w_4 \Rightarrow m.$$

The linked/convergent distinction from Chapter 3 now has an exact role: all three premises must hold for a_1 to be defensible, while a_3 provides a separate route to m [82, Sec. 3, p. 884]. Here a_2 is classified as con m . Its notation does not silently establish a different custody option.

Stipulate accepted status for w_1, w_3, w_4, w_5 , used as ordinary premises, and stated status for the assumption w_2 . All three arguments are then defensible. The proponent supplies w_1, w_4, w_5 , and w_2 if it is questioned; the opponent supplies w_3 . These burden assignments and the legal premise are fictional policy inputs, not legal doctrine. Let $\mathcal{P}(q)$ and $\mathcal{C}(q)$ denote the defensible pro and con sets for issue q , and let $\succ$ be a declared strict partial priority. The selected standards are:

$$\begin{aligned} SE(q) &\iff \mathcal{P}(q) \neq \emptyset, \\ BA(q) &\iff \exists a \in \mathcal{P}(q) \forall b \in \mathcal{C}(q) a \succ b, \\ DV(q) &\iff \mathcal{P}(q) \neq \emptyset \wedge \mathcal{C}(q) = \emptyset. \end{aligned}$$

They are scintilla of evidence, best argument and dialectical validity [82, Defs. 5–10, pp. 884–886]. These tests use existence, absence and priority, not a numerical threshold. A weak con remains a con. BA needs *one* pro above *every* con; several pros each defeating a different con do not meet that requirement.

Initially $\mathcal{P}(m) = \{a_1, a_3\}$, $\mathcal{C}(m) = \{a_2\}$, with $a_1 \succ a_2$. SE holds because a pro exists; BA holds with witness a_1 ; DV fails because a_2 remains defensible. Its lower priority does not remove it from the con set. No comparison between a_1 and a_3 is needed for these answers.

For the same fixed inputs, the standards satisfy $DV(q) \Rightarrow BA(q) \Rightarrow SE(q)$. *Proof.* If DV holds, choose a pro witness. The con set is empty, so that witness outranks every con vacuously; BA holds. If BA holds, its witness belongs to the pro set, which is exactly SE's requirement. $\square$ The existence of the witness matters: with an empty pro set all three standards fail, even when the con set is also empty. The initial case already disproves $BA \Rightarrow DV$; further converse and priority boundaries are in Exercise 10.9. This hierarchy compares definitions at a fixed evidence state, not the adequacy of legal institutions. The source does not claim that these three definitions by themselves adequately model legal standards of proof [82, p. 885, Sec. 8, p. 894].

If w_2 is questioned, its occurrence as an assumption now requires $A(w_2)$. Questioning does not prove lack of understanding. If the statement meets its own standard, a_1 can remain defensible. If it does not, only a_3 remains pro m , and BA depends on whether $a_3 \succ a_2$. A fully specified partial order can leave these arguments incomparable, making BA false. A statement giving only $a_1 \succ a_2$ may instead leave the relevant priority input unspecified. Incompleteness of our description is different from declared incomparability.

10.2 A safety exception changes another issue

The preceding comparison held defensibility fixed and varied the standard. Now hold a policy fixed and vary the evidence at a subsidiary issue. This will separate the production and acceptance burdens introduced in Chapter 4. The stages supply successive evidence states; Chapter 9's protocol does not determine their transitions. Let s mean that a stipulated safety concern is established, and let M name the proposition proposing primary custody with the mother. In this trace M labels an issue; Chapter 4 used it for an option. Its only possible pro argument A_M has accepted ordinary premises for stable housing, school continuity and primary-care history, together with the exception premise s . The status of s remains stated; its acceptability is evaluated separately.

The fictional policy assigns SE for the burden of producing support for s , but DV for making that exception acceptable. Reports r_1, r_2 are possible pro arguments for s , and c is a con. The party seeking M must meet BA; there is no direct con argument for M . Consequently A_M is defensible exactly when s is not acceptable, and losing A_M leaves M with no pro witness.

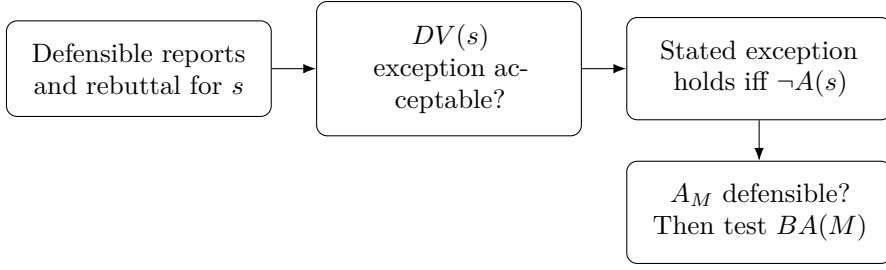


Figure 9: The safety dependency. A con to s contests the exception; it is not a direct con to M . The arrows show evaluation dependencies, not attacks or a mandatory dialogue order.

Stage	Defensible safety evidence	$SE(s)$	$DV(s)$	A_M defensible	$BA(M)$
1	No allegation: pro $\emptyset$, con $\emptyset$.	no	no	yes	yes
2	Bare allegation: sets unchanged.	no	no	yes	yes
3	Pro $\{r_1\}$, con $\{c\}$.	yes	no	yes	yes
4	Pro $\{r_1, r_2\}$, con $\{c\}$.	yes	no	yes	yes
5	Pro $\{r_1, r_2\}$, con $\emptyset$: c loses a required premise.	yes	yes	no	no

At stage 3, production has been met: r_1 supplies a defensible pro. The exception is nevertheless unacceptable under DV because c remains defensible. Stage 4 adds a second report without changing that fact. DV does not count votes or compare total strength; any defensible con prevents it. At stage 5 the rebuttal loses a required premise. The exception now has a pro and no con, so DV holds. Its stated exception premise in A_M fails, removing the only pro for M . BA fails for lack of a witness, despite the continuing absence of a direct con to M .

This explains the apparently reversed effect: removing opposition to an exception can remove support for a proposal. The bare allegation in stage 2 and the extra report in stage 4 change the narrative record without changing the corresponding evaluative result. Production, acceptance of the exception and establishment of the proposal are separate questions. No stage supplies a proof for the father option, and no calculation establishes a real person’s safety. Burden allocation and procedural consequences belong to the surrounding policy; the standards alone do not prescribe a next move [82, Sec. 2, pp. 879–880, Sec. 5, p. 887, Sec. 6, pp. 888–890].

10.3 Comparative reasons need options and foils

Chapter 4’s resource advantage could justify an arrangement without requiring the rejection of its rival. We now represent those two practical forces separately and

make the comparison target explicit. As Chapter 9’s company example showed, a consideration can count differently against an option that lacks a benefit and one that retains much of it. A comparative reason must identify both the favoured option and its alternative.

Let G be a finite set of grounds and O a finite set of options. A comparative reason is $(g, o, f) \in R_{cr} \subseteq G \times O \times O$, where $o \neq f$; f is the *foil*. A separate weighting assigns

$$w_{cr} : R_{cr} \longrightarrow \mathbb{R}_{\geq 0}^2 \setminus \{(0, 0)\}, \quad w_{cr}(g, o, f) = (jw_{cr}(g, o, f), rw_{cr}(g, o, f)).$$

Justifying weight represents force in support of permission for the favoured option; requiring weight represents the opposing demand that its foil must meet to remain permissible. For each ordered pair, sum the respective coordinates, with an empty sum equal to zero:

$$J(o, f) = \sum_{(g, o, f) \in R_{cr}} jw_{cr}(g, o, f), \quad W(o, f) = \sum_{(g, o, f) \in R_{cr}} rw_{cr}(g, o, f).$$

The selected comparison and all-alternatives rule are

$$\begin{aligned} \text{Perm}(o \mid f) &\iff J(o, f) \geq W(f, o), \\ \mathcal{O}_{\text{perm}} &= \{o \in O : \forall f \in O \setminus \{o\}, \text{Perm}(o \mid f)\}. \end{aligned}$$

These restate Definitions 3.1–3.4 of the attributed AAMAS source [83, Sec. 3.1, Defs. 3.1–3.4, PDF p. 3]. Its pair output records the two signs of $J(o, f) - W(f, o)$ and $J(f, o) - W(o, f)$; a zero margin has positive sign. Our permitted set contains exactly the options positive against every alternative. The reversed index in $W(f, o)$ and the weak inequality are essential. A repeated utterance of the same reason does not duplicate a member of R_{cr} or double its weight.

Return to a fictional dinner decision. R is dinner at the Brigade restaurant, C a meal at the café, and T takeaway eaten separately. Exactly one is to be performed; these options are feasible and exhaustive within the stipulated example. The company ground says that R provides the fullest shared meal, C a shorter shared visit and T none. A voucher makes C least expensive, while T is quickest. All three grounds are available and applicable. The following is the entire reason and weight table:

Ground	Option	Foil	JW	RW
company	R	C	4	2
company	R	T	7	5
budget	C	R	3	3
budget	C	T	2	1
speed	T	R	4	3
speed	T	C	3	3

The same company ground favours R with different force against C and T . The qualitative comparison explains why the foil matters; it does not derive these numbers. They are stipulated normative weights, not measurements, belief probabilities or values mandated by the source. Additivity is the chosen rule, not a property of all reasons. The expression “dynamic scale” concerns comparison-relative force here, not learning or temporal update.

Each ordered pair has exactly one reason, so its totals are the displayed coordinates. All six tests are needed:

Option / foil	Own J	Opposing W	Margin	Pass?
R/C	4	3	1	yes
C/R	3	2	1	yes
R/T	7	3	4	yes
T/R	4	5	−1	no
C/T	2	3	−1	no
T/C	3	1	2	yes

Thus $\mathcal{O}_{\text{perm}} = \{R\}$. Permission for C against R is insufficient: C fails against T . Likewise T passes against C but fails against R . The complete matrix shows why a single favourable comparison cannot answer the all-foil question. It supplies no theorem that every possible input has a uniquely permitted option.

10.4 Permission leaves a further choice

Change only the requiring coordinate of (g_{speed}, T, C) from 3 to 2. Its own justifying weight remains 3, so the T/C test is unchanged. The modified coordinate instead enters the opposing C/T test, changing $2 \geq 3$ to $2 \geq 2$. Equality passes. Every other test remains as before, yielding $\mathcal{O}'_{\text{perm}} = \{R, C\}$.

To choose one option, supply a further policy. If the cost order is C cheaper than T cheaper than R , a least-cost-among-permitted policy selects C . A policy giving priority to R , then C , then T , selects R from the same permitted set. These are additional stipulations, not further consequences of the permission rule. Even in the original singleton case, selection and authority to execute require their own procedural setting.

Ranking options by their total justifying weight across foils cannot recover this change: it discards the only coordinate that changed. Exercise 10.12 calculates the totals. Adding weights for different comparison questions also introduces a further aggregation rule. Permission, ranking and selection each need their own specification.

Chapter 9’s unfinished comparative target now has a precise possible interpretation: a dialogue may seek a ground, a foil-specific weight or its warrant before this fixed-input test can be run. That does not make our arithmetic a dialogue protocol. The

optional Libra study develops public disclosure and withdrawal with its own abstract comparison predicate; it is not obtained by silently adding speech moves to these six rows. Nor does an explanation certify the authority of a weight. Supplement S10.4 separates the provenance of an assignment, correctness of evaluation and the evidence needed to justify its use.

10.5 Other interpretations of support and strength

The examples also show why more arrows or numbers cannot by themselves supply a semantics. A bipolar framework $B = (A, R^-, R^+)$ records attacks and supports. For a support xR^+y , a deductive reading constrains acceptance of x toward acceptance of y ; a necessary-support reading instead makes x a condition for accepting y . These acceptance dependencies differ from the linked premises within a_1 . Their interpretations cannot be chosen from the direction of the drawn arrow alone [84, Sec. 4, pp. 53–65].

The earlier distinctions also guide a choice among numerical alternatives. A strict priority says which argument outranks another, without measuring the difference; SE/BA/DV test existence, priority and absence. Neither is a comparative weight. Supplement S10.1 instead calculates gradual and social scores from supplied bases or votes under declared propagation rules. Such a score has no probability interpretation by default. Supplement S10.2 returns to Chapter 4’s values through an audience’s ordering of what arguments promote. That order supplies neither a cardinal magnitude nor moral authority. Supplement S10.3 specifies normalized probability models: uncertainty about which graph exists differs from belief in arguments on a fixed graph.

These studies pursue additional formal choices. The main progression can continue to Chapter 11 with the question already established: by which principles should an interpretation of support and evaluation be assessed? Its answer depends on the intended object, as it did when Chapter 3 reconstructed a reason and Chapter 4 chose what to ask of it.

10.6 Exercises

An asterisk marks work using a named optional study. All exercises have complete public solutions in the collected Solutions section.

- 10.1. *[solution]* [Priority diagnosis] Keep $\mathcal{P}(m) = \{a_1, a_3\}$ and $\mathcal{C}(m) = \{a_2\}$, but stipulate $a_2 \succ a_1$ and $a_2 \succ a_3$. Evaluate all standards. If only the first comparison were specified and the remaining order left open, give completions making BA true and false. Distinguish this from a fully specified partial order with no relevant comparison.
- 10.2. *[solution]* [Representation] Use claims u : a stable care schedule; v : a shorter school

journey; w : contrary schedule evidence; m : adopt the arrangement. Construct a bipolar graph with two supports and two attacks, stating each meaning. What further input would be needed to evaluate acceptance?

- 10.3. [\[solution\]](#) [Premise diagnosis] Recover the premise roles, initial statuses and burden holders for w_1 – w_5 . Why is ordinary con premise w_3 not an exception? Give both possible consequences of questioning w_2 , depending on its own acceptability, and identify the priority needed if a_1 is lost.
- 10.4. [\[solution\]](#) * [Calculation] Under the local attenuation rule of Supplement [S10.1](#), compute the score of an argument with base 0.9 and one attacker of strength 0.4. Why is subtraction the wrong operation?
- 10.5. [\[solution\]](#) * [Calculation] Use the supplement’s smoothing rule for vote pairs $(10, 0)$, $(4, 4)$, $(1, 8)$. What has been computed, and why is no final network score yet determined?
- 10.6. [\[solution\]](#) [Comparative transfer] Take the earlier two-option custody input $(g_1, M, F) : (8, 4)$, $(g_2, F, M) : (4, 1)$, $(g_3, M, F) : (5, 2)$. The father’s $JW = 4$ is a separate stipulation from Chapter 4’s 3. Compute permissions, then replace only the second row by $(7, 3)$. Explain why the result does not supply a unique selection.
- 10.7. [\[solution\]](#) [Model comparison] Compare what the Carneades graph and a bipolar graph retain about the custody case. Distinguish a linked premise, a convergent argument, a con argument and an exception premise. Which additional information does the factor task in Exercise [10.21](#) retain?
- 10.8. [\[solution\]](#) [Conceptual transfer] For xR^+y , contrast deductive and necessary-support readings. Which acceptance dependency runs in each direction? Does either description alone specify a complete semantics for an arbitrary bipolar graph?
- 10.9. [\[solution\]](#) [Proof and boundaries] Reconstruct $DV \Rightarrow BA \Rightarrow SE$. Refute both converses and explain the empty-pro boundary. Give a partial-order example in which every con is outranked by some pro but BA nevertheless fails.
- 10.10. [\[solution\]](#) [Dependency and explanation] Recompute safety stages 4–5, including the resulting pro/con sets for M . Explain the reversal both to a reader of the fictional policy and to a checker of the formal definitions. Why do two reports not override c at stage 4, and why does failure of M not establish F ?
- 10.11. [\[solution\]](#) [All-foil calculation] Recompute every directed test for R, C, T from the six-row table, with margins and failed foils. Explain the different force of the company ground and what information a dialogue would need to preserve to retrieve it.
- 10.12. [\[solution\]](#) [Transfer and selection] Apply the one-coordinate RW change, recompute all six signs and the permitted set, and apply both stated selection policies. Explain the opposing index. Calculate each option’s justifying total by summing

across all its foils in each input. Why can the resulting ranking not implement the permission rule across these two inputs?

- 10.13. *[solution]* * [Proof] Prove the $[0, 1]$ bound for the local acyclic score calculation in Supplement S10.1. Identify the two routes by which a_3 affects m . What has not been established about cyclic networks or probabilistic dependence?
- 10.14. *[solution]* * [Derivation] For the five-node ranking graph in that supplement, derive the categoriser, product scores and complete/grounded/preferred/stable results. Prove existence and uniqueness of the categoriser valuation on every finite DAG. Why can no global upper threshold on the displayed categoriser values recover the accepted set?
- 10.15. *[solution]* * [Sensitivity] Delete only (x, z) from that graph. Recompute the rankings and acceptance results. Which changes, and why is this one example not an invariance theorem?
- 10.16. *[solution]* * [Audience comparison] Reconstruct both audience-relative defeat graphs and extension results in Supplement S10.2. Could value names simply be replaced by probabilities? What further rule would a different use of priorities need?
- 10.17. *[solution]* * [Probability] Use the finite-probability preparation in Supplement S10.3 to derive its four graph-world masses and compute $P(b \text{ accepted} \mid b \text{ present})$. Identify the independence assumption. Why do the constellation and epistemic examples both use subsets of $\{a, b\}$ but answer different questions?
- 10.18. *[solution]* * [Proof and counterexample] Prove $\text{COH} \Rightarrow \text{RAT}$ for any finite AF, including self-attacks. Refute the converse using the complete supplied distribution, and compare its threshold labels with the fixed graph's complete labels.
- 10.19. *[solution]* * [Transfer] Move mass $1/5$ from $\{a, b\}$ to $\{a\}$ in the epistemic table. Recompute normalization, marginals, COH, RAT and threshold labels. Does COH alone guarantee complete Dung labels? Give a full-distribution counterexample.
- 10.20. *[solution]* * [Evidence audit] An analyst reports identical weights in ten runs on a fixed graph and reconstruction of two documented outcomes. They conclude that the weights are uniquely correct, justify all future decisions and become certified by adding a Libra explanation. Using Supplement S10.4, identify what follows and what additional evidence each conclusion requires.
- 10.21. *[solution]* [Alternative representation] Let $\text{Case} = (F_M^+, F_M^-, F_F^+, F_F^-, \preceq, \text{outcome})$, where factor sets record considerations for and against each option. Stipulate $F_M^+ = \{\text{childPreference}, \text{fatherAway}\}$, $F_M^- = \{\text{motherLessWealthy}\}$, $F_F^+ = \{\text{fatherWealthier}\}$, and $F_F^- = \{\text{fatherAway}, \text{childPreference}\}$. Also stipulate that the paired comparisons $\text{childPreference} \succ \text{fatherWealthier}$ and $\text{fatherAway} \succ \text{motherLessWealthy}$ favour M in this case. Explain this local result and the information a challenge could contest. Why does it supply no general rule for additional or unpaired factors? Are mirrored factor descriptions independent evidence?

- 10.22. *[solution]* **[Interpretation]* In the social-score study, distinguish publicly recorded votes, private beliefs, collective acceptance and the platform’s calculated score. Explain why 5475/5929 being larger than 0.7708 establishes no gain in accuracy, and why repeated endorsement does not certify a legal premise.
- 10.23. *[solution]* *[Literature investigation]* Use one standards source, one support/strength alternative and one practical source from the reading routes. Compare their input objects, output types and restrictions. For the 2012 Carneades translation, identify the source version and added representation structure before stating what kind of correspondence is being discussed.

10.7 Further reading

Standards, burdens and their representations. Begin with the scheme discussion of premise roles and questions [21, pp. 558–560], then the legal Handbook route on standards, burdens and accrual [85, pp. 132–133, 144–146, 157–161]. The exact first kernel comes from the 2007 Carneades definitions [82, Def. 4, p. 882, Defs. 5–10, pp. 884–886]. Its contextual and acyclic-graph restrictions matter, as does the distinction between evaluation and a surrounding procedure. Van Gijzel and Prakken’s translation [86, Secs. 3–4, pp. 28–38] treats a later, stage-specific CAES and constructs additional ASPIC+ structure. It is neither an identification of every 2007 procedural feature with a bare AF nor evidence of universal non-representability. Its ambiguity-blocking discussion and Prakken–Sartor’s modular framework [44, Sec. 4] expose substantive alternatives about how issue standards and burdens should interact.

What support or strength is intended? Read the bipolar account [84, Sec. 4, pp. 53–65] with its earlier foundations [87] and the joint-support and accrual route [88, Sec. 5, pp. 188–193]. Weighted approaches use richer domains and operations than a single local score [89, Secs. 3–7, pp. 362–383]; value-based reasoning and probability supply different kinds of input [39, pp. 300–318] [90, pp. 407–410, 419–421]. The social Handbook account situates votes and public structures [19, pp. 25–37, 53–59]. Bonzon and colleagues place the categoriser among ranking approaches [91, Defs. 3 and 9–10, pp. 915–916]; the broader strength literature asks which properties such methods should satisfy [92, 93]. Prakken’s later account asks about vulnerability under expansions, a different interpretation of strength [94, Secs. 1.2–1.3 and 3–6]. Chapter 11 develops these principle questions. Its preference-reduction comparison also returns to Chapter 5’s distinction between attack and preference-resolved defeat; no unique reduction policy follows from the phrase “the preferred argument wins”.

Practical reasons and the authority of weights. The decision-making Handbook chapter offers several model families [40, pp. 380–389]. The selected comparative kernel uses only Definitions 3.1–3.4 of Yu, Cai, Markovich and van der

Torre's attributed ten-page AAMAS paper [83, PDF p. 3]; its later representation and selection developments are not assumptions of our calculation. Tucker's *The Weight of Reasons* supplies a philosophical route into justification and requirement, rather than an authority for the fictional numbers. Liga, Markovich and Yu examine how weights are supplied and evaluated [95, Secs. 4.1–4.3 and 5]. Fixed-graph weighting differs from generation of topology and weights; repeatability differs from reconstructing outcomes; neither proves normative authority. The accompanying evidence study makes these distinctions operational as reading questions. Full legal case-based methods remain a further route through [85], not a general lifting rule implied by Exercise 10.21's two stipulated comparisons.

Part III. From argumentative methods to computational use

The earlier parts developed reasons as objects of inference, public interaction and practical evaluation. Dung's attack–defense abstraction made shared semantic questions precise, while the returns to structured content, dialogue and comparative choice showed what an extension leaves open. Part III asks how to compare the resulting methods, compute their answers and explain those answers in settings that need to use them.

The first question is which commitments a method should preserve. Chapter 11 turns the familiar complete, grounded, preferred and stable semantics into objects of comparison. Existence, uniqueness and locality constrain different aspects of their answers. A compact locality proof shows when context can safely be ignored; a stable counterexample shows why a local choice need not have a whole completion. Support and preference then test whether the same requirement survives a change in the operation being compared. Public response rules require a different property, about the next transition and the responsibility left by its reply. Practical permission supplies another precise comparison: expanding the considered alternatives can exclude an old option even when its old pairwise assessments are fixed. Actual cognitive studies introduce another question: what predictions do these representations make about people's judgments? Their evidence neither replaces the formal comparison nor follows from it.

Chapter 12 fixes a semantic question and makes it computational. A proposed stable extension first becomes a checkable witness. An explicit representation then allows operation counts, a Boolean encoding turns semantic conditions into constraints, and a full reduction shows why one particular acceptance task expresses SAT. The computational-reuse framework returns from Exercise 6.2 with all its inputs and needed results supplied anew. Conditioning its components on incoming choices, and then adding one attack, makes clear what a reuse argument must retain. A verified answer, a complete enumeration and a measured speedup remain different achievements.

Computation also returns to the restaurant dialogue and dinner decision. A public trace checker can certify a legal question while reporting an unfinished reply. Choosing between offered continuations additionally uses a response policy, objective and move budget. The dinner calculation first accumulates comparative reasons and tests every foil, then applies a separately supplied selection order. Changing one requiring coordinate makes a new option permitted; the two orders need not select the same one. These are actual paper procedures with different inputs and outputs.

Chapter 13 asks which question a verified answer explains. Acceptance within one extension, credulous acceptance and guarded skeptical acceptance need different evidence. A short display can preserve a standalone answer while losing the original

reasons; a cyclic contextual certificate can retain those reasons without proving groundedness. Challenges reconnect the certificate with the interpretation of its attacks and the responsibilities of speakers. The dinner explanation identifies the decisive ground, foil and changed comparison, then explains the policy-dependent choice. A request for a weight's grounds leads to an explicit revision interface: public justification, authority to revise a named record and reevaluation of the new version each have a defined role. Classifier explanations then bring together correspondence, finite-test agreement and human evaluation without conflating their warrants.

These distinctions prepare Chapters 14–15. Integration asks how represented inputs, computations, explanations and surrounding decisions meet in a system, with the relevant assumptions and evidence still attached. The agentic outlook asks what changes when an agent acts, obtains information and revises what it relies on. A proposition's acceptability, an option's permission, its selection and its execution continue to be separate questions. The final movement therefore develops computational use without turning every argumentative activity into one universal procedure.

The main text supplies the connected argument and its selected proofs. The exercises develop calculation, proof, construction, diagnosis and transfer; coherent studies provide additional internally prepared routes, and complete public solutions are collected at the back. The Part III guides for Chapters 11–13 and 14–15 (pp. 187 and 191) distinguish independent work from guided reconstruction in shorter courses. Paper algorithms and hand traces are complete routes throughout. Annotated readings then open the broader traditions whose specialist details the introduction does not attempt to reproduce.

11 Chapter 11. Comparing argumentative methods

Complete, grounded, preferred and stable semantics ask different questions of the same attack graph. Chapter 6 made those differences visible through their outputs. Chapter 8 then asked whether a construction preserved independently stated requirements. We now turn that question on the semantic functions themselves. Which commitments do they share? Which distinguish them? Under what conditions may an answer about part of a framework ignore the rest?

A requirement acquires its force from a use. A decision aid may need an answer on every finite input; a cautious inference procedure may need a unique answer; a component calculation may need independence from downstream arguments. Once stated mathematically, each requirement has a determinate satisfaction question. An application can motivate a principle, but it cannot make a failed implication true. Conversely, one failure identifies a commitment to change or restrict; it does not settle every possible use of the method.

The comparison begins with semantic functions, then returns to two other objects developed earlier: public procedures and practical evaluation. A requirement about who must answer next constrains transitions; a requirement about permitted options constrains a result relative to its considered alternatives. Neither is a property of an extension family. Working through these objects will show how the comparison method travels while its precise requirements change.

11.1 Properties of outputs and properties of functions

The principle-based approach specifies a class of functions, states requirements on them, and studies satisfaction and relationships among those requirements [74, Sec. 1, pp. 799–800]. For a fixed argument set A , an extension semantics has type

$$\sigma : \mathcal{F}_A \longrightarrow 2^{2^A}.$$

Its output is a *family* of subsets of A . When we restrict a graph to fewer arguments, the semantics is understood over all finite AFs, with each family typed by its own argument set. An argument, an extension and an extension family remain different objects throughout.

That distinction already separates two boundary cases. The empty AF has stable family $\{\emptyset\}$: it returns one extension. A singleton self-attacker has stable family $\emptyset$: it returns none. Requiring every returned set to be conflict-free permits the latter outcome. Indeed, the function that always returns no sets satisfies any universal condition on each of its outputs. Existence is a further requirement.

Write a principle as

$$P(\sigma) \iff \forall F \in \mathcal{D} \, p(\sigma, F).$$

The domain $\mathcal{D}$ matters. A principle over every finite AF is different from one restricted to AFs with stable extensions. A change principle instead quantifies over allowed pairs (F, F') and compares their outputs. Its counterexample must give the pair, the allowed change and the failed comparison.

Consider five requirements on ordinary AF semantics. Conflict-free outputs and complete outputs constrain every returned member. Existence requires a nonempty family on every finite AF. Uniqueness here means *at most one* output; only together with existence does it mean exactly one. Directionality, defined below, compares a whole framework with an unattacked part.

Semantics	Conflict-free	Complete	Exists	At most one	Directional
Complete	yes	yes	yes	no	yes
Grounded	yes	yes	yes	yes	yes
Preferred	yes	yes	yes	no	yes
Stable	yes	yes	no	no	no

The table abbreviates universally quantified statements, not observations on one graph. Chapter 6 established that the grounded extension is complete, that finite admissible sets extend to preferred ones, and that preferred and stable extensions are complete. These results justify the positive output and existence cells. The mutual pair uRv, vRu has three complete extensions, $\emptyset, \{u\}, \{v\}$, and two preferred and stable extensions, refuting the corresponding uniqueness requirements. The self-attacker refutes stable existence [5, Example 9, p. 327, Lemma 15, p. 328, Theorem 25, p. 330]. We prove the grounded directionality cell below and give a counterexample for stable semantics. Complete and preferred directionality are established literature results; their general proofs are not reproduced here [74, Sec. 4, p. 817].

Requiring complete outputs does not characterize complete semantics: grounded and preferred semantics also meet it. A characterization would say that, within a specified function class, a method satisfies the selected principles *if and only if* it is the proposed semantics. Proving that a semantics has several properties establishes only one direction. This explains why a small table can guide a comparison without selecting a unique method.

11.2 When an upstream answer can ignore its context

Let $F = (A, R)$, $U \subseteq A$, and

$$F \downarrow U = (U, R \cap (U \times U)).$$

The subset U is *unattacked* when no attack enters it from outside: $R \cap ((A \setminus U) \times U) = \emptyset$. Its members may attack one another and may attack arguments outside it. Directionality requires

$$\sigma(F \downarrow U) = \{E \cap U : E \in \sigma(F)\} \quad \text{whenever } U \text{ is unattacked.}$$

Thus evaluating the upstream part by itself recovers exactly the projected whole answers. Noninterference asks for the same equality only when U is *isolated*, with no cross attacks in either direction. Directionality implies noninterference because every isolated set is unattacked [74, Defs. 4.6–4.7, Principles 14–15 and Observation 6, p. 816].

Grounded locality. Write $D_F(S)$ for the arguments defended by S . If U is unattacked, then for every $S \subseteq A$,

$$D_F(S) \cap U = D_{F \downarrow U}(S \cap U).$$

For an argument $x \in U$, every attacker yRx lies in U . Every possible counterattacker zRy also lies in U , by the same no-incoming condition applied to y . The whole and restricted defense tests therefore have exactly the same attackers and possible

witnesses. Restricting the defending set merely replaces $z \in S$ with $z \in S \cap U$. This proves both directions of the identity, including the case with no attackers.

Now start both grounded iterations at the empty set. If S_i is the whole iterate and T_i the restricted one, then $T_0 = S_0 \cap U$, and the identity gives $T_{i+1} = S_{i+1} \cap U$ whenever it holds at stage i . Chapter 6’s finite grounded theorem ensures that both iterations stabilize; continue a stabilized iterate unchanged if the other has not stopped. Their limits satisfy

$$G_{F \downarrow U} = G_F \cap U.$$

Since grounded semantics returns the singleton family containing its limit, this is directionality. The proof combines a local defense identity with the particular empty-start construction of grounded semantics [5, Def. 16 and Lemmas 18–19, pp. 328–329]; it does not derive locality merely from the existence of a fixed point.

Safe projection does not guarantee completion. A whole stable extension E always projects to a stable extension on an unattacked U . Conflict-freeness survives restriction. If $x \in U \setminus E$, stability supplies an attacker in E , and the no-incoming hypothesis puts that attacker in U . Hence

$$\{E \cap U : E \in \text{St}(F)\} \subseteq \text{St}(F \downarrow U).$$

The converse inclusion would require *every* local stable choice to have a whole completion. That is the extra requirement that fails [74, Principle 16 and stable-semantics discussion, p. 818].

Take

$$A = \{a, b, c\}, \quad R = \{(a, b), (b, a), (b, c), (c, c)\}, \quad U = \{a, b\}.$$

No attack enters U . Locally the stable choices are $\{a\}$ and $\{b\}$. In the whole graph, the self-attack excludes c from every conflict-free set. The only remaining candidates are $\emptyset, \{a\}, \{b\}$. The empty set covers nothing; $\{a\}$ misses c ; $\{b\}$ covers both outsiders. Thus

$$\text{St}(F \downarrow U) = \{\{a\}, \{b\}\}, \quad \text{St}(F) = \{\{b\}\}.$$

Both families exist. The local choice $\{a\}$ cannot be completed: it excludes b , while c can neither be selected nor covered. This is an ordinary-AF counterexample, without support or an empty-family artifact.

Chapter 8’s disconnected stable-product theorem has a different boundary. For genuinely disconnected components, choices combine as a product, but a component with no stable extension makes that product empty. Here an outgoing cross attack constrains which upstream choice has a completion even though stable outputs exist. In Chapter 12, components with incoming attacks will therefore be evaluated with their context, rather than by freely multiplying their isolated answers.

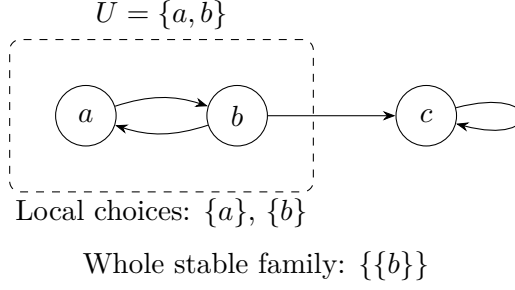


Figure 10: Stable projection can lose a local choice. No attack enters U , but the outgoing attack bRc is the only way to cover the self-attacking c . The local choice $\{a\}$ therefore has no whole completion.

11.3 Support can act at different stages

The same method of comparison applies when the representation grows richer. Chapter 10 distinguished several meanings of support. Even after a meaning has been chosen, a support relation may select among outputs, transform the attacks before evaluation, or change what counts as defense. These operations need not have the same output type.

Use the finite BAF

$$B = (A, R^-, R^+), \quad A = \{a, b, c, d\},$$

$$R^- = \{(a, b), (b, a), (c, d), (d, c)\}, \quad R^+ = \{(a, c), (c, a), (b, d)\}.$$

Only these displayed relations enter the calculation. Our domain permits attack/support overlap, as does the source's triple definition [96, Def. 1, manuscript p. 2]. Ignoring support, stability requires exactly one argument from each mutual pair: selecting both conflicts, while selecting neither leaves an outsider uncovered. The four ordinary stable sets are therefore

$$E_{ac} = \{a, c\}, \quad E_{ad} = \{a, d\}, \quad E_{bc} = \{b, c\}, \quad E_{bd} = \{b, d\}.$$

Selection retains all best existing outputs. Let $\mathcal{E}(B) = \text{St}(A, R^-)$. Internal-support selection uses

$$\text{Score}_B(E) = |R^+ \cap (E \times E)|,$$

$$\text{Sel}(B) = \{E \in \mathcal{E}(B) : \forall T \in \mathcal{E}(B), \text{Score}_B(E) \geq \text{Score}_B(T)\}.$$

The directed edges (a, c) and (c, a) count separately. The four scores, in the displayed order, are 2, 0, 0, 1, so the selected family is $\{E_{ac}\}$. Deleting just (a, c) makes E_{ac} and E_{bd} tie at 1; both must be returned. A first-found tie breaker would define a different function. This is the source's internal selection specialized to a stable base [96, Defs. 7–8, manuscript p. 8].

Selection changes neither attacks nor the stability test. It returns a subfamily of the original family, so every per-extension property of that family is inherited. A finite nonempty family has a maximum score and hence a selected member. An empty family yields an empty selected family. None of this establishes properties comparing different input frameworks: family restriction can change skeptical acceptance and locality.

One-step reduction changes the evaluated graph. Our restricted reduction is

$$R_{\text{red}}^- = R^- \cup \{(x, z) : \exists y, (x, y) \in R^+ \wedge (y, z) \in R^-\}.$$

Both right-hand relations are the original inputs. The three support–attack paths add exactly $(a, d), (c, b), (b, c)$. Of the four original pairs, E_{ad} and E_{bc} now conflict, whereas E_{ac} and E_{bd} remain stable. No singleton covers every outsider: respectively a, b, c, d miss c, d, a, a . The original mutual attacks exclude larger sets. Thus the reduced stable family is exactly $\{E_{ac}, E_{bd}\}$.

The word “one-step” is substantive. With only aR^+b, bR^+c, cR^-d , the rule adds bR^-d , but not aR^-d . The source’s full deductive reduction uses support sequences and super-mediated attacks [96, Defs. 9–10, manuscript pp. 8–9]. Its closure results are not transferred to this smaller construction.

A defense query returns defended arguments. For a supplied set S , define the permissive classroom predicate

$$\text{Def}_B^+(S, x) \iff \forall y \left(yR^-x \Rightarrow \exists z \in S [zR^-y \vee \exists w (zR^+w \wedge wR^-y)] \right),$$

with $z, w, x, y \in A$, and let $D_B^+(S)$ collect its defended targets. The intermediary w need not belong to S . From $S = \{a\}$, ordinary defense gives $\{a\}$; the additional route aR^+cR^-d also answers the attacker of c . No route answers the attackers of b or d , so $D_B^+(\{a\}) = \{a, c\}$.

This is a predicate, not a definition of extensions. Selection returns $\{\{a, c\}\}$, reduction returns $\{\{a, c\}, \{b, d\}\}$, and this query returns $\{a, c\}$ for its supplied defending set. Coinciding argument names do not identify the operations. The source’s defenses 1–3 instead strengthen ordinary defense by requiring additional support or counterattacks; the permissive predicate above is none of those definitions [96, Defs. 3–5, manuscript pp. 3–4].

The study *Mechanisms and change* (p. 214) pursues this comparison through closure, support-aware locality and finite enforcement. Its exercises ask which information each operation preserves. The main conclusion is already available: a principle must name the operation and the type of output it constrains.

11.4 Adding support, choosing a defeat policy, seeking a change

A seemingly natural principle says that adding support to a skeptically accepted target should preserve its acceptance. Fix preferred or stable semantics *after our*

one-step reduction, and use the book’s nonempty-family convention for skepticism. On arguments $\{x, y, z, w\}$, start with no support and attacks

$$R^- = \{(w, x), (x, y), (x, z), (y, w), (z, x)\}.$$

Only $\emptyset, \{z\}, \{y, z\}$ are admissible: x cannot answer w ; w cannot answer y ; y needs z ’s attack on x . Hence $\{y, z\}$ is uniquely preferred and stable, making y skeptical. Add the single support (x, y) . It composes with (y, w) to add attack (x, w) . Now $\{x\}$ answers both its attackers, w, z , and covers every outsider. The old $\{y, z\}$ remains stable. The complete admissible families, developed in the study, show that these are precisely the preferred and stable extensions. Thus the supported target loses skeptical acceptance but retains credulous acceptance. No argument is unattacked before or after, so the grounded extension stays empty. The same ordered pair (x, y) is deliberately both attack and support; excluding overlaps would exclude this example’s domain. The result is a counterexample to this specified target principle, not to every meaning of support.

A numerical target principle would need a valuation method $s_B : A \rightarrow [0, 1]$. With attacks, weights and other parameters fixed, it might require $s_{B'}(y) \geq s_B(y)$ after adding support to y . Counting internal support of an *extension* supplies no such argument valuation.

Preference also needs a policy. Chapter 10 left open how a preference becomes a defeat relation. For $A = \{a, b\}$, raw attack aRb , and strict preference $b \succ a$, consider two named policies. Reduction 1 retains xRy only when y is not strictly preferred to x . Reduction 2 additionally reverses a critical asymmetric attack: if xRy , not yRx , and $y \succ x$, add yRx to the defeat relation. Thus Reduction 1 removes the attack and gives grounded and stable extension $\{a, b\}$; Reduction 2 yields bRa and extension $\{b\}$ [51, Defs. 2.4–2.5 and 2.7, Examples 2.6 and 2.8, pp. 216–217]. The first output is conflict-free with respect to defeat, although not with respect to raw attack. These are alternative reductions, not amendments to Chapter 5’s structured defeat rule.

Enforcement reverses the question. Computing a supplied change’s consequences differs from finding a permitted change that achieves a goal. Fix arguments, grounded semantics, a target $T \subseteq A$, and deletable attacks $D \subseteq R$. Choose $\Delta \subseteq D$, evaluate $F - \Delta = (A, R \setminus \Delta)$, and minimize cost $|\Delta|$. Membership enforcement asks $T \subseteq G(F - \Delta)$; exact-target enforcement asks $G(F - \Delta) = T$. The general literature similarly makes goals and allowed operations explicit [97, Sec. 3.1, Def. 3.1, pp. 457–458].

For $A = \{p, q\}$, $R = D = \{(p, q)\}$, $T = \{q\}$, no deletion gives grounded extension $\{p\}$; deleting the edge gives $\{p, q\}$. Membership has minimum cost 1, because cost 0 fails. Exact-target enforcement is infeasible, because both permitted edits fail. The distinction is not repaired by returning the successful membership edit under the wrong specification. The study develops the exhaustive search and the difference

between inclusion-minimal and minimum-cardinality change.

11.5 Public responses and considered alternatives

A principle can constrain a procedure without selecting an extension semantics. Chapter 4's custody exchange distinguished a request being answered from the issue being established. Chapter 9 then supplied a public state with active assertions, justifications and challenges, a response ledger, and an immediate-reply field. We first ask what that procedure guarantees about the next transition, then ask what practical evaluation preserves when its alternatives change.

A pending request constrains the next public response. Write $\text{Legal}(s)$ for the legal moves at state s , and $\delta(s, m)$ for the resulting state. The pending field is either empty or a triple (i, u, σ) : the speaker required to answer, the same tagged target, and positive or negative polarity. A target is a claim label or a whole support unit, not an untyped graph edge. Reachable states are those obtained from the initial state by legal moves.

Consider the following *immediate-response requirement*: for every reachable s with pending field (i, u, σ) , every $m \in \text{Legal}(s)$ has the form

$$\text{justify}(i, \Phi \vdash^\sigma u), \quad \Phi \text{ a nonempty support unit,}$$

and the resulting state has no pending immediate reply. The corresponding ledger effect is

$$\text{DOS}' = (\text{DOS} \setminus \{(i, u, \sigma)\}) \cup \{(i, \Phi, \sigma)\}.$$

The request is discharged while responsibility for the offered support is recorded. This is a requirement on the legal transition and its public effects, not a guarantee of true premises or persuasive grounds.

Chapter 9's local rules satisfy it directly. A request under CR2 forces CR3 with the specified speaker, target and positive polarity; a request under CR4 forces the corresponding negative CR5 response. Pending-reply priority excludes every other move, and the justification effect clears the pending field and performs the displayed ledger update. These are the book's restricted public conventions. In the original Fatio account, positive justification retains the claim obligation as well as adding the argument obligation; we do not attribute our discharge equation to that source [75, Sec. 3, pp. 4–5, Sec. 4.1, pp. 6–7].

For a concrete transition, let χ be the claim that the restaurant is excellent. After $\text{assert}(A, \chi)$ and $\text{challenge}(B, A, \chi)$, the ledger is $\{(A, \chi, +), (B, \chi, -)\}$, and A's positive reply to χ is pending. The move $\text{retract}(A, \chi)$ cannot evade that request. The legal $\text{justify}(A, \{P\} \vdash^+ \chi)$, where P is the supplied claim that A had a great meal, instead produces

$$\text{DOS}' = \{(A, \{P\}, +), (B, \chi, -)\}.$$

A’s assertion and B’s challenge remain active. The immediate request has been answered; the standing responsibilities have not all disappeared.

A stronger proposed property, “every legal finite record ends with no pending reply,” fails: the two-move prefix ending at B’s challenge is legal and still awaits A. The transition requirement controls how a legal record can continue; it does not ensure that a supplied finite record has continued far enough. A completion criterion or a choice of continuation adds a different object, which Chapter 12 will compute explicitly.

Pairwise comparison and all-foil permission. Practical evaluation has another domain. Recall Chapter 10’s finite set O of feasible options and its weighted comparative reasons (g, o, f) , where $f \neq o$ is the foil. Sum justifying and requiring weights for each ordered pair to obtain $J(o, f)$ and $W(o, f)$. For a complete supplied instance, absent reasons contribute an empty sum, zero; missing assessments are not thereby supplied. The rule is

$$\text{Perm}(o \mid f) \iff J(o, f) \geq W(f, o),$$

$$P_O = \{o \in O : \forall f \in O \setminus \{o\}, \text{Perm}(o \mid f)\}.$$

The opposing requiring weight has reversed indices. These are the attributed source’s comparison definitions used in Chapter 10 [83, Sec. 3.1, Defs. 3.1–3.4, PDF p. 3].

Now compare complete instances on $O \subseteq O'$, holding both totals fixed for every ordered pair of distinct old options. This is the allowed change: new alternatives and their comparisons may be added, but old pair assessments are not revised. Each old pairwise answer remains unchanged, yet the all-foil results need only satisfy

$$P_{O'} \cap O \subseteq P_O.$$

Indeed, an old option passing against every foil in O' passes against every foil in O ; those tests have unchanged values. The reverse inclusion would require it also to pass all newly introduced comparisons. No AF directionality theorem has been used: the argument is an elementary consequence of this rule’s universal quantifier over alternatives.

The dinner makes the failed reverse inclusion concrete. Let R be a restaurant meal, C a café meal and T takeaway eaten separately. The complete pair totals from Chapter 10 give these six tests:

Option / foil	$J(o, f)$	$W(f, o)$	Pass?
R/C	4	3	yes
C/R	3	2	yes
R/T	7	3	yes
T/R	4	5	no
C/T	2	3	no
T/C	3	1	yes

On the restricted alternative set $\{R, C\}$, both options are permitted. Adding T with the displayed assessments changes neither old comparison, but C now fails against its new foil. The all-foil permitted set becomes $\{R\}$. Takeaway itself fails against R ; its role as a relevant comparison does not depend on ultimately being permitted. Holding an old pair fixed therefore differs from holding the practical answer fixed. Revising an old weight is another kind of change again; Chapters 12–13 will compute and explain the dinner’s single-coordinate revision.

11.6 What actual judgments support

Mathematical satisfaction and predictive fit answer different questions. A semantics may implement a normative requirement, or it may model judgments in a particular setting. To test the second aim we need to know how people receive the arguments, what response is requested and which population supplies it. Chapter 6’s reinstatement examples now return as actual study designs.

The following comparison reports the studies described in the Handbook’s cognitive-studies chapter [3, Vol. 2, Ch. 13, Sec. 2.2, pp. 860–865]. It supplies their designs and reported findings, rather than participant-level data or a new statistical analysis.

Confidence in verbally presented conclusions. Rahwan and colleagues’ 2010 work comprised separate studies with 20 and 47 participants approached in Dubai offices and public spaces. Participants saw one to four natural-language arguments and rated a highlighted conclusion on a seven-point false-to-true confidence scale. Confidence was highest for an unattacked conclusion, lowest when it was attacked by an unattacked argument, and intermediate after reinstatement; no simple/floating reinstatement difference was detected. The intended attack graphs were not checked as participants’ reconstructions. Thus this design relates verbal presentation to degrees of confidence, rather than directly comparing three-valued argument labels on a supplied AF.

Reconstructing attacks and then judging arguments. Cramer and Guillaume’s 2018b study involved 130 Luxembourg undergraduates. Three to five arguments appeared in news, science and calculation contexts, covering simple reinstatement, floating reinstatement and a three-cycle. Participants drew attacks, then judged arguments acceptable, rejected or undecided; both tasks included individual responses, peer discussion and revised responses. Simple reinstatement matched the major semantics. The reported CF2, stage, stage2 and SCF2 comparisons matched the majority responses across all arguments in the three AFs. Preferred and semi-stable predictions failed on the three-cycle; grounded and complete predictions failed on floating and three-cycle cases. Stable semantics was set aside where stable extensions did not exist. These names identify the compared predictions; no additional semantic calculus is needed to interpret the design.

Judging a supplied graph. Their 2019 study involved 61 undergraduates,

twelve AFs with three to eight arguments, and a fictional island/treasure setting in which attacks concerned islanders' trustworthiness. Participants were taught the attack relation and shown the intended graphs, then gave acceptable/rejected/undecided judgments. SCF2, CF2 and grounded were reported significantly better predictors than preferred, semi-stable, stage and stage2. Differences among the leading three were not significant. Stable semantics was again set aside on AFs without stable extensions.

The second and third designs do not form a single experiment with more participants. They change graph families, narrative settings and the relation between reconstruction and response. Grounded semantics' different reported fit therefore neither contradicts its mathematical properties nor isolates the cause of the changed ranking. Better prediction under one design does not establish a universally best normative semantics. The Handbook prints "SFC2" on p. 864 but "SCF2" in its definitions and subsequent account; we use SCF2 while retaining that source discrepancy.

For a proposed study, the tuple

$$Study = (\mathcal{I}, \sigma, H, P, m)$$

records presented inputs, the semantic prediction, a behavioral hypothesis, population and measurement. Our running BAF could support the hypothesis that people prefer $\{a, c\}$ because it contains two support edges. That remains hypothetical: we have supplied no observations. Before testing it, one must specify whether the task concerns cohesion, justified claims or something else, and whether support is presented as endorsement, implication or a prerequisite. A mismatch could motivate revising that model-to-task relation. It would not refute a theorem about the fixed selection function.

A finite mathematical tester has a different evidential contract again. A checked violating input refutes a universal principle; a completely checked nonempty sample with no violation establishes only success on that sample. If no violation has been checked, an incomplete or empty sample supplies neither result. Exhausting a finite domain can establish a universal claim about that domain, but the enumeration and checker then need their own correctness arguments. Runtime measurements concern yet another object: the computation used to obtain the answer. Chapter 12 takes up that question.

11.7 Exercises

All tasks admit a paper route. The named study supplies the extra definitions for the mechanism and change questions. Complete solutions, including the optional transfers, appear under Chapter 11 in the collected Solutions.

- 11.1.** [\[solution\]](#) [Calculation] Compute the ordinary stable family of the running BAF in Section 11.3, ignoring support. Justify exhaustiveness.

- 11.2. [\[solution\]](#) [Calculation] Compute every internal-support score of those stable sets. Then delete only (a, c) from support and retain every maximizer. Which skeptical statuses change?
- 11.3. [\[solution\]](#) [Calculation] Apply the one-step reduction to the running BAF. List all added attacks and justify the resulting stable family. Explain which relations the rule reads.
- 11.4. [\[solution\]](#) [Comparison] Explain why selection and reduction return different information, even when both retain $\{a, c\}$. Compare them also with $D_B^+(\{a\})$.
- 11.5. [\[solution\]](#) [Exhaustive verification] For the four-argument supported-target example in Section 11.4, list all admissible sets before and after the support addition. Use the candidate method in the mechanisms study to justify that none is missing. Give the preferred, stable and grounded outputs, the target's changed and unchanged acceptance statuses, and the attack/support overlap on which the example relies.
- 11.6. [\[solution\]](#) [Formulation] State target support monotonicity for a method assigning $s_B : A \rightarrow [0, 1]$. Specify the quantified inputs, permitted change and fixed parameters. Does the Boolean target-loss example alone refute this principle?
- 11.7. [\[solution\]](#) [Construction] Construct a grounded-enforcement problem in which deleting one attack enforces membership of a target argument. State the allowed deletions and unit cost, and certify that no cheaper edit succeeds. Any fully justified example is acceptable.
- 11.8. [\[solution\]](#) [Interpretation] Why are cognitive studies relevant to comparing principles without defining a new semantics? Using the study tuple in Section 11.6, propose a response task and measurement for the hypothetical running-BAF study. State a prediction and one alternative interpretation of a mismatch. Keep proposed observations distinct from reported observations.
- 11.9. [\[solution\]](#) [Classification] Classify the five requirements in the mechanisms study's table as output, construction/change or methodological requirements, allowing an explicitly explained combination. Explain where the methodological question enters when choosing one of these mathematical requirements for an application.
- 11.10. [\[solution\]](#) [Proof] Compute $D_B^+(\{a\})$ and ordinary defense on the running BAF. Prove that D^+ is monotone in its defending set and, with attacks fixed, in support inclusion. Explain why these results do not define an extension semantics.
- 11.11. [\[solution\]](#) [Transfer] Take $A = \{a, b, c\}$, attacks $(a, b), (b, a)$, sole support (a, c) , and $U = \{a, b\}$. Apply the support-aware locality test defined in the mechanisms study. Compute both selected families and the whole family's projection. Check its no-incoming-edge hypothesis and identify the failed inclusion. Is U isolated?
- 11.12. [\[solution\]](#) [Certificate] For $R = D = \{(p, q)\}$ and $T = \{q\}$, compare membership and exact-target grounded enforcement. Certify minimum cost or infeasibility

within the permitted edits. How does either certificate differ from a finite principle test that finds no counterexample?

- 11.13. [\[solution\]](#) [Counterfunction] Does complete-output satisfaction characterize complete semantics? Would adding existence and directionality do so? Use the mutual pair to separate the proposed functions. Then assess the everywhere-empty-family function against these three principles, distinguishing $\emptyset$ from $\{\emptyset\}$.
- 11.14. [\[solution\]](#) [Proof reconstruction] Expand the grounded-locality proof: establish both membership implications in the defense identity, and prove by induction that the two empty-start iterates agree after projection. For the chain $xRyRz$, compare the whole grounded extension with the restricted results on $U_1 = \{x, y\}$ and $U_2 = \{y, z\}$. Where is the hypothesis used?
- 11.15. [\[solution\]](#) [Changed input] In the three-node stable-directionality witness, delete only (c, c) . Compute the new whole stable family and its projection onto $\{a, b\}$. Which previously blocked completion becomes possible? Does this establish a general locality theorem for stable semantics?
- 11.16. [\[solution\]](#) [Policy comparison] Apply both named preference reductions of Section 11.4 to aRb , $b \succ a$. Verify their grounded and stable outputs. What must a claim that “the preferred argument wins” specify? Assess conflict-freeness against raw attack and against the chosen defeat graph.
- 11.17. [\[solution\]](#) [Evidence comparison] Using only the three reported designs in Section 11.6, identify the study that measures confidence, the study that elicits attack reconstruction and the study that supplies intended attacks. Explain why the 130- and 61-participant results cannot simply be pooled. Do the findings establish a unique best semantics, a cause of the different rankings, or a failure of a theorem about grounded semantics?
- 11.18. [\[solution\]](#) [Optional study transfer] Use the mechanisms study. (a) On $A = \{a, b\}$ with sole attack (a, b) , test selection’s deductive closure with sole support (a, b) , and inverse closure with sole support (b, a) . (b) Prove invariance under deletion of support internal to no ordinary stable candidate. (c) On the support chain aR^+b, bR^+c, cR^-d , compare one application of the reduction with repeated application to newly added attacks.
- 11.19. [\[solution\]](#) [Optional study algorithm] Write a paper algorithm that returns every minimum grounded-enforcement deletion or certifies infeasibility. Prove its contract for finite D . Distinguish inclusion-minimal from minimum-cardinality edits. Finally, interpret a tester’s outcomes on (i) one checked violation, (ii) six completed passes, (iii) five passes and one unevaluated record, and (iv) no records, when the principle’s domain is all finite BAFs.
- 11.20. [\[solution\]](#) [Transition property] Use the local public rules and the restaurant state in Section 11.5 after B’s challenge, with A’s positive reply to χ pending. Compare the proposed next moves $retract(A, \chi)$, $justify(B, \{Q\} \vdash^+ \chi)$, and

$justify(A, \{P\} \vdash^+ \chi)$, where P, Q are claim labels. Which is legal, and what ledger and pending field result? Establish the immediate-response requirement for every reachable state of the local protocol. Refute the stronger claim that every legal finite record has already completed its pending requests, and state what neither property would prove about the offered grounds.

- 11.21.** *[solution]* [Practical-domain transfer] Use the six dinner pair tests in Section 11.5. Compute permission on $\{R, C, T\}$, on $\{R, C\}$, and separately on $\{C, T\}$, retaining each surviving ordered-pair assessment. Prove the displayed inclusion for an arbitrary alternative-set expansion with old pair totals fixed. Identify failures of the reverse inclusion in these calculations. Does an option have to be permitted in the larger set to affect another option's permission? What changes in the proof if old pair totals may also be revised?

11.8 Annotated further reading

From selected principles to a theory of comparisons. Van der Torre and Vesic's introduction explains why function types and characterizations matter; their locality discussion gives the full setting for our restriction argument [74, pp. 799–802, 816–818]. Read it with the mutual pair and the blocked stable completion at hand. Baroni and Giacomin's foundational account develops the broader comparison programme [98]; Baroni's modularity chapter asks when local answers can be composed [99, Secs. 2–3]. These routes deepen ordinary semantic comparison. Chapter 8's structured rationality and construction questions remain distinct.

Change the support or preference mechanism. The bipolar manuscript's selection definitions are the next step from our internal edge count; its deductive reduction shows what support paths and super-mediated attacks add [96, Defs. 7–10, manuscript pp. 8–9]. Compare its defenses 1–3, pp. 3–4, with the permissive classroom query before borrowing any extension theorem. The preference chapter develops the two reductions used above and their design consequences [51, Sec. 2, pp. 216–217]. For numerical requirements, the gradual-semantics analyses connect principles to valuations rather than to extension selection [100, 93]. Agent defense supplies a source-sensitive continuation of Chapter 2's agent-labelled reasons [17]. These extensions change the objects being compared; our finite agreements do not characterize them.

From a permitted deletion to general enforcement. The enforcement chapter's explicit goals and operation classes explain why our two-node problem has two answers [97, Sec. 3.1, pp. 457–458]. Its broader algorithms and the dynamics surveys [101, 102] let the reader vary operations, semantics and preservation requirements. The finite search in the study supplies preparation for those questions, not their efficiency guarantees.

From formal predictions to human judgments. Read the three cognitive-

study accounts together, keeping presentation, population and response measure in separate columns [3, Vol. 2, Ch. 13, Sec. 2.2, pp. 860–865]. They extend the internally supplied comparison with experimental context. The proposed explanation of ranking differences on p. 866 is a question for further testing, not an isolated causal result. Neither this reading nor a small BAF calculation validates a universally appropriate semantics.

Requirements on public and practical procedures. Return to the exact locutions and effects when comparing a next-response requirement with a completion or strategic aim [75, Secs. 3–4.1, pp. 4–7]. The dialogue theory route distinguishes permitted interaction from a purpose for choosing among its moves [35, Secs. 3–4]. For practical outputs, read the comparative-reason definitions with the dinner’s alternative-set restriction in view [83, Sec. 3.1, Defs. 3.1–3.4, PDF p. 3]. The inclusion proved here fixes old pair assessments. Revising those assessments or justifying their authority asks a further question, rather than following from locality by analogy.

Literature/research exercise 11.L. *[solution]* Consult one foundational source, one Handbook passage and one further source from this guide. Write a five-sentence comparison covering each source’s question, formal or empirical object, contribution, connection to a chapter example, and limitation. External reading is the purpose of this task; the ordinary exercises above require no external source. A model comparison and criteria for alternatives appear with the Chapter 11 solutions.

12 Chapter 12. Computing a specified answer

A stable extension printed by a solver is a proposed answer. To use it, we must know which graph was supplied, which task was asked and why this set meets that task. Chapter 11 compared semantic functions by their requirements. We now fix a function and ask how to obtain its answers. The distinction between checking, finding and exhausting those answers will explain both the usefulness of computational encodings and their limits. The same question about the required output will then take us back to public dialogue and practical choice: checking moves, choosing a bounded continuation, finding permitted options and selecting one of them.

No programming language or prior complexity course is needed. Sets, Boolean assignments and traces of finite loops provide complete paper routes through the algorithms. Their costs will be stated for an explicit representation, so that a semantic definition does not quietly become a claim about implementation speed.

12.1 Start with a witness and its task

Return to the computational-reuse framework of Exercise 6.2:

$$\begin{aligned} F_{\text{reuse}} &= (A, R), \quad A = \{a, b, c, d, e\}, \\ R &= \{(a, b), (b, a), (b, c), (c, d), (d, c), (d, e)\}. \end{aligned}$$

This is different from Chapter 6's linked-cycle example F_{link} . We re-establish every result used here, rather than assuming that the earlier exercise has been solved.

Consider $E_1 = \{a, c, e\}$. It is conflict-free: its outgoing attacks reach b, d , not another member. These are exactly its outsiders, and both are attacked. Hence E_1 is stable. This one check certifies stable existence and credulous acceptance of each member. It says nothing by itself about whether another stable set omits e .

For a fixed semantics σ , write $\text{Ext}_\sigma(F)$ for its extension family. Verification receives (F, E) and asks whether E belongs to that family. Search asks for a member, perhaps containing a designated query. Enumeration asks for *all* members. The two acceptance decision tasks are

$$\text{Cred}_\sigma(F, q) = 1 \iff \exists E \in \text{Ext}_\sigma(F), q \in E,$$

$$\text{Skept}_\sigma(F, q) = 1 \iff \text{Ext}_\sigma(F) \neq \emptyset \wedge \forall E \in \text{Ext}_\sigma(F), q \in E.$$

We retain the nonempty-family guard from Chapter 6. The standard task literature also discusses skepticism without it; the difference matters when the family is empty [7, Sec. 3.2, pp. 647–648].

A certificate is an object together with a specified checking procedure. A stable set containing q certifies a positive credulous answer. A stable set omitting q refutes

skepticism, but does not refute credulity. A negative credulous answer must rule out every containing stable set. Such evidence might be an exhaustive finite calculation or a checked unsatisfiability proof for a correct encoding. Failed search and timeout supply neither conclusion.

These distinctions extend beyond acceptance. Minimizing a permitted edit adds an optimization goal to enforcement. Finding a smallest explanation combines explanation, search and optimization, once admissible explanation objects and sufficiency have been specified. Monitoring asks whether an observed sequence meets a rule as it unfolds. The output type is part of the task; an algorithm for one does not automatically solve the others.

12.2 An explicit representation and two algorithms

Number arguments $1, \dots, n$, store the m attacks as a list of ordered pairs, and represent a set by a Boolean membership array with n cells. A Boolean cell says whether the indexed argument belongs to the set. One machine word holds an argument index; reading a cell, writing a cell, comparing two indices and advancing a loop each take constant word cost. This is our declared cost model, not an assumption that every possible set representation offers constant-time membership.

Input storage is $O(n + m + 1)$ words, or $O((n + m + 1) \log(n + 2))$ bits with ordinary integer labels. A candidate set uses n membership bits, stored here in n cells. Big-O bounds ignore constant factors and describe an upper bound for sufficiently large inputs; the added 1 keeps the empty-input case meaningful. These polynomially related sizes connect our concrete operation counts to encoded-input complexity [7, p. 636 and Sec. 2.1.1, p. 637].

Checking stability. Given a well-formed candidate $E \subseteq A$, use the following loops.

```

CheckStable(n, edges, inE):
    covered[1..n] := false
    for each (u,v) in edges:
        if inE[u] and inE[v]: return false
        if inE[u]: covered[v] := true
    for v := 1 to n:
        if not inE[v] and not covered[v]: return false
    return true

```

After any processed edge prefix, if no rejection has occurred, no processed edge is internal to E , and `covered[v]` is true exactly when a processed attack from E reaches v . Initialization makes both claims true; processing one edge either detects an internal attack or updates precisely its target's coverage. At the end, the first condition is conflict-freeness, and the final loop checks every outsider. This invariant proves that the answer is true exactly for stable sets [5, Def. 13, p. 328].

Initialization and the last loop cost $O(n + 1)$; the edge scan costs $O(m + 1)$. Thus verification takes $O(n + m + 1)$ word operations and $O(n + 1)$ extra cells. For E_1 above, the marked targets are b, d , which covers every outsider. On the empty AF the empty candidate passes. For a self-attacking singleton, selecting it fails conflict-freeness and omitting it fails coverage.

Computing the grounded extension. Grounded semantics permits direct construction. First compute the defense operator on any supplied set S .

```

Defended(n, edges, inS):
    attacked[1..n] := false
    for each (u,v) in edges:
        if inS[u]: attacked[v] := true
    next[1..n] := true
    for each (b,a) in edges:
        if not attacked[b]: next[a] := false
    return next

Grounded(n, edges):
    S[1..n] := false
    repeat:
        T := Defended(n, edges, S)
        if T == S: return S
    S := T

```

The first scan marks exactly the arguments attacked by S . The second leaves a target true exactly when none of its attackers lacks a counterattack from S . An unattacked target remains true. Thus **Defended** computes $D_F(S)$, directly implementing defense [5, Defs. 6 and 16, pp. 326, 328–329]. A new array retains simultaneous updates: decisions in one round all read the same old S .

Chapter 6 proved that the empty-start iterates are admissible and increasing, and stabilize at the least fixed point, the grounded extension. Every strict increase adds an argument, so at most n strict increases are followed by one equality-detecting evaluation. Each defense evaluation and array comparison costs $O(n + m + 1)$. The total bound for this implementation is

$$O((n + 1)(n + m + 1))$$

word operations, with $O(n + 1)$ extra cells. The whole algorithm is not linear merely because each round is. The empty AF takes one equality check.

For a visible multiround trace, use the chain $xRyRz$. The successive sets are $\emptyset, \{x\}, \{x, z\}, \{x, z\}$: first x is unattacked; then its attack on y defends z . In F_{reuse} , every argument has an attacker, so the very first empty-start update is empty. Its grounded extension is therefore empty, despite its verified stable witness. Finding a fixed point from a different initial set would not justify calling that point grounded.

12.3 Encoding the same conditions as Boolean constraints

A Boolean variable has value true or false. A literal is a variable or its negation; a clause is a disjunction of literals; a formula in conjunctive normal form (CNF) is a conjunction of clauses. A total assignment satisfies a CNF formula when every clause has a true literal. SAT asks whether some such assignment exists; UNSAT means that none exists. Empty disjunction is false and empty conjunction true.

Introduce one variable x_v for each argument v , with true meaning v is selected. Define

$$K_F = \bigwedge_{(u,v) \in R} (\neg x_u \vee \neg x_v) \wedge \bigwedge_{v \in A} \left(x_v \vee \bigvee_{(u,v) \in R} x_u \right).$$

The first clauses exclude internal attacks. The second require each argument to be selected or attacked by a selected argument. An unattacked argument's coverage clause is simply x_v , forcing it in.

The representation is exact. Given a satisfying assignment, set $E = \{v : x_v \text{ is true}\}$. Every attack clause rules out an internal attack. For an outsider v , x_v is false, so its coverage clause requires some attacker in E . Hence E is stable. Conversely, a stable E satisfies every attack clause by conflict-freeness; a coverage clause holds either because its target is in E or because stability supplies a selected attacker. The two maps are inverse because there is exactly one original variable per argument. This proves an assignment/extension bijection from Dung's definition [5, Def. 13, p. 328].

Before duplicate removal, K_F has $n + m$ clauses and $n + 3m$ literal occurrences, counting repeated occurrences. It can be built with linear work in the indexed list representation. The empty AF yields the true empty conjunction and its one empty assignment. The self-attacker yields both $\neg x_a$ and x_a , and hence no assignment. The encoding preserves the empty-extension/empty-family distinction.

For F_{reuse} , the coverage clause for e is $x_e \vee x_d$, since d is its sole attacker. Assignment $x_a = x_c = x_e = 1, x_b = x_d = 0$ satisfies every clause, reproducing E_1 . The important next step is to use the correspondence for the *query* we actually want.

Task	Exact Boolean question
Stable existence	Is K_F satisfiable?
Credulous acceptance of q	Is $K_F \wedge x_q$ satisfiable?
Guarded skeptical acceptance of q	Is K_F satisfiable, and is $K_F \wedge \neg x_q$ unsatisfiable?

Each row follows by applying the bijection to existence, membership or omitted membership. The last row has two obligations. Its second question alone expresses the literature's unguarded skepticism, which is true even when K_F has no model [7,

Sec. 3.2, pp. 647–648]. A checked base assignment and a checked UNSAT argument for the omitting formula support a positive guarded answer; no polynomial-size UNSAT proof is promised.

Here $K_F \wedge x_e$ is satisfiable by E_1 . But $\{a, d\}$ is also stable: it has no internal attack and covers outsiders b, c, e by aRb, dRc, dRe . Its assignment satisfies $K_F \wedge \neg x_e$, so e is not skeptical. One omitting witness settles this question without requiring full enumeration.

12.4 Checking is not searching, and listing has a size

A finite baseline algorithm can examine every subset of A and apply **CheckStable**. After each processed prefix of candidates, its list contains exactly the stable sets in that prefix. When all 2^n subsets have been examined, the list is exactly $\text{St}(F)$. Existence, credulity and guarded skepticism are then evaluated by their definitions. This is a correct algorithm with Boolean outputs for decision questions and a list for enumeration; it need not return a separate certificate object. Its exhaustive search does not become polynomial because each individual check is inexpensive.

A SAT-based enumeration can instead find one assignment, exclude exactly that assignment and ask again. The query-and-certificate study (p. 217) derives the exact blocking clause and its invariant, including auxiliary variables and the empty AF. Correctly checking each reported set establishes soundness of a list; only exhaustion or a valid final UNSAT argument establishes its completeness.

There is also an unavoidable output-size issue. Take k disjoint mutual-attack pairs $\{u_i, v_i\}$, with no other attacks. Every stable extension chooses exactly one endpoint from each pair, and every such choice is stable. There are 2^k extensions on $n = 2k$ vertices and $m = 2k$ attacks. An explicit list of membership vectors writes $n2^k$ bits when $k \geq 1$. Merely emitting that list takes at least that many bit writes, which is superpolynomial in the ordinary explicit input length. A compact symbolic description of the family answers a different output request. When $k = 0$, the one extension is empty.

12.5 Why one acceptance problem is hard

We need a little complexity vocabulary to make a precise comparison. A decision problem is in P if a deterministic algorithm decides it in time bounded by a fixed polynomial in encoded input length. It is in NP if positive instances have polynomial-length witnesses that a deterministic polynomial-time procedure can verify, and negative instances have no passing witness. This definition says nothing about how quickly a witness can be found. The stable checker puts stable existence and stable credulous acceptance in NP: guess an n -bit set, then check stability and, for credulity, query membership.

A polynomial many-one reduction from a problem L to a problem M is a polynomial-

time map f such that $x \in L$ if and only if $f(x) \in M$. If every NP problem reduces to M , it is NP-hard; if it is also in NP, it is NP-complete [7, Secs. 2.1–2.2, pp. 637–643]. We use the foundational theorem that CNF satisfiability is NP-complete: every problem with a polynomially checkable witness has a polynomial many-one reduction to SAT. We state this theorem as our starting result, without proving the general computation encoding behind it.

To prove an argumentation task NP-hard it therefore suffices to map SAT into that task: composing the SAT reduction with our map transfers every NP problem. This use of an encoding differs from the previous section. There we encoded an AF into Boolean constraints to compute its answer. Here we encode a Boolean problem into an AF to show how much a designated argumentation query can express.

Let $\varphi = \bigwedge_{j=1}^t C_j$ be a CNF formula over variables $x_1, \dots, x_k$. Clauses have separate indices even if two are equal; remove repeated occurrences of the same literal within a clause, and put $L = \sum_j |C_j|$. Make fresh, pairwise distinct argument names: query q_φ , clause arguments c_j , and positive/negative literal arguments ℓ_i^+, ℓ_i^- . Tags distinguish these from the formula’s variable names. Define

$$\begin{aligned} A_\varphi &= \{q_\varphi\} \cup \{c_j : 1 \leq j \leq t\} \cup \{\ell_i^+, \ell_i^- : 1 \leq i \leq k\}, \\ R_\varphi &= \{(\ell_i^+, \ell_i^-), (\ell_i^-, \ell_i^+) : 1 \leq i \leq k\} \\ &\quad \cup \{(\ell_i^+, c_j) : x_i \in C_j\} \cup \{(\ell_i^-, c_j) : \neg x_i \in C_j\} \\ &\quad \cup \{(c_j, q_\varphi) : 1 \leq j \leq t\}. \end{aligned}$$

The returned instance is (F_φ, q_φ) of *stable credulous acceptance* [7, Reduction 3.6 and Credulous Acceptance, pp. 651–652]. The opposite pairs encode a truth choice. A true literal can cover its clause; selecting the query excludes every clause argument. Stability must then supply the clause coverage through literals.

A satisfying assignment gives a query-containing extension. Given a satisfying total assignment I , select q_φ and exactly the literal argument made true for each variable. Call this set E_I . It is conflict-free: no opposite pair is selected, and no clause argument is selected. Every omitted literal argument is attacked by its chosen opposite. Every clause has a true literal, so its clause argument is attacked by a selected literal argument. These are all outsiders. Hence E_I is stable and contains the query.

A query-containing extension gives a satisfying assignment. Conversely, suppose E is stable and contains q_φ . Conflict-freeness permits at most one of ℓ_i^+, ℓ_i^- . Stability requires at least one: if both were omitted, neither would have a selected attacker, because its opposite is its only attacker. Thus exactly one is selected for every variable. No clause argument belongs to E , since it attacks the selected query. Every omitted clause argument must therefore be attacked by a selected literal occurring in that clause. Define $I(x_i) = \text{true}$ exactly when $\ell_i^+ \in E$. The opposite-pair condition makes this a total assignment, and clause coverage makes

every clause true. Consequently

$$\varphi \text{ is satisfiable} \iff \text{Cred}_{\text{stable}}(F_\varphi, q_\varphi) = 1.$$

There are $2k + t + 1$ arguments and $2k + t + L$ attacks. A scan of variables, indexed clauses and literal occurrences constructs them in $O(k + t + L + 1)$ indexed word operations; writing integer names adds only a polynomial bit cost. Both directions and this size bound establish the polynomial reduction. Together with the verifier they prove NP-completeness of stable credulous acceptance. This is a worst-case classification. It neither proves $P \neq NP$ nor predicts an implementation's runtime on a chosen collection of graphs.

The proof includes boundary cases. With no clauses the query is unattacked, and the empty conjunction is satisfiable. An empty clause gives an unattacked clause argument that cannot be covered while the query is selected. An unused variable still has an opposite pair requiring a choice. A tautological clause is covered under either choice for its complementary variable.

The query cannot be dropped. For $\varphi = (x) \wedge (\neg x)$, the constructed graph has literal pair $x, \bar{x}$, clause arguments c_1, c_2 , attacks $xRc_1, \bar{x}Rc_2$, and c_1Rq, c_2Rq . Its stable extensions are $\{x, c_2\}$ and $\{\bar{x}, c_1\}$. Each chooses a literal and the uncovered clause, whose attack covers the omitted query. Neither contains q . Thus this unsatisfiable formula still produces an AF with stable extensions. Our construction proves hardness of the specified credulous query, not stable existence. The latter's separate NP-completeness result has its own proof route [7, Table 1, p. 649, Sec. 3.3].

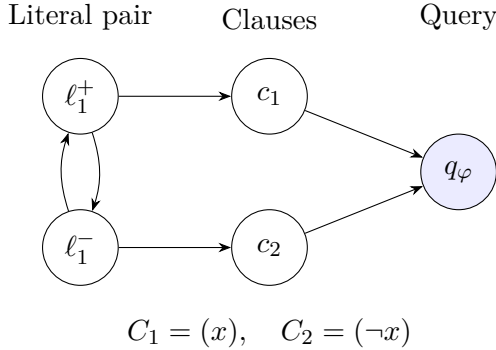


Figure 11: The reduction for $(x) \wedge (\neg x)$, using the fresh tagged names of the general construction. Opposite literals attack one another; a literal attacks each clause in which it occurs; every clause attacks the query. The two stable sets $\{\ell_1^+, c_2\}$ and $\{\ell_1^-, c_1\}$ omit q_φ . Stable existence therefore does not answer the reduced query.

12.6 A witness need not be a complete output

Preferred semantics adds maximality to admissibility. Checking that a supplied set is admissible is polynomial in our representation: check conflict-freeness, compute $D_F(S)$, then test $S \subseteq D_F(S)$. Checking preferredness also requires excluding every proper admissible superset. A loop that tries only single-argument additions is insufficient. In the directed cycle aRb, bRc, cRd, dRa , every singleton is undefended, yet $\{a, c\}$ is admissible. Starting from $\emptyset$, no admissible singleton addition exists, but the empty set is not preferred.

For a *credulous query*, maximality need not be checked on the witness. Chapter 6’s finite extension theorem says that every admissible set is contained in a preferred extension. Thus

$$\exists P \text{ preferred with } q \in P \iff \exists S \text{ admissible with } q \in S.$$

The forward direction uses preferred admissibility; the reverse extends the admissible witness while preserving q . This proves NP membership of preferred credulous acceptance using a potentially smaller witness [5, Theorem 11(2), p. 327][7, Improved Procedures, p. 651]. It does not make preferred verification the same task. The optional query study uses this distinction and supplies the additional vocabulary for a bound on guarded skepticism.

12.7 Condition components on incoming choices

An argument is reachable from another if a directed attack path leads to it, allowing a length-zero path to itself. A strongly connected component (SCC) is a maximal set of mutually reachable vertices. Collapsing SCCs gives an acyclic component graph: a directed cycle among distinct components would make their vertices mutually reachable, contradicting maximality. Incoming choices can therefore be passed forward, provided the semantic computation uses the information they carry [99, Sec. 2.2, pp. 941–945].

In F_{reuse} , the SCCs are $\{a, b\}$, $\{c, d\}$, and $\{e\}$. Cross attacks run bRc and dRe . The first pair requires exactly one of a, b : selecting both conflicts, while selecting neither leaves a uncovered. There are now three complete conditioned branches.

First choice	Consequence downstream	Stable extension
a	Choose c in the second pair. Then d is excluded and cannot cover an omitted e , so e must be included.	$\{a, c, e\}$
a	Choose d instead. Its attacks exclude and cover c, e .	$\{a, d\}$
b	The incoming attack excludes c . Omitting d would leave it uncovered, so choose d ; it excludes e .	$\{b, d\}$

When a is chosen, b is absent, so exactly one of c, d is needed to cover that pair. When b is chosen, c is forced out and d forced in. The status of e is then forced by d . This proves the three branches exhaust all possibilities; each displayed set passes conflict-freeness and coverage. We have established the entire stable family without using Exercise 6.2's answer.

Computing the isolated c, d pair and freely combining its choices with $\{b\}$ would include $\{b, c\}$, which conflicts. The component interface must transmit the incoming selection and attack information. Chapter 8's product theorem applies to disconnected components, and Chapter 11's safe projection goes from a whole extension to an upstream part. Neither licenses arbitrary downstream combinations.

One edge can change the decomposition. Add only (e, a) , keeping all arguments and stable semantics fixed. The old $\{a, c, e\}$ now conflicts; $\{a, d\}$ and $\{b, d\}$ still pass both stable tests. Rechecking those old candidates establishes survival and failure, but it does not yet exclude newly created stable extensions.

For completeness, every stable set still chooses exactly one of a, b : both conflict, and omitting both leaves b uncovered because its only attacker is a . If it chooses a , it must omit e ; covering that omitted e then forces d , excluding c . This yields only $\{a, d\}$. If it chooses b , it excludes a, c , forces d to cover the second pair and excludes e , yielding $\{b, d\}$. The updated family is exactly these two sets. In particular, e changes from credulous but not skeptical to not credulous.

Moreover, $eRaRbRcRdRe$ is now a directed cycle through every vertex. All three old SCCs have merged into one. A single-edge change can therefore invalidate the previous component structure even when some old answers survive. This finite calculation motivates dependency analysis for incremental methods [103, Secs. 2–3, p. 667 onward]; it proves no speedup from reuse and no general SCC algorithm for every semantics.

A solver portfolio similarly requires a conditional performance claim. Different graph families or tasks may favor direct search, SAT, ASP or a structural algorithm. Selecting among them can help when instance features predict useful choices and selection overhead is controlled. Establishing that help requires a declared task, representation, instance collection, hardware, configurations, time and memory measures, and comparison protocol. A hardness proof, a graph partition and a single fast run supply different evidence. The need to specify the task also governs computations whose inputs are public records or comparative reasons instead of attack graphs.

12.8 Check the public trace, then choose a continuation

Chapter 9's local public protocol makes a trace executable by retaining active assertions, active justifications, recorded challenges and a pending immediate reply. Commitments and the response ledger are derived or updated from these records;

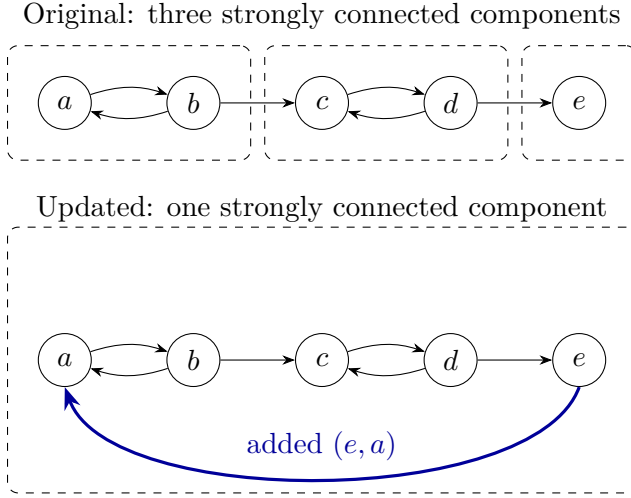


Figure 12: F_{reuse} and its single-edge update. The first decomposition supports conditioned branches, not a product of isolated choices. Adding only (e, a) creates a cycle through all five arguments. The stable family changes from $\{\{a, c, e\}, \{a, d\}, \{b, d\}\}$ to $\{\{a, d\}, \{b, d\}\}$; completeness requires the argument in the text.

they do not replace them. The rules CR1–CR6 determine the legal next move and its public effect, including CR2’s priority over CR4 when both have a witness. For a supplied well-formed initial state and finite move list, the following procedure checks each transition in order. Here **LegalRule** and **Delta** are exactly those local rules and effects from Chapter 9, “What may be said?”, as recalled in Section 11.5.

```

CheckPublic(initial, moves):
  s := initial; certificate := empty list
  for j := 1 to length(moves):
    if not WellFormed(moves[j]):
      return Malformed(j, s)
    rule := LegalRule(s, moves[j])
    if rule == none: return Illegal(j, s)
    s := Delta(s, moves[j], rule)
    append rule to certificate
  return LegalPrefix(certificate, s, s.pending)

```

After each successful iteration, s is exactly the state obtained by the checked prefix, and the certificate contains its applicable rules. The empty prefix establishes the invariant. One legal transition preserves it by the definition of δ ; the first failure identifies the earliest move to which no such transition can be applied. Thus a successful return certifies legality of the supplied prefix and reports any remaining immediate reply. It does not certify that a dialogue has ended. These are the book’s

restricted public rules, including its changed discharge convention, not a checker for every Fatio precondition [75, Secs. 3–4.1, pp. 4–7].

Use the restaurant record to see what a successful check leaves open. The agents are A, B, C ; χ claims that the restaurant is excellent; P reports A’s great meal; Q reports poor vegetarian food; and S says that this counts against excellence. Their content is opaque to the public checker. Start from the fully specified state s_3 after A’s assertion, B’s challenge and A’s positive reply:

$$\begin{aligned} & \text{assert}(A, \chi), \quad \text{challenge}(B, A, \chi), \\ & \text{justify}(A, \{P\} \vdash^+ \chi). \end{aligned}$$

Its active records are A’s assertion of χ , A’s displayed justification and B’s challenge of χ ; there are no other active assertions, justifications or challenges. The pending field is empty, $\text{Com}_A = \{\chi, P\}$, and $\text{Com}_B = \text{Com}_C = \emptyset$. Write

$$D_A = (A, \{P\}, +), \quad K_B = (B, \chi, -), \quad D_B = (B, \{Q, S\}, -).$$

The initial ledger is $\{D_A, K_B\}$. This supplies the active witnesses as well as the visible stores, so that the next legal rule can be decided.

Checking the one-move list $\text{question}(A, B, \chi)$ succeeds under CR4: it targets B’s recorded challenge, while B has no active assertion or support unit with that target to trigger CR2. The stores and ledger are unchanged, and the new pending field is $(B, \chi, -)$. If the next move is $\text{retract}(A, \chi)$, the checker rejects it at position 2. The pending opposing reply takes priority over retraction. If instead the next move is $\text{justify}(B, \{Q, S\} \vdash^- \chi)$, CR5 applies, Com_B becomes $\{Q, S\}$, the ledger becomes $\{D_A, D_B\}$, and the pending field clears. A’s active records persist. The two-move certificate is $(CR4, CR5)$.

Which offered continuation serves the stated purpose? Checking either list does not choose what A should do. For that task, retain Chapter 9’s finite experiment with exactly two offered plans: withdraw A’s assertion, or ask B about the challenge and take the supplied reply. The response policy fixes B’s reply to $\{Q, S\}$. CR5 requires an appropriate nonempty support unit; the policy, rather than the legal rule, selects this one.

Let $\text{retained}(s)$ mean that A’s assertion of χ remains active, and $\text{exposed}(s)$ mean that B has an active negative justification of χ . Observe the first state after at least one additional move at which no reply is pending and either $\text{retained} = 0$ or $\text{exposed} = 1$. This is the experiment’s stopping condition, not a requirement that the whole ledger be empty. Among offered plans reaching that condition within budget b , maximize $(\text{exposed}, \text{retained}, -n)$ lexicographically: compare exposure first, retention next and, if both tie, fewer added moves.

The paper procedure is finite. Generate the two move lists from the offered first moves and the fixed response policy. Starting afresh at s_3 for each list, use the transition checker and test the stopping condition after every successful move. Retain

a plan only if its first stopping state is reached in at most b moves. A legal prefix whose budget expires before that state is recorded as unfinished, not as a completed candidate. Compute each retained plan's score, compare the scores in their declared order, and return all maximizers with their traces; return no completed plan if none qualifies. Exhausting the two offered lists proves completeness for this offered set. It proves nothing about unoffered legal continuations.

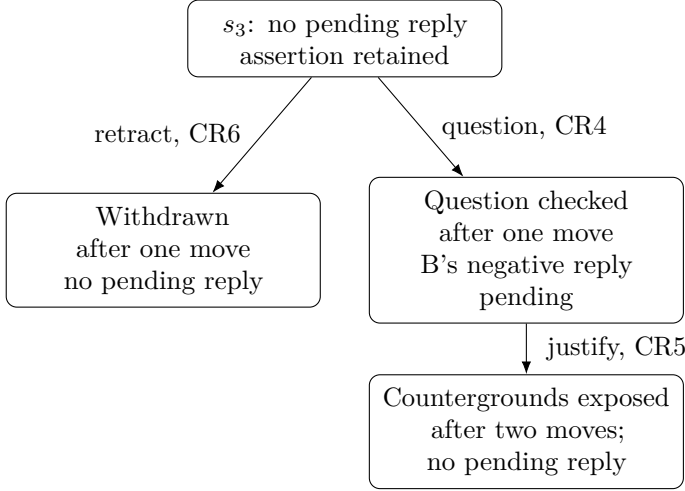


Figure 13: The offered continuations branch from the same public state. Both first moves are legal; only withdrawal already meets the observation condition after one move.

The withdrawal trace applies $retract(A, \chi)$ under CR6. A's justification and B's challenge remain, so $\text{Com}_A = \{P\}$, $\text{Com}_B = \emptyset$, and $\text{DOS} = \{D_A, K_B\}$. It stops after one move with score $(0, 0, -1)$. The inquiry trace was checked above: it stops after two moves with $\text{Com}_A = \{\chi, P\}$, $\text{Com}_B = \{Q, S\}$, ledger $\{D_A, D_B\}$, and score $(1, 1, -2)$. With budget 2, inquiry wins on exposure. With budget 1, only withdrawal completes in time. The intermediate question remains legal under either budget. Figure 13 shows why the checker and the choice procedure return different kinds of result.

No representation cost for general dialogue search has been derived here. Our finite procedure simulates at most three supplied transitions when both plans complete, then compares two scores. How expensive each state operation is depends on how active records and support units are stored. Neither the AF hardness result nor this tiny enumeration establishes a general strategic complexity classification, equilibrium or optimal policy over all public histories.

12.9 Compute permission before applying a selection policy

The practical computation receives a different input: a complete finite set of comparative reasons and their two weights. Reuse the dinner options R , restaurant; C , café; and T , takeaway eaten separately. These option names are distinct from the public agents above. The company ground favours a full shared meal over the shorter café visit or none; the budget ground reflects the café voucher; the speed ground favours takeaway. The entire input from Chapter 10 is small enough to reprint.

Ground	Option	Foil	JW	RW
company	R	C	4	2
company	R	T	7	5
budget	C	R	3	3
budget	C	T	2	1
speed	T	R	4	3
speed	T	C	3	3

All rows are available and applicable, and each distinct reason (g, o, f) occurs once. These are stipulated normative weights, not quantities inferred from the descriptions of the dinner.

For each ordered pair, Chapter 10 sums the justifying and requiring coordinates into $J(o, f)$ and $W(o, f)$. Option o is permitted exactly when $J(o, f) \geq W(f, o)$ for every other considered option f . The opposing total reverses the option and foil; equality passes [83, Sec. 3.1, Defs. 3.1–3.4, PDF p. 3]. The following loops implement those rules. They assume a validated complete reason set and a weight pair for each member; an unknown assessment must be resolved before this particular numerical evaluation can return a set.

```

Permitted(options, reasons, weights):
  J[o,f] := 0 and W[o,f] := 0 for each o != f
  for each distinct (g,o,f) in reasons:
    J[o,f] := J[o,f] + weights[g,o,f].JW
    W[o,f] := W[o,f] + weights[g,o,f].RW
  permitted := empty set
  for each o in options:
    passes := true
    for each f in options other than o:
      if J[o,f] < W[f,o]: passes := false
    if passes: add o to permitted
  return permitted

```

```

Select(permitted, policyOrder):
  for each o in policyOrder:
    if o is in permitted: return o
  return NoSelection

```

The selection input `policyOrder` is a supplied strict total order of the considered options, from most to least preferred by that policy. It is not calculated by the permission loop. After a reason prefix is processed, each cell contains exactly the corresponding coordinate sum over that prefix. The all-foil loop therefore returns precisely the permitted set. The selection loop returns its first member in the supplied order, or no selection when the set is empty. Neither procedure authorizes the selected action’s execution.

Trace the accumulation in the displayed row order. The first row makes $J(R, C) = 4, W(R, C) = 2$; the second makes $J(R, T) = 7, W(R, T) = 5$. Each remaining row fills a different pair, so its printed coordinates are also that pair’s final totals. For example, testing C against R uses $J(C, R) = 3$ but $W(R, C) = 2$, not its own requiring weight 3. The all-foil loop produces the following trace summary.

Option	First foil test	Second foil test	Failing foils
<i>R</i>	against C: $4 \geq 3$	against T: $7 \geq 3$	none
<i>C</i>	against R: $3 \geq 2$	against T: $2 < 3$	$\{T\}$
<i>T</i>	against R: $4 < 5$	against C: $3 \geq 1$	$\{R\}$

Thus the returned set is $\{R\}$. On the cost order C, T, R , **Select** skips C and T before returning R. On the priority order R, C, T , it returns R at the first membership check.

Now revise only $RW(speed, T, C)$ from 3 to 2. The accumulation changes only $W(T, C)$. This cell is read when C is tested against T, so that test changes from $2 < 3$ to $2 \geq 2$; T’s own test against C still reads $J(T, C) = 3$ and $W(C, T) = 1$. All other tests are unchanged. The new permitted set is $\{R, C\}$. The same cost order now returns C immediately; the priority order still returns R immediately. A correct implementation should preserve both the changed permission result and this dependence of the selected result on a separate policy.

For an operation count, index k options and store the two totals in arrays, with r distinct reason records carrying exact rational coordinates. The displayed loops initialize $2k(k - 1)$ cells, make $2r$ additions and $k(k - 1)$ comparisons, then perform at most k membership checks for selection. This counts arithmetic and array operations; rational arithmetic has an additional bit cost that depends on the sizes of its numerators and denominators. A completed empty reason list for a pair correctly contributes zero. A missing weight or an unfinished list is different information and cannot be filled by the initial zeros. Repeating an utterance also does not create another set member to accumulate.

These procedures establish what follows from supplied records and rules. The

next chapter asks how to explain their different outputs, including which public responsibility, comparative ground or selection policy answers the user’s particular request.

12.10 Exercises

All tasks can be solved using assignments, edge lists and paper traces. Programming is an optional implementation of that work. Complete solutions appear under Chapter 12 in the collected Solutions.

- 12.1. [\[solution\]](#) [Verification] Verify $\{a, c, e\}$ in the reprinted F_{reuse} by both stable conditions. Relate each condition to the checker invariant.
- 12.2. [\[solution\]](#) [Verification] Verify $\{a, d\}$ in the same AF, naming the attacker covering every outsider. Would checking outsider coverage alone suffice in general?
- 12.3. [\[solution\]](#) [Witness] Give and verify a certificate that b is credulously accepted under stable semantics in F_{reuse} .
- 12.4. [\[solution\]](#) [Counterexample] Is e skeptically accepted under stable semantics in F_{reuse} ? Give the smallest kind of evidence needed to settle this question, explaining why no full enumeration is necessary.
- 12.5. [\[solution\]](#) [Encoding] Write every conflict-free clause for F_{reuse} , first one per attack and then with duplicates removed. Which information about attack direction is lost from these clauses alone?
- 12.6. [\[solution\]](#) [Encoding] Write the coverage clause for c in F_{reuse} . Explain why it uses attackers rather than arguments attacked by c .
- 12.7. [\[solution\]](#) [Task diagnosis] Classify “find a smallest subgraph explaining why e is accepted” among verification, search, enumeration, optimization, explanation and monitoring. What information must be fixed before this becomes a precise computational task?
- 12.8. [\[solution\]](#) [Performance hypothesis] Give a reason a portfolio might outperform a single solver on a declared collection of tasks, and state what measurements would test that hypothesis. Does NP-completeness prove or refute the proposed performance advantage?
- 12.9. [\[solution\]](#) [Boundary] On the one-argument self-attacking AF, compare unguarded and guarded stable skepticism. Give the stable CNF and explain which Boolean query distinguishes the two conventions.
- 12.10. [\[solution\]](#) [Certificate diagnosis] What would certify that b is not credulously accepted under stable semantics? Can any valid such certificate exist for F_{reuse} ? Distinguish an omitting extension, an exhaustive argument and a search timeout.

- 12.11.** *[solution]* [Algorithm transfer] Use the four-node input from Chapter 7’s translation exercise, reprinted here:

$$H = (\{a, b, c, d\}, \{(a, b), (b, c), (c, b), (c, d)\}).$$

Trace **CheckStable** on $\{a, c\}$ and $\{a\}$, and trace **Grounded** through its equality check. Then delete only (a, b) and repeat grounded iteration. Explain the role of simultaneous updates, the empty start, the per-round cost and the total general bound.

- 12.12.** *[solution]* [Query and exclusion] For H in Exercise 12.11, write the complete stable CNF, find all satisfying assignments and decide guarded skepticism of c . Using the exact-block definition in the query study, block the first extension and justify exhaustion. After deleting (a, b) , rebuild the encoding, find all outputs and reassess credulous and skeptical acceptance of c and credulous acceptance of d . Explain why graph deletion is not in general implemented by deleting an arbitrary clause.
- 12.13.** *[solution]* [Reduction reconstruction] Construct the reduction graph for $(x \vee y) \wedge (\neg x)$, using $q, c_1, c_2, x, \bar{x}, y, \bar{y}$ as distinct argument names. Count its vertices and attacks, find all query-containing stable extensions, and recover the corresponding assignments. Diagnose $\{q, x, \bar{y}\}$. For arbitrary CNF input, reconstruct both directions and the size bound. Explain the roles of the query, fresh names, empty clauses and unused variables. Test a claimed reduction to mere stable existence on $(x) \wedge (\neg x)$.
- 12.14.** *[solution]* [Condition and update] Derive all three stable branches of F_{reuse} from its SCCs. Why can $\{b\}$ not be combined freely with the isolated second pair’s choice $\{c\}$? Add only (e, a) ; verify which old sets survive, then separately prove completeness of the updated family. Find the new SCCs and the old/new stable acceptance status of e . Which claim about incremental performance remains unproved?
- 12.15.** *[solution]* [Witness and output size] For H in Exercise 12.11, show that $\{a\}$ is an admissible witness for preferred credulous acceptance of a , although it is not preferred. Explain the theorem that licenses this witness. Refute a preferred verifier that tests only one-argument enlargements by using aRb, bRc, cRd, dRa . Finally count the stable sets of k disjoint mutual pairs and the bit writes in an explicit membership-vector listing, including the $k = 0$ boundary.
- 12.16.** *[solution]* [Optional query study] Using the definitions of coNP and DP in the query study, prove membership of guarded stable skepticism in DP. Specify the two languages and the witness for the complement of the second. What would an additional DP-hardness claim require?
- 12.17.** *[solution]* [Optional query study] Prove that the exact blocking clause excludes precisely one total argument assignment. Compare it with the clause using only

negations of selected variables. In an encoding $K_F \wedge (z \vee \neg z)$ with fresh auxiliary z , explain why blocking full assignments can repeat an extension. Prove the projected enumeration invariant and give the traces for the empty AF and the self-attacking singleton.

- 12.18.** *[solution]* [Public trace and bounded choice] Use the fully specified restaurant state s_3 in Section 12.8. Trace **CheckPublic** on (a) the question $question(A, B, \chi)$ alone; (b) that question followed by $retract(A, \chi)$; and (c) that question followed by the supplied negative reply with support $\{Q, S\}$. Give the first rejection or certificate, final public stores, ledger and pending field, as applicable. Then run the offered-continuation procedure with budgets 2 and 1, recording each completed candidate, its stopping reason and score. Explain which parts of its answer depend on the legal rules, response policy, objective and budget. Why does the procedure establish neither the best continuation over all legal histories nor the restaurant’s excellence?
- 12.19.** *[solution]* [Permission and selection trace] Run **Permitted** on the six dinner rows in Section 12.9, showing the final cells and the failing foils for each option. Apply **Select** under both supplied orders. Change only $RW(speed, T, C)$ from 3 to 2: identify the changed cell, every affected comparison, the new permitted set and the two selected outputs. Explain why neither summing only justifying weights nor counting a repeated utterance as a new reason implements the procedure. Finally distinguish a completed empty reason list from an unassessed weight, and give the displayed loops’ arithmetic-operation counts in terms of the numbers of options and distinct reasons. What cost remains unspecified by treating each arithmetic operation as one step?

12.11 Annotated further reading

Task-specific complexity. Dvořák and Dunne supply the input-size conventions and class definitions behind this chapter, then organize results by task and semantics [7, p. 636, pp. 637–643, Secs. 3.2–3.3.2, pp. 647–652]. The full credulous reduction is preparation for reading their other reductions; it does not supply the separate proofs of every table cell. Keep their unguarded stable skepticism distinct from the book’s two-obligation task. The broader hierarchy and parameterized classifications are continuations, not assumed prerequisites.

Alternative computational representations. The implementation chapter compares direct methods with encodings and empirical evaluation [104, pp. 705–706, 743–744]. Egly, Gaggl and Woltran’s ASP approach asks how a declarative program can compute different semantic tasks [105]; the logic-programming connections in Volume 3 return to the representation question developed in Chapter 7 [56, Sec. 3.2, pp. 438–440]. Read these after checking the CNF correspondence, so that extraction and query correctness remain visible. No solver installation is required for that

conceptual comparison.

What crosses a component boundary? The locality/modularity account develops semantic interfaces beyond our finite SCC trace [99, Sec. 2.2, pp. 941–945]. Volume 3’s input/output and replacement discussion asks when a component can be replaced without changing the relevant context’s answers [4, Ch. 13, Sec. 4.2, pp. 926–930]. Its account of Baroni and colleagues’ input/output approach includes a preferred-transparency failure; it should not be read as a general replacement theorem. The incremental chapter explains the different problem of reusing work after updates [103, Secs. 2–3, p. 667 onward]. Our merged-SCC example prepares the dependency question without predicting savings.

From task contracts to measured implementations. The retained IC-CMA 2023 DC-ST task and format material asks for a stable credulous decision and its specified witness behavior [106]. It is a useful comparison with our hand-checkable certificates, rather than a claim about current solver rankings. Benchmark conclusions require the declared instance family, task, configuration and evaluation protocol; a portfolio’s possible advantage remains empirical until tested on those terms.

Executable public and practical inputs. Return to the public acts and effects in Fatio [75, Secs. 3–4.1, pp. 4–7], comparing them with Chapter 9’s declared local changes before proposing another trace checker. For permission, reconstruct the two opposing sums in the attributed balancing source [83, Sec. 3.1, Defs. 3.1–3.4, PDF p. 3] and test an implementation on a zero margin and an empty pair list. These routes extend different contracts: neither the numerical source nor the public rules supply our dinner selection orders or the restaurant’s bounded objective. Studying strategies over a larger protocol requires additional assumptions about responses, objectives and observations.

Literature/research exercise 12.L. *[solution]* Read one original encoding or implementation source, one Handbook passage and one further computational source from this guide. Compare their tasks, representations, outputs, correctness warrants and limits in five sentences, connecting each to a concrete calculation above. Alternatively, use the ICCMA 2023 task/format reading for the further source and compare its witness contract explicitly. This exercise requires those external readings; implementing a solver is optional. A model comparison appears in the solutions.

13 Chapter 13. Explaining an answer in context

In Chapter 12, e belonged to the verified stable extension $\{a, c, e\}$, was credulously accepted, and was not skeptically accepted. All three statements were correct. An explanation beginning simply “ e is accepted” would leave its most consequential choice unstated: which of those questions is it answering?

An explanatory request also reaches beyond a truth value. A reader may want to see the defense of e , compare it with the alternative containing d , challenge the interpretation of an attack, or understand why an edit changed the result. These requests share a formal input but demand different information. This chapter develops explanations whose targets and retained context make that demand explicit, then compares their mathematical warrants with evidence about explanations for other AI systems. The dinner decision will also return as an explained result: a changed requiring weight changes permission, while a separate policy decides which permitted option is selected. Asking who may revise that weight will expose an additional interface between public reasons and numerical evaluation.

13.1 Fix the target before selecting its reasons

Use the framework whose complete stable calculation was established in Chapter 12:

$$\begin{aligned} F_{\text{reuse}} &= (A, R), \quad A = \{a, b, c, d, e\}, \\ R &= \{(a, b), (b, a), (b, c), (c, d), (d, c), (d, e)\}, \\ \text{St}(F_{\text{reuse}}) &= \{E_1, E_2, E_3\} = \{\{a, c, e\}, \{a, d\}, \{b, d\}\}. \end{aligned}$$

The grounded extension is empty. These results belong to the computational reuse example, not to Chapter 6’s different linked-cycle graph.

Represent an explanation target by

$$\text{Target} = (F, \sigma, q, o, \tau),$$

where F is the represented input, σ the semantics, q the query, o its observed answer, and τ its type. For an extension-relative query, q includes a particular witness E , which must be verified. A credulous query ranges over some extension; a skeptical query requires a nonempty family and membership in every extension. A system-output query instead concerns an additional declared operation, for example a choice function applied to the semantic family and other supplied inputs. Its answer need not itself be an extension.

Target	Question about e	Evidence already available
Extension-relative	Is $e \in E_1$?	Verify E_1 , then explain its accepted member.
Credulous stable	Does some stable set contain e ?	The verified E_1 is a containing witness.
Guarded skeptical stable	Does a stable set exist, and do all contain e ?	E_2 is a verified omitting counterexample, so the answer is false.

The false skeptical answer does not say that e is out in every extension. Nor does the containing witness establish that its reasons must be accepted under every semantics.

Labels also depend on a context. Relative to a complete extension E , arguments in E are in, arguments attacked by E are out, and the remaining arguments are undecided. Thus e is in relative to E_1 and out relative to E_2 , whose member d attacks it. Relative to the empty grounded extension of this framework, every argument is undecided. An argument outside an extension is not automatically out: an accepted attacker is required. In a complete labelling, an undecided argument has no in attacker and has at least one undecided attacker; otherwise all attackers would be out, the argument would be defended and completeness would put it in.

13.2 A contextual defense can be finite and cyclic

To explain e relative to E_1 , begin with its sole attacker d . The accepted argument c attacks d . If the reader asks why c can play that role, both its attackers must be considered: b is answered by a , and d by c itself. The accepted argument a in turn answers its own attacker b . The full record is therefore more than the short chain $cRdRe$.

Repeated accepted arguments close a branch by reference to an earlier occurrence. They do not erase an attack. The certificate retains every attacker of every expanded proponent and a verified counterattack from E_1 against each one. Its cycles explain admissible defense within that verified context; they are not a derivation of grounded acceptance.

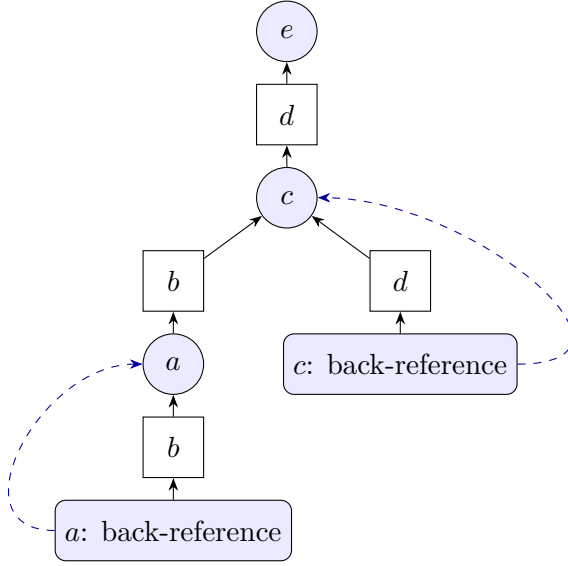


Figure 14: Explaining e within the independently verified $E_1 = \{a, c, e\}$. Circles are expanded proponents, rectangles their original attackers. Solid arrows point in the attack direction, including every response from E_1 . Dashed arrows close repeated proponent occurrences by reference to c or a ; they are not extra attacks. Both attackers of c , and the attacker of a , remain visible. Cyclic closure explains defense in the verified witness, not grounded acceptance.

Here is an exact finite procedure for this local kind of certificate. Supply a verified complete, grounded, preferred or stable extension E and an argument $a \in E$. Each such extension is admissible, so the requested defenders exist. Fix an order of argument identifiers to make choices reproducible, and start with an empty branch path.

```

ExplainAccepted(F, E, a, path):
  if a occurs in path: return BackReference(a)
  nextPath := path followed by a
  tree := Proponent(a)
  for each attacker b of a in F:
    c := first member of E that attacks b
    add Opponent(b) to tree, with response
      ExplainAccepted(F, E, c, nextPath)
  return tree

```

An expanded branch contains at most $|E|$ distinct proponent arguments before ending or referring back. Each has finitely many attackers, so the whole finite-branching expansion terminates. Every loop ranges over all original attackers, and every selected response is an actual counterattack by a member of E . These facts prove the displayed certificate contract from Dung's defense condition [5, Def. 6, p. 326]. Verification of E is a separate premise, not a conclusion obtained by recognizing a repeated node. Unfolding can duplicate shared substructures, so bounded depth alone is not a polynomial output-size guarantee.

For an out argument, choose one attacker in E , show its attack on the target, and expand that accepted attacker by the same procedure. For example, relative to E_2 , dRe explains why e is out, and the subsequent expansion explains d 's defense. If no attacker lies in E , that out-certificate is unavailable; an undecided explanation is a different task. The procedure also does not prove skepticism or compute an optimal explanation. Those targets require their additional obligations.

This is a pedagogical contextual certificate, rather than an identification with every dispute-tree or explanation formalism. Its positive content is precise: each shown accepted reason has all its original attackers answered within the verified witness, with cycle closure made inspectable.

13.3 Small enough for which question?

A substructure might be an induced subframework, an edge-deleted graph, a labelled certificate or a selected route. Before calling it sufficient, we must specify whether it is evaluated *standalone* or with omitted parts of F retained as context. We must also restrict the allowed substructures so that the query remains defined. A verified witness in the whole framework is not automatically a valid witness after a deletion.

For our local conventions, an allowed substructure $S \preceq F$ is sufficient when its declared local or contextual evaluation returns the target answer o . It is inclusion-

minimal when no proper allowed substructure remains sufficient. Minimum size additionally compares it with all sufficient alternatives under a specified size measure. These notions need not coincide, just as minimal and minimum enforcement differed in Chapter 11. They are definitions for the present tasks, not an assertion that all research accounts of explanation use the same objects.

Consider standalone credulous stable acceptance of e . Retain all vertices but only edges

$$(a, b), (b, c), (c, d), (d, e).$$

This chain has unique stable extension $\{a, c, e\}$: a is forced, which excludes b , forces c , excludes d and forces e . It preserves the yes-answer locally. But it omits the original attacks (b, a) and (d, c) , so it is not an attacker-complete contextual certificate of the original defense.

If vertex deletion is also permitted while retaining e , the smaller chain $cRdRe$ has unique stable extension $\{c, e\}$. Even the singleton $\{e\}$ with no attacks gives a sufficient local yes-answer. It is inclusion-minimal among these allowed subgraphs because the target vertex must remain and no edge is present. This tiny explanation preserves a recomputed answer while losing the original reasons for it. To explain F_{reuse} , the attacker information must instead remain in the display or in its explicit verified context.

Necessity relative to permitted deletion. Call an element necessary for the present target if its allowed deletion changes the *recomputed* answer. This is local but-for necessity for a declared edit class, not a claim that the element appears in every other notion of explanation. For credulous stable acceptance of e , the edge (c, d) is necessary under single-edge deletion. Removing it makes d unattacked. Every stable extension must include d , which excludes c, e . Exactly one of a, b is then chosen, giving precisely $\{a, d\}$ and $\{b, d\}$. The recomputed credulous answer for e is false. Looking at membership in an unchanged printed E_1 would miss the operation being tested.

The update in Chapter 12 adds (e, a) and reaches the same two stable sets by a different change. Its new cycle invalidates the former $\{a, c, e\}$ witness and merges all old SCCs. An explanation that reuses that old certificate without checking its graph version would no longer have a verified context, even if many of its individual edges were unchanged.

13.4 Contrast and interaction select another target

Why is e in under E_1 but out under E_2 ? Here the framework and semantics are fixed and the witness changes. In E_1 , the choice of c answers d , allowing e ; in E_2 , the choice of d excludes c and attacks e . That is the relevant contrast, not merely a second recital that both sets are stable. A contrastive target can be represented by two typed targets together with the declaration of what is held fixed and what varies.

In general, a contrast explains which features support the selected alternative and which defeat or exclude its foil, under the specified mechanism.

The dinner example below gives another foil-sensitive contrast. There the question concerns an option’s permission or selection, rather than a claim meeting its issue’s proof standard. Before turning to that result, consider how a request can reach the content on which a calculation depends.

A request for grounds can also question an edge’s meaning. Suppose a system shows cRd , and the user asks why the supplied content warrants that attack. Another extension of the same AF will not answer this reconstruction question. Chapter 3 distinguished requests for grounds from opposition; Chapter 9 supplied public states in which the two acts have different effects. We can use that existing protocol without treating a raw edge as a legal move target.

Let the speakers be system M and user U . Let the opaque claim label χ mean “the represented reason c attacks d ”. Let ρ name a supplied interpretation of the reasons and ω a proposed warrant for that attack. The protocol treats these as claim labels; their truth is a separate matter. Our initial state has only $assert(M, \chi)$ active, commitments $Com_M = \{\chi\}$, $Com_U = \emptyset$, ledger $DOS = \{(M, \chi, +)\}$, and no pending reply or prior challenge.

Now $question(U, M, \chi)$ is legal under Chapter 9’s CR2 because it targets M ’s active assertion. It leaves the ledger unchanged and makes M ’s immediate positive reply pending. The supplied reply

$$justify(M, \{\rho, \omega\} \vdash^+ \chi)$$

is legal under CR3. It replaces the requested obligation $(M, \chi, +)$ by the support-unit obligation $(M, \{\rho, \omega\}, +)$, clears the pending field and adds ρ, ω to M ’s commitments. The assertion remains active. A *challenge* in place of the question would additionally record U ’s opposition obligation. These are the restricted public rules already distinguished from full Fatio in Chapter 9 [75, Secs. 3–4.1, pp. 4–7].

The exchange makes responsibility visible: an active claim has received requested grounds, and those grounds can themselves become a whole-unit target. Protocol legality does not verify the warrant ω or the extracted attack. A semantic certificate, a justification’s content and a record of who owes a reply each contribute something different to answering the user.

13.5 Why the dinner result changes

Chapter 12’s two practical procedures make it possible to explain the dinner result without treating permission and selection as synonyms. Use its entire six-row input from Section 12.9, now called version v_1 . The options are restaurant R , café C and separate takeaway T . Each is available; the company, budget and speed grounds are applicable. Permission requires passing every other considered option as foil, by

comparing $J(o, f)$ with the opposing $W(f, o)$, with equality passing [83, Sec. 3.1, Defs. 3.1–3.4, PDF p. 3].

Why is R permitted? Against C, the company’s justifying force for the restaurant is 4, meeting the budget reason’s opposing requirement of 3. Against T, its justifying force is 7, meeting the speed reason’s opposing requirement of 3. These are both foils, so the two successful tests establish R’s permission. The same company ground does different comparative work: the café retains a shorter shared visit, whereas separate takeaway offers none. That difference explains the relevance of the foil; it does not derive the stipulated numbers.

Why is C excluded even though it passes against R? Its budget force 3 meets the restaurant’s opposing company requirement 2, but its budget force against T is only 2, below the speed reason’s requiring force 3. The decisive failed comparison is C against T. T itself fails against R, since its justifying force 4 falls below the company’s requiring force 5. A nonpermitted option can therefore supply the foil that excludes another option. These specific failures, together with R’s two passes, explain the original permitted set $\{R\}$.

Now compare v_1 with v_2 , which changes only the requiring coordinate of $r = (\text{speed}, T, C)$ from 3 to 2. The options, reason set, other weights and permission rule are held fixed. The explanatory contrast is the single test

$$\begin{aligned} v_1 : \quad & J(C, T) = 2 < W_{v_1}(T, C) = 3 \quad (\text{C fails against T}), \\ v_2 : \quad & J(C, T) = 2 \geq W_{v_2}(T, C) = 2 \quad (\text{C passes against T}). \end{aligned}$$

C already passed against R, so removing this only failed comparison admits C to the permitted set. R remains permitted and T still fails against R. The changed result is $\{R, C\}$. This explains why reducing a weight on a reason *for T* can admit *C*: it reduces the opposing requirement in the C/T test. T’s own justifying coordinate, and therefore its test against C, is unchanged.

The question “why C rather than R?” still has no answer until a selection policy is named. Chapter 12 supplied a cost order C, T, R and a priority order R, C, T ; both select the first permitted option in their order.

Version	Permitted set	Cost selection	Priority selection
v_1	$\{R\}$	R	R
v_2	$\{R, C\}$	C	R

In v_2 , C is selected under the cost policy because it is permitted and occurs before R. The priority policy selects R from the same set because R occurs first. Neither choice establishes that its selected option is the only permissible one or that it defeats the other in a contest. Both C and R pass their mutual permission tests in both versions. Changing the policy alone in v_1 cannot make C a permitted choice. The weight revision first makes C available to a policy that selects among permitted options.

Ask for the weight’s grounds and authority. Checking $2 \geq 2$ answers an arithmetic question. A participant may instead ask why speed should have requiring strength 2 in this comparison, who endorsed that assessment, or who may replace the recorded 3. These requests identify the context, ground, option, foil and coordinate; asking only whether “speed matters” leaves the contested target underspecified. Chapter 10’s study “Where do weights get their authority?” distinguishes this provenance and normative question from reproducible calculation.

For a concrete public request, let coordinator K assert the opaque claim η : “In this dinner context, the requiring coordinate of $(speed, T, C)$ should be 2 under the agreed deadline policy.” Let ρ be the proposed ground that both C and T can meet the agreed end time, and ω the proposed warrant that the policy assigns speed’s requiring force 2 when both options can meet that time. These are supplied claim contents, not facts or policies verified by the protocol. They make the requested justification more specific than the fact that takeaway is quickest.

Start with only K’s assertion of η active, commitments $\text{Com}_K = \{\eta\}$, $\text{Com}_U = \emptyset$, ledger $\{(K, \eta, +)\}$, no recorded challenges and no pending reply. The question $question(U, K, \eta)$ invokes CR2, requiring K’s immediate positive response while leaving the ledger unchanged. K’s supplied reply

$$justify(K, \{\rho, \omega\} \vdash^+ \eta)$$

invokes CR3, clears the pending field, adds ρ, ω to K’s commitments and replaces $(K, \eta, +)$ by $(K, \{\rho, \omega\}, +)$. The active assertion persists. This local public transition identifies who offered the ground and warrant and who is responsible for them. It establishes neither their truth nor K’s authority to change a numerical input; the separate model is still v_1 .

An explicit revision interface. To connect this discussion with reevaluation, stipulate a small revision interface. It is a pedagogical design for this example, not a rule imported from Fatio or the balancing source. Let the context identifier be d , and let v_1 name the current complete numerical input. Independently of the public utterances, the supplied authority record allows K to revise the RW coordinate of row $r = (speed, T, C)$ in context d ; U has no such authority. Authority for other records or for executing an option is not supplied.

A proposed revision carries the following fields. The right column gives one fully specified candidate, with its supplied ground and warrant records identified by ρ and ω .

Field	Candidate value
Context and base version	d, v_1
Reason row and coordinate	$r = (speed, T, C), RW$
Expected old value; proposed value	3; 2
Issuer	K
Ground and warrant records	$\rho; \omega$

The interface accepts a candidate exactly when its context and base version match the current input, its row and coordinate exist, its expected old value matches that cell, its issuer is authorized for that context, row and coordinate, its proposed value preserves the allowed nonnegative, nonzero weight pair, and the proposal supplies its identified ground and warrant records. Matching these records is not an assessment of their truth or normative adequacy. Here changing $(JW, RW) = (3, 3)$ to $(3, 2)$ meets the weight-domain condition.

On acceptance, copy the input to a new version, change exactly that coordinate and retain the proposal’s provenance. Keep every other row, coordinate and option fixed. Then run **Permitted** on the new input and **Select** with the expressly named policy; an old selection cannot simply be carried over. This candidate therefore produces v_2 , permitted set $\{R, C\}$, and either C under the cost policy or R under the priority policy. The arithmetic contrast above explains this reevaluation.

On failure of any condition, record the failed check and leave the current input and its outputs unchanged. For example, a proposal with the same numbers but issued by U fails the authority check; one naming a different base version fails the version check. A legal assertion or justification on behalf of either proposal does not repair those failures. Conversely, accepting K’s matching record supplies a procedurally authorized numerical revision under the stipulated authority relation, not a proof that the new weight is normatively best. A later request may still contest ρ , ω or the authority record itself.

The connection now has a visible boundary. The public protocol tracks claims, requests and responsibility; the interface checks an authorized change to a particular model version; the numerical rules evaluate that version. Libra’s optional public model in Chapter 9 uses its own abstract comparison predicate within agent-relative relevance and permission records, not these numerical sums. Neither that predicate nor protocol legality grants an unrecorded weight update. Keeping these roles explicit makes the dinner explanation both inspectable and contestable without identifying the three formalisms.

13.6 Explaining another system through argumentation

So far the explained object was an argumentation result. In explanation *through* argumentation, an argumentative construction instead explains a classifier, recommender or other system. It then needs a relation to the behavior it purports to explain.

For a synthetic example, suppose a classifier f approves an application. An explanatory reconstruction offers a reason a_1 from high income to approval, an opposing reason a_2 from a missing document, and a reason a_3 that a waiver defeats the missing-document objection. Such a reconstruction may help organize a contrast, but its relation to f has still to be tested. Let g be the output of the chosen argumentative surrogate. The following four pairs are stipulated synthetic test

outputs, not participant observations or values derived from a complete classifier model printed here.

Input	f	g
x_1	approve	approve
x_2	approve	refuse
x_3	refuse	refuse
x_4	refuse	refuse

For a declared nonempty finite test set X' , define sampled fidelity by

$$\text{Fidelity}_{X'}(f, g) = 1 - \frac{1}{|X'|} \sum_{x \in X'} \mathbf{1}[f(x) \neq g(x)],$$

where the indicator is 1 on disagreement and 0 otherwise. Here one of four outputs differs, giving $3/4$. This establishes agreement on the declared test set. It does not establish a correspondence for every possible input, the correctness of the classifier’s decisions, or the helpfulness of the explanation to a person.

An actual construction illustrates why these warrants must be separated. Lertvit-tayakumjorn and Toni’s AXPLR explains a pattern-based logistic-regression binary text classifier using relationships among patterns expressed in a quantitative bipolar structure. Its reported prediction correspondence belongs to that particular construction; the 2023 publisher version states it in Section 4, Theorem 2 [107, Sec. 4, Theorem 2]. A general surrogate with high sampled agreement cannot inherit that theorem.

The same work reports human evaluation with different targets. Sections 9.3–9.4 compare explanation scores with human judgments. Sections 10.1–10.4 and Table 10 study tutorials and assistance for deceptive-review detection. Assistance improved scores relative to within-method baselines, but there was no significant between-method performance difference [107, Secs. 9.3–9.4, 10.1–10.4, Table 10]. Plausibility ratings and assistance performance are different outcomes, and neither is the mathematical correspondence claim. The self-contained classifier-explanation study (p. 219) supplies the promised Chapter 3 return and an ordinary evidence-comparison task. It reports the source’s result at statement level without reproducing its construction or proof.

These comparisons connect the chapter’s two directions. A contextual AF certificate is checked against semantic conditions and a verified witness. A classifier explanation additionally needs a declared relation to the classifier. Evidence that an audience understands or benefits from either kind of explanation needs an appropriate human task and measure, as the cognitive comparisons of Chapter 11 already showed. Chapter 14 takes up how such different objects and obligations meet in a larger system. The interface must retain which question was answered and which evidence supports that answer.

13.7 Exercises

Use the displayed inputs and retained context, rather than an unstated meaning of “accepted.” Complete solutions appear under Chapter 13 in the collected Solutions. The optional study task uses only its printed evidence; no model training, programming or external reading is required.

- 13.1. [\[solution\]](#) [Certificate construction] In F_{reuse} , build the full contextual certificate for c relative to $E_1 = \{a, c, e\}$. Include every attacker branch and every needed back-reference. State the verified-witness premise.
- 13.2. [\[solution\]](#) [Status explanation] Explain why e is out relative to $E_2 = \{a, d\}$, and expand the accepted attacker with the certificate procedure. Compare this with the status of e in the grounded labelling. Why is “outside the extension” insufficient as an out-explanation?
- 13.3. [\[solution\]](#) [Recomputation] Remove only (c, d) from F_{reuse} , retaining all vertices. Compute every stable extension. Is that edge necessary for the original credulous stable acceptance of e under this allowed-deletion test? Explain why the query must be recomputed.
- 13.4. [\[solution\]](#) [Minimality and context] Allow vertex and edge deletion while retaining a . Give a sufficient standalone subgraph for credulous stable acceptance of a , find an inclusion-minimal example, and prove its sufficiency and minimality. Which original explanatory context does it lose?
- 13.5. [\[solution\]](#) [Target formulation] Formulate typed targets for “why e is in under E_1 rather than out under E_2 .” Specify what is held fixed and what varies, then give the relevant attack/defense contrast. Would an explanation of stable credulity alone answer this contrast?
- 13.6. [\[solution\]](#) [Legal interaction] Start from the fully supplied state in Section 13.4. Replace the worked question by $\text{challenge}(U, M, \chi)$, then use the same reply $\text{justify}(M, \{\rho, \omega\} \vdash^+ \chi)$. Give the legal rule, commitments, ledger and pending field after each move. Which effect differs from the question route? May M evade the pending reply by retracting χ ? Explain why neither a raw edge as target nor a new extension of the same AF would supply the requested content justification.
- 13.7. [\[solution\]](#) [Finite calculation] Compute fidelity from the four synthetic output pairs in Section 13.6. Identify the disagreement, denominator and scope of the result. What further claim would agreement on every possible input make, and is it established here?
- 13.8. [\[solution\]](#) [Comparison] Distinguish explanation for argumentation from explanation through argumentation using the contextual e -certificate and the classifier example. State the object explained and the additional relation that the second direction must establish.

- 13.9.** *[solution]* [Typed explanations] Give separate explanations of e 's extension-relative acceptance in E_1 , stable credulity and failure of guarded stable skepticism. Specify the evidence each requires and why the third does not make e out in every stable extension.
- 13.10.** *[solution]* [Cycle boundary] For the mutual-attack AF on $\{a, b\}$, verify the stable witness $\{a\}$ and construct the cycle-closing accepted certificate for a . Compute the grounded extension and labelling. Why does a back-reference not prove grounded acceptance?
- 13.11.** *[solution]* [Changed explanation] Add only (e, a) to F_{reuse} . Use Chapter 12's complete updated-family argument to explain why the old e -certificate loses its verified context. Give the new credulous status of e and guarded skeptical status of d , with evidence appropriate to each. Would the survival of two old certificates alone prove these family-level claims?
- 13.12.** *[solution]* [Procedure proof] For a finite AF, verified complete, grounded, preferred or stable extension E , and $a \in E$, prove that every defender requested by **ExplainAccepted** exists, every original attacker of an expanded node is included, and the procedure terminates. State what its back-references certify and what they do not establish about groundedness, skepticism, minimality or output size.
- 13.13.** *[solution]* [Optional evidence study] Using Study S13, match the following claims to their possible warrants: construction-specific prediction correspondence, agreement on a finite test set, agreement with human plausibility judgments, and improved assistance performance. For AXPLR, identify the targets of Theorem 2, Sections 9.3–9.4 and Sections 10.1–10.4, using the supplied card. Interpret the reported assistance finding: does it establish superiority between the compared methods? Can the synthetic surrogate inherit AXPLR's correspondence theorem? Explain why accurate prediction alone does not establish a useful explanation for every person.
- 13.14.** *[solution]* [Guard and explanation] Use the directed three-cycle rRs, sRt, tRr with no other attacks. Determine its stable family and explain the false guarded-skeptical answer for r . Contrast its certificate requirements with the false guarded skeptical answer for e in F_{reuse} . Could the same kind of omitting stable witness be supplied in both cases?
- 13.15.** *[solution]* [Practical contrast] Using the dinner versions in Section 13.5, explain why R is originally permitted, why C fails despite passing against R, and why T's own failure does not remove it as a foil. Identify the ground, coordinate and reversed comparison responsible for the change from $\{R\}$ to $\{R, C\}$. Explain the two policies' selections in both versions, stating which inputs are fixed in the weight contrast and which are fixed in the policy contrast. Does selecting C establish that C alone is permitted, or that it defeats R? Could changing only the selection order in v_1 select C under the supplied procedure? Finally

distinguish a request for the arithmetic calculation from requests for the ground, warrant and authority of the changed assessment.

- 13.16.** *[solution]* [Public grounds and authorized revision] Start from the supplied K/U state with only K's assertion of η active. Trace the question about η and K's displayed reply: give the legal rules, commitments, ledger and pending field after each move. Has either move changed the numerical input? Then consider four proposals independently against v_1 : the complete candidate in the revision table; the same candidate with U as issuer; the same candidate naming v_0 as its base version; and the same candidate with no warrant record identified. Apply every relevant interface condition, giving the accepted version and outputs or the failed check and unchanged outputs. What happens if the first candidate is submitted again after it has already produced v_2 ? Explain what the public certificate, the accepted revision and the recalculated selection each establish, and what remains open about the warrant's adequacy and authority to execute.

13.8 Annotated further reading

From contextual certificates to explanation formalisms. The ABA Handbook chapter introduces explanations through disputes and assumption-based structure [49, Sec. 5.1, p. 383, and Sec. 6]. It provides the route to Fan and Toni's 2015 *On Computing Explanations in Argumentation*. Read it after the accepted/out distinction here, asking which assumptions and defense obligations its objects expose. The formal-discussion chapter connects this question with proof and dialogue [52]; a valid move sequence and a semantic certificate still need distinct correctness claims. Volume 3's explanation discussion offers a further map of the questions [4, Ch. 13, Sec. 4.3, pp. 934–940].

Borg and Bex compare minimality, necessity and sufficiency using their own precise explanatory objects [108]. Compare those objects and quantifiers with our permitted-deletion convention before transferring a result. A different representation makes the defender–attacker–target relation primary: Liao and van der Torre's attack-defense semantics returns to the reasons for acceptance and to equivalence questions [53, Secs. 2, 4–5]. It develops the defense-first alternative named earlier in the book; it is distinct from Chapter 2's agent-indexed defense and from merely changing the layout of our certificate.

Correspondence, surrogate agreement and human evaluation. The argumentative XAI survey maps ways argumentation can explain other systems [109]. The Handbook's classifier discussion and the 2023 AXPLR publisher version provide a focused comparison [4, Ch. 9, pp. 597–598][107, Sec. 4, Secs. 9.3–10.4]. Use Study S13 to keep the theorem statement, three-dataset evaluation, plausibility judgments and assistance tasks separate. The construction is specific to its pattern-based classifier; the human results do not establish a universal advantage of

argumentative explanations. The original proof is a further technical undertaking, not an unstated premise of our ordinary evidence exercise.

Changing a model to explain an observation. The selected abduction account asks which specified candidate changes could explain an observation [79, arXiv v1, Secs. 3–5]. It gives a different explanatory direction from extracting a defense from a fixed verified extension. Causal explanation imposes further assumptions about interventions and dependence: the Handbook’s causation chapter is the appropriate next comparison [110]. Deleting an attack from an AF does not by itself identify a causal intervention in the world.

The interactive return can be pursued through Chapter 9’s supplied CONSULT/EQR study and the dialogue applications chapter [38, Sec. 4.2, pp. 251–252]. Those readings distinguish theoretical explanation properties, a limited implementation and human outcomes. The dinner interface here makes one connection explicit; Chapter 14 takes up the wider system in which such records are used. Neither a certificate nor a dialogue trace alone settles that wider question.

Explain and contest a practical assessment. Reread the two-weight comparison and all-alternatives rule [83, Sec. 3.1, Defs. 3.1–3.4, PDF p. 3] with the C/T contrast in view: which coordinate answers the opposing demand, and which additional policy accounts for selection? Chapter 10’s weight-evidence study then opens the provenance question through the fixed-graph assignment and reconstruction investigations [95, Secs. 4.1–4.3, PDF pp. 4–5]. Read their evidence for its stated target; repeatability or outcome reconstruction is not itself normative authorization. Our versioned revision interface supplies a local procedure for a contestable assessment, not an empirical result from that work or a theorem of the public protocol.

Literature/research exercise 13.L. *[solution]* Consult one foundational explanation source, one Handbook passage and one further source from this guide. In five sentences compare their targets, explanatory objects, contributions, connections to this chapter and limits. Make explicit whether each source explains an argumentation result, another system, a changed model or an audience outcome. External reading is the purpose of this task; a model comparison appears in the collected solutions.

14 Chapter 14. Argumentation as system architecture

A correct answer to a formal question becomes useful when someone can rely on it for a specified purpose. A policy adviser must distinguish a supported recommendation from a calculation made on unsupported inputs. A public-response service must distinguish a legal question from a completed exchange. A dinner recommender must say which version of the reasons and which selection policy produced its choice. These are different services. Each makes some of the book’s argumentative resources usable, without requiring every resource at every stage.

Chapter 13 taught us to ask what an explanation explains and which context it retains. We now ask who supplies that context, which component computes each result, and what a caller is entitled to do with the returned record. Reasons with content supply the object of scrutiny; a public procedure makes a person answerable for a ground; practical evaluation identifies permitted options; an explanation lets the recipient challenge a particular step. Integration preserves these contributions by giving their connections explicit meanings. It does not turn inference, dialogue and balancing into successive stages of one universal algorithm.

14.1 A recommendation with an inspectable route

Consider a fictional adviser asked whether a late filing can be accepted. This is a constructed policy example, not a rule of any legal jurisdiction. Its purpose is to return a recommendation together with the represented reasons and the checks needed for its release. The extraction component supplies four argument summaries:

$$\begin{aligned} a_1 : \textit{emergency} &\Rightarrow \textit{allow}, & a_2 : \textit{deadlineMissed} &\Rightarrow \neg \textit{allow}, \\ a_3 : \textit{medicalEvidence} &\Rightarrow \textit{emergency}, & a_4 : \textit{publicInterest} &\Rightarrow \textit{allow}. \end{aligned}$$

It also supplies the relations

$$R^- = \{(a_3, a_2), (a_2, a_1), (a_2, a_4)\}, \quad R^+ = \{(a_3, a_1), (a_4, a_1)\}.$$

In particular, the attack from a_3 to a_2 records the stipulated policy’s medical-evidence objection to applying the deadline reason. That edge is part of the supplied representation. The displayed summaries alone do not derive it by an unspecified structured-argumentation rule.

Put $A = \{a_1, a_2, a_3, a_4\}$ and $F = (A, R^-, R^+)$. Use Chapter 6’s grounded semantics on (A, R^-) , with characteristic function $\mathcal{F}$: an argument belongs to $\mathcal{F}(S)$ exactly when S attacks each of its attackers. The empty-start iteration is

$$S_0 = \emptyset, \quad S_1 = \{a_3\}, \quad S_2 = \{a_1, a_3, a_4\}, \quad S_3 = S_2.$$

Only a_3 is initially unattacked. It answers the sole attacker of a_1 and a_4 , namely a_2 , and no accepted argument answers a_3 on behalf of a_2 . Thus the grounded extension

is $E = S_2$. It is also stable: it is conflict-free and attacks its only outsider a_2 [5, Defs. 13, 16 and 20, pp. 328–329].

The service still needs a rule for the question about filing. Here the sole con argument is a_2 . Declare

$$\delta(E) = \text{allow} \quad \text{iff} \quad (\exists a \in E : \text{Conc}(a) = \text{allow})$$

and E attacks every con argument;

otherwise $\delta(E) = \text{withhold}$. Failure to recommend acceptance does not automatically recommend denial. Both a_1 and a_4 support the positive conclusion, and a_3 defeats the deadline objection, so $\delta(E) = \text{allow}$. This is a decision rule applied to accepted arguments; it is not another name for grounded membership.

The positive links in R^+ have an *explanatory* job in this design. An explanation may display a_3 's medical route and a_4 's public-interest route into a_1 . Removing those links changes neither E nor $\delta(E)$. This specifies a support-use contract. A different design could use support inside semantic evaluation, or use it in a decision rule after evaluation. Supplement S14.1 supplies a complete acyclic scoring variant: with the same graph its value for a_1 changes from 0.672 without support to 0.7730688 with support. That vector of strengths is a different output type. Our set-based δ cannot consume it without a new declared interface.

14.2 Computation, checking and release

Suppose the extractor returns exactly this graph in two cases. In the first, the supplied evidence record supports the medical premise; in the second, it does not. The grounded calculation is identical. The service's reason for releasing its result is not.

For a finite, checkable example fix input version u_1 , graph version f_1 and decision-rule version d_1 . The declared required premise records are

$$\left\{ \begin{array}{l} (a_1, \text{emergency}), (a_2, \text{deadlineMissed}), \\ (a_3, \text{medicalEvidence}), (a_4, \text{publicInterest}) \end{array} \right\}.$$

A provenance entry has fields $(u, \text{argument}, \text{premise}, \text{locator}, \text{supported})$. For the first case supply exactly one matching entry for each pair above, with locators e_1, e_2, e_3, e_4 and $\text{supported} = 1$. For the second case keep the same entries except that e_3 has $\text{supported} = 0$. The verifier checks that every required pair has a matching entry for u_1 with value 1; a missing entry fails. These are supplied evidence-check results. Comparing fields does not implement a natural-language truth detector or authenticate the person who supplied them. It makes the input responsibility visible.

A route certificate records

$$c = (u_1, f_1, \text{grounded}, d_1, S_0, S_1, S_2, S_3, E, \text{allow}).$$

The verifier binds it to the actual input, full attack graph, semantics and decision rule, checks $S_0 = \emptyset$, replays each characteristic-function step, checks the last step is a fixed point and that $E = S_2 = S_3$, and checks $\delta(E) = \text{allow}$. This certifies the stated *grounded-result route*. Conflict-freeness and defense of E alone would certify admissibility, not that grounded semantics selected E . Chapter 13's concern for retained context matters even in this tiny application: a certificate for another graph or decision rule does not explain this result.

We separate three Boolean checks. c_{prem} checks the supplied premise records just described. c_{route} replays the grounded iteration and decision against the actual inputs. c_{cert} checks that a certificate is present, well-formed, binds those same versions and names precisely the returned E and candidate decision. The required release condition is

$$\text{RequiredOK} = c_{\text{prem}} \wedge c_{\text{route}} \wedge c_{\text{cert}}.$$

An absent, malformed or mismatched certificate fails its check, even if someone has written an otherwise correct extension in an audit log.

A computation result and its caller

Computation input: the finite input and versions, extraction and relation maps, grounded evaluator, decision rule, explanation constructor, supplied evidence records and the three check procedures.

Computation output: a named record containing F , E , candidate decision, explanation, candidate certificate, each check result and audit record. A computed candidate is not yet a released recommendation.

Release policy: release a certified recommendation exactly when all three required checks pass. Otherwise withhold that recommendation. In either case retain the computation and the individual checks in the audit record. A diagnostic display may report them as diagnostics, without claiming release.

```

ComputePolicy(x, evidence):
  A := mu(x); Rminus, Rplus := rho(x, A)
  F := (A, Rminus, Rplus)
  E, steps := GroundedWithIteration(A, Rminus)
  candidate := delta(E)
  explanation := ExplainRoute(F, E, candidate)
  certificate := BindVersionsAndSteps(
    x, F, E, candidate, steps)
  checks.premise := CheckPremises(x, A, evidence)
  checks.route := ReplayRoute(
    x, F, E, candidate, certificate)
  checks.certificate := CheckBinding(
    x, F, E, candidate, certificate)
  audit := Record(x, F, E, candidate, explanation,
    certificate, checks)
  return Result(F, E, candidate, explanation,
    certificate, checks, audit)

ReleasePolicy(result):
  if every required check in result.checks is true:
    output := Certified(
      result.candidate, result.E,
      result.explanation, result.certificate)
    release := released
  else:
    output := NoCertifiedRecommendation
    release := withheld
  append (release, output) to result.audit
  return (release, output, result.audit)

```

The returned accepted arguments are available to a caller who needs to inspect the reason for the candidate. Individual failed checks are available to a caller who needs to repair the input. Logging a failed check and enforcing a release policy are different operations; the second routine performs the latter. A service with a time limit can withhold release and return a marked partial diagnostic. Switching semantics or restricting the graph on timeout would change the task and could not claim the original grounded result.

Evidence case	Candidate	$(c_{pre}, c_{route}, c_{cert})$	Release
All four supported	<i>allow</i>	(1, 1, 1)	certified recommendation
Medical entry unsupported	<i>allow</i>	(0, 1, 1)	withheld
Medical entry missing	<i>allow</i>	(0, 1, 1)	withheld
Certificate absent	<i>allow</i>	(1, 0, 0)	withheld

In the last row the graph and computed answer are retained but replay cannot use the missing certificate. The diagnostic explanation may still identify what was computed and what failed. It is not advertised as a certified route.

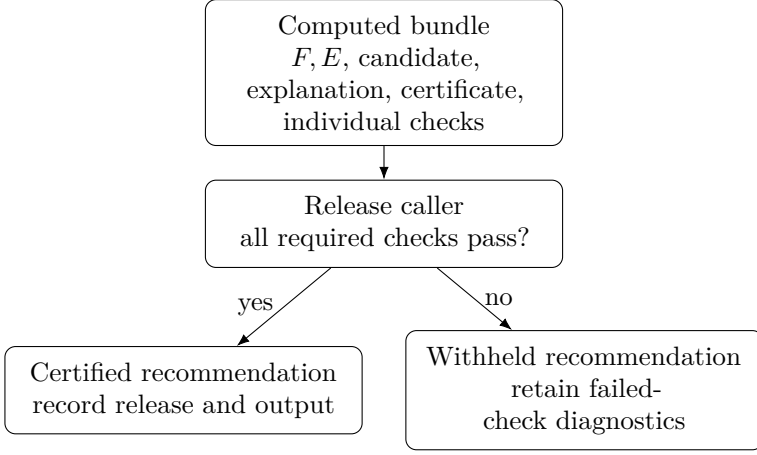


Figure 15: The same computed candidate can have different release outcomes. The audit retains both the calculation and the caller’s decision.

There is a short local guarantee: every certified recommendation released by this caller has passed every declared required check. The release branch is entered only under their conjunction, and the other branch returns no certified recommendation. This branch argument concerns the supplied checks and this caller. It proves neither that the source is true nor that another component cannot bypass the caller. Its positive contribution is precise: a recipient can recover the represented reasons, replay the selected semantics and locate the input responsibility on which this release depends.

14.3 Which claim does a check support?

An audit record can retain input identifiers, graph, E , output, explanation, certificate, versions and checks. A hash can help identify the retained bytes; it does not make their assertions true. An agency might believe that a_3 is accepted even when a later input check fails. We can write $B_i\text{accepted}(a_3)$ for “agency i believes that a_3 is accepted.” Here B_i is just this local modal reading. No modal axioms or inference from belief to truth are assumed.

A learned classifier introduces another target of checking. Suppose its output agrees with the argumentation decision in three of five supplied nearby cases and disagrees in two. The fraction of agreements is $\text{Fid} = 3/5 = 0.6$. Under a declared threshold 0.8, the evidence does not license a claim that the explanation meets that sampled-fidelity standard. It may still be a correct explanation of the argument route. Failure of the fidelity claim neither erases that route nor supplies release

permission. For this policy service, the three required checks above continue to govern release; if a particular service additionally promises classifier faithfulness, it must require the extra fidelity check for that claim. The five cases say nothing by themselves about agreement on every other input or about human understanding. Chapter 13’s fidelity and human-evidence distinctions therefore become distinct responsibilities in an implemented service.

Governance language should identify its enforcement point. A hard constraint rules out an output in the defined system; a norm prescribes conduct but may be violated; a monitor policy declares what to do on a check failure. Our caller implements the necessary condition $recommend(o) \Rightarrow exists(Cert_o)$, and the stronger condition of successful required checks. Certificate existence alone is insufficient. An advisory display policy could allow a warning labelled “unverified candidate”, but it must not also call that output a certified recommendation. Chapter 15 will distinguish such enforced checks from norms whose violation requires further institutional interpretation.

We can now summarize this design by $\mathcal{S} = (I, \mu, \rho, \sigma, \delta, \epsilon, \nu, L)$. For $x \in I$, extraction returns a finite set $A_x \subseteq U$ of argument identifiers. Relation extraction returns (R_x^-, R_x^+) with both relations contained in $A_x \times A_x$; a relation with an absent endpoint fails this interface. Evaluation σ returns E , δ consumes that output type, ϵ explains its route, ν supplies the named checks and L retains the record. The caller’s release rule uses their results. This tuple summarizes a particular design. Its notation does not require a public-response service or a numerical dinner selector to extract this kind of graph.

14.4 Public and practical services

Consider first a service commissioned to help A expose the reasons behind B’s restaurant challenge. Section 12.8 supplies its complete active state s_3 : A asserts excellence χ and has justified it by P ; B’s challenge remains recorded; no immediate reply is pending. The two offered continuations are withdrawal and inquiry. Inquiry asks $question(A, B, \chi)$, then receives the response-policy reply $justify(B, \{Q, S\} \vdash^- \chi)$. The policy supplies that reply; the legal rules do not predict what a person will choose to say.

The checker returns the inquiry certificate $(CR4, CR5)$ and a final state with A’s assertion retained, B’s countergrounds exposed and no pending reply. Its score is $(1, 1, -2)$ for the declared lexicographic objective $(exposed, retained, -moves)$. Withdrawal applies CR6, leaves no immediate reply pending and scores $(0, 0, -1)$. Thus budget 2 selects inquiry from the offered set; budget 1 leaves only withdrawal complete. The service returns the chosen *plan*, checked simulated trace, response policy, objective, budget and stopping condition together. That record tells the caller why the two-move plan serves its purpose better, and why it ceases to be available under the smaller budget.

Suppose instead the caller receives only the actual utterance $question(A, B, \chi)$. CR4 verifies that utterance, but its resulting pending field is $(B, \chi, -)$. The caller cannot report that countergrounds have already been obtained. It must await and check B's reply or report an unfinished prefix. Selecting an offered plan, checking a supplied trace and observing a completed exchange are three different achievements. The finite maximization proves no strategic optimality over unoffered continuations. It does provide an accountable recommendation about what to try under the supplied objective. The response service can explain that recommendation without computing a Dung extension or evaluating numerical dinner weights.

Now consider a dinner service that promises a versioned practical recommendation. Sections 12.9 and 13.5 already supply the reason rows, permission procedure, selection policies and authorized revision interface. At numerical version v_1 , the permitted set is $\{R\}$. A public request asks K to justify claim η that the requiring coordinate for $(speed, T, C)$ should be 2. K's reply supplies ground ρ and warrant ω ; CR2 and CR3 record the request and answer. This makes those grounds publicly attributable while leaving the numerical version unchanged.

The dinner caller next submits the distinct revision record

$$(d, v_1, (speed, T, C), RW, 3 \rightarrow 2, K, \rho, \omega).$$

The supplied authority permits K to change just that coordinate in context d . All version, old-value, domain and record checks succeed, so the interface creates v_2 . The C/T test now compares $2 \geq 2$ instead of $2 \geq 3$; C joins R in the permitted set. Running the two supplied selectors gives

$$(v_2, \{R, C\}, cost, C), \quad (v_2, \{R, C\}, priority, R).$$

The cost order C, T, R selects C; the priority order R, C, T selects R. An explanation of the first result names both the changed opposing requirement in the C/T comparison and the cost rule that places C before R. Permission and selection are not the same question.

These records also expose useful failures. If U submits the same numerical proposal, the authority check fails and v_1 remains current. If the caller replays K's v_1 proposal after v_2 is installed, the base-version and expected-old-value checks fail. A legal public assertion cannot cure either failure. Conversely, carrying the old choice R into a v_2 cost recommendation would be wrong even though R remains permitted: it would fail the named selection rule. Return version, selector, permitted set, selected option and explanatory target together, and recompute after an accepted revision.

The weight evidence in Supplement S10.4 now has an operational target. Who supplied ρ and ω , and who authorized K? Repeatedly obtaining the same weight or reproducing a recorded outcome does not answer those questions. The service preserves their records so they can be challenged. It neither verifies their normative adequacy by arithmetic nor grants authority to execute the selected dinner option.

The public protocol, numerical evaluation and revision policy retain their own objects; Libra’s distinct abstract comparison predicate is not silently inserted into this service.

14.5 Receiving a report and deciding what to admit

Before a system reasons with incoming information, it must decide what that information is allowed to contribute. Identifying a purported sender, selecting reported formulas and revising accumulated beliefs are different tasks. A small comparison makes selective admission visible.

Let p, q be independent propositional atoms and a report be a set of formulas. Consider two interpretations of the formula $p \wedge q$: a trusted white-list item or a distrusted black-list item. In the white-list interpretation, the report $\{p\}$ supplies no trusted information sufficient to admit p ; $\{p, q\}$ supplies the trusted conjunction and both reported atoms are admitted. In the black-list interpretation, $\{p\}$ can be retained without entailing the forbidden conjunction. Once q is also reported, the method retains only what survives every maximal way of avoiding that conjunction; neither reported atom survives all of them. Thus adding q enables admission of p in one mode and removes it in the other.

Supplement S14.2 supplies both exact equations, all four reports and the complete remainder derivation [111, Def. 2, p. 463, Defs. 15, 17, p. 467]. The outputs are formulas selected from the actual report, not deductively closed memories or accepted argument identifiers. These methods express two attitudes to reported content. They do not authenticate an issuer, prove the content true or prescribe a temporal belief update. Chapter 15 will explicitly supply reliable admitted observations when it studies such an update. The present comparison explains why that assumption is a substantive interface, rather than something conferred by writing “trusted” in a log.

14.6 Evidence for a system’s contribution

The preceding services are finite designs. For an implemented application, we also need evidence about which components actually exist and which user benefits have been tested. Supplement S9.2 gives the published CONSULT/EQRbot description: encoded information enters an argumentative recommendation process, while the implemented follow-up interaction uses a limited locution set [38, Sec. 4.2, pp. 251–252]. Three questions lead to three distinct investigations. If a sensor value is misencoded, inspect the input transformation. If a requested follow-up is unavailable, inspect the implemented interaction interface rather than cite a property of the larger dialogue theory. If someone claims that explanations improve patient understanding, ask for a human study with a specified task and measure. Neither the formal dialogue result nor a functioning implementation supplies that empirical conclusion.

This is a positive division of work. Formal methods can make a represented objection effective in a calculation, expose the obligations created by an utterance, or bind a recommendation to a declared policy and reason record. Implementation can make those operations available to users. Empirical work can then test whether the resulting service helps those users with its stated purpose. The system’s contribution becomes easier to assess when these claims are individually inspectable. The next chapter asks what happens when a useful service receives new information and proposes actions whose conditions may already have changed.

14.7 Exercises

Every ordinary exercise has a paper route; optional programs may check the same finite inputs. Supplements [S14.1](#) and [S14.2](#) supply the additional preparation explicitly named below. Complete answers are collected at the back of the book.

- 14.1. [\[solution\]](#) [Interfaces] For the policy service, identify the input and output of (a) its decision interface, (b) its learned-model fidelity check and (c) its audit interface. State the claim each can support.
- 14.2. [\[solution\]](#) [Graph check] Check that $E = \{a_1, a_3, a_4\}$ is conflict-free for the printed R^- . Account for every attack in that relation.
- 14.3. [\[solution\]](#) [Explanation] Why is a_2 rejected in the grounded result? Identify the accepted attacker and explain why its acceptance is not circular.
- 14.4. [\[solution\]](#) [Modelling contract] Classify R^+ in the principal service as semantic, decision or explanatory support. Test your classification by deleting both support links.
- 14.5. [\[solution\]](#) [Necessary and sufficient checks] Express “no recommendation without a certificate” as an implication. Explain why this necessary condition is weaker than the declared release policy, using an invalid but present certificate.
- 14.6. [\[solution\]](#) [Finite evidence] On five supplied comparison cases the classifier and argument route disagree twice. Compute their agreement fraction, apply threshold 0.8, and state separately what follows about route correctness, sampled faithfulness and other untested inputs.
- 14.7. [\[solution\]](#) [Status and belief] Using the local B_i reading in Section [14.3](#), write that agency i believes a_3 is accepted. Does that sentence imply the medical premise is true under any rule supplied here?
- 14.8. [\[solution\]](#) [Failure response] Choose either a missing medical-premise entry or a failed fidelity threshold. State the failed check, the output claim it defeats and a response consistent with the declared policy. If you propose a diagnostic display, give its label and explain why it is not a certified recommendation.

- 14.9. [\[solution\]](#) [Changed support] Replace explanatory support by the score method in Supplement S14.1. Which evaluation input and output change? What further contract is required before the service can recommend an option?
- 14.10. [\[solution\]](#) [Enforcement] Explain the distinction among a hard constraint, a norm and a monitor policy. Classify the no-certificate/no-recommendation requirement as used here, and locate the operation that enforces its stronger release version.
- 14.11. [\[solution\]](#) [Optional score study] Using Supplement S14.1, compute (a) an isolated argument of base 0.8; (b) a base-0.8 argument with one attacker of score 1 and no supporters; (c) $s(a_1)$ with the printed support links; (d) its score without them. (e) Explain the acyclicity requirement and why a change on overlapping attack/support roles need not obey either one-coordinate monotonicity claim.
- 14.12. [\[solution\]](#) [Repair and certify] For Section 14.2, (a) give all returned fields needed to separate computation from release; (b) compute the check vector and release when only $e_3.supported = 0$; (c) explain why conflict-freeness and defense alone do not certify groundedness, using the mutual-attack graph $\{u, v\}$ with edges both ways and candidate $\{u\}$; (d) supply its actual empty-start grounded iteration; (e) give a truthful timeout response when no complete iteration certificate is available.
- 14.13. [\[solution\]](#) [Service transfer] Use Section 14.4 and the complete procedures in Sections 12.8 and 13.5. (a) State the restaurant service’s chosen plan and achieved stopping condition for budgets 2 and 1. Why is the single legal question not a completed result? (b) Trace K’s valid dinner revision and both selectors, then U’s otherwise identical attempt on v_1 . (c) Diagnose a returned record $(v_2, \{R, C\}, cost, R)$. (d) Identify which additional authority would be needed to execute the selected option; do not invent that authority from the public reply.
- 14.14. [\[solution\]](#) [Optional admission study] Using Supplement S14.2, replace report $\{p, q\}$ by the singleton $\{p \wedge q\}$. Compute both outputs, explaining their relation to the unchanged closed theory and its three remainders. Explain why neither an AF certificate nor an issuer-identity check solves this task, and whether these observations determine a real person’s hidden trust uniquely.
- 14.15. [\[solution\]](#) [Application evidence] Using the supplied account in Supplement S9.2, allocate (a) a sensor-encoding error, (b) an unavailable follow-up locution and (c) a claim of improved human understanding to the relevant evidence. State one concrete check for each. No external paper is needed.

14.8 Reading routes

For requirements, read Gordon’s *Towards Requirements Analysis for Formal Argumentation*, pp. 145–156, asking which requirement each component of the policy service discharges [22]. Continue to the implementation chapter’s Section 6 to

connect representations and computation to usable systems [104]. The Argument Interchange Format is a historical route into communicating argumentative structure across tools [29]; our simple records do not claim to implement it.

For admission, *Revealed Epistemic Trust*, Sections 3–4, develops representation results beyond the definitions and finite case printed in Supplement S14.2 [111]. Read its distinction between an idealized complete acceptance function and partial public observations before treating a few answers as evidence of a person’s underlying attitude. Source authentication and temporal revision remain separate questions. For numerical input authority, return to Supplement S10.4: it differentiates weight assignment, consequence calculation and evidence about outcomes.

The application chapters in Handbook Volume 3 broaden the purpose of a service. Social argumentation asks whose contributions become visible; debate search asks what a query retrieves and how a user navigates reasons; legal systems ask which procedural and evidential commitments their representation retains [19, 30, 85]. Dialogue applications supply the CONSULT/EQR distinction used here; population models ask what local argumentative conduct does collectively; decision applications connect reasons to the criteria by which options are assessed [38, 112, 40]. These routes extend the design question, rather than provide interchangeable validation for the fictional service.

Choose a cross-field route when the purpose requires its additional object: conditionals for defeasible dependencies, machine learning for a learned component and its evidence, causation for intervention-sensitive claims, epistemic planning for information-changing action, or modal logic for agent-relative status [57, 33, 110, 113, 114]. Normative multiagent systems and LogiKEy offer further institutional and verification perspectives [78, 115]. The local B_i notation and fictional medical objection establish no results in those additional theories.

Literature/research task 14.L. *[solution]* Consult one requirements source, one application chapter and one cross-field or institutional route named above. In five connected sentences identify the three questions they address, an object needed for one question, the supplied result or evidence, its possible contribution to one of this chapter’s services, and a limitation of that transfer. External reading is the purpose of this task. The collected answers give a worked route and criteria for alternatives. The study guide distinguishes full-book, semester and intensive expectations.

15 Chapter 15. Argumentation in an acting and revising system

A service can give a well-supported answer and still need to reason again. A new notice may defeat the evidence on which it relied. The same conclusion may acquire a new provenance, making an old authorization stale. A permitted response may help its user while leaving the original goal open. This final chapter develops those questions as a bounded outlook within Part III. Argumentation contributes a way of retaining the reasons for a changing answer, relating them to a proposed act, and exposing what further information or authority the next step requires.

We will follow a research assistant that has one candidate paper for answering a scientific question. A received retraction changes what it can responsibly say. Its response is useful: it can explain the withdrawal, give its current source and identify the missing evidential answer. That contribution requires more than a correct inference, and less than a general theory of autonomous planning. The finite case lets us examine receipt, revision, adaptation, authorization and action without pretending to have solved every interface that a deployed agent would need.

Chapter 14's services already showed how purposes select different methods. We do not add a compulsory process that sends every input through inference, dialogue and balancing. Instead we make one particular handoff explicit and compare it with the public and practical procedures already established. The source-method question from Chapter 1 returns: what exactly is preserved when one route supplies an input to another?

15.1 A received notice changes an accepted conclusion

Let p_0 identify the sole candidate paper. Fixed metadata m_0 says it is tracked, recent and relevant. Here those are trusted supplied observations, not conclusions extracted by a language model. Write $p = tracked(p_0)$ and $q = retracted(p_0)$. The following records are synthetic inputs to the finite case:

Record	Version and locator	Supplied content
s_0	<code>status-0;</code> <code>fixture:</code> <code>status-0#p0</code>	Issuer <code>publisher-service</code> ; paper p_0 ; retracted=false.
s_1	<code>notice-1;</code> <code>fixture:</code> <code>notice-1#status</code>	Same issuer and paper; literal $retracted(p_0)$; retracted=true.

Their record identifiers are s_0, s_1 , distinct from their document versions. The initial negative snapshot is explicit. Absence of a notice is not itself evidence of absence of retraction.

Three assumptions govern the first handoff. **A1:** q describes the paper's status, and

the supplied records are correct at their respective times; it does not mean “some source asserted q ” or “citation is allowed.” **A2**: issuer identity, paper resolution and literal extraction are reliable here, with no conflicting hard observations. Field comparisons do not prove that reliability. **A3**: use only the following finite memory and declared arguments. These assumptions keep report admission distinct from both authentication and revision. The admission comparison of Section 14.5 showed why receiving content does not by itself decide what to accept. For the present case admission is stipulated under A1–A2, not inferred from an AF calculation.

The memory separates hard premises H , strict rules $\mathcal{R}_s$, defeasible rules D and observations Obs :

$$B_0 = (\{p\}, \emptyset, \{d := \neg q\}, \emptyset), \quad B_1 = (\{p\}, \emptyset, \{d := \neg q\}, \{q\}).$$

On receipt of s_1 , check its issuer, paper and literal against the supplied card, then perform $Obs := Obs \cup \{q\}$. Leave the other components fixed. The initial negative status remains defeasible; s_0 is not inserted as a hard negative observation. It will instead check the status exported to the next component.

Use distinct revision-argument identifiers

$$R_p = \langle p \rangle, \quad R_d = [\Rightarrow \neg q], \quad R_q = \langle q \rangle_{obs}.$$

Before receipt only R_p, R_d exist, both unattacked, so the grounded extension is $\{R_p, R_d\}$ and its conclusion projection is $\Gamma_0 = \{p, \neg q\}$. After receipt R_q rebuts the defeasible top rule of R_d ; the default cannot undermine the observation axiom. Attacks and defeats coincide here and consist of $R_q \rightarrow R_d$. Grounded iteration gives

$$\emptyset \mapsto \{R_p, R_q\} \mapsto \{R_p, R_q\}, \quad \Gamma_1 = \{p, q\}.$$

The default argument remains constructible and its generating rule remains in memory. It is the accepted conclusion that changes. This restricted use of firm premises and defeasible rebut follows Chapter 5’s distinctions [46, Defs. 5, 8–9, pp. 366–368].

There is a second way to compute the conclusion set for this family: close $H \cup Obs$ under the strict rules, then add the sole default conclusion $\neg q$ only when q is absent. The routes agree. If either an observation or the optional strict rule $p \rightarrow q$ supplies q , its firm argument defeats R_d ; if neither supplies it, R_d is unattacked. These two cases cover all four combinations of strict rule present/absent and observation present/absent. Supplement S15.1 prints the complete four-case table. This is a proof for that declared finite family, not a general belief-revision or representation theorem.

The strict-rule variant reveals a different phenomenon. With $p \rightarrow q$, $R_s = [R_p \rightarrow q]$ already warrants q . Receiving the observation adds R_q but leaves accepted conclusions at $\{p, q\}$. Information has been received without a change of belief at that projection. This is an abstract comparison, not a plausible claim that tracking a paper entails its retraction. The no-strict-rule case remains our principal temporal example.

15.2 Exporting a status is another reasoning obligation

The action component does not consume revision arguments. It consumes a finite context

$$C = (\text{version}, \text{paper}, \text{recent}, \text{relevant}, \text{retracted}).$$

Mapping the accepted formula q to a Boolean might look trivial, but several mistakes would be possible: using a claim about another paper, exporting a default as reliable negative evidence, or pairing a new belief with an old source record. An explicit partial adapter identifies the conditions that rule out those mistakes in this case.

A4 supplies the action side's trusted inputs: the fixed metadata, safety result *true*, authorized reviewer **reviewer-1**, checked visible warning and append-only audit log. **A5** supplies currentness: a trusted caller maintains the current admitted source-record identifier. Every new receipt gets an immutable context version, identifying both the Boolean record and its frozen provenance envelope, even if the Boolean values do not change. The adapter checks against that pointer; it does not authenticate it. Set $v(s_0) = \text{agentic-c0}$ and $v(s_1) = \text{agentic-c1}$.

Adapt(*B, s, m0, currentRecordId*):

```

  require s.id == currentRecordId
  require s.issuer == publisher-service
    and s.paper == m0.paper == p0
  require B is a declared finite revision state
  if s is a received positive notice:
    require s.literal == retracted(p0)
      and q belongs to B.Obs
  recompute Frev, Erev and
    Gamma = conclusions(Erev) from B
  require exactly one of q, not-q belongs to Gamma
  require (q belongs to Gamma) == s.retracted
  C := Context(v(s), p0, m0.recent, m0.relevant,
    q belongs to Gamma)
  P := Receipt(adapter-version, B, Frev, Erev, Gamma, m0, s, C)
  return (C, P)
  on any failed requirement: return handoff_error

```

Under the supplied assumptions this gives

$$C_0 = (\text{agentic-c0}, p_0, 1, 1, 0), \quad C_1 = (\text{agentic-c1}, p_0, 1, 1, 1).$$

The exclusive-sign test defines one Boolean; agreement with the current status record checks the intended meaning of that Boolean. Paper equality binds it to p_0 , and fixed metadata supplies the other flags. In particular, the initial default cannot manufacture reliable negative status: without s_0 the required source agreement cannot be established. A notice about another paper fails the paper/literal checks instead of being silently renamed to q .

The receipt P explains the handoff. It retains the revision graph, its accepted arguments and conclusion projection, the source record and the exported context. It is not the action exception certificate defined below, and the raw monitor does not secretly inspect it. The trusted caller associates it with the context version and checks its currentness before dispatch. This is a pedagogical interface construction. Its argument establishes agreement on the supplied records under A1–A5, not the trustworthiness of arbitrary sources.

The explanatory targets consequently differ. Why is R_q accepted? It is an unattacked axiom in the revised graph. Why will an evidential citation lose support? That will be answered in the separate action graph. Why may a marked discussion be dispatched? That additionally requires purpose-bound authority, safety and audit. Chapter 13’s contextual explanations let us retain the object and target of each answer rather than present three names for one proof.

15.3 Reasons for an action and permission to perform it

A6 fixes policy `policy-1`, authorized reviewers `{reviewer-1}`, warning-allowed actions $W = \{\text{ask}\}$, and priority $r_2 \succ r_1$. All citation failures are blocking. A proposed citation has payload

$$x = (\text{id}, \text{cite}, p_0, \text{purpose}, \text{warning}), \quad \text{purpose} \in \{\text{evidence}, \text{discussRetraction}\}.$$

The action domain also contains `ask`, a request without a citation payload. Wider search, summarization and writing tools are not monitored by this finite construction.

At C_0 , the declared runtime argument is $A_1 : \text{recent}(p_0), \text{relevant}(p_0) \Rightarrow \text{citeEvidence}(p_0)$. At C_1 all three arguments are present:

$$\begin{aligned} A_1 &: \text{recent}(p_0), \text{relevant}(p_0) \Rightarrow \text{citeEvidence}(p_0), \\ A_2 &: \text{retracted}(p_0) \Rightarrow \neg \text{citeEvidence}(p_0), \\ A_3 &: \text{retracted}(p_0) \Rightarrow \text{discussRetraction}(p_0). \end{aligned}$$

The contradictory conclusions create attacks $A_1 \leftrightarrow A_2$. Priority $r_2 \succ r_1$ leaves only defeat $A_2 \rightarrow A_1$; A_3 concerns a different action and is isolated. The grounded result therefore changes from $\{A_1\}$ at C_0 to $\{A_2, A_3\}$ at C_1 . These are the declared trees for this example, not an enumeration theorem for an arbitrary structured theory. The identifiers A_1, A_2, A_3 are different from R_p, R_d, R_q : receiving an axiom does not itself produce an action-support argument.

The support map gives $S(x) = \{A_1\}$ for evidential citation and $S(x) = \{A_3\}$ for retraction discussion, restricted to arguments present in the current runtime graph; for `ask` it gives $\emptyset$. With $E_t = \text{Ground}(F_t)$ define

$$\text{Justified}_t(x) \iff E_t \cap S_t(x) \neq \emptyset.$$

At C_1 the discussion has accepted support and evidence does not. Permission remains a separate question: the policy requires a named reviewer to authorize a retraction discussion with visible warning.

An exception certificate has fields

$$c = (id, Cver, \Pi ver, p, purpose, warning, scope, reviewer).$$

$ValidEx_t(x, c)$ holds exactly when x cites the current context's retracted paper; proposal, context, policy and paper identifiers match; both purposes are `discussRetraction`; both warning fields are true; scope is `retraction-discussion-only`; and the reviewer belongs to the current authorized set. Every conjunct is required. Versions identify immutable records; changing a record creates a new version. Comparing reviewer fields does not authenticate a reviewer. Define

$$Perm_t(x, c) = \begin{cases} 1 & x = \text{ask}, \\ 1 & x \text{ cites the context's paper as evidence} \\ & \text{and it is not retracted,} \\ 1 & ValidEx_t(x, c), \\ 0 & \text{otherwise.} \end{cases}$$

This permission predicate belongs to the citation policy. It is not the all-foil comparative permission calculation of Chapters 4 and 10. An exception does not reverse A_2 's defeat or make a retracted paper eligible evidence.

Before dispatch append the exact intent record

$$r_t = (x, C_t, \Pi_t, F_t, E_t, S_t, c)$$

to the trusted log. Let $Auditable_t(x, c)$ mean that this current record belongs to the log. Altering any component invalidates an older record for that membership check. The implementation also retains the pre-preference attacks; that is compatible with keeping attacks distinct from defeats. A later outcome record will report what the tool actually did. An intent cannot be replaced by that future outcome.

Writing $J = Justified_t(x)$, $U = Auditable_t(x, c)$, the monitor is

$$Mon_t(x, c) = \begin{cases} \text{block} & \neg Safe(x, C_t) \text{ or } \neg Perm_t(x, c), \\ \text{block} & (\neg J \vee \neg U) \text{ and } x.action \notin W, \\ \text{warn} & (\neg J \vee \neg U) \text{ and } x.action \in W, \\ \text{pass} & \text{otherwise.} \end{cases}$$

Here `Safe` is the supplied safety-check result, true in the worked traces. Its truth is not inferred from accepted support. Every passed proposal satisfies safety, permission, grounded justification and current intent-audit membership. To reach the last branch, neither first-branch failure can hold. If $\neg J \vee \neg U$ held, exactly one

of the middle branches would apply, according to membership in W . Hence all four conditions hold. This short case analysis proves the local conditional claim. It does not prove tool safety, source accuracy, authentic identity or resistance to altered logs. An unsupported `ask` may receive a warning; it is not a passed proposal with all four guarantees.

15.4 A current authorized response has a positive use

A7 restricts output to two supplied symbolic templates. The requester separately selects the purpose. An evidence proposal emits *EvidenceCitation*(p_0). A discussion emits *RetractionDiscussion*(p_0 , `RETRACTED`, $s.locator$), restricted to the withdrawal and its notice, without using the paper to support the scientific claim. Its warning is true only under the supplied visible-output check. This condition concerns the rendered output, not merely a flag the generator sets about itself.

Fix $x_e = (\text{e0}, \text{cite}, p_0, \text{evidence}, 0)$ and $x_d = (\text{d1}, \text{cite}, p_0, \text{discussRetraction}, 1)$. Let k_1 bind x_d to C_1 , `policy-1`, paper p_0 , that purpose, warning true, discussion-only scope and reviewer 1. This is a supplied new authorization, not a consequence of accepting q or A_3 .

The caller's output policy $O1$ allows citation dispatch only if the provenance receipt is current, the proposal and template fields match, and the monitor returns `pass`. Changing a proposal payload creates a fresh proposal identifier. Before monitoring the caller appends its exact current intent; after monitoring it records the decision; after an authorized dispatch it records the actual outcome. This hand-traced output policy is an additional interface, not a claim that the monitor implements arbitrary text generation.

Attempt, always with <i>Safe</i> = 1	<i>J</i>	<i>Perm</i>	<i>U</i>	Result
x_e at C_0 , no exception, fresh intent	1	1	1	pass
x_e at C_1 , no exception, fresh intent	0	0	1	block
x_d at C_1 , no exception, fresh intent	1	0	1	block
x_d at C_1 , k_1 , fresh intent	1	1	1	pass
x_d at C_1 , k_1 , no intent	1	1	0	block

The first row describes a citation before receipt. Rechecking the same proposal after receipt changes both support and permission. An accurate belief about retraction creates a reason for discussion, but not the exception needed to authorize it. The fourth row combines the accepted reason with the current authority and record needed for dispatch. The fifth shows why merely possessing a valid certificate does not complete the action contract.

A successful supplied execution outcome for the fourth row is the rendered notice "Paper p_0 is retracted; current notice: `fixture:notice-1#status`. It is not used

here as evidence.” The reader learns a relevant fact, its source and its consequence for the requested answer. The reasons and authority for the act remain inspectable. If instead the supplied tool outcome is a delivery failure, the monitor’s previous pass has not made the notice arrive. Record that failure separately.

Set the original goal to “answer the scientific question using eligible evidence”, and stipulate that p_0 is the only candidate evidence supplied. After retraction, even successful delivery of the discussion does not satisfy that goal. It satisfies a different useful subgoal: inform the user of why this candidate cannot supply the answer. Obtaining another source or asking for additional information creates a new task with new inputs. Accepted conclusion, authorized response, dispatch, execution outcome and goal achievement are distinct stages of this finite case.

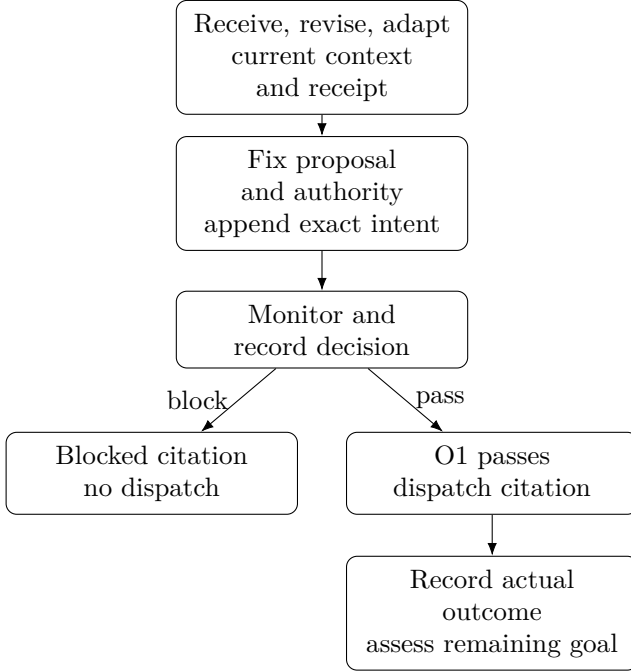


Figure 16: The citation lifecycle under O1. A passed monitor still needs the caller’s current-receipt and template checks; a dispatched act still needs an outcome record. The diagram does not prescribe a universal agent architecture.

15.5 The same belief can require new authorization

Suppose x_d, k_1 have passed at C_1 , but dispatch has not occurred. A new receipt s_2 arrives from the same trusted issuer, with the same positive literal and paper, document version `notice-2` and locator `fixture:notice-2#status`. Its new receipt identifier is s_2 . Set-valued observation update gives $\{q\} \cup \{q\} = \{q\}$. No additional observation argument is created for a second occurrence. Revision trees, accepted

conclusions, runtime graph and accepted action support are unchanged; provenance retains the two receipts separately.

A5 nevertheless assigns $C_2 = (\text{agentic-c2}, p_0, 1, 1, 1)$. Let r_1 be the exact C_1 intent with k_1 , and let k_2 be the same purpose-bound certificate with context version changed to C_2 , *supplied anew by the authorized reviewer*. The agent has not gained permission to forge a renewal. With C_2 current, $Safe = 1$ and $J = 1$ throughout, we obtain:

Certificate and intent supplied at current C_2	<i>Perm</i>	<i>U</i>	Monitor
Old k_1 , old r_1	0	0	block
Old k_1 , exact C_2 intent naming k_1	0	1	block
New k_2 , only old r_1	1	0	block
New k_2 , exact C_2 intent naming k_2	1	1	pass

The old certificate fails its context equality. Logging that stale certificate in a new intent repairs audit membership but not permission. Renewing only the certificate repairs permission but leaves the old intent unequal to the required record. Renewing both permits the discussion, subject still to O1's current receipt and template checks. The current template now uses the `notice-2` locator.

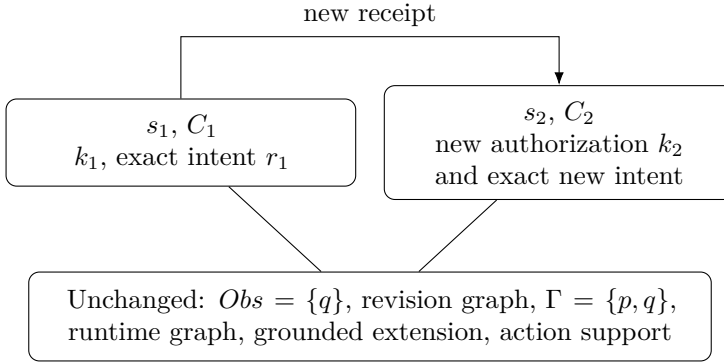


Figure 17: Semantic stability does not imply authorization stability. The new context names changed provenance even though the calculated status and reasons remain the same.

A complete old bundle is a decisive counterexample. Call the raw stateless checker again with the old tuple $(x_d, C_1, \Pi, k_1, [r_1])$. It still returns `pass`: all its fields agree, and it has no independent access to the caller's current receipt pointer. At actual stage 2, O1 rejects that handoff because $s_1.id$ is not $currentRecordId = s_2$. Thus even a valid internal certificate and exact old intent cannot establish currentness without the declared caller convention. The monitor detects mixed old/new fields supplied to it; it does not discover that an internally consistent whole bundle is obsolete. This counterexample explains why the caller's assumption is part of the

argument for dispatch. It does not show that the assumed caller cannot itself be bypassed.

Chapter 13's dinner revision already bound a change to a current model version. Here versioning has a further job: a new source receipt changes what counts as a current authorized response without changing the accepted proposition. Both examples show why a saved answer is insufficient to justify every later use. Their policies remain distinct. A weight-update authority does not authorize citation, and this citation certificate does not authorize execution of a selected dinner option.

15.6 Composition, norms and remaining work

An orchestration state can collect the objects needed for one design:

$$O_t = (B_t, G_t, F_t, E_t, x_t, \text{Trace}_t, c_t, \Pi_t).$$

Here memory and goals are distinct; F_t, E_t name the runtime graph and its result; x_t is a proposal; the trace retains receipts, intents, decisions and outcomes. A tuple is a bookkeeping choice, not a proof that its components fit. A typed requirement might say that every emitted claim has a certificate verified against its current evidence and policy context. To discharge it, one must specify what claims can be emitted, what their certificates verify and which caller prevents an unchecked release.

The supplied reasoning-alignment manuscript motivates this discipline by comparing direct and explanatory routes with common input/output types, distinguishing exact, partial and approximate alignment [11, Sec. 2, pp. 3–5]. The four-instance revision agreement above is one fully supplied restricted comparison. Its agentic composition discussion is an outlook, not a theorem that covers our receipt, adapter and output policy [11, Sec. 4.3, pp. 17–18]. A generated rationale is another object again: it can record proposed thought and action without proving equality of routes or authority to dispatch.

There are already richer public and practical options in the book. If a user asks why the discussion is allowed, Chapter 12's checker can verify an actual public request and response under its supplied rules; Chapter 13 can distinguish an inquiry about a source from an inquiry about authority. To connect that exchange to this action policy, a designer would have to say which public record is supplied as which evidence field. A legal utterance alone does not renew k_1 . Similarly, a system choosing among information-seeking alternatives can use the permission-then-selection procedure already developed for dinner, provided it supplies the new alternatives, reasons and selection policy. The research case has no such data and therefore does not manufacture a balancing calculation. The methods contribute different capabilities without losing their identities at the interface.

Norms ask another question about conduct. In local notation

$$N : \text{retracted}(X) \Rightarrow O(\neg \text{citeEvidence}(X))$$

says that evidential citation of a retracted paper is forbidden; O reads “it is obligatory that”, with no further deontic axioms assumed. If an agent knows that p_0 is retracted and nevertheless uses it as evidence, that supplied knowledge and act establish a knowing violation of N . A visibly marked, authorized discussion has a different purpose and does not violate this narrower evidential norm. It is an exception to the broader citation policy. The finite checker does not infer knowledge or intention from the emitted words.

Institutional responses need still more facts. A knowing violation to expose an exclusionary rule can be offered as civil disobedience and a reform signal; a knowing violation intended to mislead suggests a different response. These interpretations are not exhaustive disjoint outputs of the monitor. Consider an accessibility assistant that knowingly submits an alternative-format form contrary to a supplied paper-only norm, openly records the violation and asks that the exclusionary requirement be revised. Whether the institution accepts that justification is a further normative issue. The example locates reasons for reform; it does not grant our citation monitor power to sanction or reform an institution.

Finally, a sequence of unobjectionable local steps may leave every substantive goal open. A designer may add a bounded progress check, using a stated window and observable events such as acquiring eligible evidence or recording a request for missing information. Such a warning must not override a safety or permission block, and an attempted request must not be counted as a delivered answer. Exercise 15.11 supplies a finite design task. A proof of long-run liveness or a general planning calculus would require much more.

The outlook is consequently constructive. Represented reasons let us explain which status changed and why; separate authority and currentness records make an act reviewable; public procedures make grounds answerable; practical methods can assess supplied alternatives; outcome and goal records identify what to ask next. Their contribution is strongest when the claims remain exact enough to challenge and revise. The book’s recurring question survives at this larger scale: what has this particular argumentative method enabled us to conclude, say, choose or do, under which retained assumptions?

15.7 Exercises and capstone

All ordinary tasks are solvable on paper using this chapter and the named internal studies. Programming may reproduce the same finite checks but is optional. Numbered subparts must be answered separately; collected complete answers include a worked design for each open construction.

- 15.1. *[solution]* [Attack and defeat] At C_1 , construct all attacks among A_1, A_2, A_3 , state the rule priority and derive the defeat relation. Explain the different role of A_3 .
- 15.2. *[solution]* [Permission and support] For an evidential citation at C_1 with $Safe = 1$

and exact current intent, compute J , $Perm$ and the monitor result. Would supplying k_1 change it?

- 15.3. *[solution]* [Explain a block] Give the full runtime defeat graph, grounded extension, evidential support set and failed conditions for Exercise 15.2. Explain why “the paper is rejected” loses distinctions needed by this answer.
- 15.4. *[solution]* [Conduct and interpretation] Suppose an agent knows q and deliberately emits an evidential citation. Compare it under norm N with a correctly authorized marked discussion. State what is known about violation and what further facts would be needed to call the first act a reform attempt or an attempt to mislead.
- 15.5. *[solution]* [Four revision cases] Using $H = \{p\}$, $D = \{\Rightarrow \neg q\}$ and the declared trees in Supplement S15.1, compute accepted arguments and projected conclusions (a) before and (b) after receiving q , with strict rule $p \rightarrow q$; (c)–(d) repeat without that rule. Compare the direct route and state whether the default is deleted in any case.
- 15.6. *[solution]* [Intent and outcome] Write the lifecycle for x_d, k_1 at C_1 with current receipt, matching template, $Safe = 1$ and exact intent. Give (a) a supplied successful delivery and (b) a supplied delivery failure after dispatch. Also show where the no-certificate variant stops. Which events establish that the user received the notice?
- 15.7. *[solution]* [Typed design] Construct an orchestration state for Chapter 14’s policy adviser, using its printed graph, versions and release policy. State one output-verification obligation with typed inputs and output. Explain why listing the tuple does not prove the obligation.
- 15.8. *[solution]* [Cumulative comparison] Explain the distinct jobs of Chapters 14 and 15 through one of their actual cases. Identify an earlier method reused and a new question asked on return. Both chapters belong to the systems movement of Part III.
- 15.9. *[solution]* [Complete binding] List every field and condition of the discussion exception at C_1 . Using the full mutation inputs in Supplement S15.1, compute (a) a wrong context, (b) a wrong paper and (c) missing intent. All have $Safe = 1$ and the valid discussion proposal; (a)–(b) have fresh intents recording their supplied invalid certificate. Explain why that logging does not repair permission.
- 15.10. *[solution]* [Institutional construction] Give a finite civil-disobedience example using the alternative-format filing case of Section 15.6. Supply a norm, knowledge of the norm, the act that violates it and an explicit reform purpose. Distinguish identifying those stipulated facts from endorsing the act or inferring intentions.
- 15.11. *[solution]* [Bounded progress] Let the goal be an eligible-evidence answer. In the last three recorded steps an agent has (1) received s_1 , (2) attempted a permitted

marked discussion, and (3) recorded delivery failure. It has obtained no new eligible source or acknowledged information request. Design a three-step warning condition that counts a newly acquired eligible source or an acknowledged request for missing information as progress. Evaluate it on this trace; extend the trace with an acknowledged request and evaluate the new last-three-step window. Keep citation blocking prior to the progress response. Give a counterexample showing that this progress test does not prove goal completion or general liveness.

- 15.12.** *[solution]* [Receive, revise, adapt, act] For B_0, s_0, s_1 and O1, (a) compute the before/after revision graphs and conclusion projections; (b) construct C_0, C_1 and identify every adapter check that succeeds; (c) compute the evidence and discussion outcomes at C_1 with fresh intent, first without and then with the matching exception; (d) explain why purpose, reviewer authority and source reliability are separate inputs. What fails if s_1 instead names another paper?
- 15.13.** *[solution]* [Unchanged conclusion, changed context] After x_d, k_1, r_1 pass but before dispatch, receive s_2 as in Section 15.5. (a) Identify unchanged and changed objects. (b) Reconstruct every row of the current- C_2 renewal table. (c) Evaluate the raw monitor on the complete old bundle and assess dispatch under O1. (d) State whether a renewed and successfully delivered discussion completes the original goal.
- 15.14.** *[solution]* [Nonprogramming capstone] Use the fully supplied research-assistant case $m_0, s_0, s_1, B_0, A1-A7, O1, x_e, x_d, k_1$ and the successful discussion outcome. The goal is the eligible-evidence answer and p_0 is its only candidate. Produce:
- (a) an input/output architecture naming the caller and each typed handoff;
 - (b) the revision and runtime argumentation objects;
 - (c) their evaluations before and after receipt;
 - (d) one contextual explanation of the changed action result;
 - (e) a precise verification or monitoring condition and its bounded guarantee;
 - (f) a failed variant with its exact failed conditions;
 - (g) one further update, using s_2 , and the required response;
 - (h) a reasoned account of which perspective—inference, dialogue or practical evaluation—dominates this supplied system, what useful act it achieves and what remains of its goal.

No external facts or software are required. An alternative design may use another fully supplied book case, but must declare any extra local policy, keep output types distinct, give a complete finite trace and meet the same eight obligations. Do not infer missing empirical facts. External domain investigation is the separately labelled research variant below.

15.8 Annotated outlook and reading routes

The supplied *Reasoning Alignment for Agentic AI* manuscript provides the exact route for the comparison used here: Section 2, pp. 3–5, distinguishes relations between typed routes; Section 4.3, pp. 17–18, proposes agentic composition as outlook [11]. Its separate base-revision result in Section 3.4 is not proved by our four cases. The published version is a distinct continuation [14]; do not carry the supplied manuscript’s page locators over to it without checking the version.

The A-BDI metamodel returns to the book’s relation among balancing, dialogue and inference [34]. Read it asking what each perspective contributes to an agent’s purpose, and what an actual interface must still specify. Volume 3’s concluding challenge chapter, Sections 4.3–4.4 and conclusion, pp. 934–947, places those connections in the broader programme [1]. The epistemic-planning chapter is a route into action that changes knowledge [113]; the present window-based exercise is not a planning or liveness theory.

The *Handbook of Normative Multiagent Systems* extends the institutional question behind *N* [78]. Norm representation, compliance, permission, violation and institutional response demand distinctions that a Boolean monitor does not settle. The disobedience route asks how reasons for challenging a rule differ from ordinary compliance or evasion [116]. Neither citation makes intentions observable in our trace; the construction exercises must supply them.

ReAct supplies a useful contrasting interface: language steps and environment actions are interleaved with observations [117, Sec. 2, pp. 3–4, arXiv v3]. Read its trajectory as generated interaction, then ask which additional relation an alignment claim would need and which current authority a dispatch needs. A generated rationale, a proved finite route agreement and a checked action contract are distinct evidence. None alone establishes empirical superiority, general safety or a user’s understanding.

Literature/research task 15.L1. [\[solution\]](#) Read ReAct, arXiv:2210.03629v3, Section 2, pp. 3–4, and the supplied RAD manuscript, Section 2, pp. 3–5 and Section 4.3, pp. 17–18. For a transcript that receives the notice and proposes marked discussion, explain what the sequence shows, what an exact alignment claim additionally requires, and which evidence in this chapter warrants dispatch. External source comparison is the task’s purpose; no ordinary exercise depends on it.

Literature/research task 15.L2. [\[solution\]](#) Choose the supplied RAD route, one Handbook route and one further route above. In five connected sentences compare their question, formal object, result or proposed contribution, relation to the finite handoff and a remaining limitation. A research capstone may instead investigate a chosen domain, clearly separate source-supported facts from stipulations, and then meet the eight ordinary capstone obligations. Complete answer guidance and criteria for legitimate alternatives appear with the collected solutions.

Reading routes and supplied studies

A first reading should establish how one question leads to the next. When an example returns, record its input, the new question, the answer and the reason that answer is justified. Comparing those records makes the purpose of a return visible: a graph used to distinguish semantics can later support a computation, while a practical comparison can return to explain a changed choice. Keep distinct examples and output types separate even when they use the same letters.

The opening chapters contain their own reading guidance. Chapter 0 prepares the mathematical habits; Chapter 1 develops source-method comparison; Chapter 2 develops agency and aggregation. Their ordinary exercises and the two Chapter 2 studies are complete paper routes. In Chapters 3–4, reconstruct the act and the intended question before calculating with a formal object. These early uses prepare later precision; they need not be replaced by a preview of every specialist definition.

The following guides give more detailed routes through the construction, semantic, public, practical and systems developments. They distinguish the full book's depth from the independent work a semester or intensive course can select. None makes the whole bank a weekly assignment. A difficult proof can be followed with guidance on one route and reconstructed independently on another without changing its place in the book's argument.

On a later reading, choose a comparison whose assumptions you can state. For example, ask whether two source routes agree, whether an extension can be completed after an input change, or whether a recorded dialogue satisfies the supplied response policy. Do not require every example to instantiate structured inference, public dialogue and balancing at once. The service's purpose determines which connection is useful.

The annotated chapter readings provide the next literature step. Read a selected source with its formal object, hypotheses and output type beside the book's treatment. Distinguish the original result from a later survey and a restricted teaching construction from the source's full formalism. Keep the cited version and locator when comparing passages. This makes further reading a way to extend or challenge a particular argument, rather than a quota of references to collect.

The studies listed after these guides are included in full before the collected answers. Each can be reached directly from the directory or the task that uses it. They provide substantive mathematics, inputs or evidence; they are not pointers to external material needed to finish an ordinary task.

Optional study guide for Part II

The chapters' main argument runs from construction through semantics to representation and design, then outward to dialogue and balancing. This guide collects practical reading and exercise advice so that those choices do not interrupt that argument. It is a guide to routes, not a requirement to assign every exercise in a course.

Dependencies and recurring examples

Chapters 5–8 form a cumulative sequence. Chapter 5 supplies structured arguments and their evaluated defeats; Chapter 6 supplies extension semantics; Chapter 7 adds a source method and an extraction to compare whole outputs; Chapter 8 tests postulates and constructions across inputs. The later dialogue and balancing developments draw on the attack–defence calculus while adding their own objects and operations. The computational and explanatory developments of Chapters 12–13 return to this calculus and to the later dialogue and balancing settings; Chapter 7's full representation discussion need not be treated as a prerequisite for every computational route.

Example	Distinct jobs in Chapters 5–8
Scot	Construct arguments and local defeats; select stable sets; compare extracted outputs with an independent priority procedure.
Linked cycles	Separate complete, preferred, stable and grounded semantics within one graph; interpret the corresponding labellings.
Grounded chain	Explain reinstatement and separate the iteration's admissibility, termination and leastness obligations.
Shared default	Test how a common rule changes a comparison; then encode the chosen AF exactly, including its unwanted source output.
Professor	Transfer a priority pattern and show why closure and consistency do not choose the purpose of a policy.
Sherlock	Separate adding a route from replacing evidence, and compare a graph calculation with an attack-assignment condition.
PDL witness	Use three related inputs to connect exact output-family representation, local attack restrictions and context independence.

The five-node computational-reuse framework in Exercise 6.2 has a different relation from the linked-cycle graph in the main narrative. Keep the displayed edge set with

each calculation. Similar argument names do not make two examples the same input.

Choosing exercise work

The exercise roles name the intellectual activity: calculation, conceptual diagnosis, proof, transfer, alternative formalism, computation, design or research extension. They do not rank those activities by importance. A substantial proof or an unfamiliar formalism can belong in an exercise because reconstructing it is the intended learning, not because it is dispensable mathematical content. Starred tasks offer a further route; different courses may select different routes without narrowing the public exercise set.

For the finite semantic proof route, work through Exercises 6.4–6.6 in order. The first supplies invariants and the Fundamental Lemma; the next uses them to connect the semantics; the last distinguishes a terminating iteration from a justified grounded construction. The AA2LP proof in Exercise 7.3 is another complete proof route, using only the fact-only reduct defined in the main chapter. The optional normal-program supplement develops the extra positive-body machinery and provides the separate exercises S7.1–S7.2.

For transfer, follow the rank change from Exercise 5.1 to Exercise 7.1, or follow the professor from Exercise 5.5 to Exercise 8.2. The former changes an input while holding a source policy fixed; the latter asks what the output tests can and cannot decide. Exercise 5.6 explicitly requires the later chapter and need not interrupt a first reading of Chapter 5.

For breadth, the existing exercises develop deductive arguments, DeLP construction, ABA, ADF models and alternative preference reductions. The annotated further-reading sections offer further routes into semantics, other representations and construction principles. A topic named in a literature guide is an invitation to further work, not a claim that its full development is already an exercise or a prerequisite of the main sequence.

Working with complete solutions

All solutions are public and collected at the end of the book under their chapter and exercise numbers. For a calculation, record both the selected objects and a reason why no others satisfy the test. For a proof, identify the hypothesis used at each step before consulting the complete argument. For a comparison, state the two output types before asking whether they agree. These practices make it easier to locate a disagreement with a solution than a bare comparison of final answers.

A cumulative example notebook can keep four short entries: the informal question, the exact formal input, the computed output, and what changed when the example returned. Keep alternative outputs separate rather than combining them into one

body of beliefs. A useful final check is to ask which distinction made the calculation possible: raw attack versus defeat, inclusion versus cardinality, an argument versus its conclusion, a fixed source versus a changed construction, or no output versus an empty output. The literature guides then show where the small calculation leads and which of its assumptions the fuller theory changes.

Part III study guide: comparison, computation and explanation

A useful reading route follows three changes of question: which properties should the method preserve, what must a computation establish, and what must an explanation retain? The Chapters 11–13 main sequence supplies all three, including its selected proofs. Courses can choose routes through the larger exercise bank without treating that selection as the limits of the book’s argument.

Preparation and recurring objects

The required preparation is Chapter 0’s finite sets, quantified statements, Boolean logic, proofs and iteration, together with Chapter 6’s semantic definitions and finite grounded and preferred-extension results. Chapter 11 returns to Chapter 8’s distinction between output and construction principles; Chapter 12 supplies its own computational representation and complexity vocabulary. Its SAT reduction therefore requires no previous complexity course. The public and practical returns across all three chapters use Chapter 9’s local state rules and Chapter 10’s comparative reasons, weights and distinction between permission and selection. Their worked inputs and required consequences are recalled at each new task; no earlier exercise answer is a prerequisite. Claim acceptance under an issue’s proof standard remains a separate target.

Keep the input with each example. Chapter 6’s F_{link} discriminates among semantics. The different F_{reuse} of Exercise 6.2 is reprinted in Chapter 12, where its three stable outputs and its changed outputs are derived. Chapter 13 uses those established results for target, deletion and context questions. The four-node graph from Chapter 7’s translation exercise is separately reprinted in Exercise 12.11 for algorithm and query transfer. Similar letters do not make those graphs interchangeable.

The restaurant public record has its own recurring job. Chapter 11 uses a forced reply to test a transition property; Chapter 12 checks a complete state and compares offered continuations under an explicit budget. The dinner’s comparative reasons instead test a considered-alternative property, feed permission and selection procedures, and then support a practical explanation and authorized revision. The public claim of restaurant excellence, the practical option R and the agents in a dialogue remain distinct objects even when their topics overlap.

Three levels of study commitment

The table describes capabilities to select, not measured workloads or a promise that a fixed number of meetings suffices.

Development	Book route	Semester selection	Intensive selection
Principles and locality	Understand the function comparison, reconstruct grounded locality, and diagnose stable failure.	Independently solve a principle/counterexample transfer and reconstruct the locality argument.	Guided comparison and one restriction or failed-completion calculation.
Mechanisms and evidence	Compare support operations, preference policies, enforcement and all three cognitive designs.	One substantial mechanisms/change transfer and the supplied cognitive comparison.	Guided typed comparison, a short change example and interpretation of the actual evidence.
Computation	Follow checker correctness and cost, exact queries, the full reduction and conditioned update.	Independently trace algorithms and queries; reconstruct the reduction with guidance and perform the component/update transfer.	One exact checker/query trace; prepared or follow-up guided reduction and one conditioned branch/update.
Public and practical methods	Test response and alternative-domain properties; compute bounded continuations, permission and selection.	Independently test one property and trace both a public continuation and a practical update.	Guided pending-reply check, one changed foil comparison and the two-policy contrast.
Explanation	Retain targets, context, cyclic defense, practical grounds and authorized revision, and the classifier evidence contrast.	Independently construct a contextual certificate, recompute a deletion, explain a practical revision and compare the evidence accounts.	One contextual explanation, the decisive dinner contrast and a guided evidence comparison.

A selected proof's place in the book does not by itself make memorizing the whole proof an examination objective. Independent mastery, guided reconstruction and informed comparison are different promises. A course that places computation before principles can supply the local semantic and cost preparation first, then return to Chapter 11's comparison; this adoption choice does not change the book's progression.

Choosing connected exercise routes

For the ordinary-semantic argument, connect Exercises 11.13–11.15: first separate properties from characterization, then reconstruct the locality proof and change the blocked stable branch. The support transfer begins with Exercises 11.1–11.4 and continues through target loss, defense and support-aware locality in 11.5, 11.10 and 11.11. Study S11 supplies the extra definitions and complete candidate method; its optional 11.18–11.19 extend closure, change and certificate reasoning. Preference and actual cognitive evidence have their own comparison tasks, 11.16–11.17. Exercises 11.20–11.21 make the public and practical principles transferable: distinguish a guaranteed reply transition from completion, then test what happens to permission when the considered alternatives change while old pair assessments stay fixed.

In Chapter 12, the first ten tasks distinguish basic witnesses, clauses, guards and task claims. Exercises 12.11–12.12 change the graph’s job from Chapter 7’s representation question to explicit computation and SAT queries. Exercise 12.13 reconstructs the selected full reduction, including the query-versus-existence counterexample; 12.14 follows the incoming-conditioned branches and update. Exercise 12.15 separates a useful decision witness from a complete semantic output. Study S12 prepares exact blocking, auxiliary-variable projection and the optional DP-membership route in 12.16–12.17. It does not require access to a SAT solver. Exercises 12.18–12.19 trace the public and practical procedures on their supplied inputs. The first separates checking a legal prefix from choosing a completed offered plan within budget. The second follows the two weight arrays, the reversed comparison and the later policy choice. The restaurant branching diagram locates the intermediate state that makes the budget difference visible.

For explanation, begin with 13.1–13.5 to move among accepted defense, out status, recomputed necessity, local minimality and contrast. Exercise 13.6 uses the fully supplied public state to distinguish a challenge from a question. The update and procedure proofs in 13.11–13.12 test what retained context licenses. Study S13 and Exercise 13.13 compare the AXPLR evidence with synthetic fidelity; 13.14 separates failure of universal membership from absence of any stable output. Exercises 13.15–13.16 explain the dinner’s changed permission and the two selections, then trace a public request and independently check proposed revisions. A missing warrant, unauthorized issuer or stale version raises a different question from an incorrect sum. The three chapter literature exercises are explicitly external investigations. Optional ordinary studies instead supply their prerequisites inside the book.

Paper computation and complete answers

A set can be represented by a row of membership marks; an attack list can be scanned with a separate coverage row. A Boolean query can be solved by propagating the printed clauses or exhaustively trying small assignments. An update

calculation should identify changed constraints and prove whether its returned family is complete. These are full nonprogramming routes, not preparation that requires a later software installation to become meaningful. Code can implement them as an optional comparison, retaining the same input, query and output contract.

For a public trace, retain active records, the ledger and the pending field in separate columns; a legal next move may leave an experimental observation unfinished. For practical computation, accumulate two ordered-pair arrays, mark each option's failing foils, and only then scan the supplied selection order. A proposed revision identifies the one changed cell and its version before those calculations are repeated. These paper records make the distinctions inspectable without software.

Complete solutions are collected under Chapters 11–13 at the back. For self-study, first record the exact query, the supplied input and the first justified step. Then compare both the result and its warrant with the answer. A correct family without exhaustiveness does not answer a request for all outputs; a successful edit without excluding cheaper ones does not certify minimum cost; a plausible explanation without its target does not identify what was explained. These distinctions prepare the later systems and agentic chapters without assuming that solving a finite exercise validates a deployed system or establishes human understanding.

Part III study guide: systems and the bounded agentic outlook

Read Chapters 14–15 as a connected change of question. First ask what a particular service should return and which evidence licenses its use. Then ask what changes when information is received and a proposed action must be current. The policy, restaurant, dinner and research-assistant cases retain different identities. Comparing their interfaces is more instructive than forcing them into one story.

Route	Expected work
Full book	Follow all main service traces and the revision–adapter–action hand-off. Reconstruct the release branch argument, monitor guarantee and complete-old-bundle counterexample. Use S14.1–S14.2 and S15.1 for internally prepared score, admission and mutation work. The complete capstone has a paper route and a worked answer.
Semester	Independently reconstruct a release failure and the received-notice handoff, then diagnose the unchanged-belief/new-context case. Select one trust, score or documentary transfer instead of requiring all advanced extensions in the same tutorial. A capstone needs written or small-group defense, or separate presentation time.
Intensive	Use a guided policy/dinner service comparison and one live currentness/remaining-goal transfer. Assign complete inputs in advance. The full remainder derivation, score reconstruction and broader literature are selective preparation or follow-up; the final joint explanation/integration session cannot also promise mastery of every study and a full capstone.

A useful self-study sequence is to attempt Exercises [14.12](#) and [14.13](#), then [15.12](#) and [15.13](#), before consulting their answers. The inherited shorter tasks isolate the necessary distinctions. Exercise [15.14](#) then asks for a coherent design with eight connected outputs. Substantial construction does not require programming: a graph, trace, checked table and precise explanation can supply the complete answer. The source-reading tasks are visibly literature/research work. Their assigned passages provide material for comparison, rather than missing premises for an ordinary calculation. Optional code can reproduce finite calculations; it does not by itself establish source truth, general assurance or a benefit to users. These routes state intended capabilities, not measured workload or empirical teaching effectiveness.

Directory of supplied studies

The sixteen studies below belong to the book. Optionality concerns the reading route; where an exercise calls on a study, its supplied definitions, data and assumptions are part of that task's preparation. Study exercises keep their own identifiers, including 2.S1–2.S2 and S7.1–S7.2, and have complete answers in the chapter-ordered collection.

Owner	Study and function	Page
Chapter 2	Coalition witnesses: inputs, enumeration and minimality	193
Chapter 2	Grounded comparison of the majority graphs, with its own iteration	196
S3.1	Evaluating supplied extraction data	198
Chapter 7	Stable models of normal programs: reduct and closure	199
S9.1	Comparative disclosure and withdrawal	203
S9.2	From dialogue theory to application evidence	207
S10.1	Scores and rankings	208
S10.2	Values and audiences	210
S10.3	Uncertainty about graphs and beliefs	211
S10.4	Where weights get their authority	213
S11	Mechanisms and change	214
S12	Queries, projections and certificates	217
S13	Evidence for a classifier explanation	219
S14.1	When support changes a score	221
S14.2	Selecting the content of a report	223
S15.1	Finite revision and action handoff	225

Chapter 2 study. Enumerating coalition witnesses

A single witness shows that collective defense is possible. Enumeration asks for every inclusion-minimal way of supplying it. The extra work is worthwhile when we want to distinguish indispensable sources from alternative coalitions, or to explain why a smaller answer does not make every larger answer redundant. This study supplies the finite procedure and all data for Exercise 2.8; its answers remain in the collected solutions.

Fix a finite argument set A , attack relation R , source set S , and association relation $\text{src} \subseteq A \times S$. The defending arguments $E \subseteq A$ and target $x \in A$ are part of the input. Put

$$\text{Att}(x) = \{b : (b, x) \in R\}, \quad S_x = \{i : (x, i) \in \text{src}\}, \quad A_i = \{d : (d, i) \in \text{src}\}.$$

For enumeration assume that S_x is finite and explicitly listed. The general source framework requires finite A ; it need not make the entire source universe finite.

For each $i \in S_x$, form its coverage set

$$U_i = \{b \in \text{Att}(x) : \exists d \in E \cap A_i, (d, b) \in R\}.$$

A coalition witness is $C \subseteq S_x$ with $\bigcup_{i \in C} U_i = \text{Att}(x)$. It is inclusion-minimal if no proper subset has that coverage. These are precisely the derived certificates of Chapter 2's collective predicate [15, underlying predicate: supplied-proof Def. 6, p. 5], not a new agent semantics.

For a complete worked coverage table, reuse the five-node input

$$\begin{aligned} A &= \{a, b_1, b_2, c_1, c_2\}, \\ R &= \{(b_1, a), (b_2, a), (c_1, b_1), (c_2, b_2)\}, \\ S &= \{\text{Alice}, \text{Bob}, \text{Cayrol}\}, \quad E = \{c_1, c_2\}, \end{aligned}$$

target a , and

$$\begin{aligned} \text{src} &= \{(a, \text{Alice}), (a, \text{Bob}), \\ &\quad (b_1, \text{Cayrol}), (b_2, \text{Cayrol}), \\ &\quad (c_1, \text{Alice}), (c_2, \text{Bob})\}. \end{aligned}$$

All unlisted pairs are absent. Here $S_a = \{\text{Alice}, \text{Bob}\}$, $U_{\text{Alice}} = \{b_1\}$ and $U_{\text{Bob}} = \{b_2\}$. The four coalition candidates are empty, Alice alone, Bob alone, and their pair. Only the pair covers both attackers. Neither member can be deleted, so the pair is minimal.

A successful certificate lists a source and a qualifying counterargument for each attacker. To refute a candidate, give one attacker that none of its sources can cover through E . Minimality needs a failure for proper subsets as well as success for the whole coalition. It is enough to check each one-agent deletion: any proper subset

is contained in one such deletion, and coverage cannot increase when sources are removed.

A finite paper algorithm. Compute the coverage sets, then consider subsets of S_x in increasing cardinality, including the empty set:

```

accepted := empty list
for each C subset of S_x, in increasing cardinality:
    if an accepted D is a proper subset of C:
        skip C
    else if union of U_i over i in C equals Att(x):
        append C to accepted
return accepted

```

Considering smaller sets first does not mean optimizing cardinality. A larger incomparable witness must still be considered. The skip removes supersets of an already accepted witness, since they cannot be minimal. In the unattacked case the first candidate, the empty set, covers the empty attacker set and rules out every nonempty candidate as nonminimal. There can be $2^{|S_x|}$ candidates; this transparent procedure is not an efficiency claim for large source sets.

Complete inputs for the transfer exercise

In both cases below the target is x , and every attack and source pair not listed is absent. The tables specify inputs, not the coalition answers.

Transfer: inclusion need not minimize cardinality.

$$\begin{aligned}
 A &= \{x, b_1, b_2, d_1, d_2, e_1, e_2\}, \\
 R &= \{(b_1, x), (b_2, x), (d_1, b_1), (d_2, b_2), (e_1, b_1), (e_2, b_2)\}, \\
 S &= \{\text{Ann}, \text{Bob}, \text{Cara}\}, \quad E = \{d_1, d_2, e_1, e_2\}, \\
 \text{src} &= \{(x, \text{Ann}), (x, \text{Bob}), (x, \text{Cara}), (d_1, \text{Ann}), (d_2, \text{Ann}), (e_1, \text{Bob}), (e_2, \text{Cara})\}.
 \end{aligned}$$

Co-owned defender.

$$\begin{aligned}
 A &= \{x, b, d\}, & R &= \{(b, x), (d, b)\}, \\
 S &= \{\text{Alice}, \text{Bob}\}, & E &= \{d\}, \\
 \text{src} &= \{(x, \text{Alice}), (x, \text{Bob}), (d, \text{Alice}), (d, \text{Bob})\}.
 \end{aligned}$$

An argument associated with two sources is still one argument. The coverage calculation asks which source certificates it can supply, not how many different argument objects it becomes.

Study exercise

2.S1. *[solution]* Prove that the finite algorithm returns exactly all inclusion-minimal witnesses, independently of the order among equal-size candidates. Your proof must justify both accepted-set minimality and completeness. Explain why checking one-agent deletions suffices for minimality, and identify the separate termination argument. Use the transfer input above to explain why stopping after the first successful cardinality would be incorrect for the stated task. This is an ordinary internal proof exercise; a program is optional. Its complete answer is collected with the Chapter 2 solutions.

Chapter 2 study. A grounded comparison of the majority graphs

Chapter 2 used the stable test to compare aggregation before and after evaluation. The comparison also matters under a different policy: begin by accepting what can be defended without relying on any accepted arguments, then use those results to support further defense. This study supplies the entire finite calculation. No Chapter-6 definition or result is required to perform it.

For a finite attack graph $F = (A, R)$, a set $E \subseteq A$ defends x when every attacker of x is itself attacked by some member of E . Define the defense operator

$$D_F(E) = \{x \in A : \forall b \in A ((b, x) \in R \Rightarrow \exists d \in E (d, b) \in R)\}.$$

Start from $E_0 = \emptyset$ and compute $E_{n+1} = D_F(E_n)$ until equality. For finite graphs this is the grounded construction; Chapter 6 proves its general leastness and its relation to the other semantics [5, Defs. 6 and 20, pp. 326, 329]. Here all four traces will be displayed and justified directly. Unlike Chapter 0's successor-addition update, this formula does not explicitly union its input into its output.

The shared argument set is $A = \{a, b, c\}$, and the complete relations are

$$\begin{aligned} R_1 &= \{(a, b), (b, c)\}, & R_2 &= \{(b, c), (c, a)\}, \\ R_3 &= \{(c, a), (a, b)\}, & R_{\text{maj}} &= \{(a, b), (b, c), (c, a)\}. \end{aligned}$$

All unlisted edges, including loops, are absent. An edge belongs to R_{maj} when at least two of the first three relations contain it. This is the same Handbook profile as in the chapter [16, Vol. 2, Ch. 4, Sec. 3, pp. 261–262].

In the first graph, a has no attacker, so the empty set defends it. The empty set does not defend b against a or c against b . Thus $D_1(\emptyset) = \{a\}$. Now a attacks the attacker b of c , while nothing can attack a to defend b . Therefore $D_1(\{a\}) = \{a, c\}$, and applying D_1 again changes nothing. Rotating the argument names gives the next two traces:

	E_0	E_1	E_2	E_3
F_1	$\emptyset$	$\{a\}$	$\{a, c\}$	$\{a, c\}$
F_2	$\emptyset$	$\{b\}$	$\{a, b\}$	$\{a, b\}$
F_3	$\emptyset$	$\{c\}$	$\{b, c\}$	$\{b, c\}$
F_{maj}	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

In the cycle each node has an attacker and the empty set attacks none, so $D_{\text{maj}}(\emptyset) = \emptyset$. It is already a fixed point.

Call the resulting single grounded sets $G_1, G_2, G_3, G_{\text{maj}}$. A strict majority of membership in G_1, G_2, G_3 gives $\{a, b, c\}$, because each argument belongs to two input outputs. Evaluating the majority graph instead gives $G_{\text{maj}} = \emptyset$. Here both operations return a single set, and those sets differ. The membership-majority set also fails conflict-freeness in the aggregate graph.

The empty grounded set is an output, not the absence of an output. The aggregate's grounded-extension family is $\{\emptyset\}$. Its stable-extension family, as the chapter checked, is $\emptyset$: no set satisfies the stable test. Keeping single outputs and output families distinct prevents an empty notation from hiding a substantive difference between the two policies. Neither finite comparison is a theorem against every possible aggregation rule.

Study exercise

2.S2. *[solution]* Use only the operator and finite iteration defined above. Let $A = \{x, y, z\}$ and $R = \{(x, y), (y, x), (y, z)\}$, with no other edges. (a) Compute the iteration from the empty set until equality. (b) Now add one new argument t and the sole new edge (t, y) . Recompute every stage, explaining the defense or failed defense of each argument at the fixed point. (c) Give the single grounded output and the one-member grounded-output family in each case. (d) Explain why this calculation does not prove that the operator is inflationary on every input; supply a particular set in the original three-node graph witnessing the failure. The complete solution appears with the Chapter 2 solutions.

Supplement S3.1. Evaluating a supplied extraction

A relation score evaluates a representation against a declared reference; it does not decide which reconstruction should serve an application. To see the distinction quantitatively, fix R_0 from Chapter 3 as the reference, assume the seven claim vertices were supplied correctly, and score exactly these eight candidate ordered pairs. The labels in the last column are stipulated predictions, not measured system outputs.

Pair	Reference	Prediction P
(b, a)	support	support
(e, a)	support	support
(g, a)	support	support
(c, a)	attack	attack
(f, a)	attack	attack
(d, c)	attack	support
(g, f)	attack	attack
(d, a)	none	none

Each pair receives exactly one label. Relation error is the number of mismatches divided by eight; accuracy is the number of matches divided by eight. Here they are $1/8$ and $7/8$, respectively. These are not word-detection or corpus-level scores. Adding every unlisted pair as a correct negative would change the task and its denominator.

For the downstream query, the reference answer is $\{d, g\}$, while the predicted graph returns $\{g\}$. Define recall on this nonempty reference answer as the proportion of reference defenders retrieved. It is $1/2$. The single wrong relation therefore removes one of two required response witnesses, although seven of eight relation labels agree. The labels do not tell us which error will matter to a user; the application requirement does.

The alternative R_1 is a defensible reconstruction for another declared purpose. It is not the graph P , and its defensibility does not erase an error in a task explicitly scored against R_0 . Conversely, conformity to R_0 would not prove that it is uniquely faithful to the paragraph. Annotation, specification verification and requirements assessment remain separate judgments; the natural-language Handbook route supplies the broader evaluation context [26, Sec. 2, pp. 584–585]. The complete calculation and interpretation are revisited in Exercise 3.11.

Chapter 7 mathematical supplement: Stable models of normal programs

Supplementary reading ★. Chapter 7's AA2LP programs have no positive body conditions: after default negation is evaluated, every surviving rule is a fact. Normal logic programs allow a surviving rule to depend on other atoms still to be derived. This supplement develops that additional step, including the least-model proof, the complete Nixon calculation and two transfer exercises with complete public solutions. It is not a prerequisite of the chapter's fact-only correspondence.

Deriving the least positive model

Fix a finite propositional atom universe U . A finite normal program is a finite set of rules

$$h \leftarrow b_1, \dots, b_s, \text{not } c_1, \dots, \text{not } c_t, \quad s, t \geq 0,$$

where all displayed names are atoms in U . The head is h and the body is conjunctive. An empty body gives a fact; a rule or program without `not` is positive [56, Sec. 2, Def. 1, pp. 431–432]. The word `not` will test absence from a candidate interpretation. It is distinct from classical negation, and an atom called *neg* p is still just an atom. Its relation to p must be supplied by the rules, not inferred from its name.

First consider a positive program Q . Write $h \leftarrow B$ for a rule whose positive body atoms form the set B . A set $J \subseteq U$ is a model of Q when

$$\text{for every } (h \leftarrow B) \in Q, \quad B \subseteq J \implies h \in J.$$

The full universe is always a model, but it may contain many atoms for which the program supplies no derivation. We want the *least* model: a model contained in every model, denoted $\text{LM}(Q)$. Merely finding a set closed under the rules does not establish this leastness.

Facts begin the derivation. Other rules contribute their heads when all their prerequisites have already been obtained. Formally, put

$$C_Q(S) = S \cup \{h : (h \leftarrow B) \in Q, B \subseteq S\}, \quad S_0 = \emptyset, \quad S_{i+1} = C_Q(S_i).$$

Bodies are tested against the old S_i , so the displayed rounds are simultaneous and do not depend on the order in which rules are printed. The empty-body test always succeeds. Continue until the first equality $S_{k+1} = S_k$.

Proposition S7.1. For every finite positive program over finite U , this iteration terminates and returns its unique least model.

Proof. Each round retains the previous set. A strict increase adds at least one new atom, so there can be at most $|U|$ strict increases, followed by an evaluation detecting equality. This includes the empty universe.

At the stopping point, if a rule body B is contained in S_k , its head belongs to $S_{k+1} = S_k$. Thus the returned set is a model. To prove leastness, let J be any model of Q . We show by induction that $S_i \subseteq J$ for every i . This holds for $S_0 = \emptyset$. If it holds for S_i , every retained atom remains in J ; every new head has a rule body contained in S_i , hence in J , and the model condition puts that head in J . Therefore $S_{i+1} \subseteq J$. In particular, the returned model is contained in every model. Two least models would each contain the other and so be equal. $\square$

This is the elementary finite construction of the least positive model used by the stable-model test [5, Sec. 4.3.1, p. 344, paragraph before Thm. 49]. The induction is its substantive step: termination alone would show only that a fixed point had been reached, not that no unsupported atoms had been introduced.

For example, $Q = \{p \leftarrow p\}$ has both $\emptyset$ and $\{p\}$ as models. Starting from $\emptyset$ derives nothing, so the least model is $\emptyset$. Starting instead from $\{p\}$ would preserve an assumption that the program never establishes. A positive cycle may transmit support, but cannot supply its own beginning.

A candidate determines a reduct, not a starting point

Definition S7.2. For a normal program P and $M \subseteq U$, form the reduct P^M in two steps:

1. Delete every rule containing not c with $c \in M$.
2. In each remaining rule, delete all default-negated conditions, retaining its positive body.

The result is positive. The candidate M is a stable model of P exactly when

$$M = \text{LM}(P^M)$$

[5, Sec. 4.3.1, p. 344, paragraph before Thm. 49].

The candidate determines which rules survive. It does not seed the positive derivation, which starts again from $\emptyset$. Nor are default conditions tested afresh at intermediate closure rounds: they have already been resolved using M . Equality requires every guessed atom to be derived and every derived atom to have been guessed. Failure can occur in either direction.

For the positive loop $p \leftarrow p$, both candidates give the same reduct, whose least model is empty. Thus only $\emptyset$ is stable. This contrasts with the default-negative self-loop $p \leftarrow \text{not } p$, which has no stable model: omitting p produces its fact, while including p removes its only rule.

The Nixon program and its complete output family

Let q mean that Nixon is a Quaker, r that he is a Republican, and p that he is a pacifist. Use an ordinary atom neg_p for the contrary conclusion. The program is

$$P_N = \left\{ \begin{array}{l} q \leftarrow, \\ r \leftarrow, \\ p \leftarrow q, \text{ not } neg_p, \\ neg_p \leftarrow r, \text{ not } p \end{array} \right\}, \quad U = \{q, r, p, neg_p\}.$$

Every reduct retains the two facts. Hence every stable model contains q, r , and only four candidates remain to be considered.

For $M_p = \{q, r, p\}$, the rule for p survives and loses its default condition; the rule for neg_p is deleted. The reduct is

$$P_N^{M_p} = \{q \leftarrow, r \leftarrow, p \leftarrow q\}.$$

Its successive positive-closure sets are

$$\emptyset, \quad \{q, r\}, \quad \{q, r, p\}, \quad \{q, r, p\}.$$

The candidate equals the least model and is stable. The opposite candidate $M_{\neg p} = \{q, r, neg_p\}$ retains only $neg_p \leftarrow r$ beyond the facts and is stable by the symmetric calculation.

The other two candidates fail for opposite reasons. If neither contested atom is guessed, both rules survive, and closure derives both heads. If both are guessed, both rules are deleted, leaving only the facts. The complete audit is

Candidate M	$LM(P_N^M)$	Stable?
$\{q, r, p\}$	$\{q, r, p\}$	yes
$\{q, r, neg_p\}$	$\{q, r, neg_p\}$	yes
$\{q, r\}$	$\{q, r, p, neg_p\}$	no
$\{q, r, p, neg_p\}$	$\{q, r\}$	no

Thus the stable-model family is exactly $\{\{q, r, p\}, \{q, r, neg_p\}\}$. The exclusion of a stable model containing both contested atoms is a consequence of these rules. There is no general consistency axiom attached to the spelling of neg_p .

To compare this source answer with a small argumentative route, construct the two derived arguments $\alpha = [q; q \Rightarrow p]$ and $\beta = [r; r \Rightarrow \neg p]$, keeping the facts inside their construction, and stipulate mutual rebut as the evaluated relation. The stable extensions are $\{\alpha\}$ and $\{\beta\}$. Their conclusion outputs $\{p\}$ and $\{\neg p\}$ agree with the program after the facts are suppressed and neg_p is read as $\neg p$. This is an instance comparison on the displayed construction.

It is not an identification of P_N with the AA2LP translation of that graph. The latter has only the argument-naming rules

$$x_\alpha \leftarrow \text{not } x_\beta, \quad x_\beta \leftarrow \text{not } x_\alpha.$$

The Nixon program instead retains factual prerequisites in positive bodies. Their two output comparisons illustrate distinct routes; neither supplies a general translation from arbitrary normal programs to AFs. That reverse direction receives its own construction in the Handbook [56, Sec. 3.3, pp. 440–448].

Checking and enumerating

The definitions give a direct finite procedure: for each $M \subseteq U$, construct P^M , run the positive iteration from $\emptyset$, and retain M exactly when equality holds. The procedure finds all stable models because it tests every candidate. Its inner iteration terminates after at most $|U|$ strict increases and one equality test; its outer search still has $2^{|U|}$ candidates. A terminating, easily checked candidate test is not by itself an efficient enumeration method. Chapter 12 takes up the distinction between verification, search and enumeration. The present account covers finite propositional normal programs, not variables, disjunctive heads, constraints or optimization directives.

Exercises ★

Complete solutions to S7.1–S7.2 are in the collected Solutions section, under [Chapter 7](#).

- S7.1.** [\[solution\]](#) Add $u \leftarrow p$ and $v \leftarrow u, r$ to the Nixon program, extending its universe by u, v . Does the old candidate $\{q, r, p\}$ remain stable? Compute its reduct and every simultaneous closure round through the first equality. Find every stable model of the enlarged program and justify exhaustiveness. An implementation instead begins positive closure at the candidate itself. Would it accept $N = \{q, r, \text{neg_}p, u, v\}$? Is N actually stable?
- S7.2.** [\[solution\]](#) Return to the original Nixon rules, but initially omit the fact $r \leftarrow$, retaining the universe $\{q, r, p, \text{neg_}p\}$. Compute all stable models, then add that fact and compare the output families. Under the nonempty-family skeptical convention, is p lost as a consequence? Make the corresponding change in the two-derived-argument construction above, constructing an argument only when its factual prerequisite is available. Keep the two defaults and their mutual-rebut policy fixed. Why is this a source-information change, not an implicit priority for one default?

Supplement S9.1. Comparative disclosure and withdrawal

Fixed comparative inputs make permission calculable, but a deliberation may introduce a rival, request a comparison's grounds and withdraw a disclosed premise. *Libra* connects these different public acts through agent-relative comparison targets and Boolean explanatory equations [81, Defs. 3.1–4.13, pp. 4–9]. Its comparison predicate *Bal* remains abstract. Neither Fatio's immediate-reply rules nor Chapter 10's numerical permission test applies by default.

Public objects and disclosed dependencies

Options belong to O , with ownership tokens $Opt(a)$. Grounds split as $G = G_{exo} \dot{\cup} G_{qry}$: fixed exogenous grounds cannot be questioned or retracted, while queryable grounds can. Weight atoms such as $J(r, a, b) = high$ and $R(r, a, b) = low$, for $a \neq b$, use the ordered scale $low \prec mid \prec high$. The truth value of such an atom is Boolean; it is not the numerical weight itself. A contrastive element $\rho = \langle g, J(r, a, b) = w_J, R(r, a, b) = w_R \rangle$ has body the conjunction of those three atoms. A same-pair package conjoins its elements' bodies.

The public state records the live options Ω , token ownership, agent-relative relevance $Rel(i, a) \subseteq O \setminus \{a\}$, outstanding targets, and each agent's disclosed Boolean structural-equation fragment. An option is live when at least one agent owns its token. Relevance can retain a rival that is no longer live. The public fragment is not an agent's complete private belief model or an agreed collective model.

For example, $p := g \wedge j \wedge r$ makes p depend on three Boolean parents. With all three equal to one, $p = 1$. An intervention $[g \leftarrow 0]$ replaces the equation for g by $g := 0$, leaving other equations in place for reevaluation. More generally, equation replacement removes old equations with the new left-hand sides and inserts the new ones [81, Sec. 2.3, p. 4, Defs. 4.4–4.5]. The examples below use only constants and conjunctions in finite acyclic fragments, so direct substitution suffices. No causal-identification theory is being imported.

There are three kinds of target. A broad target for owner i 's option a asks for $Perm_i^{(a,b)} = 1$ for every currently relevant b . A focal target asks only for one such comparison; an element target asks for a ground or weight atom $\chi = 1$. A broad answer requires an adequate same-pair package for each relevant comparison, a focal answer one for its pair, and an element answer a finite ground package. The fixed predicate $Bal(a, b, \Gamma)$ determines package adequacy for comparisons; high and low labels do not alone prove it. Public permissibility tests all current relevant pairwise variables in the owner's fragment. It does not additionally test closed burdens, another agent's agreement, factual truth or execution [81, Defs. 4.6–4.12].

The source updates needed for our two cases are as follows [81, Def. 4.13, pp. 8–9]:

Assert.

If i does not already own $Opt(a)$, add that ownership and live option, initialize $Rel(i, a)$ by a fixed *InitRel* policy, and set its relevant permission variables to one. This opens no target.

Question.

A different agent can question an owned option, queryable ground or weight atom. Open its current broad or element target; keep ownership, relevance and equations unchanged.

Challenge.

For distinct agents and options, challenge owner i 's a against b . Give the challenger ownership of $Opt(b)$, make it live, add b to $Rel(i, a)$ and open that focal target. A new owner of b gets relevance $InitRel(j, b) \cup \{a\}$ and corresponding permission variables set to one; an existing owner adds a and resets that pair. Also set the challenged owner's focal permission variable to one.

Justify.

Answer an outstanding target with an admissible package. Replace/add its induced equations; disclose each queryable ground or weight atom by setting it to one and adding its ownership; remove the answered target. A comparison equation uses its package body; an element answer installs $\chi := \bigwedge \Gamma$.

Retract.

An owner may withdraw a token. Withdrawing $Opt(a)$ removes that ownership, sets the owner's existing permission variables for a to zero and removes the source-specified option/focal targets; remove a from the live set only when no owner remains. Withdrawing a ground or weight removes that ownership and any element target for it, and sets its atom to zero in that agent's fragment. It does not automatically prune relevance, delete dependent equations, reopen every burden or change another agent's fragment.

This bounded account retains the source's distinct public components. Our cases have no exogenous grounds, one outstanding target at each justification and well-formed acyclic equations with defined parents. No unanswered broad target is carried across an enlargement of relevance. These are properties of the examples, not a repair or completeness claim for all possible *Libra* histories.

The source's six-move comparison

Use agents A, B and options *Save1*, *Bystander*, *Cafe*. In the source scenario, *Save1* rescues a child from a burning building at the cost of severe burns, *Bystander* watches, and *Cafe* keeps a promise to meet a friend. The queryable ground g says the child is in danger [81, Sec. 2.2, p. 3, Sec. 5, pp. 9–10]. The scenario does not determine a moral verdict or the assumed package comparisons. Initially live

options, ownership, actual relevance, targets and equation fragments are empty. The used policy entries are $InitRel(A, Save1) = \{Bystander\}$ and $InitRel(B, Cafe) = \{Save1\}$.

Abbreviate the permission variables as $p = Perm_A^{(Save1, Bystander)}$, $q = Perm_A^{(Save1, Cafe)}$, and $z = Perm_B^{(Cafe, Save1)}$. Let j_1, r_{1w} be the respective high-justifying and low-requiring atoms for reason r_1 on $(Save1, Bystander)$, and j_2, r_{2w} the corresponding atoms for reason r_2 on $(Save1, Cafe)$. The packages are $\Gamma_1 = \{g, j_1, r_{1w}\}$ and $\Gamma_2 = \{g, j_2, r_{2w}\}$. Supply exactly the two source assumptions $Bal(Save1, Bystander, \Gamma_1)$ and $Bal(Save1, Cafe, \Gamma_2)$. In the move column only, abbreviate *Save1* as *s* and *Cafe* as *c*.

Step	Move	Consequential public update	(p, q, z)
0	Start	All components empty.	$(-, -, -)$
1	$assert(A, s)$	A owns its option; $\Omega = \{Save1\}$, relevance $\{Bystander\}$; install $p := 1$.	$(1, -, -)$
2	$question(B, A, Opt(s))$	Open broad target $\{p = 1\}$; equations unchanged.	$(1, -, -)$
3	$justify(A, \Gamma_1)$	A owns g, j_1, r_{1w} , each set to one; replace p by $g \wedge j_1 \wedge r_{1w}$; close target.	$(1, -, -)$
4	$challenge(B, A, s, c)$	B owns <i>Cafe</i> , now live. A's relevance becomes $\{Bystander, Cafe\}$, B's $\{Save1\}$. Open A's focal $\{q = 1\}$; install $q := 1, z := 1$.	$(1, 1, 1)$
5	$justify(A, \Gamma_2)$	A also owns j_2, r_{2w} ; set them and g to one, replace q by $g \wedge j_2 \wedge r_{2w}$; close target.	$(1, 1, 1)$
6	$retract(B, Opt(c))$	Remove B's option ownership and its last live occurrence; $\Omega = \{Save1\}$. Set $z := 0$; both relevance sets and A's equations remain.	$(1, 1, 0)$

A dash means a variable has not been introduced, not that its value is zero. Unmentioned components persist. At each justification there is exactly one target and the corresponding stipulated adequate package; every introduced parent is defined. The broad question is answered before relevance expands. The finite substitutions are therefore determined without choosing among competing targets or evaluating a cycle.

The values alone would conceal several distinctions. p is already one after assertion,

before any disclosed explanation; asking the question opens a target without changing that value. *Bystander* is relevant without ever being live. The challenge temporarily makes both q and z one, so it is not a zero-sum adjudication. It opens A's focal target, not an automatic reciprocal target for B. After *Cafe*'s withdrawal, A still evaluates against it because relevance has not been pruned. Both of A's relevant variables remain one, so its public-permissibility test passes. The singleton live set and empty target set define no collective winner, agreement, moral correctness, selected action or termination theorem.

Withdrawing a disclosed ground

A short pedagogical continuation exposes the dependencies more sharply. Starting after step 6, *B* questions A's owned atom j_1 , opening $\{j_1 = 1\}$. A answers with the ground package $\{g\}$, replacing the constant equation by $j_1 := g$, retaining $g := 1$ and closing the target. A then retracts g , removing only its ownership of that ground and replacing its equation by $g := 0$. The relevant dependencies are

$$p := g \wedge j_1 \wedge r_{1w}, \quad q := g \wedge j_2 \wedge r_{2w}, \quad j_1 := g.$$

They imply $(p, q, z, j_1, j_2, r_{1w}, r_{2w}, g) = (0, 0, 0, 0, 1, 1, 1, 0)$. A still owns its option and j_1 , although both permission variables and the truth of j_1 are now zero. The live set, relevance and closed target set remain unchanged. The update has neither deleted an AF attack nor recalibrated a numerical weight. Public withdrawal of g is also not evidence that the child is no longer endangered.

Exercise 9.15 reconstructs all bookkeeping, while Exercise 9.14 checks the source case. The source leaves broader persistence, soundness and complexity questions for further work [81, Sec. 7, p. 11]. General target selection, changing relevance under an unanswered broad request, and unrestricted equation packages require further care. Nothing in these bounded acyclic cases settles those questions or adds a strategy, collective outcome or causal identification result.

Supplement S9.2. From a dialogue theory to an application

A formal account, an implementation and a demonstrated benefit to users support different claims. The Handbook's description of CONSULT and EQRbot offers a compact example [38, Sec. 4.2, pp. 251–252]. It reports that CONSULT represents wearable-sensor input, clinical guidelines, patient preferences and health-record information as arguments. An ASPARTIX-based engine produces recommendations; textual templates are filled with acceptable arguments, and stored outputs feed EQRbot. EQRbot begins with an explanation containing the recommendation and responds to follow-up questions.

The same account explicitly describes the implementation as limited relative to the full locution set. It separately attributes exhaustivity, selectivity, transfer of understanding and contextuality to the formal EQR account, together with soundness and completeness relative to AF admissibility. Reporting those attributions does not establish that the limited implementation realizes the entire theory. These pages also do not establish measured patient understanding, clinical benefit or learning. The theory's exact theorem domain would have to be checked before importing its result; no such theorem is reproduced here.

This gives three distinct questions for an application reader. Which inputs and transformations support a recommendation? Which follow-up moves does the actual implementation handle? What evidence shows that its users benefit? Downstream acceptability answers none of these by itself. A recommendation based on an ungrounded sensor interpretation requires scrutiny of encoding and provenance; a missed follow-up requires scrutiny of the dialogue interface; a claim about patient understanding requires human evidence. Exercise 9.16 separates these warrants. Chapters 13 and 14 return to explanation and system interfaces. This study reconstructs an evidence boundary, not a clinical recommendation or a new implementation.

Supplement S10.1. Scores and rankings

A numerical valuation can make one reading of support and opposition explicit without reproducing an issue standard or a practical permission rule. Begin with a local bipolar example using familiar custody labels: $A = \{a_1, a_2, a_3, m\}$, supports $R^+ = \{(a_1, m), (a_3, m)\}$, and attacks $R^- = \{(a_2, m), (a_3, a_2)\}$. These edges are a new stipulated representation. They do not carry the earlier premise roles or standards.

The graph is acyclic: evaluate a_1, a_3 , then a_2 , then m . With bases $\tau(a_1) = 0.8, \tau(a_2) = 0.6, \tau(a_3) = 0.7$, use attack attenuation and the following support aggregation:

$$\begin{aligned} s(a_1) &= 0.8, & s(a_3) &= 0.7, \\ s(a_2) &= 0.6(1 - 0.7) = 0.18, \\ S^+(m) &= 1 - (1 - s(a_1))(1 - s(a_3)) = 0.94, \\ \text{score}(m) &= S^+(m)(1 - s(a_2)) = 0.7708. \end{aligned}$$

The first operation retains a fraction of a_2 's base; it does not subtract a_3 's strength. The next combines the unfilled fractions of the supporters, and the last attenuates by the remaining attack. Thus a_3 affects m both directly as support and indirectly by weakening its attacker. Whether both routes are appropriate to some evidence is a modelling question, not something the arithmetic decides.

These local operations keep the values in $[0, 1]$. Their proof uses closure under complements and finite products, in the declared evaluation order; Exercise 10.13 supplies the full argument. The bound establishes no existence or convergence theorem for cyclic networks. Nor has a rule turning a score into acceptance been supplied. A value in the unit interval is not thereby a probability.

When votes supply the bases

Suppose a fictional platform records positive/negative vote pairs $(8, 1), (3, 6), (5, 0)$ for a_1, a_2, a_3 . Set $\tau(x) = (V^+(x) + 1) / (V^+(x) + V^-(x) + 2)$. This declared smoothing rule returns $1/2$ for no votes and stays strictly between zero and one for finite nonnegative counts. It neither measures voter competence nor removes coordinated voting. Keeping the graph and propagation rule fixed gives

$$\begin{aligned} \tau(a_1) &= \frac{9}{11}, & \tau(a_2) &= \frac{4}{11}, & \tau(a_3) &= \frac{6}{7}, \\ s(a_2) &= \frac{4}{11} \left(1 - \frac{6}{7}\right) = \frac{4}{77}, \\ S^+(m) &= 1 - \frac{2}{11} \frac{1}{7} = \frac{75}{77}, \\ \text{score}(m) &= \frac{75}{77} \left(1 - \frac{4}{77}\right) = \frac{5475}{5929}. \end{aligned}$$

The higher final value, approximately 0.923, reflects changed bases under the same rule. It establishes no increase in accuracy. Votes are public endorsement records; their relation to private belief, expertise or collective acceptance requires a further account. This numerical example illustrates one local teaching rule, not every social-argumentation semantics [19, pp. 25–37, 53–59].

A named ranking answers a different question

For a finite directed acyclic AF, the categoriser assigns each argument x the value

$$\text{Cat}(x) = \frac{1}{1 + \sum_{y \in x^-} \text{Cat}(y)}, \quad x \succeq_{\text{Cat}} y \iff \text{Cat}(x) \geq \text{Cat}(y),$$

where x^- is the set of attackers and an empty sum is zero. The equations and induced ranking are the named categoriser method [91, Defs. 9–10, p. 916]; we restrict our worked construction to finite DAGs. In a topological order, attackers are evaluated before their targets, yielding a unique value in $(0, 1]$ at each step. Comparison of real values induces a total preorder. The complete proof in Exercise 10.14 uses no claim about cyclic convergence.

Consider $A = \{u, v, w, x, z\}$ with attacks from u to each of v, w, x , and from each of those three to z . There are no support edges. Compare the categoriser with the local attack product $s(a) = \tau(a) \prod_{b \in a^-} (1 - s(b))$, using $\tau(a) = 1$ everywhere and empty product one:

Vertex	Categoriser	Local product	Complete Dung label
u	1	1	in
v, w, x	$1/(1 + 1) = 1/2$	0	out
z	$1/(1 + 3/2) = 2/5$	1	in

The grounded iteration is $\emptyset \mapsto \{u\} \mapsto \{u, z\} \mapsto \{u, z\}$. The unique complete, preferred and stable extension is also $\{u, z\}$. Nevertheless the categoriser ranks each of v, w, x above z , while the product ranks u and z equally above those three. The categoriser accumulates the three attenuated attacks on z ; Dung defense asks whether the accepting set counters those attackers, which u does.

No upper threshold, strict or weak, on these categoriser values can select exactly $\{u, z\}$: including z includes the larger-valued middle arguments. Positivity cannot mean acceptance either, since every categoriser value is positive. The product's agreement with acceptance on this example is also limited to this graph and these bases. Exercises 10.14 and 10.15 supply the exhaustive finite argument and a deletion transfer. Neither valuation is a belief probability: no such sample space or interpretation has been defined.

Supplement S10.2. Values and audiences

A value attached to an argument can identify what its acceptance promotes, while an audience supplies a priority among values. This is an ordinal comparison, not a strength measurement. In the selected value-based reading, a raw attack succeeds for an audience unless the target’s value is strictly preferred to the attacker’s value [39, Defs. 2.1–2.7, pp. 305–306].

Take two mutually attacking arguments a, b , with $val(a) = \textit{privacy}$ and $val(b) = \textit{convenience}$. An audience that strictly prefers privacy keeps $a \rightarrow b$ and rejects $b \rightarrow a$; one that strictly prefers convenience does the reverse. The respective grounded, preferred and stable extensions are $\{a\}$ and $\{b\}$. This calculation uses a finite irreflexive graph, the declared attack-to-defeat policy and opposite strict total orders on two values. It proves neither uniqueness for all value-based frameworks nor the moral correctness of either audience.

The example exposes two choices that a statement such as “the preferred argument wins” conceals: what the ordering compares, and what policy turns that ordering into evaluation. Different named reductions can use the same priority input differently; Chapter 11’s reading route compares that question with Chapter 5’s structured defeat construction. Replacing privacy and convenience by probabilities would not preserve this model. A value ordering prioritizes ends for an audience; a probability needs events and normalized masses. Exercise 10.16 reconstructs the two outputs and their limits.

Supplement S10.3. Uncertainty about graphs and beliefs

Probabilistic argumentation includes two importantly different questions. We may be uncertain which arguments and attacks are present, then evaluate each possible graph. Or we may keep a graph fixed and represent beliefs about its arguments. A distribution over subsets can serve either purpose, but the interpretation of a subset changes.

Finite-probability preparation

A finite probability model consists of a nonempty set Ω of possible worlds and masses $w(\omega) \geq 0$ whose sum is one. An *event* is a subset $E \subseteq \Omega$, with probability

$$P(E) = \sum_{\omega \in E} w(\omega).$$

Thus an impossible event has probability zero, and the complement of an event of probability p has probability $1 - p$. For independent binary presence choices, obtain a world's mass by multiplying one factor for each object: its presence probability if present, and one minus that probability if absent. Independence is an assumption about these choices, not a rule for arbitrary events.

To condition on an event H with $P(H) > 0$, restrict attention to its worlds and renormalize their masses. The probability of E given H is

$$P(E \mid H) = \frac{P(E \cap H)}{P(H)}.$$

This formula is undefined when $P(H) = 0$. In the study below, being present and being accepted select different sets of worlds. Use these definitions to derive the world masses and conditional probability in Exercise 10.17; no further probability prerequisite is needed.

Possible graphs

Potential arguments are $\{a, b\}$, with potential attack $a \rightarrow b$. Include a with probability $3/5$ and b with probability $1/2$, independently. Whenever both are present, include the attack; thus every world is a full induced subgraph with no further edge uncertainty.

Present arguments	Probability	Grounded extension
$\emptyset$	$1/5$	$\emptyset$
$\{a\}$	$3/10$	$\{a\}$
$\{b\}$	$1/5$	$\{b\}$
$\{a, b\}$	$3/10$	$\{a\}$

The masses sum to one. Sum the worlds whose unique grounded extension contains the queried argument: $P(a \text{ accepted}) = 3/5$ and $P(b \text{ accepted}) = 1/5$. But $P(b \text{ present}) = 1/2$. Absence from a world is not an **out** label assigned to a present argument. This finite constellation example follows the distinction and independence questions in [90, Defs. 4.2–4.6, pp. 419–421]. Independence is a supplied modelling assumption, not a consequence of the drawn attack.

Beliefs on a fixed graph

Now keep $A = \{a, b\}$ and the attack $a \rightarrow b$ fixed. A probability function $P : 2^A \rightarrow [0, 1]$ assigns normalized mass to belief interpretations. Its marginal is $P(x) = \sum_{Y \subseteq A, x \in Y} P(Y)$ [90, Defs. 3.1–3.2, pp. 407–408]. Use this different table:

Belief interpretation	$\emptyset$	$\{a\}$	$\{b\}$	$\{a, b\}$
Mass	1/5	3/10	1/10	2/5

It gives $P(a) = 7/10$ and $P(b) = 1/2$. No independence is assumed. These are beliefs about acceptability, not probabilities of membership in the grounded extension of this fixed graph. Normalization alone does not require the beliefs to conform to a graph semantics.

Two postulates make possible constraints explicit:

$$\begin{aligned} \text{COH}_F(P) &\iff \forall(a, b) \in \text{Att}, \quad P(a) \leq 1 - P(b), \\ \text{RAT}_F(P) &\iff \forall(a, b) \in \text{Att}, \quad P(a) > \frac{1}{2} \Rightarrow P(b) \leq \frac{1}{2}. \end{aligned}$$

These are coherence and rationality in the selected epistemic account [90, p. 410]. They return Boolean judgments about a distribution. Here RAT holds: $7/10 > 1/2$ and $P(b) = 1/2$ satisfies the consequent. COH fails because $7/10 \not\leq 1 - 1/2$. The table remains a normalized distribution; adopting COH as a normative requirement would exclude it. For every finite AF, COH implies RAT, but this example refutes the converse; the general short proof belongs to Exercise 10.18.

The same account defines epistemic labels by whether a marginal is above, below or equal to $1/2$: **in**, **out**, **undec**, respectively [90, Def. 3.3, p. 408]. Our table therefore labels a **in** and b **undec**. The complete Dung labelling is instead a **in**, b **out**: b has an in attacker. Agreement on the in set alone does not identify the full labellings. Exercise 10.19 changes the distribution to satisfy COH, then tests whether that postulate could generally recover complete labels. Neither distribution can be obtained merely by renaming the local score in the preceding study a probability.

Supplement S10.4. Where do weights get their authority?

A fixed-input evaluation can be exact even when the grounds for its weights remain disputed. Liga, Markovich and Yu distinguish assignments on fixed argument graphs from generation of both topology and weights, and investigate repeated assignments and reconstruction of selected outcomes [95, Secs. 4.1–4.3, PDF pp. 4–5]. Their one-step measurement in Section 4.3 differs from the recursive Euler-based evaluation defined earlier and applied to averaged weights in Section 5. These are distinct operations; this reading imports neither as a replacement for the comparative-permission rule.

Four questions separate the evidence needed for a weight-based claim:

Question	Evidence to inspect
Where did the weights come from?	Source passages, representation choices, elicitation or prompting method, version, and who may revise or endorse them.
Was the consequence calculated correctly?	The exact graph, weights and evaluation rule, with a check of that calculation.
What did the evaluation establish?	Whether the test concerns repeatability, outcome reconstruction, robustness or another declared target, on which cases.
Is the conclusion warranted and contestable?	Reasons for the representation and weights, an identifiable contested target, and an accountable response procedure.

These are questions to ask of a study, not claims that the cited experiment answers all four. Consistently repeating an assignment does not make it uniquely correct or normatively justified. Matching a recorded outcome does not show that every future decision will be warranted. Different representations or weights can reconstruct the same output.

An explanatory dialogue can make particular commitments and dependencies contestable. It cannot supply authority simply by exposing them. The Libra study has its own public state and abstract comparison predicate; it does not certify the weights fed to another model. Chapter 14 develops the larger question of accepting reports and their sources. Here the solved audit in Exercise 10.20 stays with what the stated weighting evidence can and cannot establish.

Study S11. Mechanisms and change

The three support mechanisms in Chapter 11 make different things sensitive to an edge: a candidate’s score, the graph being evaluated, or the witnesses available in a defense test. This study follows that distinction through requirements and changes. It supplies all additional conventions for Exercises 11.5, 11.9, 11.11, 11.18 and 11.19; it uses only the chapter’s BAF definitions and Chapter 6’s ordinary semantics.

What an output requirement says

For a returned extension E , deductive closure requires $u \in E \wedge uR^+v \Rightarrow v \in E$. Inverse closure requires $v \in E \wedge uR^+v \Rightarrow u \in E$. The first fits an implication-like reading of support, the second a prerequisite-like reading [96, Introduction, manuscript pp. 1–2]. They constrain every returned extension once that interpretation has been selected. Maximizing the number of internal support edges does not itself enforce either implication. Exercise 11.18 tests this on the smallest useful inputs.

Requirement	Precisely compared objects
Deductive closure	Each returned E contains the supported consequences of its members.
Inverse closure	Each returned E contains the supporting prerequisites of its members.
Skeptical target support monotonicity	Across a single support addition to y , guarded skeptical acceptance of y persists.
Support-aware locality	On a subset with no incoming attack or support, local selected family equals projected whole selected family.
Inactive-support deletion	Deleting support internal to no ordinary candidate leaves the selected family unchanged.

These mathematical conditions may implement different methodological choices. The interpretation of an arrow motivates a requirement; it does not prove that the selected mechanism satisfies it.

For support-aware locality, restrict *both* relations to U , writing $B \downarrow U$, and assume

$$((A \setminus U) \times U) \cap (R^- \cup R^+) = \emptyset.$$

The requirement is

$$\text{Sel}(B \downarrow U) = \{E \cap U : E \in \text{Sel}(B)\}.$$

This is our explicitly defined bipolar analogue of ordinary AF directionality, not the Handbook’s definition with an unmentioned support relation. An outgoing support

can change an extension's global score even when nothing enters U . Exercise 11.11 asks whether this permits an upstream choice to depend on downstream context.

A sufficiently strong meaning of an *inactive* support edge is that it is internal to none of the ordinary candidate extensions. If attacks stay fixed, deleting such an edge changes neither the candidate family nor any candidate score. This observation suggests a general preservation proof. Deletion of (a, c) in the running BAF has a different status: that edge is internal to E_{ac} , and its removal creates a tie.

Exhaustiveness in the supported-target example

The attack graph on $\{x, y, z, w\}$ before adding support is

$$\{(w, x), (x, y), (x, z), (y, w), (z, x)\}.$$

After adding support (x, y) , its one-step reduction adds only (x, w) . The undirected conflicts have not changed: x conflicts with each other argument, and y conflicts with w . Thus the complete conflict-free candidate list, in both graphs, is

$$\emptyset, \{x\}, \{y\}, \{z\}, \{w\}, \{y, z\}, \{z, w\}.$$

There is no conflict-free triple. Self-defense now gives the full enumeration.

Candidate	Before	After	Reason relevant to admissibility
$\emptyset$	yes	yes	No members require defense.
$\{x\}$	no	yes	Initially it cannot answer attacker w ; the new attack answers w , and xR^-z answers its other attacker.
$\{y\}$	no	no	Attacker x is unanswered.
$\{z\}$	yes	yes	It counterattacks its sole attacker x .
$\{w\}$	no	no	It cannot counter attacker y .
$\{y, z\}$	yes	yes	z answers x , the attacker of each member.
$\{z, w\}$	no	no	The attack yR^-w remains unanswered.

Before the edit the sole maximal admissible set is $\{y, z\}$; afterward $\{x\}$ is a second maximal set. Both cover their outsiders in the changed graph. The unchanged family member keeps y credulous, while the newly available alternative removes its skeptical status. Adding a route for a supporter therefore need not increase the supported target's status under every evaluation policy.

The defense query has a different preservation argument. If the defending set grows, an old defender remains available. If only support grows, every old direct or indirect answer remains available. Exercise 11.10 turns these witness observations into the two quantified inclusions. Neither argument requires, or defines, a family of extensions for D^+ .

Searching permitted edits

For a fixed AF $F = (A, R)$, deletable set $D \subseteq R$, target T , and grounded semantics, evaluate each $\Delta \subseteq D$ against the stated membership or exact-target goal. A complete paper procedure lists subsets in increasing cardinality. At the first cardinality with a success, it finishes that level and returns *all* its successes. If no level succeeds, it reports infeasibility within the supplied operations. The space contains $2^{|D|}$ edits; no efficient general enforcement algorithm is claimed by this enumeration.

Every returned edit satisfies the goal because it was tested. Every cheaper edit failed; completing the successful level ensures that no equally cheap solution was omitted. If all levels fail, the finite space has been exhausted. These are the three distinct obligations behind validity, optimality and completeness. Stopping at the first successful edit loses the last obligation when ties are requested.

An edit is *inclusion-minimal* if none of its proper subsets succeeds. It is *minimum-cardinality* if no successful edit has smaller size. Under unit deletion cost, a minimum edit is inclusion-minimal, since a successful proper subset would be cheaper. The reverse implication is not valid for arbitrary success conditions: if the successful minimal sets were $\{r_1\}$ and $\{r_2, r_3\}$, both would be inclusion-minimal, but only the first would be minimum. This is an order-theoretic illustration, not a claim that those success sets occur in our two-node AF.

What a finite test returns

A principle tester takes a finite list of domain-valid records and a correct instance checker. A record can be a single BAF, a changed pair, a target or a restriction. The checker may return true, false or unevaluated. Report 0 with the first checked violation; report 1 if every record in a nonempty list is evaluated and passes; otherwise report ?. Keep the sample size and evaluation counts with the answer.

A 0 carries a counterexample to the universal claim. A 1 describes the evaluated sample, unless a separate argument proves that the list covers the whole claimed domain. A ? records incomplete evidence. This remains true whether the records were checked on paper or by a program. In contrast, the enforcement enumeration has a declared finite space whose exhaustion can certify optimum cost or infeasibility. The quantifier over which the calculation is complete determines what the result establishes.

Study S12. Queries, projections and certificates

A Boolean encoding is useful only when its models can be read back into the required answers. Chapter 12 proved a bijection for the stable CNF K_F with exactly one variable per argument. This study asks how to retain that contract when repeatedly querying a solver, adding auxiliary variables or combining positive and negative decision obligations. It supplies the extra preparation for Exercises 12.12, 12.16 and 12.17. Everything can be checked with total assignments on paper.

Exclude one answer, then ask again

For a total assignment with selected set $E \subseteq A$, define

$$B_E = \bigvee_{a \in E} \neg x_a \vee \bigvee_{a \in A \setminus E} x_a.$$

All its literals are false at the recorded assignment. Any different total assignment changes at least one argument variable, making its corresponding literal true. Thus B_E excludes exactly the recorded assignment. For example, on arguments $\{a, b\}$, the assignment selecting just a is excluded by $\neg x_a \vee x_b$. The assignment selecting both satisfies this clause. In contrast, $\neg x_a$ alone excludes both assignments, so it is not an exact-assignment block in general. This is a Boolean claim about assignments; constraints in a particular application may already exclude some supersets.

Begin with $J = K_F$. If J is satisfiable, read its total assignment as a stable set E , report that set and replace J by $J \wedge B_E$. Repeat until a correct UNSAT decision is obtained. The invariant is that models of J are exactly the original stable assignments not yet reported. The correspondence establishes it initially; exact exclusion preserves it. A final UNSAT result establishes that nothing remains. At most $2^{|A|}$ successful calls occur, but this says nothing about polynomial delay between outputs or practical solver runtime.

The empty AF initially has its one empty assignment. After reporting it, $B_\emptyset$ is an empty disjunction and hence false; the next query is UNSAT. An initially UNSAT encoding reports no extensions. These are respectively one empty answer and no answer.

Auxiliary variables and projected answers

An optimized encoding may introduce auxiliary variables for intermediate conditions. Then different full assignments can describe the same argument set. Its contract should be stated through projection: satisfying assignments restricted to the argument variables yield exactly the stable sets, possibly with several assignments per set. Soundness requires every projected set to be stable; completeness requires each stable set to have at least one satisfying full assignment.

To enumerate distinct extensions, use the total assignment to the original argument variables in B_E . This removes all auxiliary realizations of the same extension while retaining assignments for every other extension. Blocking a whole assignment including auxiliaries removes only one realization. Exercise 12.17 uses a deliberately irrelevant fresh variable to expose this difference without requiring another solver formalism.

Two obligations and a complexity upper bound

Chapter 12 defined NP through polynomially checkable positive witnesses. The complement of a decision problem exchanges its yes and no instances. A problem is in coNP when its complement is in NP: its negative instances have polynomially checkable witnesses. This does not assert that enumerating all possible candidates is the only available algorithm.

Define

$$\text{DP} = \{L_1 \cap L_2 : L_1 \in \text{NP}, L_2 \in \text{coNP}\}.$$

Here a decision problem is identified with its set of yes-instances, and intersection requires both conditions on the same input. This is a class definition, not a statement that NP and coNP are disjoint or that every such problem has a known lower bound [7, Sec. 2.1.2, pp. 638–639].

Guarded stable skepticism on a well-formed input (F, q) , $q \in A$, requires a stable extension to exist and requires no stable extension to omit q . A positive witness for the first condition and a negative witness for the second have different roles. Exercise 12.16 asks the reader to use the stable checker to place these two languages in the required classes. The resulting DP membership is an upper bound only. A completeness claim would need an additional reduction establishing hardness; the SAT-to-credulous proof concerns a different target problem.

The same task distinction explains why a proposed preferred extension needs more than an admissibility check, while an admissible set can certify a preferred credulous query. For verification, an admissible proper superset witnesses failure of maximality; for credulity, the finite extension theorem supplies the preferred superset without requiring it to be guessed. One must identify which fact the certificate is supposed to establish before judging whether it is complete.

Study S13. Evidence for a classifier explanation

An argument graph can organize an explanation of a classifier, but at least three different claims may then be made: the graph reproduces the classifier, its displayed reasons fit human judgments, or using the explanation improves a task. This study supplies the promised optional return to Chapter 3’s explanation-through-argumentation example. Exercise 13.13 asks what evidence would warrant each claim; it requires only the account printed here.

The construction being explained

AXPLR, described by Lertvittayakumjorn and Toni, concerns a pattern-based logistic-regression binary text classifier. Its input includes a text and the fitted model’s patterns. A quantitative bipolar structure represents relationships among patterns, and gradual evaluation connects those patterns to the classifier’s prediction and an explanation. This is a particular classifier construction, not a method that automatically turns any black-box model into a faithful argument graph.

The Handbook reports prediction correspondence for this construction, evaluation on three datasets and two human studies [4, Ch. 9, p. 597]. The original source used here is the 2023 publisher version, *Argument & Computation* 14(2), pp. 163–234, DOI 10.3233/AAC-220004 [107]. Its Section 4, Theorem 2, is the statement-level locator for the reported mathematical correspondence. The construction’s assumptions and theorem proof are not reproduced here; we use the report to distinguish kinds of warrant, rather than importing a new theorem for arbitrary classifiers.

What the evaluations ask

Sections 9.3–9.4 of that source compare explanation scores with human judgments. This asks whether the explanation’s assessment relates to what people find plausible. It is a question about a response measure, not a proof that the classifier and argumentation construction coincide.

Sections 10.1–10.4 and Table 10 ask a different question through tutorials and assistance for deceptive-review detection. Assistance improved scores relative to within-method baselines. There was no significant between-method performance difference. Thus a reported improvement relative to a method’s baseline does not establish that it outperformed the other compared methods [107, Secs. 9.3–9.4, 10.1–10.4, Table 10]. The narrower original-results account matters when interpreting a general summary of plausibility and helpfulness.

Compare this with Chapter 13’s four synthetic test pairs. Their 3/4 fidelity is an exact finite agreement calculation, with no correspondence for other inputs and no human observations. Even a fidelity score of 1 on that test would change

neither boundary. Conversely, a theorem establishing prediction correspondence for a specified construction would still leave open how a person understands or uses its explanation.

Question	Evidence relevant to answering it
Do construction and classifier predictions correspond on the stated domain?	A mathematical argument with the construction's exact assumptions.
Do two output functions agree on a specified test set?	The paired outputs, denominator and disagreement calculation.
Do explanation assessments fit human judgments?	A specified presentation, population, response measure and comparison.
Does assistance improve a person's task performance?	A study with the relevant task, baseline, comparison condition and measured outcome.

These questions can support one another in a system evaluation while remaining logically distinct. Faithfulness to a prediction does not certify that the prediction is correct, that all important reasons have been exposed, or that the explanation is useful for every audience. Equally, an explanation that people like can misrepresent the model. The useful comparison keeps the explained object and the evidence measure attached to each claim.

Supplement S14.1. When support changes a score

In the policy assistant, support links explain a decision already selected using attacks. This study changes that contract explicitly. Its scoring rule is a pedagogical construction for a finite acyclic graph, not a canonical bipolar semantics, a probability model or a legal proof standard. It returns strengths, leaving the score-to-decision question open.

Let A be finite and require the directed union $R^- \cup R^+$ to have no cycle. Supply a base $b(x) \in [0, 1]$ for every x and a parameter $\lambda \in [0, 1]$. A topological order puts every predecessor before its target. Such an order exists: every nonempty finite acyclic graph has a vertex with no incoming edge, since otherwise following predecessors would eventually repeat a vertex and give a cycle. Remove that vertex and repeat. Evaluation in that order defines

$$s(x) = \left(1 - (1 - b(x)) \prod_{zR^+x} (1 - \lambda s(z)) \right) \prod_{yR^-x} (1 - s(y)).$$

An empty product is 1. Consequently an isolated argument retains its base, and an argument with no supporters has value $b(x) \prod_{yR^-x} (1 - s(y))$. The factor containing $b(x)$ matters: using only $1 - \prod (1 - \lambda s(z))$ as the positive term would assign zero to every argument without supporters, however large its base.

The computation is well-defined and lies in $[0, 1]$. Induct on the topological order. Once predecessor scores belong to $[0, 1]$, every factor in either product belongs to that interval. The positive term and attacking product therefore do too. Their product defines a unique $s(x) \in [0, 1]$. Each node's value depends only on already determined predecessors, so choosing another topological order does not change the result. This establishes a recursive evaluation on this domain; it supplies no convention for cyclic fixed points.

There are useful one-coordinate comparisons. Hold the base, parameter and all other input coordinates at a target fixed. Increasing one supporter score weakens its factor $1 - \lambda s(z)$, cannot increase the supporter product, and therefore cannot decrease the positive term or $s(x)$. Increasing one attacker score weakens its factor $1 - s(y)$ and cannot increase $s(x)$. The claims concern separate coordinates. If one predecessor both attacks and supports x , changing its score moves both coordinates; the two influences compete. A propagated change can similarly arrive along different paths with different roles. Neither local comparison by itself determines such a total effect.

For a complete numerical instance use

$$A = \{a_1, a_2, a_3, a_4\}, \quad b(a_i) = 0.8, \quad \lambda = 0.5, \\ R^- = \{(a_3, a_2), (a_2, a_1), (a_2, a_4)\}, \quad R^+ = \{(a_3, a_1), (a_4, a_1)\}.$$

Order a_3, a_2, a_4, a_1 is topological. On paper evaluate each row after its predecessors:

$$\begin{aligned}
 s(a_3) &= 0.8, \\
 s(a_2) &= 0.8(1 - 0.8) = 0.16, \\
 s(a_4) &= 0.8(1 - 0.16) = 0.672, \\
 s(a_1) &= (1 - 0.2(1 - 0.5 \cdot 0.8)(1 - 0.5 \cdot 0.672))(1 - 0.16) \\
 &= (1 - 0.2 \cdot 0.6 \cdot 0.664)0.84 = 0.7730688.
 \end{aligned}$$

If the support relation is emptied, the first three values are unchanged and $s(a_1) = 0.8(1 - 0.16) = 0.672$. The two links now affect the evaluated strength, whereas in the main example their only role was explanatory presentation. This change has preserved the input graph's identities but changed the question asked of it. Before using its result to recommend acceptance of a filing, a service would need a stated map from score vectors to its decisions. The original rule tests membership of accepted arguments and cannot be applied to a vector by treating a positive score as membership without saying so. Exercise [14.11](#) consolidates the boundary and arithmetic.

Supplement S14.2. Selecting the content of a report

This study gives the complete preparation for Chapter 14's admission comparison. It imports two definitions and a finite case from *Revealed Epistemic Trust* [111, Def. 2 and examples, pp. 463–464, Defs. 15, 17 and Ex. 5, p. 467]. The output selects formulas from a report. It does not authenticate the sender, establish the report's truth or revise an accumulated belief state.

Work in classical propositional logic with exactly two independent atoms p, q . For a formula set X , $\text{Cn}(X)$ is the set of all formulas true in every valuation satisfying X . A report A is a set of formulas. Fix a set T of trusted formulas, not assumed consequence-closed. The white-list mode is

$$\alpha_T(A) = \text{Cn}(T \cap \text{Cn}(A)) \cap A.$$

First determine which trusted formulas the report entails; close those under consequence; finally return only formulas actually in the report. T contains formulas, not agents or numerical weights.

For a set D of distrusted formulas, define a remainder family

$$B \perp D = \{X \subseteq B : \text{Cn}(X) \cap D = \emptyset, \\ \text{no } X \subsetneq Y \subseteq B \text{ has } \text{Cn}(Y) \cap D = \emptyset\}.$$

Each remainder is an inclusion-maximal part of B avoiding every distrusted consequence. The black-list mode is

$$\beta_D(A) = \bigcap (\text{Cn}(A) \perp D) \cap A.$$

Remainders are taken over the *closed theory* $\text{Cn}(A)$, not merely over the report's few displayed formulas. The final intersection again selects reported formulas. In every case below the remainder family is nonempty, so no empty-family convention is needed.

Fix $T = D = \{p \wedge q\}$ to compare two alternative interpretations of the same formula. This does not say that a single agent must both trust and distrust it. The four valuations provide all the classical preparation:

Valuation	p	q	$p \wedge q$	$p \leftrightarrow q$
00	0	0	0	1
01	0	1	0	0
10	1	0	0	0
11	1	1	1	1

We obtain

Report A	$\alpha_T(A)$	$\beta_D(A)$
$\emptyset$	$\emptyset$	$\emptyset$
$\{p\}$	$\emptyset$	$\{p\}$
$\{q\}$	$\emptyset$	$\{q\}$
$\{p, q\}$	$\{p, q\}$	$\emptyset$

For the white-list column, neither singleton entails the trusted conjunction. The intersection with T is then empty, whose consequences are tautologies; neither

independent atom is a tautology. The paired report entails $p \wedge q$, whose consequences include both reported atoms. Intersection with an empty report always gives empty output.

For the black-list singletons, $\text{Cn}(p)$ does not entail $p \wedge q$, since valuation 10 satisfies p without satisfying the conjunction. The whole theory therefore avoids D and is its unique maximal remainder. Similarly $\text{Cn}(q)$ avoids D using valuation 01. The paired report is more instructive:

$$\text{Cn}(p, q) \perp \{p \wedge q\} = \{\text{Cn}(p), \text{Cn}(q), \text{Cn}(p \leftrightarrow q)\}.$$

There are three remainders. To see completeness, first observe that any maximal remainder is closed. Adding a missing consequence of X would not change $\text{Cn}(X)$ or introduce a distrusted consequence, but would enlarge X inside the already closed $\text{Cn}(p, q)$.

A closed subtheory of $\text{Cn}(p, q)$ has a model set containing 11. To avoid entailing $p \wedge q$, it must also have at least one of 00, 01, 10. Maximality as a theory means minimality of that enlarged model set: choose exactly one extra valuation. This is exhaustive because, over these two atoms, every set of valuations is definable by a formula (take the disjunction of its complete truth-value descriptions). Two or more extra valuations permit a strictly stronger still-avoiding theory by keeping just one of them. The three minimal model sets are $\{11, 10\}$, $\{11, 01\}$ and $\{11, 00\}$, which define the three displayed theories. Their intersection contains neither p nor q , establishing the last table entry.

The calculation is a useful return to Chapter 1's source-method question and Chapter 7's extraction discipline. Logically equivalent presentations can have the same closed theory while asking the output operation to retain different reported formulas. Exercise 14.14 asks for the singleton report $\{p \wedge q\}$. Its setup is entirely here; the answer belongs with the collected solutions.

Merely removing the written conjunction from a theory would not avoid it: retaining both atoms rederives it. The remainder operation handles that inferential dependence. It still gives no temporal rule for what the reader should believe after successive reports. Nor do these four observations uniquely reveal an actual person's trust. The source distinguishes idealized complete acceptance functions from partial public traces. Its general representation results are further reading, not conclusions proved by this finite table.

Supplement S15.1. A reference for the finite revision and action handoff

The main chapter works the consequential receipt, action and refresh cases. This study supplies a compact reference for reconstructing the complete restricted family and its secondary mutations. All tasks are finite paper calculations. The construction's agreement and monitor guarantees remain conditional on the assumptions printed at their use; no general revision or agent-assurance theorem is imported.

The four revision instances. In every instance $H = \{p\}$ and $D = \{d : \Rightarrow \neg q\}$. The optional strict-rule set is either $\emptyset$ or $\{p \rightarrow q\}$; the observation set is either $\emptyset$ or $\{q\}$. The complete declared trees are $R_p = \langle p \rangle$, $R_d = [\Rightarrow \neg q]$, optional $R_s = [R_p \rightarrow q]$ when the strict rule is present, and optional $R_q = \langle q \rangle_{obs}$ when the observation is present. Hard premises and observations are axioms. Only a conclusion attacking the defeasible top of R_d yields rebut here; the default cannot rebut a strict top or undermine an axiom. Attacks coincide with defeats in this restricted family.

Strict rule	Observation	Defeats	Grounded arguments	Conclusions
Absent	Absent	$\emptyset$	$\{R_p, R_d\}$	$\{p, \neg q\}$
Absent	Present	$R_q \rightarrow R_d$	$\{R_p, R_q\}$	$\{p, q\}$
Present	Absent	$R_s \rightarrow R_d$	$\{R_p, R_s\}$	$\{p, q\}$
Present	Present	$R_s \rightarrow R_d, R_q \rightarrow R_d$	$\{R_p, R_s, R_q\}$	$\{p, q\}$

For each row, the listed grounded set is exactly the set of unattacked arguments. It defends no further argument: when R_d is excluded, an unattacked attacker of R_d cannot itself be attacked. Thus the first characteristic-function application from empty is already fixed. In the direct route, close $H \cup Obs$ under the optional strict rule, then add $\neg q$ just if q is absent. This yields the same last column in all four instances. A defeated default remains in D , and the strict rule is not an extra conclusion node.

Receipt and adapter reference. The main temporal instance uses the first two rows only. Its records are s_0 with false retraction status and s_1 with positive literal *retracted*(p_0), both with trusted issuer, paper and locator as printed in Section 15.1. The positive observation update changes only *Obs*. The adapter of Section 15.2 recomputes the revision graph and conclusion set, requires exactly one status sign, checks agreement with the current source record and exports the corresponding Boolean context. A receipt retains those objects. It is distinct from an exception certificate.

Here is the complete assumption allocation. A1 binds q to the actual paper status and supplies correctness at the respective times. A2 supplies reliable identity, paper

resolution and extraction without conflicting hard observations. A3 restricts the finite memory and declared trees. A4 supplies fixed metadata, safety, reviewer identity, checked warning and the append-only log. A5 supplies a trusted current-record pointer and immutable versions binding provenance as well as Boolean payload. A6 fixes priority, policy and warning-allowed actions. A7 restricts output to the two templates. They have different jobs; agreement of two route outputs cannot prove any omitted reliability assumption.

For s_2 , duplicate receipt of the positive literal leaves $Obs = \{q\}$ and the second revision row unchanged, but creates new context version **agentic-c2** with the new locator. The current caller tests the receipt pointer before dispatch. A raw stateless monitor supplied the complete old bundle has no independent access to that pointer and still passes it. The main text’s four-row renewal table is therefore a statement about calls at *current* C_2 , not a universal stale-data detector.

Secondary certificate mutations. Use discussion proposal $x_d = (\text{d1}, \text{cite}, p_0, \text{discussRetraction}, 1)$ at C_1 , policy $\Pi = (\text{policy-1}, \{\text{reviewer-1}\}, \{\text{ask}\})$ and $Safe = 1$. The matching certificate is

$$k_1 = (\text{d1}, \text{agentic-c1}, \text{policy-1}, p_0, \text{discussRetraction}, 1, \\ \text{retraction-discussion-only}, \text{reviewer-1}).$$

For the wrong-context case change only its context field to **agentic-c0**. For the wrong-paper case change only its paper to $p_{other} \neq p_0$. In both cases append the exact fresh intent recording the supplied mutated certificate. For the missing-intent case keep k_1 but supply an empty log. Each input is now fully specified for Exercise 15.9. More generally, the exception conjunction checks proposal, context, policy and paper, both purposes, both warnings, scope and authorized reviewer; mutation of a required equality cannot be repaired by logging the invalid object.

The revision target R_q -in, runtime target A_1 -out and permission target x_d -dispatch remain distinct contextual questions. To explain an outcome, identify which target is being answered and retain its whole relevant graph or policy record. This study is a reference for those distinctions, not a replacement for the main text’s complete successful handoff and remaining-goal comparison.

Solutions

Complete public solutions for Chapters 0–15 are collected here in chapter and exercise order. The exercise statements remain at the ends of their chapters; exercises attached to optional studies are identified separately.

Chapter 0

Exercise 0.1. The predecessors of b are a, c , since $(a, b), (c, b) \in R$. There is no pair with first component b , so b has no successor. The relation image $R[A]$ is $\{b\}$. A set records its member once even though two pairs have that second component.

Exercise 0.2. The sequence a, b, c is a path from a to c . The sequence a, b, c, a is a cycle of length three. None of $(a, a), (b, b), (c, c)$ occurs, so the set of self-loops is empty. A returning path of length three is not a self-loop.

Exercise 0.3. A model of p must make $p = 1$, and a model of $\neg q$ must make $q = 0$. At those values $p \rightarrow q$ is false. Thus no valuation satisfies the three formulas, and the set is inconsistent. For the separate entailment claim, the complete valuation $p = 0, q = 0$ satisfies $p \rightarrow q$ and falsifies q . Therefore $\{p \rightarrow q\} \not\models q$.

Exercise 0.4. Every model of p and $p \rightarrow q$ makes $q = 1$, regardless of its value for r . Formally, assume $\Gamma \vdash \varphi$ and $\Gamma \subseteq \Delta$. Every model of Δ is a model of Γ , so it satisfies φ ; hence $\Delta \vdash \varphi$. If Δ is inconsistent it has no model, and there is still no countermodel to the entailment. Explosion may make the output uninformative, but it does not violate monotonicity.

Exercise 0.5. One answer is $T = \{p, p \rightarrow q, s\}$, witnessed by $p = q = s = 1$. The only omitted formula is $\neg q$; adding it makes the set inconsistent because $p, p \rightarrow q$ entail q . There is consequently no consistent proper superset of T within K . Other valid answers are $\{p, \neg q, s\}$, with valuation $(p, q, s) = (1, 0, 1)$, and $\{p \rightarrow q, \neg q, s\}$, with valuation $(0, 0, 1)$. In either case adding the sole omitted formula creates the same three-formula conflict. Only one such certificate was requested.

Exercise 0.6. Give an edge $(x, y) \in R$ with both endpoints in E . Here (a, b) is such an edge and refutes conflict-freeness. A nonedge tests only one ordered pair; another pair of members may still be an edge. Verification of conflict-freeness must exclude every internal edge, including self-loops.

Exercise 0.7. The two family sizes are respectively 0 and 1, and both unions are empty. Neither family has a member containing a , so the existential assertion is false in each. The universal assertion is true for $\mathcal{E}_0$ by empty-domain truth and false for $\mathcal{E}_1$, whose member $\emptyset$ does not contain a . The guarded skeptical assertion is false for both: the first fails the nonempty-family guard and the second fails the membership test.

Exercise 0.8. The first statement is true using d_1 for b_1 and d_2 for b_2 . The second is false: d_1 misses b_2 , while d_2 misses b_1 . Their negations, respectively, are

$$\exists b \in B \forall d \in D \neg(dQb), \quad \forall d \in D \exists b \in B \neg(dQb).$$

With available set $\{d_1\}$, choose b_2 for the first failure certificate; its only possible answer d_1 does not answer it. With $B = D = \emptyset$, the first original statement is true because no task needs an answer; the second is false because there is no answer to choose.

Exercise 0.9. Initially both $\{a\}$ and $\{b, c\}$ are minimal and maximal. There is no least or greatest member because the two are incomparable. The unique maximum-cardinality member is $\{b, c\}$. After adding $\emptyset$, it is the unique minimal and least member. The two original members remain maximal, there is still no greatest member, and $\{b, c\}$ still uniquely maximizes cardinality. Leastness compares containment, not size alone.

Exercise 0.10. The trace is $\{a\}, \{a, b\}, \{a, b\}$. Every inserted node is reachable from a : the invariant holds at the start and each insertion extends a path by one edge. The update only adds nodes, so on this finite set it terminates. The result contains a and is successor-closed because each of a, b points to the other and neither points to c . If Y is any successor-closed set containing a , it also contains b ; thus $\{a, b\} \subseteq Y$, proving leastness. Equivalently induction gives $X_n \subseteq Y$ at every stage. The constant map $H(X) = \{a\}$ is monotonic: for any $X \subseteq Y$, $H(X) = H(Y)$. It is not inflationary on 2^A , since $\{b\} \not\subseteq H(\{b\}) = \{a\}$.

Exercise 0.11. If $u \in X \cap (Y \cup Z)$, then $u \in X$ and either $u \in Y$ or $u \in Z$. In the first case $u \in X \cap Y$; in the second $u \in X \cap Z$. Thus u belongs to their union. Conversely, a member of $(X \cap Y) \cup (X \cap Z)$ belongs to one of these two intersections. Either way it belongs to X and to at least one of Y, Z , hence to $X \cap (Y \cup Z)$. The two inclusions prove equality, including when some sets are empty.

Literature exercise 0.L1. *[prompt]* A model comparison is as follows. Makinson's three-pivot account asks how a consequence operation is obtained by using assumptions, valuations or rules; it suggests asking what remains fixed in the $p, p \rightarrow q$ example. The Handbook's abstract-framework passage makes the graph and the tests on sets precise; it supplies an application for the distinction between an edge, a path and a conflict-free candidate. The computational-problems passage distinguishes questions asked about such objects, including verification and acceptance tasks; it develops the distinction between checking a proposed set and finding or enumerating sets. These are complementary contributions: a source policy is not a graph definition, and a graph definition is not an algorithm or a complexity result. A satisfactory response identifies the assigned passage and an actual mathematical object in each, connects it to a chapter example, and states a limit of the comparison. It need not make a new claim about the full theories in those sources.

Chapter 1

Exercise 1.1. The repairs are $S_+ = \{p, q\}$ and $S_- = \{\neg p, q\}$. The valuations $(p, q) = (1, 1)$ and $(0, 1)$, respectively, satisfy them. Each omits only the opposing literal, whose addition is inconsistent. For completeness, no consistent subset contains both $p, \neg p$; any subset missing q can consistently add it, and a subset containing neither literal can add one. Thus a maximal consistent subset contains q and exactly one of the two literals.

Exercise 1.2. The table is

	p	$\neg p$	q
fixed pivot	yes	yes	yes
cautious MCS	no	no	yes
credulous MCS	yes	yes	yes

The fixed pivot is inconsistent and therefore entails every query. Both repairs entail q . The repair S_+ entails p ; S_- entails $\neg p$. The model $(p, q) = (0, 1)$ of S_- refutes cautious p , while $(1, 1)$ of S_+ refutes cautious $\neg p$. These same repairs provide the positive credulous witnesses.

Exercise 1.3. The repairs differ on $p, \neg p$ and agree on q . The highest-priority difference is p , contained in S_+ . Hence S_+ is uniquely preferred. Its closure contains p, q but not $\neg p$, as witnessed by $(p, q) = (1, 1)$. The new cautious support for p depends on the stated priority.

Exercise 1.4. There are eight subsets of K_0 . The two containing both p and $\neg p$ are inconsistent; the other six are

$$\emptyset, \{p\}, \{\neg p\}, \{q\}, \{p, q\}, \{\neg p, q\}.$$

Each nonempty listed set is closure-irredundant: removing a literal loses that literal as a consequence, as a valuation with its opposite value and the other retained literal shows. The empty set has no proper subset. Put $P_1 = \{p\}, P_2 = \{p, q\}, N_1 = \{\neg p\}, N_2 = \{\neg p, q\}$. The attacks are exactly

$$\begin{aligned} P_1 \rightarrow N_1, P_1 \rightarrow N_2, P_2 \rightarrow N_1, P_2 \rightarrow N_2, \\ N_1 \rightarrow P_1, N_1 \rightarrow P_2, N_2 \rightarrow P_1, N_2 \rightarrow P_2. \end{aligned}$$

A P_i entails the negation of the opposing premise $\neg p$, and an N_i entails $\neg p$; no consistent listed argument entails $\neg q$. The empty base and $\{q\}$ yield no attacks.

The stable sets are

$$E_+ = \{\emptyset, \{q\}, P_1, P_2\}, \quad E_- = \{\emptyset, \{q\}, N_1, N_2\}.$$

Each is conflict-free and attacks its two outsiders. Every stable set must include the unattacked empty base and $\{q\}$. It cannot mix a P_i with an N_j , and cannot

omit both sides, since there would be no attack on those outsiders. If it includes only one member of a side, the other same-side member is an unattacked outsider. Therefore it includes that whole side, proving completeness. The corresponding source bases are S_+ and S_- ; their base unions recover them in this instance. The general hard-premise readback still uses source-labelled pairs.

Exercise 1.5.

Route	selected object		common-case out-put	cautious p ?
Default assumptions	maximal	consis- tent background subsets	$\{S_+, S_-\}$	no
Lexicographic	preferred	repair family	$\{S_+\}$	yes
Base argumentation	stable sets		$\{E_+, E_-\}$, read as $\{S_+, S_-\}$	no

The default and base routes align under the specified source readback. Lexicographic selection refines the source repair family to one member. It does not show that the original two-output method already had a unique answer.

Exercise 1.6. The only subset of K is $\emptyset$, and its union with P is inconsistent. There is no compatible background subset and hence no maximal one. The input violates $\text{Cons}(P)$; the guarded repair query awards no cautious or credulous support. Classically $\text{Mod}(P) = \emptyset$, so $\text{Cn}(P) = \text{Th}(\emptyset) = L$. This classical empty-model consequence is a different definition, not a repair that retained no background formulas.

Exercise 1.7. Abbreviate $f = p \rightarrow q$, $g = q \rightarrow r$, and $h = \neg r$. Every maximal repair contains s : it involves a separate atom and can be added to any consistent subset. With hard premise p , the set $\{f, g, h\}$ is inconsistent, since the first two derive r against h . Each pair is consistent. The repairs and models of their union with P are

repair	p	q	r	s	q entailed?	r entailed?	s entailed?
$\{f, g, s\}$	1	1	1	1	yes	yes	yes
$\{f, h, s\}$	1	1	0	1	yes	no	yes
$\{g, h, s\}$	1	0	0	1	no	no	yes

Each repair omits one of f, g, h ; adding it creates the three-formula conflict. A subset containing fewer than two of f, g, h extends to one of the displayed consistent pairs. This proves both maximality and completeness.

Thus only s is cautious among q, r, s . The last row's complete valuation refutes cautious q ; the middle row's refutes cautious r . All background pairs remain

consistent even together with P , as the displayed pair models show; pairs involving s also extend to a displayed model. A graph with only pairwise inconsistency attacks therefore has no edges. It cannot expose the conflict requiring all three of f, g, h with p , and its stable test would accept the full background-node set instead of the three repairs.

Exercise 1.8. The repair candidates remain S_+, S_- . Now the highest-priority difference is $\neg p$, so S_- is uniquely preferred. One paper algorithm lists the consistent subsets, keeps the inclusion-maximal ones, then compares their membership vectors in the order $\neg p, p, q$. They are $(0, 1, 1)$ for S_+ and $(1, 0, 1)$ for S_- , so the latter wins. Neither the MCS definition nor the base arguments, attacks and unprioritized stable test contains the priority parameter. Their outputs therefore remain unchanged. A program may implement these same finite tests, but the calculation is the complete nonprogramming route.

Exercise 1.9. (a) With all ranks zero, no valuation is strictly preferred to another. Every premise-satisfying valuation is minimal. Hence

$$\text{Best}(P) = \{v_{10}, v_{11}\}, \quad \text{Best}(P') = \{v_{10}\},$$

and their theories are respectively $\text{Cn}(\{p\})$ and $\text{Cn}(\{p, \neg q\})$.

(b) The initial preferential output no longer includes q : v_{10} is a countermodel. There is no initially supported q to withdraw. The unchanged MCS outputs are $\text{Cn}(\{p, q\})$ and $\text{Cn}(\{p, \neg q\})$, so they still withdraw q . The initial outputs now disagree.

(c) A valuation satisfies $K \cup P$ iff it satisfies K and P ; therefore $\text{Mod}(K \cup P) = \text{Mod}(K) \cap \text{Mod}(P)$. Taking the theory of either side proves $\text{Cn}(K \cup P) = C_{\text{Mod}(K)}^{\text{fix}}(P)$; the same proof applies to P' . This equality concerns a fixed model restriction. The representation theorem concerns MCS source sets and the specified stable construction. Neither identifies arbitrary valuation preferences with repairs.

(d) The empty subset of $K = \{f\}$ is consistent with P' , whereas adding f is not. It is the sole maximal compatible subset, so the family is $\{\emptyset\}$. For $B = \{f, p, \neg q\}$, the three maximal consistent sets are its three pairs. The hard-premise filter retains only $S = \{p, \neg q\}$. Its pair is

$$(S, \rho(S)) = (\{p, \neg q\}, \{\emptyset, \{p\}, \{\neg q\}, \{p, \neg q\}\}).$$

The singleton bases and the pair are consistent and closure-irredundant; the empty base is included by definition. The correspondence gives the stable set. Projection $S \cap K$ is empty, but its union with P' has the consequence set $\text{Cn}(\{p, \neg q\})$, as required.

Chapter 2

Exercise 2.1. The argument set is $A = \{a, b_1, b_2, c_1, c_2\}$; the attack relation is $R = \{(b_1, a), (b_2, a), (c_1, b_1), (c_2, b_2)\}$. The agents are Alice, Bob and Cayrol. The source pairs are

$$(a, \text{Alice}), (a, \text{Bob}), (b_1, \text{Cayrol}), (b_2, \text{Cayrol}), (c_1, \text{Alice}), (c_2, \text{Bob}).$$

Thus $\text{Att}(a) = \{b_1, b_2\}$ is a set of arguments, whereas $S_a = \{\text{Alice}, \text{Bob}\}$ is a set of agents. Cayrol is a source of the attackers, not of a .

Exercise 2.2. For attacker b_1 , choose $c_1 \in E$, with $(c_1, b_1) \in R$. For attacker b_2 , choose $c_2 \in E$, with $(c_2, b_2) \in R$. These are all attackers, so ordinary defense holds. This certificate needs no source association.

Exercise 2.3. Alice's available associated set is $E \cap A_{\text{Alice}} = \{c_1\}$. It does not attack b_2 , so her fixed-agent predicate fails. Bob's corresponding set is $\{c_2\}$, which misses b_1 . These are the only agents associated with a , so $\text{IndDef}_F^{\exists}(E, a)$ is false. Choosing Alice for one attacker and Bob for the other would change the quantifier order.

Exercise 2.4. Use (Alice, c_1) against b_1 and (Bob, c_2) against b_2 . Each agent belongs to S_a , each argument belongs to E and to that agent's A_i , and the required counterattack is present. Collective defense therefore holds.

Exercise 2.5. The pair covers both attackers. The empty coalition covers neither; Alice alone misses b_2 , and Bob alone misses b_1 . These exhaust the proper subsets and also every other candidate in the original S_a , proving uniqueness. After adding Dana only as a source of a , her coverage set is empty. Any witness still needs Alice to cover b_1 and Bob to cover b_2 . The pair remains the unique minimal witness; its union with Dana is a witness containing that proper witness subset and is therefore nonminimal. The unchanged Cayrol still is not a source of a .

Exercise 2.6. With $E = \{c_1\}$, no member of E attacks b_2 . Ordinary and collective defense fail. Alice misses b_2 ; Bob has no available associated argument in E , so also fails, and existential individual defense is false. The definition quantifies over E , not over all arguments that happen to exist. Using c_2 would change the input.

Exercise 2.7. In both cases $S_x = \emptyset$, so individual defense is false: no target-associated agent exists. In the attacked case, $\text{Att}(x) = \{b\}$. No agent in S_x can supply a defender, so collective defense is false and there is no coalition witness. The presence of $d \in E$, which ordinarily answers b , does not supply the missing target-source association. In the unattacked case the collective universal condition has no attacker to check and is true. The sole subset of $S_x = \emptyset$ is $\emptyset$, which is a witness and has no proper subset. Thus its minimal-witness family is $\{\emptyset\}$, rather than the empty family of the first case.

Exercise 2.8. In the transfer input the attackers are b_1, b_2 , and

$$U_{\text{Ann}} = \{b_1, b_2\}, \quad U_{\text{Bob}} = \{b_1\}, \quad U_{\text{Cara}} = \{b_2\}.$$

Ann uses d_1, d_2 ; Bob uses e_1 ; Cara uses e_2 . The minimal witnesses are $\{\text{Ann}\}$ and $\{\text{Bob}, \text{Cara}\}$. The singleton covers both attacks and its only proper subset, the empty set, covers neither. The pair covers both, but deleting Bob leaves b_1 uncovered and deleting Cara leaves b_2 uncovered; its empty subset also fails. For completeness, any coalition containing Ann and anyone else contains the proper witness $\{\text{Ann}\}$. Without Ann, both Bob and Cara are required. This exhausts all candidates.

In the co-owned case, both Alice and Bob are target sources and each is associated with the same defender $d \in E$. Thus $U_{\text{Alice}} = U_{\text{Bob}} = \{b\}$. The two singleton coalitions are the inclusion-minimal witnesses; deleting their sole member leaves b uncovered. The empty coalition fails and the pair contains a proper singleton witness. Therefore the pair is a witness but not a minimal one. The prompt asks for every *inclusion-minimal* witness in both cases. Minimum cardinality would discard the transfer pair $\{\text{Bob}, \text{Cara}\}$, because Ann alone suffices, although neither member of that pair can be removed.

Exercise 2.9. The numbers of in ballots for a, b_1, b_2, c_1, c_2 are respectively 2, 1, 1, 1, 1. A strict majority of three requires at least two, so only a satisfies **MajIn**. For each other argument the indicator says false; it does not choose between out and undecided. The profile was not required to satisfy a semantic labelling condition, and no such condition was checked for the output. Counting these reports therefore supplies neither a total three-valued collective labelling nor a proof of its semantic legality.

Exercise 2.10. Initially $S_x = \{\text{Alice}\}$. The available d attacks the only attacker b , so ordinary defense is true. But Alice has no defender in E , and Bob is not associated with x . Individual and collective agent defense are false; neither the empty coalition nor Alice alone covers b , so the minimal-witness family is empty.

After adding (x, Bob) , the target-source set becomes $\{\text{Alice}, \text{Bob}\}$. Ordinary defense is unchanged. Bob's fixed-agent predicate now succeeds using d ; Alice's still fails. Hence existential individual and collective defense are true. The unique minimal witness is $\{\text{Bob}\}$: it covers b , while the empty set and Alice alone do not; the pair is a nonminimal superset. A paper implementation computes $U_{\text{Alice}} = \emptyset$ and, after the mutation, $U_{\text{Bob}} = \{b\}$, then tests the four subsets in increasing size. These calculations are the full nonprogramming certificate.

Exercise 2.11. (a) Each of $a \rightarrow b, b \rightarrow c, c \rightarrow a$ has two supporters. The reverse edges and all three loops have zero. The majority relation is exactly the three-cycle.

(b) In F_1 , an unattacked a must be in a stable set, which excludes b ; then c must be in because its only attacker b is absent. Thus $\{a, c\}$ is the unique candidate, is conflict-free, and attacks its sole outsider. The same argument in F_2 , starting from

unattacked b , gives $\{a, b\}$, and in F_3 , starting from unattacked c , gives $\{b, c\}$. In the majority graph, $\emptyset$ fails to attack all three outsiders. The singletons $\{a\}, \{b\}, \{c\}$ respectively miss outsiders c, a, b . The pairs $\{a, b\}, \{b, c\}, \{a, c\}$ respectively contain attacks $a \rightarrow b, b \rightarrow c, c \rightarrow a$. The full set also contains these attacks. Those are all eight subsets, so the stable family is empty.

(c) Every argument belongs to two individual outputs; membership majority therefore returns $\{a, b, c\}$. For example, $a \rightarrow b$ is an internal attack, refuting conflict-freeness.

(d) Edge majority has failed to preserve acyclicity of these three inputs. Evaluating each input by the stated stable policy and taking membership majority also fails to produce a stable candidate for the aggregate graph; aggregating first leaves no stable candidate. This is a finite counterexample concerning these operations, not a general aggregation impossibility result. No source association, target-source restriction or explicit defending set was part of the graph-majority input. Individual graph reports therefore do not determine an AAF collective-defense answer without those further data.

Study exercise 2.S1. *[prompt]* Coverage is monotonic in the coalition: if $C \subseteq C'$, every defender supplied by a source in C is still available through C' . Suppose a covering candidate C has a proper subset D that is also a witness. That smaller subset was considered earlier. If it was accepted, C would be skipped. If it was skipped, it contained an already accepted witness D' , which is also a proper subset of C , again causing C to be skipped. It cannot have failed coverage, since it is a witness. Thus every appended candidate is minimal and, by the explicit test, covers all attackers.

Conversely, let C be any inclusion-minimal witness. No accepted proper subset can exist, since accepted sets are witnesses. Therefore C is not skipped and passes the coverage test when considered. Every minimal witness is returned. Proper inclusion implies smaller cardinality, so the argument does not depend on the order among tied sizes.

If every one-agent deletion fails, any proper subset D also fails: choose a member $i \in C \setminus D$; then $D \subseteq C \setminus \{i\}$, whose failed coverage cannot improve by removing more sources. The empty coalition has no deletions to test; when it covers the attacker set it is minimal directly.

Termination is a separate finite argument: there are $2^{|S_x|}$ candidates, finitely many attackers and defenders, and each test inspects finite sets. It proves the procedure ends, not by itself that the returned family is correct. In the transfer input Ann alone succeeds at size one. Stopping there would miss the incomparable minimal pair Bob–Cara at size two. Inclusion-minimal enumeration retains both; minimum-cardinality search need not.

Study exercise 2.S2. *[prompt]* (a) In the original graph every node has an attacker: x is attacked by y , y by x , and z by y . The empty set attacks none, so $E_0 = \emptyset$, $E_1 = D(\emptyset) = \emptyset$. The iteration has already reached equality.

(b) After adding $t \rightarrow y$, the only unattacked node is t . Thus $E_0 = \emptyset$, $E_1 = \{t\}$. The argument t attacks y , which is the attacker of both x and z , so $E_2 = \{t, x, z\}$. The node t remains defended without a counterattack, and x, z remain defended against y . The attackers of y are now x, t ; no argument attacks t , so y cannot be defended by this set. Hence $E_3 = \{t, x, z\} = E_2$.

(c) The original single output is $G = \emptyset$, with family $\{G\} = \{\emptyset\}$. The changed output is $G' = \{t, x, z\}$, with family $\{\{t, x, z\}\}$. Each family has one member.

(d) A trace starting at $\emptyset$ says nothing by itself about every other possible input to D . In the original graph, take $E = \{z\}$. The argument z attacks nothing, so it defends none of the three attacked nodes. Thus $D(\{z\}) = \emptyset$, and $\{z\} \not\subseteq D(\{z\})$. The operator is not inflationary on all subsets of this graph.

Chapter 3

Exercise 3.1. The main proposal is a , supported by b . The concession is c , and d limits the force of its use against the proposal. One unit is $(\{b\}, a)$. The words expressing harm are an occurrence; b is their normalized content; the pair records their reconstructed role. Neither e, f, g nor expert testimony is in the paragraph. The first three belong to the larger stipulated debate; the expert premises are supplied by a scheme reconstruction. The policy goal and means premise are also supplied warrants, even when they are plausible interpretations. The concession and response are not automatically further premises of $(\{b\}, a)$. Asserting a sponsors the proposal; asking why a requests support without opposing it. Offering c against a supplies an opposing consideration; answering with b offers support. These acts can concern the same proposal without making the same contribution. Withdrawing e removes its public endorsement, not the earlier utterance or the separate reason b , and asserts no contrary claim.

Exercise 3.2. (a) Reasoning from expert testimony (expert opinion) needs relevant expertise and reliable testimony: ask whether the source actually asserted p in the relevant domain. (b) Inferring a condition from a symptom (argument from sign) needs a reliable association: ask whether the symptom has another likely explanation. (c) Negative consequences require both the consequence connection and its adverse practical relevance: ask whether the harm occurs, can be mitigated or is outweighed. (d) Transferring a judgment between similar cases (analogy) needs a relevant similarity: ask whether a difference defeats the transfer. (e) Practical reasoning needs the goal and means connection: ask whether the action is feasible or a better means exists. Classifying the pattern organizes the questions; it supplies none of their answers. The descriptive classifications suffice; knowing the conventional

names is not required.

Exercise 3.3. For expert opinion, ask for evidence of domain expertise, the actual statement and independent support for it. These request premises or their warrants. Asking whether another expert disputes the statement searches for a competing reason. For practical reasoning, ask whether the actor has the goal and whether the action is feasible and promotes it; these probe premises or applicability. Ask also about adverse consequences or alternative means; these can elicit competing practical reasons. Several classifications are possible if the intended answer type is explicit. An unanswered request adds an open item and leaves its requested support unsubstantiated under the chapter's convention. It supplies neither a contrary fact nor an automatic defeat.

Exercise 3.4. Use vertices a – g , supports $(b, a), (e, a), (g, a)$, and attacks $(c, a), (f, a), (d, c), (g, f)$. Harm, the hospital claim and mitigation are recorded for the ban; liberty and lost revenue against it; the limiting norm responds to liberty, and mitigation to lost revenue. These are seven vertices and seven typed edges. Replacing only (d, c) by support (d, a) produces R_1 . The moderator should be told that this changes what the limiting norm is recorded as doing, not what its words say. The implementer needs the reconstruction version, edge types and provenance; otherwise an answer cannot be attributed to a determinate annotation. Reply order records who responded when, which need not identify either a support relation or an attack on the immediately preceding claim.

Exercise 3.5. For R_0 , the supporter walks are $b \rightarrow a, e \rightarrow a, g \rightarrow a$, attackers are c, f , and defender witnesses are $d \rightarrow c \rightarrow a, g \rightarrow f \rightarrow a$. For R_1 , add supporter d via $d \rightarrow a$ and remove the first defender witness. For P , supporters and attackers are the same as R_0 , while the only defender is g . The support edge $d \rightarrow c$ followed by attack $c \rightarrow a$ is not an all-support walk.

A finite procedure starts with all one-edge support walks ending at the query target. Record their starting vertices and walks, then prepend each incoming support edge to each current walk, stopping at length k . Return the union of the starting vertices with at least one witness each. For $k = 0$, return the empty set before expanding. Induction on length shows that extension constructs exactly the required walks; the finite length bound ensures termination even with cycles. Attackers are a scan of incoming attack edges; defenders are a second incoming-attack scan from each attacker, retaining the corresponding two-edge witness. In the support cycle, $\text{Supporters}_2(x) = \{y, x\}$: witnesses are $y \rightarrow x$ and $x \rightarrow y \rightarrow x$. A simple-path restriction would wrongly remove x under this contract.

Exercise 3.6. The initial score is $5/10 = 0.5$; changing evidence to 2 gives $7/10 = 0.7$. Evidence 0 is supported by the explicit absence of sources. Relevance 2 is defensible because the answer addresses the ban. Dialectical engagement 0 is contestable: acknowledging liberty engages one salient counterconsideration, even though cost and alternatives are ignored. Under a rubric awarding partial credit

for that engagement, choose $Dial = 1$, giving vector $(2, 0, 1, 2, 1)$ and score 0.6. Alternatively, retaining zero requires explaining why mere acknowledgment does not count as engagement. The original vector remains the stipulated arithmetic input; this second judgment exposes what must be specified to score a new answer. A paper algorithm initializes a total to zero, adds each of the five supplied components once, and returns the total divided by ten. Optional code is `return sum(scores)/10`. The assumptions assign five integer judgments equal additive influence; they do not establish that these judgments measure a validated construct, or that score differences are calibrated across users. There is no sample space, probabilistic model or truth-verification procedure. Scoring a new answer requires a stated rubric for each component and textual or evidential reasons for the assigned values; reporting the total without them hides the evaluation decisions.

Exercise 3.7. The claimed forty-percent reduction has no supplied evidence. A transparent revision is: “A forty-percent reduction has been suggested, but this number is unsupported in the present record. If reliable evidence supports a reduction, it would supply a reason for the ban; costs and liberty objections still need consideration.” The unsupported factual content is explicitly labelled, so it meets the stipulated support-status requirement. The label neither establishes the number nor makes the conclusion immune to objections. Deleting the number is another valid repair; inventing a source is not.

Exercise 3.8. Start with an empty set of identifiers. Inspect each entry of the supplied dictionary once and add its identifier exactly when its value is "open"; return the resulting set. This paper algorithm suffices. An optional implementation for valid inputs is:

```
def open_critical_questions(unit):
    questions = unit["critical_questions"]
    return {
        qid for qid, status in questions.items()
        if status == "open"
    }
```

The empty dictionary returns $\emptyset$. A dictionary with "credibility": "open" and "domain": "answered" returns only "credibility". Every key is included exactly when its recorded status is open. The code never inspects an answer's content and therefore cannot certify its adequacy or truth.

Exercise 3.9. Initially the request is open and u unsubstantiated. No contrary has been asserted or derived. In the changed strict theory, modus ponens gives $\neg u$, the contrary of the testimony premise. It does not give $\neg e$: the valuation $r = \text{true}, u = \text{false}, e = \text{true}$ satisfies both strict premises and falsifies $\neg e$. Evidence that a source did not say something does not by itself show that the thing is false.

Exercise 3.10. The unanswered question is a request with no supplied cost evidence. Adding f and the defeasible scheme constructs a counterargument for $\neg a$; it does not establish its acceptance, especially with g as a response to the cost. In the changed strict theory, f and $f \rightarrow \neg a$ yield $\neg a$ by modus ponens. Without that new strict bridge, $f = \text{true}, a = \text{true}$ is a model of $\{f\}$ in which $\neg a$ is false. The strict derivation thus depends on a changed warrant; it does not upgrade the original defeasible scheme.

Exercise 3.11. Only (d, c) differs from the reference, so error is $1/8$, accuracy $7/8$. The reference defender answer is $\{d, g\}$; the prediction retrieves only g , giving recall $|\{g\} \cap \{d, g\}|/2 = 1/2$. Unlisted pairs are outside the fixed candidate set; adding them changes the denominator and task. R_1 supplies a different argumentative target and is not P . More fundamentally, defensibility under a different purpose does not satisfy a benchmark whose reference was explicitly fixed as R_0 . The arithmetic assesses that convention, not unique interpretive correctness or real-world performance.

Exercise 3.12. One route combines Walton–Reed–Macagno, the Handbook’s natural-language chapter and Wachsmuth’s quality study. The first supplies scheme patterns and critical questions; the second organizes annotation and extraction; the third distinguishes quality dimensions and investigates their annotation. They respectively require an explicit warrant/instance, a declared annotation task and empirical evidence about the chosen judgments. Whether e ’s supporting testimony is reliable is answered by none of the three local graph queries. Returning every incoming attack is a mathematical contract; classifying d ’s function is an annotation judgment; claiming that an interface improves understanding is an empirical claim. Alternative source selections are adequate when these objects and evidence requirements are kept equally distinct.

Chapter 4

Exercise 4.1. Use $w_6 \Rightarrow m$. Its resulting argument A_6 conflicts with A_2 , which concludes $\neg m$. For the attack from A_6 , the local defeat rule needs to know whether $A_2 \succ A_6$; for the reverse attack it needs whether $A_6 \succ A_2$. These comparisons are not supplied by $A_1 \succ A_2$. Even after supplying them, acceptance would require its own semantics over the complete relevant graph.

Exercise 4.2. After t_2 the mother has answered the request for grounds of m and remains responsible for w_1, w_2 ; the father’s opposition still awaits grounds if requested. After t_4 he has answered that request with w_3 , for which he remains responsible. Neither answer settles the disagreement. Withdrawing the mother’s assertion when no reply is waiting removes that sponsorship but leaves her separately offered pair and the father’s contribution. It does not reopen the answered requests. The mother may offer w_5 and r_4 to explain the relevance of the child’s wish; the judge

may then ask whether the stipulated rule applies. Another plausible continuation supplies the same relevance answer and asks whether the child's understanding w_2 is disputed. Neither is a formal legality certificate for this further exchange. The local fragment specifies the earlier assertion/challenge responses, but lacks the move language, speaker/target conditions and updates for the judge's individual-premise relevance request. Those additions would determine its discharge and next response. Allocating the mother responsibility for evidence about w_2 does not say how much evidence establishes understanding: producing a relevant item and meeting the issue's acceptance standard are different tests. An answer may satisfy the first and fail the second.

Exercise 4.3. Originally $J(M, F) = 8 + 5 = 13$, $R(M, F) = 4 + 2 = 6$, $J(F, M) = 3$ and $R(F, M) = 1$. Thus $13 \geq 1$ permits M , while $3 < 6$ excludes F , giving $\{M\}$. The changed tests are $13 \geq 1$ and $7 \geq 6$; both M, F are permitted. The wealth reason's increased justifying force now suffices for F . Its requiring force against M is still 1, so that test is unchanged. This is precisely why the two coordinates have different roles. A stipulated policy preferring M among permitted options selects M ; a policy preferring F selects F . Both respect the permission set. The permissibility rule supplies neither policy, much less an execution.

Exercise 4.4. The inferential example retains premises, rule use, contrary conclusions and the displayed priority, but not turn order or weights. The public fragment retains acts, participants, targets and local responsibilities; the judge's further request lacks its own complete legality conditions. The practical comparison retains alternatives, goals, grounds and an assessment of their force, made numerical in Exercise 4.3, but not derivation structure or a protocol. A suitable procedure first fixes the requested output, lists the chosen method's required fields, and checks whether each is supplied. If so, it constructs the encoding and returns only the scoped result; otherwise it returns the missing fields. It also records relevant case information not represented by that encoding. For a numerical option query the weights matter; for a public-obligation query the speaker, request, target and protocol matter. To a requester, the limitation is that a score cannot tell them who must answer a question. An absent field is not evidence that its value is zero or that no obligation exists.

Exercise 4.5. The supplied rules require an inferential representation and a stated acceptance/output policy. Contesting an input requires a public protocol, roles, permissible challenges and effects on the record. Comparing schedules requires feasible options, the authorized resource criteria and a comparison rule, plus a selection policy if one schedule must be chosen. A combined interface is appropriate because all three tasks were stipulated, not because all medical applications must use a particular hybrid. Keep the clinical rule result, status of an input challenge, scheduling permissions and selected schedule separately identified. No clinical rule, weight or medical recommendation is derived from the hypothetical description.

Exercise 4.6. A faithful explanation says: “Given the stipulated priority, the attack from A_1 defeats A_2 and the reverse attack does not. Under the separate weight table, $13 \geq 1$ passes for M , while $3 < 6$ fails for F . Acceptance semantics and the legality of the judge’s further request remain unspecified.” The first result depends on an argument priority, the second on comparative weights; neither calculation derives the other’s inputs. The supplied public fragment establishes that the mother answered with w_1, w_2 , the father with w_3 , and each remains responsible for those grounds. If a dialogue answer changes a weight, the interface must specify who may authorize that change, which record it replaces and when reevaluation occurs. Merely uttering a new number does not automatically change the practical model.

Exercise 4.7. Initialize the four totals $J(a, b), R(a, b), J(b, a), R(b, a)$ to zero. For each record, add its two weights to the totals for its ordered pair. Return the two truth values $J(a, b) \geq R(b, a)$ and $J(b, a) \geq R(a, b)$. This paper algorithm suffices; an optional implementation for valid finite inputs is:

```
def two_option_permissibility(options, reasons):
    a, b = options
    rows = list(reasons)
    def total(x, y, weight):
        return sum(row[weight] for row in rows
                    if row[1] == x and row[2] == y)
    return {a: total(a, b, 3) >= total(b, a, 4),
           b: total(b, a, 3) >= total(a, b, 4)}
```

Each sum includes exactly the records for its ordered pair once. The original table returns $M : \text{true}, F : \text{false}$; the changed weight makes both true. An empty list has four zero totals and permits both. One $(JW, RW) = (1, 1)$ record in each direction gives a nonzero double tie and also permits both. For a single equality with failure in reverse, use $(J(a, b), R(b, a)) = (1, 1)$ and $(J(b, a), R(a, b)) = (1, 2)$, realized by a $(1, 2)$ record for a and a $(1, 1)$ record for b . Only a passes. The code implements the stipulated two-option test, not a general decision or legal procedure.

Exercise 4.8. Missing weights prevent the requested numeric comparison: identify those weights and their source as needed input. An explicitly empty reason list is a complete input for the local test; both options pass because $0 \geq 0$. Two permitted options with no selection policy yield a complete permission set but no unique selection. The missing item is now a policy, not another weight or a further run of the same test.

Exercise 4.9. Gordon asks whether an implementation meets independently specified stakeholder requirements. A-BDI compares complementary perspectives on argumentation. A dialogue alternative such as Prakken’s distinguishes the protocol from strategy and outcomes. They share concern with the task a formalism serves,

while a requirement, legal move and strategic choice are different outputs. In the return reading, an evidential module may export whether an issue meets a specified standard to a legal-rule module that uses it as an input. The interface must preserve issue identity, standard, stage and applicability conditions. Such an interface need not identify the two semantics, and a semantic account of their combination is not evidence that a practical system has been deployed or validated.

Chapter 5

Exercise 5.1. Every argument has premise set $\{\top\}$. The subargument sets are $\{P\}$, $\{P, A\}$, $\{P, A, B\}$, and $\{P, C\}$, respectively. The top rules are undefined, d_1, d_3, d_2 . The only rule enabled by a is d_3 , and none is enabled by b or $\neg b$; the two rules from $\top$ and this one continuation exhaust construction. Only B, C have conflicting conclusions at eligible targets, giving rebuts (B, C, C) and (C, B, B) , where the third entry is the attacked subargument. There are no ordinary premises or conclusions contrary to rule names. Changing the rank changes none of these objects. It makes B 's weakest selected rank $\min(4, 3) = 3$, its last rank still 3, and C 's rank still 2. Both defeat relations are now $\{(B, C)\}$.

Exercise 5.2. Originally, the local test compares 2 with 1 and succeeds; the whole-target test compares 2 with 5 and wrongly rejects the defeat. After the change, the local test compares 2 with 3 and fails; the whole-target test compares 2 with 1 and wrongly accepts it. These are, respectively, a false negative and a false positive. The raw attack by U still targets A . In neither case does T attack U : b is not contrary to $\neg a$, and the attacker clause does not search T 's subarguments for a different initiating conclusion.

Exercise 5.3. (a) N undermines the ordinary-premise argument $\langle q \rangle$. It does not undermine the axiom $\langle p \rangle$, although $\neg p$ is also declared incompatible with p . The attack on q is symmetric, so it is preference-dependent. No preference comparison involving this ordinary premise has been supplied; the rule-only calculation cannot fill that gap.

(b) There is no attack from N to T . Its strict top is not a rebut target; its only defeasible-top subargument concludes a , not b ; and its premise is an axiom. No rule-name conflict is supplied. Nevertheless T is defeasible, since it uses d , and $\text{LastDefRules}(T) = \{d\}$. (c) The argument for p contrary-undermines $\langle \eta \rangle$. The attack is asymmetric and hence a defeat independently of preference. No comparison of its rank with a premise rank is needed. (d) Both arguments for $\neg b$ rebut the defeasible-top argument for b . Their selected defeasible-rule sets are empty under both policies, so neither is strictly weaker than $\{d\}$. Both attacks succeed. In the reverse direction, the premise is an axiom and the other top step is strict, so neither is a rebut or undermining target for the argument for b .

Exercise 5.4. For nonempty selected sets Γ, Δ , a witness $r \in \Gamma$ below every member of Δ has a rank at least as high as the minimum rank in Γ , and lower than the minimum in Δ . Conversely, if the first minimum rank is lower than the second, a rule attaining the first minimum supplies the required witness. Thus the quantified and minimum-rank comparisons agree.

The empty left set cannot supply the existential witness, so $\emptyset \not\prec_{\text{Eli}} \{r\}$. The nonempty left set supplies r , and the universal condition over an empty right set is true, so $\{r\} \prec_{\text{Eli}} \emptyset$. Finally $\{r\} \not\prec_{\text{Eli}} \{r\}$, since $r < r$ is false. Giving an empty selected set minimum rank ∞ reproduces these comparisons, including two empty sets, since $\infty < \infty$ is false. This is a convention for the stated ranked fragment, not a new lifting for arbitrary orders. Equal selected strengths therefore make neither attacker strictly weaker; both directions of an otherwise preference-dependent symmetric rebut survive. In the repeated-rule case both functions give $\{r\}$ on both arguments. The derivations differ in depth and subarguments even though their conclusion and selected rule sets coincide.

Exercise 5.5. The complete argument set is

$$\begin{aligned} S_0 &= \langle \text{snore} \rangle, & P_0 &= \langle \text{professor} \rangle, \\ S_1 &= [S_0 \Rightarrow \text{misbehaves}]_{r_1}, & P_1 &= [P_0 \Rightarrow \neg \text{accessDenied}]_{r_3}, \\ S_2 &= [S_1 \Rightarrow \text{accessDenied}]_{r_2}. \end{aligned}$$

Only S_2, P_1 mutually rebut. Their weakest ranks are 1 and 2, and their last ranks are 3 and 2. Weakest link gives $P_1 \rightarrow S_2$; last link gives $S_2 \rightarrow P_1$, with no other defeats. The disputed chains preserve Scot's 1/3 versus 2 pattern, but use two distinct fact arguments instead of a shared $P = \langle \top \rangle$; there are five arguments rather than four. The scenario is a pedagogical transfer of the source's priority pattern [45, pp. 2, 5–6]. The ranks alone do not specify whether the application wants an order of rule application, a comparison of whole derivations, or another policy requirement. That requirement must be obtained before the request has a determinate answer.

Exercise 5.6. The complete source computations and semantic comparisons are in Section 7.2 and the solution to Exercise 7.1. On the original input simple weakest link agrees with PDL, whereas last link does not; after the rank change both agree. A defeat relation is not yet an output family: semantics and extraction are still needed. Two successful inputs cannot establish equality throughout a domain of theories. The later chapter also separates the raw-rank disjoint comparison from the separately reported classical/SP construction, with different source assumptions and a transformed priority order; its claim status is explained in Section 7.3.

Exercise 5.7.* The support $\{p, p \rightarrow q\}$ entails q by classical modus ponens and is consistent, as witnessed by $p = q = \text{true}$. Deleting p permits $p = q = \text{false}$; deleting $p \rightarrow q$ permits $p = \text{true}, q = \text{false}$. Thus neither deletion entails q , and neither does any smaller support. The singleton $\{\neg q\}$ is consistent and entails

$\neg q$; the empty set does not, as a valuation with $q = \text{true}$ shows. Both pairs are arguments. The full K is inconsistent and also contains a proper subset already entailing q , so (K, q) fails both conditions. An uninterpreted rule-name atom need not be true exactly when $p \rightarrow q$ is true: setting that atom and p true with q false refutes the substituted entailment. An inference-rule construction would need its own definition. No attack relation or selection semantics has yet been supplied for these support pairs.

Exercise 5.8.* Each singleton subset of Δ , together with the fact $\text{bird}(t)$, derives its rule's head and not its contradictory. The empty defeasible support derives neither head, so each singleton is minimal. Using both rules derives opposing literals and does not give a consistent support. The two resulting argument structures counterargue at their contrary conclusions. Neither derivation alone establishes warrant: an evaluation policy must determine which counterarguments prevail and how their conflict affects acceptance. No such policy is supplied here. This identifies the missing evaluation work without requiring the names of DeLP's technical stages. For further reading, DeLP develops defeat comparison and acceptable dialectical-line/tree analysis [50, Sec. 4, pp. 417–422]. Neither a DeLP defeasible arrow nor the displayed explicit negation is Chapter 7's default-negation operator or a Reiter normal default.

Exercise 5.9. The relevant arguments are

$$T = [\langle p \Rightarrow q \rangle_t, \quad U = [\langle z \Rightarrow \neg ok_t \rangle_u.$$

The conclusion of U is contrary to the name of T 's defeasible top rule, so U undercuts T on T . No displayed contrary entry makes this a rebut on q or an attack on either axiom. The undercut is preference-independent and therefore a defeat under both link policies, despite $1 < 9$. Rejecting it because of that inequality would apply a rebut-style strength test to an objection to the inference permission itself. The objection need not conclude $\neg q$; it challenges whether this route to q may be used.

Chapter 6

Exercise 6.1. An admissible set cannot contain c : answering $e \rightarrow c$ requires d , but $c \rightarrow d$ would be internal. It cannot contain e : answering $d \rightarrow e$ requires c , but $e \rightarrow c$ would be internal. If it contains d , the only remaining way to attack c is to include b . Since a, b conflict, only $\emptyset, \{a\}, \{b\}, \{b, d\}$ remain; all defend their members. Their characteristic images, in that order, are

$$\emptyset, \quad \{a\}, \quad \{b, d\}, \quad \{b, d\}.$$

The fixed points give the displayed complete family. The two inclusion-maximal admissible sets are $\{a\}$ and $\{b, d\}$. Only the latter attacks every outsider. Choosing largest cardinality would wrongly remove the incomparable preferred extension $\{a\}$.

For the all-undecided labelling every node has an undecided attacker and no in-attacker. For in-set $\{a\}$, b is out because a attacks it, and a 's sole attacker is out. Each of c, d, e has an undecided attacker and no in-attacker; in particular c 's other attacker b is out. For in-set $\{b, d\}$, every attacker of an in-node is out, and every out-node is attacked by b or d . Thus all three labellings are legal. The preferred extensions witness credulous a, b, d ; each also has a preferred counterexample omitting it. Neither c nor e occurs in any preferred extension. The single stable extension gives both stable acceptance statuses exactly to b, d .

Exercise 6.2. The complete conflict-free family is

$$\begin{aligned} & \emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \\ & \{a, c\}, \{a, d\}, \{a, e\}, \{b, d\}, \{b, e\}, \{c, e\}, \{a, c, e\}. \end{aligned}$$

The attacks prohibit simultaneous a, b , b, c , c, d , or d, e , which also shows that the list is exhaustive. The admissible family is

$$\{\emptyset, \{a\}, \{b\}, \{d\}, \{a, c\}, \{a, d\}, \{b, d\}, \{a, c, e\}\}.$$

For the five omitted conflict-free candidates, $\{c\}$ and $\{c, e\}$ leave the attacker b of c unanswered; $\{e\}$, $\{a, e\}$, $\{b, e\}$ leave the attacker d of e unanswered. Each retained set defends its members. Among those retained sets, $\{b\}$ additionally defends d , and $\{a, c\}$ additionally defends e . All others are fixed points. Therefore

$$\begin{aligned} \text{Ext}_{\text{co}}(F_{\text{reuse}}) &= \{\emptyset, \{a\}, \{d\}, \{a, d\}, \{b, d\}, \{a, c, e\}\}, \\ \text{Pref}(F_{\text{reuse}}) = \text{Stable}(F_{\text{reuse}}) &= \{\{a, d\}, \{b, d\}, \{a, c, e\}\}, \\ \text{Ground}(F_{\text{reuse}}) &= \emptyset. \end{aligned}$$

The three maximal admissible sets each attack every outsider; Theorem 6.1 excludes any further stable set. Every node is attacked, so the grounded iteration stops immediately.

For part (b), $c^- = \{b, d\}$: a answers b , and c answers d . Neither member of $\{a, c\}$ answers both attackers. The negated defence condition is

$$\exists y \in A \quad (yRx \wedge \forall z \in S : \neg(zRy)).$$

For target e and $S = \{a, e\}$, choose $y = d$; neither member attacks it. One unanswered attacker suffices for failure.

For part (c), the in/out/undecided triples are

$$(\{a\}, \{b\}, \{c, d, e\}), \quad (\{b, d\}, \{a, c, e\}, \emptyset).$$

In the first, a 's attacker is out, b has in-attacker a , and the other nodes have an undecided attacker and no in-attacker. In the second, b, d have only out-attackers, and b, d together attack all three out-nodes. Every argument is stable-credulous:

a, d occur in $\{a, d\}$, b in $\{b, d\}$, and c, e in $\{a, c, e\}$. None is skeptical: $\{b, d\}$ omits a, c, e , $\{a, d\}$ omits b , and $\{a, c, e\}$ omits d .

After adding $e \rightarrow a$, the preferred and stable families are both $\{\{a, d\}, \{b, d\}\}$. The former $\{a, c, e\}$ now has an internal attack. The new admissible candidates are exactly $\emptyset, \{b\}, \{d\}, \{a, d\}, \{b, d\}$: the remaining candidates involving a cannot answer e without d . The two maximal sets attack every outsider. This changed calculation concerns F_{reuse} , not the similarly labelled linked-cycle graph.

Exercise 6.3. For the empty AF, both families are $\{\emptyset\}$. For the self-attacker, $\emptyset$ is the only conflict-free candidate, so it is preferred; it is not stable. In F_3 , only the empty set and singletons are conflict-free. Each singleton leaves its own attacker unanswered, so the only admissible and hence preferred set is $\emptyset$. It attacks no outsider. Thus both nonempty frameworks have preferred family $\{\emptyset\}$ and stable family $\emptyset$. When the stable family is empty, both guarded acceptance queries are false and the no-output diagnostic is true. This reports neither a stable rejection witness nor acceptance by vacuity. For F_4 , the grounded iterates are

$$\emptyset, \quad \{w\}, \quad \{w, y\}, \quad \{w, y\}.$$

The first selected argument is unattacked; its attack on x then defends y . Neither x nor z is defended.

Exercise 6.4. Suppose $S \subseteq T$. Every counterattacking witness in S is still available in T , so $\mathcal{F}_F(S) \subseteq \mathcal{F}_F(T)$. This is monotonicity; it does not assert $S \subseteq \mathcal{F}_F(S)$ for arbitrary S .

Next suppose S is conflict-free. If $x, y \in \mathcal{F}_F(S)$ and xRy , defence of y gives $s \in S$ with sRx . Defence of x then gives $t \in S$ with tRs , contradicting conflict-freeness of S . The argument also permits $x = y$, so it covers self-attacks. Hence $\mathcal{F}_F(S)$ is conflict-free. If S is admissible, $S \subseteq \mathcal{F}_F(S)$ by definition. Monotonicity then gives

$$\mathcal{F}_F(S) \subseteq \mathcal{F}_F(\mathcal{F}_F(S)).$$

Together with the conflict-freeness just proved, this says that $\mathcal{F}_F(S)$ is admissible.

For the Fundamental Lemma, the case $x \in S$ is immediate. Otherwise, no $s \in S$ attacks x : defence of x by S would give $t \in S$ attacking s , an internal attack. Conversely, x cannot attack any $s \in S$, because admissibility of S would then supply a member of S attacking x , which we have just excluded. Finally, xRx is impossible: defence of x would again require a member of S attacking x . These three checks prove conflict-freeness of $S' = S \cup \{x\}$. Every member of S' was already defended by S , and monotonicity preserves those witnesses in S' . Thus S' is admissible. The same monotonicity preserves defence of y . This proves both parts of the lemma [5, Lem. 10, p. 327, Lems. 18–19, p. 329].

Exercise 6.5. Fix an admissible S . The finite family

$$\mathcal{A}_S = \{T \subseteq A : S \subseteq T, T \text{ admissible}\}$$

is nonempty because it contains S . Choose a member P of largest cardinality within that family. Any proper admissible superset Q of P would also contain S , and would be a larger member of $\mathcal{A}_S$, a contradiction. Thus P is inclusion-maximal among all admissible sets and is preferred. Taking $S = \emptyset$ proves preferred existence. Maximum cardinality is used to find *one* maximal member of this finite upward family; it does not exclude other incomparable preferred sets of smaller size [5, Thm. 11(2) and Cor. 12, p. 327].

Now let P be preferred. Its admissibility gives $P \subseteq \mathcal{F}_F(P)$. If some $x \in \mathcal{F}_F(P) \setminus P$ existed, the Fundamental Lemma would make $P \cup \{x\}$ an admissible proper superset, contrary to maximality. Hence $P = \mathcal{F}_F(P)$; it is already conflict-free, so it is complete [5, Thm. 25(1), p. 330].

Finally let E be stable. An attacker y of a member $x \in E$ lies outside E , by conflict-freeness. Stability supplies $z \in E$ attacking y , so every member of E is defended. Moreover, any proper superset adds an outsider attacked by E , creating an internal conflict. Thus E is admissible with no admissible proper superset: it is preferred, and therefore complete [5, Lem. 15, p. 328].

The converse complete-to-preferred fails in F_2 . Its empty set defends neither node and is complete, but each singleton is an admissible proper superset. The converse preferred-to-stable fails for the self-attacker's preferred set $\emptyset$. For the query equivalence, a preferred extension containing q is itself an admissible witness. Conversely, the existence result extends any admissible witness containing q to a preferred extension still containing it. Repeatedly adding only defended outsiders proves no such maximality: starting from $\emptyset$ in F_2 , it stops immediately at a complete but nonpreferred set.

Exercise 6.6. On the chain, $\mathcal{F}_{\text{chain}}(\{y\}) = \{x\}$, so an arbitrary starting set need not be contained in its next iterate. On the self-attacker, $\mathcal{F}_{\text{self}}(\{u\}) = \{u\}$, yet $\{u\}$ is not conflict-free and therefore not complete. A fixed-point equation by itself establishes neither completeness nor leastness.

Start instead with $S_0 = \emptyset$, which is admissible. Exercise 6.4 proves inductively that every S_i is admissible and $S_i \subseteq S_{i+1}$. Each strict inclusion adds at least one of the $n = |A|$ arguments. There can be at most n such increases, so for some $k \leq n$ we have $S_{k+1} = S_k$. Testing equality after each update detects this within $n + 1$ evaluations, including the final equality check. For $n = 0$ the first evaluation returns $\emptyset$, as the bound requires. Set $G = S_k$. Equality makes G a fixed point; the admissibility invariant makes it complete.

For leastness, take any fixed point T , without assuming it conflict-free. Inductively $S_i \subseteq T$: the base is empty-set inclusion, and the step is

$$S_{i+1} = \mathcal{F}_F(S_i) \subseteq \mathcal{F}_F(T) = T.$$

Thus $G \subseteq T$ for every fixed point. Leastness implies uniqueness by antisymmetry of inclusion. Every complete extension is a fixed point, so G is also the least complete extension. This proves all the stated finite guarantees [5, Lems. 18–19 and Def. 20, p. 329, Lem. 24 and Thm. 25(2), p. 330].

The weakened Fundamental Lemma fails on the same chain. $S = \{y\}$ is conflict-free and defends the unattacked x , but $S \cup \{x\}$ contains $x \rightarrow y$. Self-defence of S , not merely conflict-freeness, is essential.

Exercise 6.7.* The consequence sets of $\emptyset, \{p\}, \{q\}, \{p, q\}$ are respectively

$$\emptyset, \quad \{p, y\}, \quad \{q, x\}, \quad \{p, q, x, y\}.$$

The empty set attacks nothing and has no attacking assumption set, so it is admissible. The set $\{p\}$ derives the contrary y of q , and attacks exactly the sets containing q . Its attackers are exactly those sets containing q , since only q derives x , the contrary of p . It therefore counterattacks all its attackers and is admissible. The argument for $\{q\}$ is symmetric. The full set attacks itself and is not admissible. All four candidates have been considered: the admissible family is $\{\emptyset, \{p\}, \{q\}\}$, and the preferred family is $\{\{p\}, \{q\}\}$. Each singleton attacks its sole omitted assumption, so these are also exactly the stable assumption sets. Their consequences are $\{p, y\}$ and $\{q, x\}$.

After deleting $x \leftarrow q$, only $\{p\}$ is stable. It still attacks omitted q ; $\{q\}$ cannot attack omitted p ; the full set still attacks itself through y ; and the empty set attacks neither outsider. The convenient graph of singleton assumptions suffices for this particular flat instance. Neither this enumeration nor its one-rule change proves a correspondence for arbitrary ABA frameworks.

Exercise 6.8.* Every model makes a true. The equation for b permits either truth value. If b is false, c is false and d true, giving $\{a, d\}$. If b is true, c is true and d false, giving $\{a, b, c\}$. These two cases exhaust all valuations agreeing with the conditions. After changing conjunction to disjunction, a 's truth forces c true in both cases, giving $\{a, c, d\}$ and $\{a, b, c\}$. The parent links into c alone would not distinguish the joint requirement $a \wedge b$ from $a \vee b$. The self-dependence of b permits its two values under this model test; no claim about a different ADF semantics follows.

Exercise 6.9.* Reduction 1 deletes the only raw attack. In the resulting edgeless graph both nodes are unattacked, giving grounded set $\{a, b\}$ and stable family $\{\{a, b\}\}$. This set is conflict-free for defeat but not for the original raw attack $a \rightarrow b$. Reduction 2 gives the graph with sole defeat $b \rightarrow a$. Its grounded set is $\{b\}$, and its stable family is $\{\{b\}\}$; this selection is also raw-attack conflict-free. The same preference therefore produces different answers under the two declared policies.

A claim about which argument wins must specify the preference orientation, reduction rule, relation used to test conflict, and semantics. Here preference is between whole abstract nodes. Chapter 5 instead specifies structural attack kinds,

preference-independent exceptions, and a comparison with the attacked subargument. Replacing that clause with an abstract reduction would be a change of model, not a computational simplification.

Chapter 7

Exercise 7.1. With d_1 at rank 4, PDL must select it first. This enables rank-3 d_3 , which is selected before rank-2 d_2 . The resulting b blocks d_2 . There are no tied choices, so the complete output family is $\{\{a, b\}\}$. The arguments and structural rebuts are unchanged. The weakest rank of B is now $\min(4, 3) = 3$, its last rank is 3, and C 's rank is 2. Since B, C share no defaults, the disjoint comparison also favors B . Each graph has exactly $B \rightarrow C$ and the unique stable extension $\{P, A, B\}$: its three members are unattacked, and they attack the sole outsider.

The change matters procedurally because it enables d_3 before d_2 is used, and argumentatively because it removes the weak link in B . Two matching instances still do not prove an all-input claim. The request for a “best” policy lacks a criterion: fidelity to PDL’s application order and the stipulated outcome preference in the original case select different answers. One must obtain that requirement rather than infer it from the numerical ranks or the story alone.

Exercise 7.2. Each of $\{u\}, \{v\}$ is consistent, their union is inconsistent, and the empty subset is not maximal. These are exactly the repairs. The complete base-argument set is $\{e, U, V\}$, where $e = \emptyset$, $U = \{u\}$, $V = \{v\}$. The singletons are irredundant because their formulas are not tautologies. Premise-contradiction attack gives exactly $U \leftrightarrow V$, since $u \vdash \neg v$ and $v \vdash \neg u$. The empty argument is isolated. Every stable extension contains e and exactly one of U, V , so the family is $\{\{e, U\}, \{e, V\}\}$. Taking unions of their premise sets recovers $\{u\}$ and $\{v\}$. This simple readback is valid for this input; the general hard-premise construction in Chapter 1 retains source-labelled pairs $(S, \rho(S))$, including when source premises are redundant or tautological.

The repair intersection is empty, giving $\text{Cn}(\emptyset)$. Both repairs imply q , so $\text{Cn}(\{q\})$ is contained in both closures. Conversely, any valuation satisfying q satisfies u if p is true and v if p is false. Every formula entailed by both repairs is therefore entailed by q . This proves the second equality. The difference is strict because q is not a tautology.

For the general inclusion, $\mathcal{M} \neq \emptyset$ under the stated assumptions. For each $S \in \mathcal{M}$, $P \cup \bigcap \mathcal{M} \subseteq P \cup S$. Monotonicity of classical consequence puts the closure of the smaller set in each closure on the right, hence in their intersection. With fresh r , every maximal repair contains r and exactly one of u, v . The repairs are $\{u, r\}, \{v, r\}$; closing their intersection gives $\text{Cn}(\{r\})$, whereas intersecting their closures gives $\text{Cn}(\{q, r\})$, by the same split on p . Preserving the repair family does not require these subsequent extraction operations to agree.

The set $\Gamma = \{u, r\}$ is consistent and irredundant for its whole closure. Deleting u loses consequence p : the valuation $p = q = 0, r = 1$ satisfies the remaining premise but not p . Deleting r loses consequence r , witnessed by $p = q = 1, r = 0$. Every proper subset lies in one of these deletions, so none has the same closure. Thus Γ is a base argument. The pair (Γ, q) , however, is not a conclusion-minimal classical argument: $\{u\}$ already entails q , making r unnecessary for that conclusion. Irredundancy of an entire base and minimal support for one conclusion are different requirements.

Exercise 7.3. Fix $E \subseteq A$. By injectivity, a default condition $\text{not } \iota(b)$ fails at $\iota[E]$ exactly when $b \in E$. The rule for a therefore survives exactly when $\text{Att}(a) \cap E = \emptyset$. Every surviving rule becomes a fact, and there are no other rules. Its set of fact heads is its least model, proving equation (7.2) for arbitrary E .

First suppose E is stable. For $a \in E$, conflict-freeness gives $\text{Att}(a) \cap E = \emptyset$, so $\iota(a)$ belongs to the reduct's least model. For $a \notin E$, stable outsider coverage supplies an attacker in E , so the rule for a is deleted. Thus the least model contains exactly $\iota[E]$, which is therefore a stable model. No stable extension has been lost.

Conversely, let $M \subseteq U$ be a stable model and put $E = \text{ExtOut}(M)$. Because $U = \iota[A]$, every atom in M names an argument, so $M = \iota[E]$. The stable-model equality and equation (7.2), together with injectivity, give

$$E = \{a \in A : \text{Att}(a) \cap E = \emptyset\}.$$

An internal attack would put its target in the left side but not the right, so E is conflict-free. For any $a \notin E$, equality implies $\text{Att}(a) \cap E \neq \emptyset$, so E attacks every outsider. It is stable; there are no spurious stable models.

Finally, injectivity gives $\text{ExtOut}(\iota[E]) = E$, and the exact image universe gives $\iota[\text{ExtOut}(M)] = M$. These are inverse maps on the entire power sets and hence on their stable subfamilies. Nothing assumed a nonempty framework or distinct endpoints for an attack. The empty case has the single output $\emptyset$ on each side; a self-attack is handled by exactly the same membership conditions. $\square$

Exercise 7.4. The original program is

$$a \leftarrow, \quad b \leftarrow \text{not } a, \text{not } c, \quad c \leftarrow \text{not } b, \quad d \leftarrow \text{not } c.$$

Any stable extension contains the unattacked a , excludes b , and must then contain c , whose only attacker is b . This excludes d . The resulting $\{a, c\}$ attacks both outsiders and is uniquely stable. Its reduct contains exactly facts a, c , so the theorem gives the same unique stable model.

Omitting $\text{not } a$ replaces the rule for b by $b \leftarrow \text{not } c$. Now b, c are alternative choices: c gives $\{a, c\}$, while b gives $\{a, b, d\}$. Neither choosing both nor choosing neither can reproduce the reduct's facts. Thus these are all stable models. The additional model's reduct is exactly the facts a, b, d , but its readback violates conflict-freeness

in the original AF because $a \rightarrow b$. After deleting that edge, the altered program is the correct translation: a is isolated, and the mutual pair b, c gives precisely those two stable extensions. The extension $\{a, b, d\}$ witnesses the loss of skeptical acceptance of c . An attack on c 's opponent was withdrawn; no new attack on c was needed.

For the separate added-edge case, only the original fact for a changes, becoming $a \leftarrow \text{not } d$. The other three original rules remain. If c is selected, b, d are excluded and a is required, giving $\{a, c\}$. If c is omitted, its only attacker b must be selected; d must then be selected since its only attacker c is absent, and d excludes a . This gives $\{b, d\}$. Both sets are conflict-free and attack all outsiders, and these cases exhaust the possibilities. Their reducts contain exactly their respective facts. The stable-model family is therefore $\{\{a, c\}, \{b, d\}\}$. The original step forcing unattacked a is no longer available; in particular a is no longer skeptically accepted.

Exercise 7.5. The stable-extension family of H is $\{\{u\}, \{v\}\}$. Its program is $x_u \leftarrow \text{not } x_v$, $x_v \leftarrow \text{not } x_u$, with stable-model family $\{\{x_u\}, \{x_v\}\}$. Conclusion extraction maps both extensions to $\{p\}$, so its family has only one member, $\{\{p\}\}$. The witness for p is u in one extension and v in the other. Neither argument belongs to both extensions. This is $\forall E \exists a$, not $\exists a \forall E$.

For the empty AF the program and atom universe are empty. The sole candidate $\emptyset$ equals its reduct's least model; the extension, model and conclusion families are all $\{\emptyset\}$. There is no supporting argument for p . For a singleton self-attacker w , the program $x_w \leftarrow \text{not } x_w$ has no stable model: the empty candidate leaves fact x_w , and candidate $\{x_w\}$ deletes the rule. All three output families are empty, and the nonempty-family guard fails. Neither case makes p credulous or skeptical, but their failures are different.

In the final AF, s is unattacked and t has attackers s, t . Its program is $s \leftarrow$, $t \leftarrow \text{not } s, \text{not } t$. It has the unique stable model $\{s\}$, matching the extension that excludes and attacks the self-attacker. A self-attack somewhere in a framework is not by itself an obstruction to every stable extension.

Exercise 7.6. The two matchings exhaust the possible bijections. In M_1 , both missing pairs (m_1, w_2) and (m_2, w_1) block: each of their participants prefers the proposed partner. In M_2 , the missing pairs are (m_1, w_1) and (m_2, w_2) ; neither blocks, since everyone already has their first choice. The complete source-solution family is $\{M_2\}$.

The attack relation is exactly $\{(A_{M_2}, A_{M_1})\}$. The unattacked A_{M_2} is forced into every stable extension, excludes A_{M_1} , and attacks it. Hence the only stable extension is $\{A_{M_2}\}$, and its extracted union is M_2 . The complete extracted family agrees with the source family on this instance. Its member is a matching, not a formula or a skeptical consequence. A general theorem would need a declared input domain, a construction and extraction on every input, and proof of both preservation directions. This finite whole-matching policy does not supply Dung's general individual-pair

construction or its proof.

Exercise 7.7. The complete argument set is

$$P = \langle \top \rangle, \quad A = [P \Rightarrow a]_{d_1}, \quad B = [P \Rightarrow b]_{d_2}, \quad C = [A \rightarrow \neg b]_s.$$

PDL first selects d_2 . Adding d_1 would derive a and then $\neg b$, contradicting b , so it is not a permitted consistency-preserving addition. The complete domain-output family is $\{\{b\}\}$.

The sole structural rebut is C against B . Although C contains a default, its top rule is strict: B cannot rebut that top step, and C 's defeasible-top subargument A concludes a . The rebut by C compares $\{d_1\}$ with $\{d_2\}$; since $1 < 2$, it does not survive. The defeat relation is empty. Its unique stable extension contains all four arguments, because any omitted argument would be an unattacked outsider. Its domain-output family is $\{\{a, b, \neg b\}\}$, which differs from PDL's. Distinct ranks therefore do not suffice. This construction uses only the specified directional strict rule, not full classical inference; it lies outside the recorded classical/SP construction.

Exercise 7.8.* In the original Scot theory, d_1, d_2 are initially available from $\top$. The higher-ranked d_2 is chosen first. Rule d_3 still lacks a , so d_1 is next; its argument for a then makes d_3 available. The enumeration and order are

$$(d_2, d_1, d_3), \quad d_2 \succ_{SP} d_1 \succ_{SP} d_3.$$

PDL accepts only (d_2, d_1) : the already accepted $\neg b$ blocks d_3 . The free enumeration has no consistency or already-contrary test, so the argument for its prerequisite still licenses the third enumeration step.

In the shared-rule theory only d_1 is initially available. It makes both d_3 and d'_2 constructible; the higher-ranked d_3 precedes d'_2 . Thus

$$(d_1, d_3, d'_2), \quad d_1 \succ_{SP} d_3 \succ_{SP} d'_2.$$

PDL stops after (d_1, d_3) , since b blocks the default for $\neg b$. Again, free constructibility and permitted consistent application are different tests. Both enumerations contain every default, as required on these inputs.

The orders do not supply the rest of the representation construction. That construction also uses full classical strict inference, all finite arguments, and full conclusion extraction, together with the stated domain assumptions. Merely reordering the four displayed arguments' defaults leaves those additional obligations unmet. The calculations explain the SP mechanism; they do not establish a general correspondence.

Mathematical supplement: S7.1–S7.2

Exercise S7.1. At the old candidate $\{q, r, p\}$, the reduct contains the facts q, r and the positive rules $p \leftarrow q$, $u \leftarrow p$, $v \leftarrow u, r$. Its simultaneous rounds are

$$\begin{aligned} S_0 &= \emptyset, \\ S_1 &= \{q, r\}, \\ S_2 &= \{q, r, p\}, \\ S_3 &= \{q, r, p, u\}, \\ S_4 &= \{q, r, p, u, v\}, \\ S_5 &= S_4. \end{aligned}$$

The least model contains more than the old candidate, so that candidate is not stable. The enlarged set $M^+ = \{q, r, p, u, v\}$ produces the same reduct and is stable. The opposite candidate $\{q, r, \text{neg_}p\}$ leaves the facts, $\text{neg_}p \leftarrow r$, and the added positive rules. Since p cannot be derived, neither can u or v ; its least model is just the candidate.

These are all stable models. The facts force q, r . Guessing neither contested atom makes the reduct derive both; guessing both deletes both of their rules. Exactly one must therefore be guessed. If it is p , the added rules force u, v ; if it is $\text{neg_}p$, no rule derives p , hence none derives u or v from an empty start. The two cases give exactly the displayed models.

For the proposed implementation, N 's reduct contains $q \leftarrow$, $r \leftarrow$, $\text{neg_}p \leftarrow r$, $u \leftarrow p$, and $v \leftarrow u, r$. Starting from N retains its unsupported u and v , and no rule adds anything new. The erroneous procedure accepts it. Starting correctly from $\emptyset$ gives the least model $\{q, r, \text{neg_}p\}$, so N is not stable. The false acceptance is exactly what the leastness invariant prevents.

Exercise S7.2. Initially no rule has head r , so no positive reduct derives it. Consequently no reduct can derive $\text{neg_}p$ from its prerequisite r . In a stable model both atoms are absent. The rule for p therefore survives, and the fact q derives p . The unique stable model is $\{q, p\}$, verified by the reduct containing $q \leftarrow$, $p \leftarrow q$ and no rule deriving r . Adding the fact $r \leftarrow$ restores the original Nixon program and its two models $\{q, r, p\}$, $\{q, r, \text{neg_}p\}$. Both families are nonempty, but p belongs to every model only before the addition. It has lost skeptical status.

Initially only α can be constructed, so the AF has the unique stable extension $\{\alpha\}$. Adding factual r enables β ; the resulting mutual-attack AF has extensions $\{\alpha\}$, $\{\beta\}$. Their extracted contested conclusions agree with the program calculation at both stages. The defaults and attack policy did not change, and neither default was ranked above the other. The new fact enabled a competing argument. This establishes the displayed nonmonotonic change and the two instance agreements, not a general source-program-to-AF theorem.

Chapter 8

Exercise 8.1. In the first row the successive sets are

$$\{p\}, \quad \{p, q\}, \quad \{p, q, t\}, \quad \{p, q, t\}.$$

The initial set is not strictly closed; both consistency tests pass. In the second row the first update adds $\perp$, giving $\{p, r, \perp\}$, and the next update changes nothing. The initial set is directly consistent but neither strictly closed nor indirectly consistent. In the third row no rule can add anything. The set is strictly closed but contains the contrary pair $p, \neg p$, so both consistency tests fail.

Thus the second row refutes direct consistency implying either closure or indirect consistency. The third refutes closure implying either consistency condition. The first refutes indirect consistency implying closure.

For the first general implication, $C \subseteq \text{Cl}_s(C)$. Any contrary pair already in C therefore remains in its closure; if that closure is consistent, so is C . For the second, strict closure says $\text{Cl}_s(C) = C$, so direct consistency of C is exactly consistency of its closure. These prove the stated implications without adding classical inference to the supplied rules.

For the finite procedure put $C_0 = C$ and repeat

$$C_{i+1} = C_i \cup \{h : (B \rightarrow h) \in \mathcal{R}_s, B \subseteq C_i\}.$$

Stop at the first equality. The body B is a finite set; if it is empty its condition holds at every stage. All sets remain inside U , and $C_i \subseteq C_{i+1}$. Every strict increase adds at least one element of $U \setminus C$, so there are at most $|U \setminus C|$ strict increases before an equality is reached. At equality, every rule whose body is present already has its head present, so the result is closed.

For leastness, let D be any closed superset of C . Induction gives $C_i \subseteq D$: the base is $C \subseteq D$; if $C_i \subseteq D$, every rule applied at the next step has its body in D , hence its head in D . The fixed point is therefore contained in every such D , and is the least closed superset. The procedure uses only the supplied rules.

These arguments concern formulas, not derivation identities. In the professor case S_2 contains S_1 , which contains S_0 . A proposed selection retaining S_2 but omitting S_1 violates subargument closure even if another selected argument concludes *misbehaves*. A successful closure or consistency test cannot repair that missing dependency.

Exercise 8.2. Under weakest link the only edge is $P_1 \rightarrow S_2$. Every stable set contains the unattached S_0, S_1, P_0, P_1 . It must exclude S_2 , and the resulting E_{WL} attacks that sole outsider. This proves existence and uniqueness. With last link, S_0, S_1, S_2, P_0 are unattached; their set excludes and attacks P_1 . Thus E_{LL} is uniquely stable.

Their conclusion sets are

$$\begin{aligned} C_{WL} &= \{snores, misbehaves, professor, \neg accessDenied\}, \\ C_{LL} &= \{snores, misbehaves, professor, accessDenied\}. \end{aligned}$$

The selected S_1 retains S_0 ; S_2 , when selected, retains both; and P_1 , when selected, retains P_0 . Hence subargument closure holds. With no strict rules, each conclusion set equals its strict closure. Neither contains the sole relevant contradictory pair, so both are directly and indirectly consistent. All four tests permit both policies. The application must still say how priorities are to affect the decision—for example, whether the vulnerable early classification or the final defeasible rule should determine this conflict.

Exercise 8.3. In F_0 , unattacked A_2, N_2 are obligatory. They exclude A_1, N_1 , and N_2 attacks both. The unique stable extension is $\{A_2, N_2\}$. The same argument works in F_1^{good} , where N_2 also excludes and attacks N'_1 .

In F_1^{bad} , A_2 is still obligatory. Split on N_2 . If selected, it excludes all of A_1, N_1, N'_1 , giving the same stable extension. If omitted, its only attacker N'_1 must be selected. No selected argument can then attack an omitted A_1 or N_1 , whose only attacker is N_2 ; both must also be selected. The resulting $\{A_1, A_2, N_1, N'_1\}$ is conflict-free and attacks its sole outsider N_2 . The two cases exhaust the family.

Conclusion extraction gives

$$\mathcal{O}_0 = \mathcal{O}_1^{good} = \{\{i_2, \neg i_1\}\}, \quad \mathcal{O}_1^{bad} = \{\{i_2, \neg i_1\}, \{i_1, i_2, \neg i_2\}\}.$$

The query $\neg i_1$ is credulous in every graph and skeptical exactly in the first two. The second bad output contains the contrary pair $i_2, \neg i_2$, although its extension is conflict-free in the stipulated relation.

Different skeptical answers imply different nonempty output families. The converse inference is invalid: distinct families may have the same intersection, or agree on this one formula. This finite test also covers only stable outputs, whereas the source redundancy postulate additionally compares complete belief-set families. Nothing in the exercise derives the stipulated relation from a corrected ASPIC+ ordering or reconstructs all source arguments.

Exercise 8.4. The attacker-strengthening clause requires $N_1 \rightarrow A_2$ whenever $N'_1 \rightarrow A_2$, so the two stipulated edge facts violate it. The target-weakening clause requires $N_2 \rightarrow N'_1$ from $N_2 \rightarrow N_1$. This concerns the structural weakening relation, not a numerical comparison of the arguments' last rules.

The additional edge in the previous exercise was $N'_1 \rightarrow N_2$, not $N'_1 \rightarrow A_2$. A pair of edge facts diagnoses the displayed implication but does not supply a whole AF and all its extensions. Inserting $N'_1 \rightarrow A_2$ into the bad-family graph would change its input and require a new calculation. Nor does the structural replacement remove i_s from the evidence: both the original fact-based argument and the new defeasible route remain available in the story.

Exercise 8.5. Suppose E is stable in $F \sqcup H$. An internal attack in $E \cap A$ would also be an internal attack in E , so $E \cap A$ is conflict-free. For $a \in A \setminus E$, stability supplies an attacker in E . With no cross edges, that attacker lies in A , hence in $E \cap A$. Thus $E \cap A$ attacks every outsider of F and is stable there. The same argument gives stability of $E \cap B$ in H .

Conversely, let E_F, E_H be stable in their components. Their union has no internal attack: neither component contributes one, and cross edges are absent. An outsider belongs to exactly one component, whose extension supplies an attacker. The union therefore attacks every outsider and is stable. These two directions prove the stated family equality.

Every projection of a combined stable extension is consequently a member of $\text{Stable}(F)$. If $\text{Stable}(H) \neq \emptyset$, choose E_H from it. Each $E_F \in \text{Stable}(F)$ then occurs as the projection of $E_F \cup E_H$, proving the reverse inclusion. No nonemptiness assumption on $\text{Stable}(F)$ is needed for this equality of families: if it is empty, so is the combined family. A skeptical comparison supported by existing outputs, however, needs the old family to be nonempty as well.

For the two attack-free blocks the component families are $\{\{P, Q\}\}$ and $\{\{R, S\}\}$. Their union has family $\{\{P, Q, R, S\}\}$, and the old conclusion set is $\{p, q\}$ before and after projection. Query q is both credulous and skeptical, with no no-output flag. The singleton self-attacker has no stable extension, so the second combined family is empty. Both Boolean statuses for q are then false and the no-output flag is true. The old component still has its original output; the combined method produces none.

The proposition assumes disconnected *constructed* graphs. Separate source vocabularies do not establish that premise when, for instance, strict classical inference can create cross-vocabulary arguments by explosion. The result is not a general proof about structured noninterference or crash resistance.

Exercise 8.6. (a) After d_1 produces a in T , newly active rank-2 d_3 must precede rank-1 d_2 ; its $\neg b$ blocks the latter. Beginning with d_2 symmetrically activates d_4 and blocks d_1 . Deleting d_4 permits (d_2, d_1) in T_1 ; deleting d_3 permits (d_1, d_2) in T_2 . Both sequences yield S_3 .

(b) The inclusion from PDL outputs into stable outputs requires a preimage of S_3 in each fragment. Each conclusion has one argument there, so that preimage is E_3 . Its single outsider in T_1 is A_3 , which P6 allows only A_2 in E_3 to attack. The forced edge is $A_2 \rightarrow A_3$. In T_2 , the corresponding edge is $A_1 \rightarrow A_4$.

(c) The first pair uses $\{d_1, d_2, d_3\}$, with the same priorities in T_1, T ; the second uses $\{d_1, d_2, d_4\}$, unchanged in T_2, T . P2 transports both edges. P6 prohibits every internal attack in E_3 , while those edges cover its two full-theory outsiders. Its stability violates the other inclusion, from stable outputs into PDL outputs, because S_3 is not a source output of T .

(d) Sensibility requires consistent evidence variants with a fixed rule system, not

rule-deletion fragments. Their membership is an additional explicit premise. P6 restricts attack positions and requires undercuts; it does not require all six possible contradictions. The proof begins with an arbitrary assignment and uses only the stated commitments. A failed calculation for one link policy would not exclude all the others.

Exercise 8.7. No individual domain-literal query detects the extra output:

$$S_1 \cup S_2 = \{\top, a, b, \neg a, \neg b\}, \quad S_1 \cap S_2 = \{\top\}.$$

Adding $S_3 = \{\top, a, b\}$ changes neither set. Every domain literal is credulous and none is skeptical under either nonempty family. The full families nevertheless differ. This observation concerns individual literal queries, not all richer query languages.

Every edge of Q_0, Q_1, Q_2 is one of the permitted subargument contradictions. No strict argument is targeted, and there are no undercuts requiring other edges, so P6 holds. Under Q_0 , every stable extension contains isolated P and exactly one of A_3, A_4 . They cannot both occur because they attack each other; they cannot both be omitted because only they attack either of those two nodes. Selecting A_3 excludes A_2, A_4 . It then forces A_1 , since the absent A_4 would be the only attacker of that outsider. This gives E_1 , which attacks both outsiders. Selecting A_4 symmetrically gives E_2 . The complete full-input family is therefore $\{E_1, E_2\}$.

Under Q_1 , unattacked P, A_1 must be selected; the mutual pair A_2, A_3 contributes exactly one member. This gives $\{E_1, E_3\}$. Under Q_2 , unattacked P, A_2 are compulsory and the mutual pair A_1, A_4 contributes exactly one member, giving $\{E_2, E_3\}$. Both conflict-freeness and outsider coverage hold for each selection. Extraction therefore recovers the exact PDL families on the three inputs.

However, $A_2 \rightarrow A_3$ belongs to Q_1 but not Q_0 , and $A_1 \rightarrow A_4$ belongs to Q_2 but not Q_0 , despite unchanged arguments and relevant priorities. These are the failed P2 comparisons. We have verified three context-dependent graphs, not defined an assignment on all consistent evidence changes or proved capture throughout a sensible domain.

Exercise 8.8. (a) This weakens the preservation demand. The extra-output example shows that individual literal answers may agree while full-family representation fails. A query-preservation theorem would have to state its query language, guard convention and input domain.

(b) This drops context independence. The exercise proves exactness and P6 on three inputs, but a domain-wide construction and proof remain necessary. It has removed this obstruction without establishing the desired general representation.

(c) This changes the domain and excludes the tied witness, but does not prove exactness of any construction on the remaining inputs. In particular, it does not restore a general raw-DWL claim. Excluding the tied witness proves no positive representation result. The classical/SP full-conclusion equality claim is independently obstructed under its recorded construction (Section 7.3).

(d) This changes the prescribed construction or the selection semantics. The present negative proof no longer answers exactly that new question. One must define the new arguments and extraction, or the new semantic family, and establish the revised all-input preservation statement.

The professor’s alternative consistent answers still require a modelling purpose, and the Sherlock question still asks whether a controlled input change should matter. Resolving one incompatibility does not settle every output requirement or robustness principle.

Chapter 9

Exercise 9.1. Initially every store is empty. Assertion creates $\text{Com}_A = \{\chi\}$ and obligation $(A, \chi, +)$. Challenge retains that state, adds $(B, \chi, -)$ and forces A’s positive reply. The reply adds premise P to A’s commitments, replaces $(A, \chi, +)$ by $D_A = (A, \{P\}, +)$, and clears the pending field. C’s question targets B’s active challenge, changing no commitment or DOS entry but forcing B’s negative reply. That reply gives $\text{Com}_B = \{Q, S\}$, replaces $(B, \chi, -)$ by $D_B = (B, \{Q, S\}, -)$, and clears the field. Retraction removes A’s active assertion; $\text{Com}_A = \{P\}$ remains because the active justification still supplies P. Final DOS is $\{D_A, D_B\}$, C’s store is empty throughout, and the pending field is empty except after turns 2 and 4. The complete certificate is CR1, CR2, CR3, CR4, CR5, CR6.

Exercise 9.2. Combination rules constrain the next speaker, target, polarity and move; CR5, for example, forbids an unrelated assertion after the request to B. The public projection records commitments and structured obligations such as D_A , not merely rule names. The source’s axiomatic account additionally refers to nested beliefs and desires absent from this state: for example, whether others believe that the speaker wants them to believe its belief. The public record supplies no values for these attitudes. The source’s positive justification retains the claim obligation and adds the argument obligation; the local convention first discharges the requested claim obligation. The certificate therefore shows compliance with the restricted formal rules, not sincerity, correct reasons, uniquely correct reconstruction or the full Fatio axioms. Hiding private states alone would not explain the altered obligation effect. The source’s printed CR6 justification branch is positive; its semantic discussion separately extends the treatment by analogy to negative justification and argument retraction. The absence of a negative branch from the printed rule therefore does not establish that the full source forbids it [75, Sec. 3, p. 5, Sec. 4.1, pp. 6–7]. The local rule explicitly permits symmetric withdrawal of active whole support units of either polarity, with no immediate reply pending. Its active-target and pending-reply conditions also qualify the source’s printed “at any time following” wording. It also makes target typing, active witnesses and CR2 priority explicit. These conventions are part of the checked construction, not consequences of deleting private-state components from the source.

Exercise 9.3. The active assertion licenses the question under CR2. Only $justify(A, \Phi \vdash^+ \chi)$ may follow, under CR3, with a nonempty support unit Φ . Speaker, target, polarity and immediacy are forced; support content is not. $assert(A, P)$ violates the pending reply, although an assertion could be legal at another state.

Exercise 9.4. A must again immediately supply $justify(A, \Phi \vdash^+ \chi)$. Challenge additionally inserts $(B, \chi, -)$ in DOS, making B publicly responsible for opposition if requested. A question creates the immediate positive reply but no opposition obligation for its speaker. Neither move itself supplies an inferential counterargument.

Exercise 9.5. (a) The first three moves satisfy CR1, CR2, CR3. Move 4 fails CR2/CR4: A supplied the whole unit $\{s, f\}$, not an independent active assertion of s, and no challenge witness exists. Membership in the commitment store is insufficient to create this target. (b) Move 3 violates CR3: A's assertion is active, but retraction is unavailable while its positive reply is pending. The first failure concerns target identity; the second concerns ordering even for an otherwise valid target.

Exercise 9.6. Use $assert(A, \chi); question(B, A, \chi); justify(A, \{P\} \vdash^+ \chi)$, with certificate CR1, CR2, CR3. The final commitment is $Com_A = \{\chi, P\}$, DOS is $\{(A, \{P\}, +)\}$, and no reply is pending. Replacing the third move by $assert(A, Q)$ gives the first failure at move 3 under CR3. The result assumes the proposed speech-act reconstruction.

A checker first validates record types, recognized moves, required fields, atom versus whole-unit targets, nonempty support sets and polarity. For valid types, initialize empty active records, DOS and pending field. At each move first check a pending reply; otherwise select its active CR witness, giving CR2 priority over CR4. Return the first failed move and condition. On success apply the public effects and recompute commitments from active records. Finally return the rule sequence and complete state, including any pending reply. This separates malformed input from illegal transition, and a valid prefix from termination. It checks neither private BDI conditions nor inference validity.

Under the encoding, "P" is one atom, ["P"] a whole singleton unit, and "Q&S" one opaque atom rather than a parsed conjunction. Units are sets, so list order and repeated entries do not change identity. These distinctions determine the active witness needed for requests and retractions.

Exercise 9.7. After turn 3, $retract(A, \{P\})$ satisfies CR6. It removes the active justification and D_A ; A's assertion remains, giving $Com_A = \{\chi\}$ and DOS $\{(B, \chi, -)\}$. A's previously discharged claim obligation is not recreated. Retracting atom P would need an active assertion of P. After turn 5, $retract(B, \{Q, S\})$ removes B's negative justification and D_B , leaving B's commitments empty, A's

$\{\chi, P\}$ and DOS $\{D_A\}$. B’s recorded challenge remains; its discharged obligation is not recreated. Symmetric negative-unit retraction is local; the report’s printed CR6 justification branch is positive. Any premise still contributed by another active record remains in the commitment union; retraction does not recursively erase every use.

Exercise 9.8. CR4 licenses *challenge*(C, B, χ). It adds $(C, \chi, -)$ and forces *justify*($B, \Phi \vdash^- \chi$) under CR5. With $\Phi = \{Q, S\}$, B’s old opposition obligation is replaced by D_B , while C’s new one remains. If B also has a CR2 assertion/unit witness for the same target, CR2 takes priority and forces a positive CR3 response. The witness, not the request’s name alone, determines polarity.

Exercise 9.9. Full certificates are $(CR1, CR2, CR3, CR6)$ for withdrawal and $(CR1, CR2, CR3, CR4, CR5)$ for inquiry. Withdrawal gives $\text{Com}_A = \{P\}$, empty B/C stores, DOS $\{D_A, (B, \chi, -)\}$, no pending reply, observation $(0, 0)$ and reason “withdrawn”. Inquiry gives $\text{Com}_A = \{\chi, P\}$, $\text{Com}_B = \{Q, S\}$, empty C store, DOS $\{D_A, D_B\}$, no pending reply, observation $(1, 1)$ and reason “countergrounds disclosed”. The intermediate question has a pending $(B, \chi, -, CR5)$ response and fails the stopping condition. Objective tuples are $(0, 0, -1)$ and $(1, 1, -2)$, so inquiry is preferred with budget two. With budget one only the offered withdrawal finishes. CR4 remains legal; completion within budget excludes the two-move route. Neither terminal record closes all standing support obligations or proves excellence.

Exercise 9.10. Frontiers are $\emptyset, \{d_0\}, \{d_1\}, \emptyset, \{d_2\}, \{d_3\}, \emptyset$. Keep the target assertions at times 1 and 4. All occurrences below have speaker A and context D1:

ID/time	Content	Target	Source/version
$o_g/2$	company	General(q_R)	company/v1
$o_h/3$	invitation	GroundsFor(o_g)	invitation/v1
$o_{rc}/5$	company	Compare(R, C)	company/v1
$o_w/6$	policy row	ComparisonWarrantFor(o_{rc}, R, C)	policy-RC/S7-base

Demand d_0 opens at 1 and is answered by o_g ; d_1 opens at 2 and is answered by o_h ; d_2 opens at 4 and is answered by o_{rc} ; d_3 opens at 5 and only o_w answers it at 6. Each demand retains owner A, opening event and answer link. o_g, o_{rc} share content and source, but differ in occurrence identity, time and target. The invitation answers grounds for o_g , not a comparative policy demand, and cannot close d_3 .

Both L_5, L_6 have occurrence projection $(\{o_g, o_{rc}\}, \emptyset)$ and content projection $(\{g_{company}\}, \emptyset)$. Their frontiers differ because the new answer is documentary, not a primary occurrence. Changing the foil to T requires a new occurrence and matching RT demand; the RC policy row is mismatched. The supplied RT row is $(7, 5)$, a different comparative input rather than new attendance evidence. One row is not global permission, and repeating an occurrence does not double a set member’s weight.

Exercise 9.11. P:e has O:b and O:d children, exhausting its attackers. The first has P:a and ends; the second has P:c, followed by its only attacker O:b and reply P:a. Both leaves are unattacked, every O node has exactly one P reply, and every P node covers all attacks. No path repeats a P argument. The two a occurrences belong to different branches and are allowed. The path $[e, b, a]$ alone omits d and is no strategy certificate. For f, $[f, b, a]$ wins a play, but O's initial choice a gives $[f, a]$, terminated with O winning. Every full strategy for f would have to include that unanswerable child.

Exercise 9.12. After deleting (a, f) , grounded iteration is

$$\emptyset \mapsto \{a\} \mapsto \{a, c, f\} \mapsto \{a, c, e, f\},$$

then fixed. Now f has only attacker b, answered by unattacked a, so $[f, b, a]$ covers every O choice and is a complete winning strategy. In the separate two-cycle, O replies h to g. P's only counterattacker would be g, already played at a P-position, so it is forbidden. Thus $[g, h]$ terminates with O winning; the grounded extension is empty. This is P's path-local repetition rule, not a global tree or O repetition ban.

Exercise 9.13. (a) CR4-legal, but incomplete for the experiment because CR5 is pending. (b) Inquiry's $(1, 1, -2)$ exceeds withdrawal's $(0, 0, -1)$, within budget. (c) No: f's winning play omits O's refuting a choice. (d) No: d_3 remains open and the projection discarded it. (e) No: majority membership is $\{a, b, c\}$, conflicting in the aggregate cycle, whose grounded extension is empty and whose stable family has no members. These are procedural, purposeful, semantic-strategy, documentary and aggregation outputs. An empty grounded extension differs from absence of stable extensions; neither aggregation operation is a dialogue outcome rule.

Exercise 9.14. At 1 A does not own *Save1*: assertion adds ownership and live option, initializes relevance to Bystander and installs $p:=1$, without opening a target. At 2 B differs from A and questions its owned token, opening $\{p=1\}$ with unchanged equations. At 3 the unique target and supplied *Bal* fact license Γ_1 : A owns g, j_1, r_{1w} , sets them one, replaces p by their conjunction and closes the target. At 4 distinct B introduces *Cafe*; A's relevance gains *Cafe*, B's becomes $\{Save1\}$, and the live set becomes $\{Save1, Cafe\}$. Open A's focal target $\{q=1\}$ and set q,z one, with no reciprocal target for B. At 5 the second package supplies q's conjunction; A gains j_2, r_{2w} , retains g and closes the sole target. At 6 B may retract its owned *Cafe*: remove its ownership and last live occurrence and set z zero. A's equations, ownership, both relevance sets and the empty target set remain.

The values are initially all unintroduced, then $(1, -, -)$ at 1–3, $(1, 1, 1)$ at 4–5 and $(1, 1, 0)$ at 6. Permission first holds provisionally at 1; its disclosed conjunction appears at 3. Bystander is relevant but never live; *Cafe* remains relevant after withdrawal. A's two current comparisons still evaluate one, so its public-permissibility test passes. High/low labels do not derive the stipulated *Bal* facts. Empty targets and one live option define no collective winner, moral verdict, execution or

termination theorem.

Exercise 9.15. A owns j_1 , so B's question opens the element target $\{j_1 = 1\}$. The ground answer $\{g\}$ is admissible for that sole target, replacing $j_1 := 1$ by $j_1 := g$, retaining $g := 1$ and its ownership, and closing the target. A may then retract its owned g , removing only that ownership and replacing its equation by $g:=0$. No burden reopens.

After the answer A has equations $g := 1, j_1 := g, r_{1w} := 1, j_2 := 1, r_{2w} := 1, p := g \wedge j_1 \wedge r_{1w}$ and $q := g \wedge j_2 \wedge r_{2w}$; B has $z:=0$. After retraction only g 's equation changes. The value vector goes from $(1, 1, 0, 1, 1, 1, 1, 1)$ to $(0, 0, 0, 0, 1, 1, 1, 0)$. A retains $Opt(Save1), j_1, r_{1w}, j_2, r_{2w}$, losing only g . The live set is still $\{Save1\}$, relevance still $Rel(A, Save1) = \{Bystander, Cafe\}$, $Rel(B, Cafe) = \{Save1\}$, and targets empty. A owns a live option whose permissibility fails, and a weight atom whose Boolean value is zero. Ownership, truth, relevance and burdens differ. This is Boolean disclosure withdrawal, not numerical reweighting, attack deletion or evidence that the child is no longer in danger.

Exercise 9.16. All three claims overreach. Semantic acceptability does not establish coverage of follow-up requests; that requires dialogue targets and supported locutions. A theory's soundness/completeness relative to admissibility is not a measure of understanding, and the reported implementation has a limited locution set. An ungrounded sensor interpretation is an encoding or provenance problem which a correct downstream evaluation cannot repair. The respective claims need a coverage specification and implementation trace, appropriate human evidence, and a justified input transformation. The reported account supplies no general clinical or human-benefit guarantee.

Exercise 9.17. A public protocol asks whether moves meet conditions and how commitments change. SGG asks whether a strategy answers every opponent branch in a fixed AF. A strategic account also needs aims and information; an application needs an implemented interface and evidence appropriate to its claimed use. The abduction route adds a specified family of framework changes and an observation to support. A fixed AF supplies neither that hypothesis space nor a minimality policy. Omitting **pre-con** from **Done** changes what an encoding can certify about performed actions; reporting that boundary prevents an implementation result from being mistaken for satisfaction of every source precondition.

Chapter 10

Exercise 10.1. SE holds because pros exist. Neither pro outranks a_2 : asymmetry of the strict order and the supplied comparisons rule out both reverse edges. BA therefore fails, and DV fails because the con set is nonempty. If only $a_2 \succ a_1$ is fixed, the completion $a_3 \succ a_2 \succ a_1$ makes BA true, whereas $a_2 \succ a_3 \succ a_1$ makes it false. Both satisfy SE and fail DV. A fully specified order containing only $a_2 \succ a_1$

has a_3 incomparable to a_2 ; BA is then determinately false, not unknown.

Exercise 10.2. One supplied annotation is $A = \{u, v, w, m\}$, $R^+ = \{(u, m), (v, m)\}$, $R^- = \{(w, u), (w, m)\}$. The schedule and shorter journey support the arrangement; contrary schedule evidence challenges feasibility and the proposal. All endpoints belong to the vertex set. The graph alone provides no support propagation or acceptance semantics. An evaluation must specify the intended meaning of support and the rule connecting these relations to its output; neither the safety policy nor a numerical rule is implicit in this annotation.

Exercise 10.3. w_1, w_3, w_4, w_5 are ordinary premises with accepted status; w_2 is a stated assumption. The proponent supplies w_1, w_4, w_5 , and w_2 if challenged; the opponent supplies w_3 . Initially all holding conditions pass. A con argument can have an ordinary premise: its opposition to m does not make its premise an exception within another argument. Questioning w_2 requires its acceptability. If it is acceptable, a_1 survives; otherwise it does not. With only a_3 pro, BA requires $a_3 \succ a_2$. The question alone supplies neither that comparison nor proof of $\neg w_2$.

Exercise 10.4. The remaining fraction is $1 - 0.4 = 0.6$, so the score is $0.9 \cdot 0.6 = 0.54$. Subtraction would yield 0.5, a different operation. The stipulated rule scales the base by the unattenuated fraction; it does not subtract the attacker's numerical value directly.

Exercise 10.5. The three smoothed bases are $11/12, 5/10, 2/11$. They are derived separately from $(V^+ + 1)/(V^+ + V^- + 2)$. No graph or propagation targets were supplied, so no further network score is determined. A large positive vote share does not establish a premise's truth or an issue's proof standard.

Exercise 10.6. Initially $J(M, F) = 13, W(M, F) = 6, J(F, M) = 4, W(F, M) = 1$. The tests $13 \geq 1$ and $4 < 6$ permit only M . After the changed row, $J(F, M) = 7, W(F, M) = 3$, while the other totals remain fixed. Now $13 \geq 3$ and $7 \geq 6$ both pass, so both options are permitted. Comparing 13 directly with 7 would rank justifying totals rather than apply the declared rule. A further selection policy would be needed to pick one. The differing initial father's weight in Chapter 4 identifies a distinct input, not an arithmetic inconsistency.

Exercise 10.7. Carneades retains statement nodes, argument nodes, premise occurrences, roles, statuses and issue standards; a bipolar graph displays selected attack/support topology. Linked premises must all hold within one argument. Convergent arguments are separate routes to a conclusion. A con argument opposes an issue, while an exception premise in stated or questioned status blocks one argument when that condition is acceptable; accepted/rejected statuses have the table's explicit overrides. These differences are visible in the safety dependency and are not supplied by an untyped list of arrows. The factor object additionally records which case features favour or disfavour options and the stipulated priority comparisons among them. A topology or factor list still needs an evaluation rule

for a nonstipulated outcome.

Exercise 10.8. A deductive reading takes acceptance of x as constraining acceptance of y . A necessary reading makes x required for y , so accepting y requires accepting x . These are opposite directions of acceptance dependency for the same drawn edge. Neither description specifies all interaction with attacks, cycles or other supports. A complete semantics must supply those further choices.

Exercise 10.9. DV supplies a pro witness and no cons; universal comparison with the empty con set yields BA. BA supplies a pro witness, yielding SE. The initial custody case has BA without DV. The reversed-priority case of Exercise 10.1 has SE without BA. If the pro set is empty, no existential witness exists for any standard, even with an empty con set. For the quantifier boundary, use pros $\{p_1, p_2\}$, cons $\{c_1, c_2\}$, and a strict partial order consisting only of $p_1 \succ c_1$ and $p_2 \succ c_2$. Each con has a superior pro, but neither pro exceeds both cons. Thus $\forall c \exists p$ holds while BA's $\exists p \forall c$ fails. This is a complete declared order, not an incomplete description requiring a tie-break.

Exercise 10.10. At stage 4, $\mathcal{P}(s) = \{r_1, r_2\}$, $\mathcal{C}(s) = \{c\}$. SE holds; DV fails. Hence $A(s)$ is false under its assigned DV standard, the stated exception premise holds, and A_M is defensible. The final sets are $\mathcal{P}(M) = \{A_M\}$, $\mathcal{C}(M) = \emptyset$, so BA holds. At stage 5, c loses its required premise and leaves the con set. SE and DV for s both hold; the exception premise fails and removes A_M . Both sets for M are empty, so BA fails.

In ordinary terms, two reports do not establish this exception while a supported rebuttal remains under the stipulated policy. Once that rebuttal loses support, the exception blocks the proposal's only argument. The formal account identifies precisely the changed set and holding condition. Counting reports would replace DV with a different rule. No argument for F was constructed, so failure of the proposal for M cannot be reported as proof of F .

Exercise 10.11. In order $R/C, C/R, R/T, T/R, C/T, T/C$, compare $4 - 3, 3 - 2, 7 - 3, 4 - 5, 2 - 3, 3 - 1$. The margins are $1, 1, 4, -1, -1, 2$, with signs $+, +, +, -, -, +$. Only R passes every foil. C fails against T , and T fails against R . The company ground has pair $(4, 2)$ for (R, C) and $(7, 5)$ for (R, T) : the café retains part of the shared-meal benefit, while separate takeaway does not. Those facts motivate distinct comparisons but do not derive the numbers. A dialogue must preserve the ground, favoured option, foil and current weighting input; the general statement that the restaurant is good is not enough.

Exercise 10.12. The changed requiring weight of (g_{speed}, T, C) appears in the C/T test. Its margin becomes $2 - 2 = 0$, while the other five remain $1, 1, 4, -1, 2$ in their previous positions. The full signs are now $+, +, +, -, +, +$, yielding $\{R, C\}$. The least-cost policy selects C ; the stated priority policy selects R . T/C still compares 3 with 1; changing T 's requiring coordinate does not change its own

justifying comparison. Across both inputs the aggregate justifying totals are $R : 4 + 7 = 11$, $C : 3 + 2 = 5$, $T : 4 + 3 = 7$, ranking R above T above C . They discard the only changed coordinate, so cannot recover the changed permission set; aggregating across foils also introduces a separate rule.

Exercise 10.13. Every input lies in $[0, 1]$; so does its complement. A finite product of numbers in that interval remains in it, and subtracting that product from one remains in it. Apply these facts first to the source strengths, then to attenuation of a_2 , then to $S^+(m)$, and finally to their product defining the issue score. This proves the bound for the stated acyclic computation. a_3 raises the support aggregate directly and attenuates a_2 , thereby weakening opposition to m . Whether these two routes appropriately represent dependent evidence is not decided by boundedness. No cyclic existence, uniqueness or convergence claim, or probability/independence model, has been proved.

Exercise 10.14. u is unattacked, so its categoriser and product values are one. Each of v, w, x has only attacker u , giving categoriser $1/2$ and product zero. Thus $\text{Cat}(z) = 1/(1 + 3/2) = 2/5$ and $s(z) = (1 - 0)^3 = 1$. The rankings are $u \succ_{Cat} v \sim_{Cat} w \sim_{Cat} x \succ_{Cat} z$ and $u \sim_s z \succ_s v \sim_s w \sim_s x$.

No admissible set contains a middle vertex: u attacks it and has no attacker. A set containing z must contain u to counter its attackers. The admissible sets are consequently exactly $\emptyset, \{u\}, \{u, z\}$. Only $\{u, z\}$ contains all it defends; it is also the unique maximal admissible set and attacks every outsider. The complete, preferred and stable families are all $\{\{u, z\}\}$, and grounded iteration gives $\{u, z\}$. Including value $2/5$ by any upper threshold necessarily includes the larger $1/2$ values, so no such threshold returns exactly the accepted set.

For the general DAG result, a nonempty finite DAG has an unattacked vertex: following incoming edges indefinitely would otherwise repeat a vertex and form a cycle. Repeated removal gives a topological order. Each categoriser equation then uses only earlier values, assigning one uniquely determined positive value at most one. Induction establishes existence and uniqueness on the finite acyclic domain. Any other solution must agree at the first vertex and each successive vertex. Real comparison is reflexive, transitive and total, proving that the induced ranking is a total preorder. This proof does not analyze iteration on cyclic graphs.

Exercise 10.15. After deleting (x, z) , u still has value one and all three middle vertices still have categoriser $1/2$. Now $\text{Cat}(z) = 1/(1 + 1/2 + 1/2) = 1/2$, so it ties with them. The product remains one for u, z and zero for v, w, x . The admissible-set argument is unchanged: u excludes the middle vertices and counters both remaining attackers of z ; all four semantics still give $\{u, z\}$ as before. One ranking changes while these other outputs do not. This is a finite sensitivity comparison under one deletion, not a proof of general deletion invariance.

Exercise 10.16. For the privacy-first audience, $a \rightarrow b$ survives since convenience is not preferred to privacy; $b \rightarrow a$ is blocked. The single edge gives $\{a\}$ as grounded, unique preferred and unique stable extension. Reverse the audience order to retain only $b \rightarrow a$, with corresponding extension $\{b\}$. Value names cannot simply be treated as probabilities: the orders express priority among promoted values and supply neither a sample space nor normalized masses. A different use of priorities must state its reduction/evaluation rule and whether conflict is assessed using raw attack or defeat. No audience's moral correctness follows from this calculation.

Exercise 10.17. Use Supplement S10.3's independent-choice product rule. It gives probabilities $(2/5)(1/2) = 1/5$, $(3/5)(1/2) = 3/10$, $(2/5)(1/2) = 1/5$, $(3/5)(1/2) = 3/10$ in the table's order. b is present in worlds $\{b\}$ and $\{a, b\}$, with total mass $1/2$, but accepted only in $\{b\}$, with mass $1/5$. The event of its acceptance is contained in the event of its presence, so their intersection also has mass $1/5$. The positive conditioning mass permits the ratio $(1/5)/(1/2) = 2/5$. Correlated presence with the same marginals could change these world masses. Constellation subsets specify which graph exists before semantic evaluation; epistemic subsets are belief interpretations on a fixed graph. Neither interpretation is supplied merely by writing $2^{\{a,b\}}$.

Exercise 10.18. For any attack (a, b) , COH gives $P(b) \leq 1 - P(a)$. If $P(a) > 1/2$, then $P(b) < 1/2$, and hence $P(b) \leq 1/2$, as RAT requires. The arbitrary-attack argument also covers self-attacks and therefore proves the implication on the stated finite domain. For the converse, the complete distribution $(1/5, 3/10, 1/10, 2/5)$ is normalized and yields marginals $P(a) = 7/10$, $P(b) = 1/2$. RAT holds; COH would require $7/10 \leq 1/2$, which is false. Threshold labels are $a : \text{in}, b : \text{undec}$; the unique complete Dung labelling is $a : \text{in}, b : \text{out}$. The in sets agree, but b 's epistemic label is not a legal complete label because it has an in attacker.

Exercise 10.19. The changed masses, in the same order, are $(1/5, 1/2, 1/10, 1/5)$. They remain normalized, with marginals $P'(a) = 7/10$, $P'(b) = 3/10$. COH holds at equality, RAT holds, and the threshold labels in, out now agree with the complete labelling. This does not establish sufficiency of COH. On the same graph put all mass on $\emptyset$. Both marginals are zero; COH holds and RAT has a false antecedent. Thresholding labels both arguments out, but an unattacked a cannot be out in a complete labelling. The full distribution supplies a counterexample without assuming independence.

Exercise 10.20. The report supports repeatability for ten stated trials and agreement on two selected output comparisons, conditional on correct calculation. It supplies no uniqueness proof, coverage of future cases or normative justification of the representation and weights. Audit graph construction and the exact evaluation separately, including whether a one-step measure was confused with recursive evaluation. A Libra explanation may expose a public commitment or dependency to challenge; its abstract comparison predicate does not certify weights. Identify the

contested source, weight, relation, rule or authority and what counts as answering it. Actual empirical claims would require the data and evaluation method; the fictional task creates neither data nor a benchmark result.

Exercise 10.21. The two supplied pairs favour M : child preference outranks the wealth factor and father absence outranks the adverse wealth description. Calling the stipulated local result $M \succ_{case} F$ summarizes those two comparisons; it is not a conclusion of an unstated general lifting rule. A challenge can contest a factor's presence, its direction or its priority. Additional or unpaired factors require a rule for combining or comparing sets which was not supplied. Mirrored descriptions of the same circumstance are not independent evidence and should not be counted as corroboration. The object distinguishes factor content from issue standards, audience values and cardinal weights without implementing a legal case-based reasoner.

Exercise 10.22. The vote records are public observations in a stipulated platform; they need not reveal a voter's private belief. A collective acceptance decision would need its own aggregation rule, while the displayed score is the output of smoothing and propagation. The two numerical examples use different bases, so their differing results do not establish which is more accurate; no external correctness target was measured. Endorsement can be coordinated, uninformed or directed at something other than a legal premise's applicability. A score therefore supplies neither accepted status in the Carneades context nor normative authority for a practical weight.

Exercise 10.23. One comparison uses Carneades 2007, the bipolar Handbook chapter and the selected AAMAS practical definitions. Their inputs are respectively a statement/argument graph with context, an attack/support structure with an interpretation, and comparative grounds/options/weights. Their outputs are issue acceptability, the output chosen by a support semantics, and all-foil permission. Domain and interpretation restrictions are part of each claim. The 2012 translation concerns a later stage-specific CAES and constructs ASPIC+ arguments; a correspondence must be stated over that source domain and representation. It neither exports every procedural feature of the 2007 model to a bare AF nor establishes that all issue and option evaluations are interchangeable. Other selections are adequate when their precise objects, outputs and source restrictions are equally explicit.

Chapter 11

Exercise 11.1. Ignoring support leaves two independent mutual-attack pairs. Conflict-freeness permits zero or one choice from each, but zero leaves that pair uncovered. Exactly one must therefore be chosen from each. The four products are E_{ac} , E_{ad} , E_{bc} , E_{bd} . In every case each chosen argument attacks its omitted rival. This proves both membership and exhaustiveness. Adding support before this calculation would answer a different question.

Exercise 11.2. Count directed support edges, not supported vertices. E_{ac} contains both (a, c) and (c, a) , so its score is 2. E_{bd} contains only (b, d) , so its score is 1. Neither mixed pair contains a support edge, giving scores 0 and 0. The selected family is therefore $\{E_{ac}\}$. Removing (a, c) lowers only that candidate's score and creates a tie with E_{bd} . Both maximizers must then be returned. Both selected families are nonempty. Initially a, c belong to every selected set; after the deletion the two sets are disjoint, so a, c lose skeptical acceptance. Arguments b, d remain non-skeptical. No tie-breaking rule was supplied.

Exercise 11.3. Compose each original support with attacks leaving its target. The support (a, c) meets (c, d) , giving (a, d) ; (c, a) meets (a, b) , giving (c, b) ; (b, d) meets (d, c) , giving (b, c) . There are no other supports and each of their targets has only that original outgoing attack. Thus these are exactly the three added edges. As proved in the running calculation, singles cannot cover all outsiders, the two mixed pairs conflict, and E_{ac}, E_{bd} are stable. Do not feed the new edges back into the composition.

Exercise 11.4. Selection first computes $\mathcal{E}(B)$ using only R^- , then uses support to retain a subfamily. Reduction first changes R^- , so both conflict-freeness and outsider coverage use the new graph. On this input selection returns only E_{ac} , while reduction returns E_{ac} and E_{bd} . Even the shared E_{ac} has different justifications: a maximum score among original stable sets versus stability in a transformed AF. Neither result is a claim that a numerical strength or a preference-filtered defeat relation has been computed. The defense query instead returns the set of defended arguments for a supplied defending set: $D_B^+(\{a\}) = \{a, c\}$. It is not an extension family.

Exercise 11.5. Before the addition, the admissible sets are exactly $\emptyset, \{z\}, \{y, z\}$. The largest is $\{y, z\}$, which attacks both outsiders x, w , so it is uniquely preferred and stable. Adding support (x, y) adds only attack (x, w) . Afterward $\{x\}$ defends itself against w, z and attacks y, z, w . The admissible sets are exactly $\emptyset, \{x\}, \{z\}, \{y, z\}$; their maximal members are $\{x\}, \{y, z\}$, both stable. The supported target y loses skeptical acceptance and retains credulous acceptance. Grounded remains empty because neither graph has an unattacked argument. The complete conflict-free candidate list is $\emptyset, \{x\}, \{y\}, \{z\}, \{w\}, \{y, z\}, \{z, w\}$; x conflicts with every other argument and y conflicts with w , excluding all further sets. Before the change $\{x\}$ fails against w ; afterward it answers both attackers. The other excluded candidates still fail: y cannot answer x , while any candidate containing w in this list cannot answer y . This establishes exhaustiveness of both admissible lists. The edge (x, y) is deliberately both attack and support; excluding such overlaps would exclude this witness, not refute its calculation.

Exercise 11.6. Fix a class of BAFs and a function assigning $s_B : A \rightarrow [0, 1]$ to each BAF. For every permitted addition $(x, y) \notin R^+$, set $B' = (A, R^-, R^+ \cup \{(x, y)\})$, keeping arguments, direct attacks, initial weights and all other parameters

unchanged. The proposed target requirement is $s_{B'}(y) \geq s_B(y)$. It constrains a specified method; it does not define one. A stronger requirement could demand this inequality for all arguments, but that is not the target-only principle asked for. The Boolean counterexample and extension counts supply no numerical valuation without an additional rule.

Exercise 11.7. Use pR^-q , target $T = \{q\}$, grounded semantics and $D = \{(p, q)\}$. The membership goal is $T \subseteq G(F - \Delta)$. With no deletion, $G = \{p\}$, so cost 0 fails. Deleting the edge produces $G = \{p, q\}$, so cost 1 succeeds. There is no other allowed operation or cheaper successful choice. This supplies a minimum-cardinality certificate and preserves the inherited example. If acceptance were meant to require the exact extension $\{q\}$, this answer would not succeed.

Exercise 11.8. A semantics gives a formal prediction on a represented input. A study also fixes a population, a presentation, a response task and a measurement. Comparing observed choices with that prediction can test whether the interpretation of support is useful for the task. It can motivate changing the model or the preferred principles. It does not alter the definition of the original semantics, disprove its mathematical properties, or uniquely identify a replacement from one mismatch. For example, present the four candidates as coalitions, describe support as endorsement, and ask which coalition is most cohesive. Let the measurement be the proportion choosing E_{ac} and predict that it is the most frequently chosen candidate. A mismatch could reflect a preference for balanced endorsements rather than their total count, or a different reading of support. Changing the task to justified belief would test a different hypothesis. These are proposed design choices, with no observed outcome for the running BAF.

Exercise 11.9. Deductive and inverse closure are output-level conditions on every returned extension. Skeptical support monotonicity is a construction/change requirement evaluated by comparing target status across output families. Directionality-style locality similarly constrains restriction/projection across inputs. Inactive-support deletion is a change requirement whose conclusion is equality of selected families. None of the five table rows is, by itself, an empirical validation criterion. Whether closure is desirable for a particular use of support is a methodological question; the corresponding implication on E is a mathematical output condition. Merely labelling all support properties construction-level would conceal these distinctions.

Exercise 11.10. From $S = \{a\}$, a attacks b , so it defends a ordinarily. It does not attack d , but its support for c , which attacks d , indirectly answers the sole attacker of c . The attackers of b and d remain unanswered. Thus $D(S) = \{a\}$ and $D_B^+(S) = \{a, c\}$. For $S \subseteq T$, reuse the same existential witnesses z, w in every defended case; z remains in T . For support inclusion, reuse each old direct answer or old supporting edge. These witness-preservation arguments prove both general inclusions. They supply neither an extension-selection rule nor the fundamental

lemma for the source's different defenses.

Exercise 11.11. In the whole BAF, unattacked c must belong to every stable extension, and exactly one of a, b must belong. Thus the ordinary family is $\{\{a, c\}, \{b, c\}\}$. Its scores 1 and 0 select $\{a, c\}$. In the restriction to U , the same mutual attack yields $\{\{a\}, \{b\}\}$, and both score 0. Projection of the whole selected family yields only $\{\{a\}\}$, unlike the local selected family. The local-to-projected-whole inclusion fails at $\{b\}$, whereas projected-whole-to-local inclusion holds. No attack or support enters U from c ; the only cross edge leaves U . Hence the stated support-aware directionality-style requirement fails. This is not a failure of the weaker isolated-component condition, whose no-cross-edge hypothesis is absent.

Exercise 11.12. The permitted changes are $\Delta = \emptyset$ and $\Delta = \{(p, q)\}$, with grounded outputs $\{p\}$ and $\{p, q\}$. The membership goal succeeds only at cost 1, establishing the minimum. The exact-target goal fails on both outputs, establishing infeasibility for this finite allowed-edit class. These certificates exhaust the task's specified search space. A principle tester reporting 1 on a finite sample instead says only that its evaluated records did not violate the stated condition. If the principle ranges over arbitrary-size BAFs, the sample does not exhaust that domain. A validated 0, by contrast, provides one counterexample sufficient to refute the universal requirement.

Exercise 11.13. Complete-output satisfaction does not characterize complete semantics: grounded returns one complete extension. On the mutual pair complete semantics returns $\{\emptyset, \{u\}, \{v\}\}$, grounded returns $\{\emptyset\}$, and preferred returns $\{\{u\}, \{v\}\}$. Grounded also satisfies existence and directionality, so adding those two requirements still does not characterize complete semantics. The function $\sigma_0(F) = \emptyset$ satisfies complete outputs vacuously. It also satisfies directionality: both the local family and projected whole family are empty. It fails existence. In contrast, the family $\{\emptyset\}$ has one member and satisfies existence, though that member is not complete on every AF. A characterization must specify its function class and establish both directions of the equivalence, not merely collect properties.

Exercise 11.14. Let $x \in D_F(S) \cap U$. Every local attacker of x is a whole attacker, and is answered by some $z \in S$; since the attacked argument lies in U , no-incoming puts z in U . Thus the same witness belongs to $S \cap U$ and gives local defense. Conversely, if $x \in D_{F \downarrow U}(S \cap U)$, every whole attacker of x is in U and hence occurs in the local test. Its local defender is a whole defender in S . This gives whole defense. If there are no attackers both universal conditions hold.

For the iterates, $T_0 = \emptyset = S_0 \cap U$. If $T_i = S_i \cap U$, then

$$T_{i+1} = D_{F \downarrow U}(T_i) = D_{F \downarrow U}(S_i \cap U) = D_F(S_i) \cap U = S_{i+1} \cap U.$$

Finite empty-start iteration reaches the grounded extensions, proving the limit equality. Both uses of the hypothesis in the defense identity are necessary: it controls attackers of the target and defenders of those attackers. On the chain the

whole iterates are $\emptyset, \{x\}, \{x, z\}, \{x, z\}$. On U_1 , they are $\emptyset, \{x\}, \{x\}$, agreeing with the whole projection; no edge enters U_1 . On U_2 , the local grounded extension is $\{y\}$ but the whole projection is $\{z\}$. The incoming xRy violates the hypothesis, so this unequal comparison does not refute directionality.

Exercise 11.15. After deleting the self-loop, the conflict-free candidates are $\emptyset, \{a\}, \{b\}, \{c\}, \{a, c\}$. The empty set covers nothing; $\{a\}$ misses c , and $\{c\}$ misses a, b . The sets $\{b\}$ and $\{a, c\}$ cover every outsider, so they are exactly stable. Their projections are $\{b\}, \{a\}$, recovering the local family. Selecting c now completes the old blocked choice $\{a\}$. This repairs one instance. Stable directionality remains a universally quantified property refuted by the original instance.

Exercise 11.16. Reduction 1 drops aRb , because its target is strictly preferred. Both arguments are unattacked in the resulting defeat graph, so grounded iteration reaches $\{a, b\}$; stability forces both unattacked arguments and gives the same unique extension. Reduction 2 reverses the asymmetric critical attack to bRa . Its grounded iteration is $\emptyset, \{b\}, \{b\}$; stability forces b and excludes a , so the unique stable set is $\{b\}$. The first output conflicts under raw R but is conflict-free under its empty defeat relation; the second is conflict-free under both. The quoted claim must specify the preference relation and its orientation, reduction policy, relation used for evaluation, semantics, and intended meaning of winning. The preference alone does not select between these two results.

Exercise 11.17. Rahwan and colleagues' separate 20- and 47-participant studies use a seven-point confidence scale; their intended graphs were not checked as participants' reconstructions. The 130-participant study elicits attack graphs and then three-valued judgments, with discussion and revision. The 61-participant study teaches attack and supplies the intended graphs before three-valued judgments. Pooling the latter two as if only sample size changed would ignore different graph families, sizes, settings and reconstruction tasks. In the later comparison differences among SCF2, CF2 and grounded were not significant, so even its reported ranking does not identify a unique leading method. Neither predictive fit establishes normative superiority nor the cross-design difference isolates a cause. A theorem about grounded semantics on its defined input remains a mathematical claim; the study concerns how its predictions relate to people in that design.

Exercise 11.18. (a) The sole attack aRb forces the unique stable extension $\{a\}$. With sole support aR^+b , that candidate scores 0 and remains selected, but contains the supporter and omits the target, violating deductive closure. With sole support bR^+a , the same unique candidate scores 0 and violates inverse closure: it contains the target but not its prerequisite. The first case permits attack/support overlap, as declared. (b) Deleting support changes no ordinary candidate because attacks are fixed. If the deleted edge belongs to no candidate's internal edges, every score is unchanged, including ties. The defining maximum comparisons and therefore the entire selected family are unchanged; this also covers an empty base. (c) One-step

composition uses the original sole attack $cR^{-}d$ and adds $bR^{-}d$. It cannot use that new attack on the same right-hand side, so it does not yet add $aR^{-}d$. A further pass using the enlarged attack relation composes $aR^{+}b$ with $bR^{-}d$, adding $aR^{-}d$. No further attacks follow on this input. This is an explicit distinction between one step and repeated closure, not an assertion that the latter alone is the source's full deductive reduction.

Exercise 11.19. For each integer $j = 0, \dots, |D|$, form an initially empty list L . For every $\Delta \subseteq D$ with $|\Delta| = j$, compute the grounded extension of $F - \Delta$ and append Δ to L iff it meets the stated goal. After finishing the level, return L if it is nonempty. If all levels fail, return infeasible. There are finitely many edits and each grounded computation terminates, so the procedure terminates. Tests establish success of returned edits, earlier levels exclude every cheaper one, and completing the successful level collects all ties. Exhausting all levels proves infeasibility in the allowed space. No unlisted operation is part of that certificate.

A minimum-cardinality successful deletion is inclusion-minimal, since any successful proper subset would cost less. The converse fails for general success conditions, as the study's incomparable successes of sizes 1 and 2 show; inclusion compares subsets, whereas minimum cost compares all edits. The tester returns respectively (i) 0 with the violating record, (ii) 1, passing on six records only, (iii) ?, and (iv) ?. Only the first refutes the arbitrary-size universal principle. Six passes neither exhaust all finite BAFs nor prove its general satisfaction.

Exercise 11.20. Only $justify(A, \{P\} \vdash^{+} \chi)$ is legal. Retraction is forbidden while a reply is pending; B's proposed positive justification has the wrong speaker. The matching A reply removes $(A, \chi, +)$, adds $(A, \{P\}, +)$, and clears the pending field. Thus the ledger is $\{(A, \{P\}, +), (B, \chi, -)\}$. A's assertion and B's challenge remain active, and the justification contributes P to A's commitments.

For the general requirement, a reachable nonempty pending field arose from CR2 or CR4. Under CR2, pending priority permits only CR3 with that speaker, target and positive polarity; under CR4 it permits only the matching negative CR5 response. Both require a nonempty support unit. The common justification effect clears the pending field, removes the matching requested obligation and adds the support-unit obligation. This proves the stated transition property for all such states. It does not assert that a finite record contains the next transition. The legal prefix $assert(A, \chi), challenge(B, A, \chi)$ ends with the positive reply pending and refutes the stronger completion claim. A longer record that contains the required reply may complete this immediate request while standing support and opposition responsibilities remain. Neither legality nor completion establishes truth, validity, adequacy or persuasive success of the opaque offered grounds. The discharge effect is the book's local convention, not Fatio's unchanged claim-obligation effect.

Exercise 11.21. On $\{R, C, T\}$, R passes against both foils, C fails against T, and T fails against R, giving $\{R\}$. On $\{R, C\}$, both surviving tests pass, so both are permitted. On $\{C, T\}$, C fails against T while T passes against C, giving $\{T\}$. These are restrictions of the same pair assessments, not newly assigned weights.

For any old option $o \in P_{O'} \cap O$, every old foil $f \in O \setminus \{o\}$ is also in $O' \setminus \{o\}$. Its test passes in the larger domain and has exactly the old values, so it passes in the old instance. Thus $o \in P_O$, proving the inclusion. In the expansion from $\{R, C\}$ to all three, C is a witness against the reverse inclusion; in the expansion from $\{C, T\}$, T is one. An option need not itself be permitted to affect another: T is not permitted in the full instance but its comparison excludes C. If old totals can also change, the step identifying each new test with its old test fails. For example, on two options X, Y use one X/Y reason with weights (1, 0) and one Y/X reason with weights (1, 2). Initially X fails its sole comparison $1 \geq 2$, while Y passes $1 \geq 0$. Revising only the X/Y justifying coordinate from 1 to 2 makes X pass. The alternative set is unchanged, yet the new permitted set is not contained in the old one. Thus an inclusion allowing arbitrary old-total revisions would already fail with no domain expansion.

Literature/research exercise 11.L. *[prompt]* One model route begins with Baroni and Giacomin's comparison of semantic requirements, uses van der Torre and Vesic's Handbook account to locate precise function-level locality principles, and takes the dynamics survey of Doutre and Mailly as its further source. Its objects are semantic functions, restriction/projection equalities and changes to AFs. The contribution relevant to this chapter is the separation between satisfying an output property and preserving behavior under a change. The blocked stable completion and the permitted-deletion problem supply different chapter connections. Neither a property table nor a dynamics taxonomy makes every allowed update efficient or appropriate for a particular application. A different trio is acceptable if the answer states what each source adds, uses the external reading accurately and retains its assumptions; listing three titles without that comparison does not complete the task.

Chapter 12

Exercise 12.1. The attacks from $\{a, c, e\}$ are aRb, cRd , so none is internal. The outsiders are exactly b, d , both covered by these attacks. During the checker scan no internal-edge test rejects, and the final marked set is $\{b, d\}$. Its prefix invariant says precisely that marks record attacks from members and that no processed attack is internal. The final loop therefore accepts. Both properties are needed for stability.

Exercise 12.2. Neither aRd nor dRa is an attack, and there are no self-attacks, so $\{a, d\}$ is conflict-free. Its outsiders are b, c, e , covered respectively by aRb, dRc, dRe . Thus it is stable. Coverage alone is insufficient: in the mutual pair, choosing

both endpoints leaves no outsiders and satisfies coverage vacuously, yet violates conflict-freeness. The self-attacking singleton with its sole argument selected gives another counterexample.

Exercise 12.3. Use $E = \{b, d\}$. Neither selected argument attacks the other. Outsiders a, c, e are covered by bRa, bRc, dRe , respectively (dRc also covers c). Thus E is stable and contains b , certifying the existential query. No maximality or full-family calculation is additionally required for this stable witness.

Exercise 12.4. No: the verified stable extension $\{a, d\}$ omits e . One such counterexample is enough to make the universal membership condition false; it also witnesses that the family is nonempty. The extension $\{a, c, e\}$ establishes credulity, which is a different question. Enumeration would be correct additional evidence but is unnecessary to refute skepticism here.

Exercise 12.5. In edge-list order, the six clauses are

$$\begin{array}{lll} \neg x_a \vee \neg x_b, & \neg x_b \vee \neg x_a, & \neg x_b \vee \neg x_c, \\ \neg x_c \vee \neg x_d, & \neg x_d \vee \neg x_c, & \neg x_d \vee \neg x_e. \end{array}$$

After duplicate removal they are

$$(\neg x_a \vee \neg x_b) \wedge (\neg x_b \vee \neg x_c) \wedge (\neg x_c \vee \neg x_d) \wedge (\neg x_d \vee \neg x_e).$$

Each clause excludes joint selection of the endpoints, regardless of attack direction. Thus it does not distinguish uRv from vRu , nor one from both. Direction is retained in the coverage clauses, whose disjunctions list attackers of each particular target.

Exercise 12.6. The attackers of c are b, d , so its clause is $x_c \vee x_b \vee x_d$. If c is omitted, at least one of those attackers must be selected to cover it. Listing only the argument attacked by c , namely d , would miss the valid coverage supplied by b and confuse outgoing with incoming attacks.

Exercise 12.7. This is an explanation task requiring search and size optimization. It is not merely verifying a supplied candidate, enumerating all extensions or monitoring an observed trace. One must fix the AF, semantics and acceptance mode; the target query and observed answer; which subgraphs or deletions are allowed; whether evaluation is standalone or retains context; the sufficiency test; and the size measure. Without these choices, an isolated query vertex might preserve a local answer but omit all of its original reasons. Chapter 13 develops that distinction. Alternatives to this list are acceptable if they provide an equally precise optimization specification.

Exercise 12.8. For example, suppose a collection mixes graphs whose component structure makes a structural procedure effective with graphs favoring a SAT encoding. A selector using those features might save more solving time than its own selection overhead. Test that hypothesis on a declared instance collection and task using

fixed hardware and configurations, measuring time and memory against a specified single-solver baseline, counting failed or timed-out runs rather than silently dropping them. The answer is conditional: the selector can also choose badly or cost more than it saves. NP-completeness is a worst-case decision classification; it neither proves nor refutes this advantage on the chosen instances and does not establish that all algorithms must be slow on each one.

Exercise 12.9. For $A = \{a\}$, $R = \{(a, a)\}$, the attack clause is $\neg x_a \vee \neg x_a$, equivalent to $\neg x_a$, and coverage is $x_a \vee x_a$, equivalent to x_a . Hence K_F is UNSAT and there is no stable extension. Unguarded skepticism asks $\text{UNSAT}(K_F \wedge \neg x_a)$, which is true. Guarded skepticism additionally requires $\text{SAT}(K_F)$, which is false, so its answer is false. This is an empty family, not the empty AF's one empty extension.

Exercise 12.10. A negative stable-credulous certificate must exclude every stable set containing b . A complete subset analysis or an independently checked UNSAT proof for $K_F \wedge x_b$ would do so on an appropriate input. An extension omitting b does not exclude a different containing one; a timeout leaves the search incomplete. For F_{reuse} , the verified $\{b, d\}$ is a containing witness, so the proposed negative answer is false and no correct certificate for it exists. The general certificate specification and its applicability to this instance must be distinguished.

Exercise 12.11. For $E = \{a, c\}$, processing (a, b) marks b , processing (b, c) does nothing because $b \notin E$, processing (c, b) keeps b marked, and processing (c, d) marks d . No internal attack occurs, and b, d are all outsiders, so the candidate passes. For $E = \{a\}$, only b is marked. The final loop rejects at its first uncovered outsider, c ; d is also uncovered. Thus the candidate fails.

The grounded trace is

$$\emptyset, \quad \{a\}, \quad \{a, c\}, \quad \{a, c\}.$$

Initially only a is unattacked. With $\{a\}$, the attacked set is $\{b\}$, which defends c ; the attackers of b include the unanswered a , and d 's attacker c is not attacked. With $\{a, c\}$, the attacked set is $\{b, d\}$, and the next defended set is unchanged. After deleting (a, b) , the trace is $\emptyset, \{a\}, \{a\}$: isolated a attacks nobody, so it cannot defend c against b . Each round reads an unchanged input array; updating it while scanning would not implement this simultaneous defense operator. The empty start, together with admissibility and leastness, licenses groundedness. Each round and comparison costs $O(n + m + 1)$; at most n strict increases plus one equality check give $O((n + 1)(n + m + 1))$. The storage beyond the input is $O(n + 1)$ cells.

Exercise 12.12. Removing the duplicate conflict clause gives

$$\begin{aligned} K_H = & (\neg x_a \vee \neg x_b) \wedge (\neg x_b \vee \neg x_c) \wedge (\neg x_c \vee \neg x_d) \\ & \wedge x_a \wedge (x_b \vee x_a \vee x_c) \wedge (x_c \vee x_b) \wedge (x_d \vee x_c). \end{aligned}$$

The unit x_a forces a , then conflict excludes b , coverage forces c , and conflict excludes d . This proves that the unique model has true set $\{a, c\}$. Thus the base is SAT

and $K_H \wedge \neg x_c$ is UNSAT, establishing guarded skepticism of c . The exact block is $\neg x_a \vee x_b \vee \neg x_c \vee x_d$. It is false on the sole model, so adding it makes the formula UNSAT and certifies exhaustion.

After deleting (a, b) , rebuild the formula as

$$K' = (\neg x_b \vee \neg x_c) \wedge (\neg x_c \vee \neg x_d) \\ \wedge x_a \wedge (x_b \vee x_c) \wedge (x_c \vee x_b) \wedge (x_d \vee x_c).$$

Now a is forced independently. Exactly one of b, c is selected. If c , then d is false, yielding $\{a, c\}$; if b , then c is false and coverage forces d , yielding $\{a, b, d\}$. Both are models, and these cases exhaust the choices. Consequently c remains credulous but loses guarded skepticism, and d becomes credulous. Deleting an attack changes its conflict clause and the target's attacker list in a coverage clause; duplicates may also remain from other attacks. Arbitrary clause deletion does not implement that coordinated change.

Exercise 12.13. There are seven vertices. The nine attacks are

$$(x, \bar{x}), (\bar{x}, x), (y, \bar{y}), (\bar{y}, y), (x, c_1), (y, c_1), (\bar{x}, c_2), (c_1, q), (c_2, q).$$

Any query-containing stable set omits both clauses, chooses exactly one literal from each pair and covers both clauses. The second clause forces $\bar{x}$; then the first forces y . Thus the unique such extension is $\{q, \bar{x}, y\}$, giving $x = \text{false}, y = \text{true}$. Its outsiders $x, \bar{y}, c_1, c_2$ are covered by $\bar{x}, y, y, \bar{x}$, respectively, and it has no internal attack. The candidate $\{q, x, \bar{y}\}$ leaves c_2 uncovered and is not stable.

For general CNF, a satisfying assignment selects the query and its one true literal per variable. Opposite literals cover their omitted partners, true literals cover all omitted clauses, and no selected nodes attack one another. Conversely, a stable query-containing set must select exactly one literal per variable: both conflict and neither leaves the pair uncovered. It omits every clause node because each attacks the query. Coverage then supplies a selected literal in every clause, yielding a satisfying total assignment. The graph has $2k + t + 1$ nodes and $2k + t + L$ attacks; scanning the input takes $O(k + t + L + 1)$ indexed operations and polynomial bit cost. Fresh tags prevent accidental identifications, and clause indices distinguish repeated clauses. An empty clause has no literal attacker, blocking query selection; an unused variable still requires a paired choice. With an empty conjunction the unattacked query can be selected, and a tautological clause is covered under either choice of its complementary variable.

For $(x) \wedge (\neg x)$, exactly one literal is again selected. Choosing x forces c_2 , excludes c_1, q , and gives $\{x, c_2\}$; choosing $\bar{x}$ symmetrically gives $\{\bar{x}, c_1\}$. Each is conflict-free and covers every outsider. These are all stable sets and neither contains q , despite stable existence. Thus dropping the query breaks the SAT equivalence. Stable credulous NP membership follows from the polynomial checker; the reduction and the stated SAT starting theorem give its NP-hardness and hence NP-completeness, not the distinct existence proof.

Exercise 12.14. Initially the SCCs are $\{a, b\}, \{c, d\}, \{e\}$. Stability forces one of a, b . With a , choose exactly one of c, d : choosing c forces e , giving $\{a, c, e\}$; choosing d excludes e , giving $\{a, d\}$. With b , its attack excludes c , whose absence forces d , and d excludes e , giving $\{b, d\}$. Each passes the two stable conditions, and these forced cases are exhaustive. The free combination $\{b, c\}$ is forbidden by bRc ; the component must receive its incoming attack context.

Adding (e, a) makes the first old set conflict. The other two still have no internal edge and still cover all outsiders. To exclude new sets, observe that at least one of a, b is necessary because otherwise b is uncovered, and both cannot coexist. With a , e must be omitted, so d must cover it and excludes c , yielding only $\{a, d\}$. With b , a, c are excluded, d is forced and e excluded, yielding only $\{b, d\}$. The cycle $eRaRbRcRdRe$ passes through all arguments, making one SCC. Initially e is credulous but not skeptical; afterward it is not credulous and hence not skeptical, with a nonempty stable family in both cases. No asymptotic or measured incremental speedup, nor confinement of the update to one old component, follows from these finite results.

Exercise 12.15. In H , a is unattacked, so $\{a\}$ is conflict-free and self-defending. The proper superset $\{a, c\}$ is admissible: a defends c by attacking its attacker b . Thus $\{a\}$ is not preferred. It nonetheless witnesses preferred credulity of a , because every finite admissible set extends to a preferred set preserving its members. In the four-cycle, the empty set is admissible and no singleton defends itself, but $\{a, c\}$ is admissible: c attacks d , the attacker of a , and a attacks b , the attacker of c . So one-argument tests do not establish maximality.

Each of k disjoint mutual pairs contributes exactly one independent choice, giving 2^k stable sets. For $k \geq 1$, $n = 2k$, and an explicit vector list writes $n2^k$ membership bits, hence has that lower bound on bit writes even before delimiters. The output is superpolynomial in the standard explicit input length. For $k = 0$, there is one empty extension; its zero-length vector still represents an output record. The formula for positive-length bit writes is not a claim of zero work to report that record.

Exercise 12.16. On well-formed (F, q) , let L_1 be stable existence and L_2 be absence of a stable extension omitting q . An n -bit stable set is a polynomial witness for L_1 , checked by **CheckStable**, so $L_1 \in \text{NP}$. An n -bit stable set omitting q is a polynomial witness for the complement of L_2 , so $L_2 \in \text{coNP}$. Guarded skepticism is exactly $L_1 \cap L_2$, hence belongs to DP by the supplied definition. This proves membership only. A hardness claim needs a polynomial reduction from a known DP-hard problem (or an equivalent argument transferring all DP problems), with its own both-direction correctness and size proof. The stable-credulous reduction does not provide it.

Exercise 12.17. At the assignment for E , every selected-variable negation and every unselected-variable positive literal in B_E is false. Any different total assignment flips some original variable; its corresponding literal becomes true. Thus exactly one argument assignment is excluded. The shorter clause $\bigvee_{a \in E} \neg x_a$ is also

false at every assignment selecting a superset of E . For instance, with original variables x_a, x_b and $E = \{a\}$, it excludes both $\{a\}$ and $\{a, b\}$, whereas the exact clause $\neg x_a \vee x_b$ excludes only the first. If $E = \emptyset$, the shorter clause is already false on all assignments.

In $K_F \wedge (z \vee \neg z)$, each stable argument assignment has two auxiliary realizations. Blocking its full assignment with $z = 0$ leaves the $z = 1$ realization, so the same extension can be reported twice. Blocking only its original variables removes both realizations at once. Initially the projected models are all stable sets by the encoding contract. After reporting E , its block removes exactly that projection and none of the others, establishing inductively that the remaining projections are precisely the unreported stable sets. Final UNSAT therefore establishes exhaustion. For the empty AF, the initial true formula reports the empty assignment once; its empty block makes the next query UNSAT. For the self-attacking singleton, $x_a \wedge \neg x_a$ is UNSAT initially and no set is reported. With an auxiliary variable on the empty AF, projection still reports the empty extension once and the empty argument block removes all auxiliary realizations.

Exercise 12.18. For (a), B’s recorded challenge supplies the CR4 witness. B has neither an active assertion of χ nor a matching active support unit, so CR2 does not take priority. The checker returns a legal prefix with certificate (CR4). Its state has $\text{Com}_A = \{\chi, P\}$, $\text{Com}_B = \text{Com}_C = \emptyset$, ledger $\{D_A, K_B\}$, and pending $(B, \chi, -)$. All previously active assertions, justifications and challenges remain; the question opens the pending reply. For (b), the question again passes, but the retraction at position 2 fails: only B’s immediate negative justification of χ is now legal. The returned failure state is exactly the state in (a); no retraction has been applied. For (c), the nonempty support $\{Q, S\}$, correct speaker, target and negative polarity satisfy CR5. The certificate is (CR4, CR5). The final commitments are $\text{Com}_A = \{\chi, P\}$, $\text{Com}_B = \{Q, S\}$, $\text{Com}_C = \emptyset$; the ledger is $\{D_A, D_B\}$, and no reply is pending. B’s new justification is active, and the old assertion, justification and challenge persist.

With budget 2, withdrawal passes CR6 and reaches the first observation state after one move. A’s assertion is removed but its justification remains: $\text{Com}_A = \{P\}$, $\text{Com}_B = \text{Com}_C = \emptyset$, $\text{DOS} = \{D_A, K_B\}$, and no reply is pending. The stopping reason is “withdrawn”, and its score is $(0, 0, -1)$. Inquiry reaches the state in (c) after two moves. It cannot stop after its first move because B’s reply is pending. Its stopping reason is “countergrounds disclosed”, and its score is $(1, 1, -2)$. Inquiry wins because its first coordinate is larger; its extra move does not override the exposure priority. With budget 1, withdrawal is the only completed candidate. The question produces the legal state in (a), but the inquiry plan has not met the stopping condition and cannot complete within that budget. It is not assigned a completed-plan score or compared as if its reply had occurred.

The public rules determine legality, forced speaker, target and polarity, and the state effects. The response policy supplies B’s particular support content. The

observation condition determines completion for this experiment, the objective orders the completed observations, and the budget limits which offered plans can finish. Only two plans were offered, so exhausting them establishes no optimum over all legal histories. The checker treats the claims and supplied grounds as opaque labels, so neither trace legality nor the chosen score establishes excellence.

Exercise 12.19. Each reason fills a distinct ordered pair; after all six records are processed, the cells and their uses are as follows.

Pair (o, f)	$J(o, f)$	$W(o, f)$	Permission test for o/f
(R, C)	4	2	$4 \geq 3$
(R, T)	7	5	$7 \geq 3$
(C, R)	3	3	$3 \geq 2$
(C, T)	2	1	$2 < 3$
(T, R)	4	3	$4 < 5$
(T, C)	3	3	$3 \geq 1$

Each test reads the requiring cell with reversed indices. R has no failing foil, C fails only against T, and T fails only against R. Thus the result is $\{R\}$. In order C, T, R , selection checks C and T unsuccessfully then returns R. In order R, C, T , it returns R immediately.

The revision changes only $W(T, C)$ from 3 to 2; the row's justifying coordinate remains 3. The only affected test is C against T, now $J(C, T) = 2 \geq W(T, C) = 2$. C's failing-foil set becomes empty; R's stays empty and T's stays $\{R\}$. In particular, T against C still compares $3 \geq 1$. The new permitted set is $\{R, C\}$. The cost order returns C at its first check, and the priority order returns R at its first check. No new execution authority follows from either output.

Summing only justifying weights loses the requiring coordinate responsible for the change. Indeed the across-foil justifying totals remain 11 for R, 5 for C and 7 for T in both versions. They cannot distinguish the two permitted sets. A repeated utterance of an existing (g, o, f) does not add a distinct reason to the input set; adding its weights again would evaluate a different, multiplicity-sensitive rule. A completed empty list for a pair contributes zero to both sums. An unassessed coordinate is missing input, not a reason with zero weight; the validated-input precondition has not been met.

For k options and r distinct reason records, the loops initialize $2k(k-1)$ cells, perform $2r$ additions and $k(k-1)$ numerical comparisons, then make at most k membership checks in the supplied selection order. Here $k = 3, r = 6$, so these numbers are 12 initialized cells, 12 additions and 6 numerical comparisons. The original cost and priority selections use 3 and 1 membership checks; the revised selections each use 1. Counting an arithmetic operation as one step leaves its bit cost unspecified: exact rational addition and comparison depend on the sizes of the represented numerators and denominators. These counts also presuppose the stated

indexed arrays and validated input, rather than including an unspecified elicitation or duplicate-removal process.

Literature/research exercise 12.L. *[prompt]* A model comparison uses Egly, Gaggl and Woltran’s ASP approach, the Dvořák–Dunne Handbook task analysis, and the retained ICCMA 2023 DC-ST specification. ASP supplies a declarative implementation route whose models must be read back as the requested argumentation outputs. The Handbook separates verification, existence and acceptance, making clear why a witness for one task need not certify another. DC-ST supplies an external task and format contract against which the $\{b, d\}$ stable-credulous witness can be compared. Their different correctness and performance questions should not be collapsed into a claim that one representation is always fastest. A successful alternative response accurately compares the chosen external passages, their inputs and outputs, a concrete chapter calculation and a stated limit; source names without task comparison are insufficient.

Chapter 13

Exercise 13.1. First verify $E_1 = \{a, c, e\}$: it has no internal attack and covers outsiders b, d by aRb, cRd . It is therefore an admissible stable witness. Root $P : c$ has both original attackers b, d . On the $O : b$ branch, use defender $a \in E_1$, since aRb . Expanding $P : a$ lists its attacker $O : b$ and responds with a back-reference to that same $P : a$, because aRb . On the root’s $O : d$ branch, use cRd and a back-reference to root $P : c$. Thus the branches are

$$P : c ; O : b ; P : a ; O : b ; \text{Back}(P : a),$$

$$P : c ; O : d ; \text{Back}(P : c).$$

Every arrow used as a response is an attack in the original AF. Both attacker branches and both cycle closures are retained. The display explains membership within the separately verified witness; it does not establish that c belongs to the grounded extension.

Exercise 13.2. The set $E_2 = \{a, d\}$ is conflict-free and covers b, c, e , so it is stable. Since $d \in E_2$ and dRe , e is out. Expand the accepted attacker $P : d$: its only attacker is $O : c$, answered by dRc , which closes with a back-reference to $P : d$. The resulting out-certificate retains dRe and the cycle $P : d, O : c, \text{Back}(P : d)$. The grounded extension is empty because no vertex is unattacked and the first defense iteration remains empty. Consequently e is undecided in that labelling, as no accepted argument attacks it. Mere absence from an extension covers both out and undecided cases; an out-explanation must identify an attacker in the accepted set.

Exercise 13.3. Deleting (c, d) leaves d unattacked, so every stable set includes d . Its attacks exclude c, e . Exactly one of a, b must then be selected: both conflict, neither covers a or b . Thus the only possible stable sets are $\{a, d\}$ and $\{b, d\}$, and both cover their outsiders. Neither contains e , whereas the original stable $\{a, c, e\}$ does. The credulous stable answer changes from true to false, proving necessity under the stated single-edge deletion test. A fixed-list membership check would keep reading $e \in E_1$ and would not test whether E_1 remains a valid stable witness. The permitted operation requires semantic recomputation, not an edit to a printed answer.

Exercise 13.4. Take the singleton subgraph with vertex a and no attacks. Its unique stable extension is $\{a\}$: the empty set fails to cover a , and the singleton is conflict-free with no outsiders. It is therefore locally sufficient for credulous stable acceptance of a . No proper allowed subgraph exists that retains the query vertex, so it is inclusion-minimal (and also minimum by vertex/edge count in this particular class). It loses the original attacker b , the counterattack aRb , the surrounding choices and their consequences. It cannot explain the original mutual defense merely by reproducing a yes-answer in a new standalone graph. Larger sufficient subgraphs are acceptable first constructions, but the minimality part requires its own justification.

Exercise 13.5. For example, let $q_{e,E}$ ask for e 's label relative to the verified stable extension E . The two targets are

$$(F_{\text{reuse}}, \text{stable}, q_{e,E_1}, \text{in}, \text{extension}),$$

$$(F_{\text{reuse}}, \text{stable}, q_{e,E_2}, \text{out}, \text{extension}).$$

Pair them with the declaration that the graph, semantics and target argument are fixed while the extension varies. Alternatively use membership queries with true/false answers and retain the accepted attacker establishing out. In E_1 , c attacks the only attacker d of e ; in E_2 , d is selected, excludes c and attacks e . The choice between c and d therefore explains the contrast. A containing witness alone establishes only that some stable extension includes e ; it does not explain the specified alternative E_2 .

Exercise 13.6. Initially M has the active assertion χ , commitments $\{\chi\}$, ledger $\{(M, \chi, +)\}$, and no pending reply; U has no commitments or active challenge. The challenge $\text{challenge}(U, M, \chi)$ uses CR2, because distinct speaker U targets M 's active assertion. Commitments stay $\text{Com}_M = \{\chi\}$, $\text{Com}_U = \emptyset$. The ledger becomes $\{(M, \chi, +), (U, \chi, -)\}$, the challenge is recorded, and the pending field requires M 's immediate positive justification of χ .

The supplied justification uses CR3 with nonempty support unit $\Phi = \{\rho, \omega\}$. It clears the pending field, records the active justification, removes the requested $(M, \chi, +)$ obligation and adds $(M, \Phi, +)$. Hence

$$\text{Com}_M = \{\chi, \rho, \omega\}, \quad \text{Com}_U = \emptyset, \quad \text{DOS} = \{(M, \Phi, +), (U, \chi, -)\}.$$

M's assertion and U's challenge remain active. The question route has the same forced positive reply but adds no $(U, \chi, -)$, so that is the ledger difference. M cannot retract to evade the pending reply: CR3 is the only permitted next form, whereas CR6 requires no pending reply.

The protocol's targets are tagged claim labels or whole support units with an active witness. A raw ordered pair (c, d) has not been given that type or active record; χ provides them explicitly. Computing a new extension still assumes the same supplied attack and does not justify its reconstruction from content. The legal support response makes the proposed grounds and responsibility explicit, while their truth or validity needs a further content check.

Exercise 13.7. Only x_2 disagrees: the classifier approves while the surrogate refuses. The indicator sum is 1, the denominator is the four stipulated inputs, and fidelity is $1 - 1/4 = 3/4 = 0.75$. This is exact agreement arithmetic on a synthetic finite test, not a population estimate or a human outcome measurement. Agreement on every possible input in a declared domain would give equality of the two output functions on that domain, a stronger correspondence claim. The four pairs do not establish it; here they already contain one disagreement even on the finite test.

Exercise 13.8. The contextual certificate explains a result of an argumentation semantics: why e is defended relative to the verified E_1 in F_{reuse} . Its relations and membership evidence are checked against that AF and witness. The classifier example uses argumentative reasons to explain the output of another model. It therefore also needs a declared construction or extraction connecting those reasons and their evaluation to the classifier's behavior. Sampled fidelity tests one limited form of that relation; a construction-specific correspondence theorem can establish another. Neither direction alone guarantees audience understanding or validates the content represented by an attack.

Exercise 13.9. For extension-relative acceptance, verify E_1 and give the contextual accepted certificate: e 's attacker d is answered by c , whose attackers b, d are answered by a, c , with a 's own attacker b answered by a . Back-references close the two cycles. For stable credulity, the verified containing E_1 already witnesses existence; the same defense display can elaborate its reason. For failure of guarded skepticism, the verified $E_2 = \{a, d\}$ omits e , refuting universal membership and showing that the family exists. These last two witnesses coexist, so failure of skepticism plainly does not make e out in every stable extension. The acceptance mode determines which part of this evidence is logically sufficient.

Exercise 13.10. In the mutual pair, $\{a\}$ is conflict-free and attacks its sole outsider b , hence is stable. Its certificate expands $P : a$ to $O : b$ and answers with aRb , returning a back-reference to $P : a$. Neither vertex is unattacked, so grounded iteration starts and ends at $\emptyset$. Both arguments are undecided: neither is accepted and neither is attacked by the empty extension. The back-reference records cyclic defense relative to a separately verified stable witness. It does not

create an empty-start defense derivation or prove membership in the least fixed point.

Exercise 13.11. The added attack eRa is internal to the old E_1 , so that set is no longer conflict-free and cannot remain the stable context of the old certificate. Chapter 12’s exhaustive branch argument gives exactly $\{a, d\}$ and $\{b, d\}$ after the update. Neither contains e , so e is not credulous. The family is nonempty and both sets contain d , so d is guarded-skeptical. These are family-level claims: verifying only two surviving old sets would not exclude a new set containing e or omitting d . The separate completeness argument is what licenses them. The old certificate’s local edge fragments cannot compensate for losing its verified witness and graph version.

Exercise 13.12. Each allowed input extension is admissible. If an expanded $a \in E$ has attacker b , admissibility supplies some $c \in E$ with cRb , so the ordered choice cannot fail. Every recursive proponent is such a member of E , preserving this premise. The loop visits every original attacker, so each expanded node is attacker-complete, and every response is an actual counterattack by an accepted member.

Along a branch, a new expansion appends a previously absent member of E to the path. After at most $|E|$ such expansions, any continued recursion must revisit a member and return a back-reference; branches can also end at unattacked arguments. The finite graph gives finite branching, hence a finite tree overall. Back-references name ancestor occurrences of the same proponent and retain their cyclic dependence. They do not establish groundedness, which additionally concerns the least empty-start fixed point; skepticism, which quantifies over the extension family; or minimality, which compares allowed explanatory objects. Nor does the depth bound alone make the expanded tree polynomial-size: different branches may duplicate large shared parts. The contract assumes prior verification rather than deriving it from loops in the display.

Exercise 13.13. Construction-specific correspondence requires a mathematical result with the construction’s hypotheses. The AXPLR report locates that warrant at the 2023 publisher version’s Section 4, Theorem 2, for its pattern-based classifier and quantitative bipolar construction. Finite-test agreement instead requires paired model outputs on a specified set and a disagreement calculation; our four synthetic pairs give precisely that limited evidence. Human plausibility requires a response design and judgments, the target of AXPLR’s Sections 9.3–9.4. Assistance performance requires a task, baseline and measured outcome, addressed by Sections 10.1–10.4 and Table 10 for deceptive-review detection.

Reported improvement relative to within-method baselines does not establish superiority between methods: the latter performance differences were not significant. The synthetic surrogate has not been shown to satisfy AXPLR’s construction hypotheses, so it cannot inherit its correspondence theorem; its finite fidelity would

not establish such a theorem even if it were 1. Accurate prediction alone establishes neither that the prediction is correct nor that the displayed explanation is complete, understandable or useful for every person. Those questions require their own objects and measures. The supplied card reports evidence and its limits; it does not reproduce a theorem proof, dataset analysis or a new human study.

Exercise 13.14. In the directed three-cycle, the conflict-free candidates are the empty set and the three singletons; each two-element set contains an attack. The empty set covers no outsiders. Each singleton attacks only its successor and misses its other outsider, so none is stable. The stable family is therefore empty. Guarded skepticism of r is false because the existence condition fails. Exhausting these candidates proves that failure; there is no stable witness omitting r , since there is no stable witness at all. In F_{reuse} , $\{a, d\}$ is an omitting stable witness for e , which refutes universal membership while establishing existence. The same Boolean answer has different reasons and hence different available certificates in the two cases.

Exercise 13.15. R passes against C because the company reason’s justifying weight 4 meets the budget reason’s opposing requiring weight 3. It passes against T because the company reason’s justifying weight 7 meets the speed reason’s opposing requiring weight 3. These exhaust R’s foils. C passes against R by $3 \geq 2$, but its budget force against T fails the opposing speed requirement: $2 < 3$. T fails against R by $4 < 5$. The all-foil rule quantifies over every considered alternative, not only those that are themselves permitted, so T’s failure does not remove the test that excludes C. The complete original result is $\{R\}$.

The changed record is (speed, T, C) , and only its requiring coordinate changes from 3 to 2. Its effect is reversed in the permission test: $W(T, C)$ opposes $J(C, T) = 2$. This changes C/T from a negative margin -1 to the passing margin 0. C’s comparison against R already passed; R’s two tests and T’s failing test against R stay fixed. Hence the new permitted set is $\{R, C\}$. T’s own C-foil test still uses $J(T, C) = 3$ and $W(C, T) = 1$; its status has not improved.

In v_1 , both orders select the only permitted option R. In v_2 , the cost order C, T, R selects C, while the priority order R, C, T selects R. In the weight contrast, the option and reason sets, applicability, all other weight coordinates and the permission rule remain fixed; to explain a changed selected output, also hold the named policy fixed. In the policy contrast within v_2 , the entire numerical model and permitted set remain fixed while the selection order changes. Selection of C under the cost policy establishes neither unique permission nor C’s defeat of R. Both are permitted and both pass their mutual comparisons. Changing only a selection order in v_1 cannot make the supplied procedure return C: C is not in its permitted-set input.

The arithmetic request asks which stored totals are compared and why a zero margin passes. The ground request asks for the relevant practical consideration, such as the proposed claim ρ that C and T both meet the agreed end time. The warrant

request asks why that circumstance licenses the assessment, here the proposed policy claim ω assigning requiring force 2. The authority request asks who endorsed that policy or may revise this row and coordinate in this context. Checking the subtraction does not establish any of those further claims. The stipulated authority relation licenses K's record procedurally; it is not inferred from K's assertion or the numerical outcome.

Exercise 13.16. The initial active assertion supplies the CR2 witness for $question(U, K, \eta)$. After the question, $Com_K = \{\eta\}$, $Com_U = \emptyset$, and the ledger remains $\{(K, \eta, +)\}$; the pending field is $(K, \eta, +)$. The supplied $justify(K, \{\rho, \omega\} \vdash^+ \eta)$ satisfies CR3: the speaker, target and positive polarity match, and the support unit is nonempty. Afterwards $Com_K = \{\eta, \rho, \omega\}$, $Com_U = \emptyset$, the ledger is $\{(K, \{\rho, \omega\}, +)\}$, and no reply is pending. K's assertion and new justification are active; no challenge has been added. The public certificate is $(CR2, CR3)$. Neither move invokes the revision interface, so the numerical input is still v_1 .

For the complete candidate, context d and base v_1 match; row $(speed, T, C)$ and coordinate RW exist; the expected old value 3 equals the current value; K has the independently supplied authority for this context, row and coordinate; the new pair $(3, 2)$ is nonnegative and not $(0, 0)$; and both identified ground and warrant records are supplied. Every condition passes. The interface creates v_2 , changes exactly that coordinate and retains the provenance. Reevaluation returns $\{R, C\}$; cost selection is C and priority selection is R.

The other three candidates are each checked independently against v_1 , so they do not inherit the first candidate's update. Changing only the issuer to U fails the authority condition. Naming v_0 fails the base-version match. Identifying no warrant record fails the supplied-record condition, even if the proposed number is unchanged and the issuer is K. The other displayed fields do not repair any of these failures. Each rejected proposal leaves v_1 , its permitted set $\{R\}$, and both selected outputs R unchanged.

If the original accepted candidate is submitted again after v_2 has become current, its base v_1 no longer matches. Its expected old value 3 also differs from the current 2. It is rejected rather than silently reapplied, and v_2 and its outputs remain unchanged. A further revision would need a new matching proposal under the stated interface, not merely another public utterance.

The public certificate establishes the local rules followed and the resulting responsibilities; it verifies neither support content nor authority. The accepted revision establishes a particular authorized, version-matched change under the independently stipulated authority relation, preserving other coordinates. Reevaluation establishes the permission result and the named policy's choice for that new input. None proves that ρ is true, that ω is normatively adequate, that the authority arrangement ought to be accepted, or that any participant may execute the selected option. Those are further questions with their own grounds and authorization requirements.

Literature/research exercise 13.L. *[prompt]* A model route takes Fan and Toni’s ABA explanation work through the Handbook’s ABA chapter, then the AXPLR classifier paper as its further comparison. The first route explains argumentation outcomes through structured assumptions and disputes, extending the accepted-defense question beyond our abstract certificate. The Handbook locates those explanatory objects and their relation to formal discussion rather than making every dialogue a proof. AXPLR instead explains a particular text classifier, adding a construction-specific prediction relation and separate human evaluation targets. Comparing these routes with e ’s contextual defense and the synthetic fidelity table shows why an explanation object’s semantic guarantee cannot replace evidence about an audience’s task performance. Alternative source choices are acceptable when their actual external passages are compared accurately and the explanatory target, object, chapter connection and limit are explicit.

Chapter 14

Exercise 14.1. (a) The decision interface receives a set of accepted argument identifiers E , the conclusion map and the declared con arguments, and returns *allow* or *withhold* using δ . On the supplied graph it returns *allow*. It establishes satisfaction of that rule, not institutional release. (b) The fidelity interface receives paired classifier and argument-route outputs on the same five comparison inputs and returns their agreement fraction and its threshold comparison. It supports a claim about those paired outputs on those cases. (c) The audit interface receives versions, graph, accepted set, candidate, explanation, certificate and individual checks; the caller adds its release decision and output. It returns a retained record that can be inspected and replayed. Recording unsupported information does not certify its truth, and the log alone does not enforce withholding.

Exercise 14.2. The three attacks are (a_3, a_2) , (a_2, a_1) , (a_2, a_4) . In the first the target is outside E ; in the other two the source is outside E . Thus no attack has both endpoints in E , which is exactly conflict-freeness. The support links are irrelevant to this check under the declared contract.

Exercise 14.3. a_3 attacks a_2 and has no attacker. Therefore the first grounded iteration accepts a_3 without relying on a_1 or a_4 . No set reached by that iteration attacks a_3 , so a_2 is never defended against it. The later acceptance of a_1, a_4 uses a_3 ’s attack on their attacker a_2 ; it is not a circular reason for a_3 ’s own status.

Exercise 14.4. The links are explanatory support. Evaluation uses only (A, R^-) and δ uses the accepted set, conclusions and attacks on the con argument. Deleting R^+ therefore leaves the iteration, E and *allow* unchanged, while removing two links the explanation could display. Semantic support would change evaluation; decision support would enter the decision rule. Neither occurs in the principal service.

Exercise 14.5. Write $recommend(o) \Rightarrow exists(Cert_o)$. If recommendation occurs, a certificate must exist; the implication does not say that existence suffices. For example, supply a certificate naming graph version f_0 while the actual graph is f_1 . The object exists, but its binding check fails. The caller therefore withholds certified release. Its stronger implication is $released(o) \Rightarrow (c_{prem} \wedge c_{route} \wedge c_{cert})$ under the stated caller, with all checks evaluated on the actual returned record.

Exercise 14.6. There are $5 - 2 = 3$ agreements, so $Fid = 3/5 = 0.6 < 0.8$. The sampled faithfulness threshold fails. This comparison does not by itself show an error in the argument-route computation: the two methods may answer different models, and either could be responsible for disagreement with an intended target. A separately checked route can remain correct for its stated graph and rule. The five comparisons establish no agreement fraction on untested inputs and no claim about human understanding. Our three-check policy release remains a different decision; a service promising the sampled fidelity standard cannot attach that failed claim to its output.

Exercise 14.7. $B_i accepted(a_3)$ expresses the specified agency belief about status. No supplied rule derives truth from B_i , and accepted status is itself relative to the extracted graph and chosen semantics. Even a correct belief that $a_3 \in E$ does not prove that its medical premise is true in the world.

Exercise 14.8. One complete answer chooses the missing medical-premise entry. The required lookup has no matching supported record, so $c_{prem} = 0$. With a valid unchanged route and certificate the remaining checks are 1, but the conjunction fails and the caller returns no certified recommendation. An allowed display can say “Diagnostic only: this graph computed *allow*; medical-premise support is missing; certified recommendation withheld.” It reports the failed responsibility rather than releasing advice under another label.

Alternatively, choose fidelity 0.6 below 0.8. The sampled-faithfulness claim fails, while the route explanation and the separate required policy checks must be assessed on their own evidence. A labelled display such as “Argument-route explanation; sampled classifier faithfulness threshold not met” avoids the failed claim. It cannot serve as permission to bypass a failed premise, route or certificate requirement. A valid response names the particular claim affected and follows the declared policy; saying merely “warn” without specifying the output’s status is incomplete.

Exercise 14.9. Evaluation now receives the support relation as well as attacks, bases and λ , and returns a function $s : A \rightarrow [0, 1]$ instead of $E \subseteq A$. For the given acyclic graph the numerical calculation is internally defined. To issue a recommendation, specify a map from that score vector and any additional declared policy inputs to *allow/withhold* (or another explicit output type). A threshold could be one new policy, but none is prescribed by the study. Substituting s into the old membership rule or silently treating all positive scores as accepted invents an interface.

Exercise 14.10. A hard constraint excludes a disallowed output by the system’s definition. A norm prescribes conduct that can nevertheless be violated, leaving a question of detection and response. A monitor policy maps specified check results to permitted handling of outputs. Here no-certificate/no-recommendation is treated as a monitor-policy requirement implemented by the separate `ReleasePolicy` caller. That caller enforces the stronger conjunction of successful premise, route and certificate checks. The computational routine returns diagnostic information; recording a failure there is not the enforcement step. The same sentence could be used as a norm in a different institutional design, but that is not its assigned role here.

Exercise 14.11. (a) Both products are empty and equal 1, so the isolated score is $(1 - (1 - 0.8)) = 0.8$. (b) With no supporters the positive term is 0.8; the attacker factor is $1 - 1 = 0$, giving score 0. (c) The topological calculations are $s(a_3) = 0.8$, $s(a_2) = 0.16$, $s(a_4) = 0.672$, and

$$s(a_1) = (1 - 0.2 \cdot 0.6 \cdot 0.664) 0.84 = 0.92032 \cdot 0.84 = 0.7730688.$$

(d) Without the two support links the first three scores do not change and $s(a_1) = 0.8 \cdot 0.84 = 0.672$. (e) Acyclicity provides an order in which every required predecessor score is already known; the procedure specifies no solution rule for mutually recursive cyclic equations. For overlapping roles, a single changed score can both increase the positive term and decrease the attacking product. For example with $b = 0$, $\lambda = 1$ and one predecessor occupying both roles, the target score as a function of its predecessor’s score t is $t(1 - t)$: it rises from 0 to $1/4$ as t goes from 0 to $1/2$, then falls to 0 at 1. Each separate-coordinate monotonicity claim remains correct; neither applies to this joint change.

Exercise 14.12. (a) The named result contains F , accepted arguments E , candidate decision, explanation, candidate certificate, each check result and audit record. The caller separately adds its release state and output. Omitting E would omit the accepted arguments promised to the caller; returning *allow* without its release status would conceal the distinction being tested. (b) Only the premise condition fails: $(c_{pre}, c_{route}, c_{cert}) = (0, 1, 1)$. The returned candidate remains *allow*, with $E = \{a_1, a_3, a_4\}$, but release is withheld. The diagnostic record identifies e_3 as the failing input. (c) On the mutual pair, $\{u\}$ is conflict-free and defends u against v by attacking v . Thus it is admissible. (d) Neither argument is unattacked, so $\mathcal{F}(\emptyset) = \emptyset$ and the empty-start iteration is $\emptyset, \emptyset$. Its grounded extension is empty, not $\{u\}$. A grounded certificate must check the selected empty-start route and fixed point, not merely an admissibility property of a candidate. (e) Return “No certified recommendation: computation timed out before a complete checked grounded route”, with whatever partial iteration and checks are actually available marked partial. Withhold the certified output. A different graph or semantics would require a new declared task and cannot repair the missing original certificate by relabelling its result.

Exercise 14.13. (a) Budget 2 permits both offered stopping traces; inquiry has score $(1, 1, -2)$ and wins over withdrawal's $(0, 0, -1)$ on exposure. Its response policy supplies B's negative justification, with certificate $(CR4, CR5)$; A's assertion is retained, countergrounds are exposed and no reply is pending. With budget 1 only CR6 withdrawal meets the stopping condition. The single question is legal under CR4 but creates pending $(B, \chi, -)$ and has neither withdrawn the assertion nor exposed the response. It is an unfinished prefix, not a completed lower-cost inquiry. The chosen plan does not prove B has actually uttered its policy-supplied reply. (b) K's record matches d, v_1 , row $(speed, T, C)$, RW , expected 3, proposed 2 and the supplied authority/ground/warrant/domain conditions. Create v_2 with only that coordinate changed, rerun permission to obtain $\{R, C\}$, then select C under order C, T, R and R under order R, C, T . U's otherwise identical attempt on current v_1 fails authority. No update occurs; permission stays $\{R\}$ and both selectors return R. (c) The record names cost selection but returns R when C is permitted and precedes R in that order. It is not enough that R is permitted. Recompute selection to return $(v_2, \{R, C\}, cost, C)$ or change the stated task and identify the independently chosen priority policy honestly. (d) Execution needs an explicit authorization for the selected act in its actual context, including who can commit the resources or issue the order. The present authority is confined to one weight coordinate. Public justification records grounds and responsibility, not execution permission. No such further authorization is supplied, so this task does not dispatch.

Exercise 14.14. The report $A' = \{p \wedge q\}$ has the same closure as $\{p, q\}$. For white-list acceptance, $T \cap Cn(A') = \{p \wedge q\}$, whose closure includes the reported conjunction. Thus $\alpha_T(A') = \{p \wedge q\}$, not the different output set $\{p, q\}$. For black-list acceptance, the same three remainders are $Cn(p)$, $Cn(q)$ and $Cn(p \leftrightarrow q)$. Each has a model falsifying the conjunction (10,01,00 respectively). No remainder contains that sole reported formula, so their intersection followed by extraction gives $\beta_D(A') = \emptyset$. No AF or extension is an input to either operation, and identity checks do not compute consequence or remainders. Nor does identifying a purported issuer prove the report true. The few displayed acceptance values do not uniquely identify a real person's hidden trust; the source's idealized complete acceptance functions and representation results have a different domain. These calculations also supply no temporal update policy.

Exercise 14.15. (a) A sensor-encoding error concerns the mapping from measured/input data to the represented propositions. Compare the actual supplied measurement and its intended unit/meaning with the encoded value and argument premise; a correct solver cannot repair the wrong premise by itself. (b) An unavailable follow-up locution concerns implementation coverage. Compare the user's intended move with the implemented locution set described in S9.2, rather than assume every move in the formal EQR account has been implemented. Test whether the interface accepts that move and produces its specified state change. (c)

Improved understanding is an empirical claim. Specify a population, explanation task and measure of understanding, then inspect a suitable human comparison study. The supplied account supports the theory/implementation distinction; it does not supply that benefit claim. Reporting the missing evidence is a complete answer, not a reason to invent a study.

Literature/research task 14.L: worked route and criteria. *[prompt]* One route uses Gordon’s requirements chapter, the Handbook dialogue-application chapter and the epistemic-planning chapter. A five-sentence synthesis is: “Requirements analysis asks what a service must represent, compute and expose; dialogue applications ask how argumentative interaction is realized in an application; epistemic planning asks how actions alter information. A policy-service release record and a dialogue state therefore answer different interface questions, while an information-changing action requires a representation of its effects. The CONSULT/EQR discussion distinguishes the described implementation from properties of its broader dialogue account. These sources suggest separate requirements for the policy recommendation, the response service and a later request for missing information. Neither the implemented dialogue subset nor the fictional release calculation proves patient understanding or a general plan for obtaining correct evidence.” A legitimate alternative identifies actual passages from its named versions, attributes only their supported result or proposal, names a concrete transfer and explains the extra assumptions needed. A catalogue of source titles, unsupported empirical benefit or unqualified transfer of a theorem does not meet the task.

Chapter 15

Exercise 15.1. The complete pre-preference attack relation is

$$\{(A_1, A_2), (A_2, A_1)\}.$$

Both conclusions concern evidential citation and are contradictory. The priority of r_2 over r_1 prevents the lower-priority A_1 attack from succeeding as defeat, leaving exactly (A_2, A_1) . A_3 concludes permission-relevant grounds for a different action, *discussRetraction*(p_0); that conclusion is not the contrary of either other conclusion. It has no attack or defeat edge. Its acceptance supplies support for discussion, not permission without the certificate.

Exercise 15.2. At C_1 , $E = \{A_2, A_3\}$ and evidential support is $S(x_e) = \{A_1\}$. Their intersection is empty, so $J = 0$. Retraction rules out the evidential permission case, and an evidential purpose cannot satisfy the discussion exception, so $Perm = 0$. With $Safe = 1$, $U = 1$ the first branch already blocks. Supplying k_1 does not change this: it names a different proposal and a discussion purpose, and even a certificate with an evidential proposal’s id would still fail the required discussion-purpose conditions. It also does not change the graph or J .

Exercise 15.3. The full runtime graph has vertices $\{A_1, A_2, A_3\}$ and sole defeat $A_2 \rightarrow A_1$. From empty the characteristic function returns the unattacked $\{A_2, A_3\}$, then the same set. A_1 is not defended against unattacked A_2 . The evidence support set $\{A_1\}$ has empty intersection with this grounded result. That establishes failed justification; retraction and the evidential purpose separately establish failed permission. The supplied safety and exact intent pass. “The paper is rejected” conflates a source object, an argument status, an evidence-purpose restriction and a possible authorized discussion. The latter still has accepted support in A_3 .

Exercise 15.4. The stipulated retraction and emitted evidential citation violate the norm’s required $\neg \text{citeEvidence}(p_0)$. Because knowledge and deliberate conduct are supplied, this is a knowing intentional violation. A matching authorized marked discussion does not use the paper as evidence and does not violate this norm; it satisfies an exception to the broader citation policy. To interpret the violation as a reform attempt, further facts must establish a protest or reform purpose and the claimed reason for challenging the rule. To interpret it as misleading conduct, further facts must support that intention instead. The monitor establishes none of those mental or institutional facts from the citation alone, and the categories need not be disjoint or exhaustive.

Exercise 15.5. (a) With the strict rule and no observation, declared arguments are R_p, R_s, R_d and the sole defeat is $R_s \rightarrow R_d$. The grounded result is $\{R_p, R_s\}$, projecting to $\{p, q\}$. (b) Receipt adds axiom R_q and defeat $R_q \rightarrow R_d$. The grounded result is $\{R_p, R_s, R_q\}$ and the same conclusion set $\{p, q\}$. (c) Without either strict rule or observation, R_p, R_d are unattacked; the grounded result is both, projecting to $\{p, \neg q\}$. (d) Without the strict rule but with the observation, the sole defeat is $R_q \rightarrow R_d$ and grounded returns $\{R_p, R_q\}$, projecting to $\{p, q\}$. In all rows the listed extension is reached in one step from empty and then fixed. The direct route’s strict closure contains q in (a),(b),(d), so it does not add the default conclusion; in (c) it closes to $\{p\}$ and adds $\neg q$. The results agree on this family. The default rule is never deleted; in three rows only its argument’s conclusion fails acceptance. The strict rule likewise remains a rule, not a separate member of the conclusion set.

Exercise 15.6. First fix x_d, C_1, Π, k_1 , compute the runtime graph and extension and append $r_1 = (x_d, C_1, \Pi, F_1, \{A_2, A_3\}, S_1, k_1)$. Run the monitor with $\text{Safe} = 1$: accepted support, permission and exact audit hold, so record **pass**. O1 also checks the current receipt and template, then permits dispatch. (a) A supplied successful outcome says the visibly marked current notice was delivered, including its locator. Append that result; together with the stipulated delivery evidence it establishes the user received the notice. (b) A supplied tool failure instead gives an outcome record saying delivery failed. Dispatch and prior pass still occurred, but receipt by the user did not. A no-certificate variant can append an intent recording the absence, but permission is 0, so record **block** and stop before dispatch. An intent, monitor pass or dispatch request alone establishes no successful delivery.

Exercise 15.7. One complete policy-adviser state is

$$O = (B, G, F, E, x, Trace, c, \Pi),$$

where B retains input u_1 and the four premise/provenance records, G is a supported late-filing recommendation, F is graph f_1 from Chapter 14, $E = \{a_1, a_3, a_4\}$, x is candidate *allow* under d_1 , $Trace$ is the computation/check/release audit, c is the bound empty-start iteration certificate, and Π requires the three successful checks. The output type is either *Certified(allow, E, explanation, c)* or *NoCertifiedRecommendation* with audit. Require that every output of the first type has $c_{prem} = c_{route} = c_{cert} = 1$ on the retained u_1, f_1, d_1 inputs. Replay and binding checks have the graph, set, rule and certificate as inputs and Boolean results as outputs. The caller branch proves this local obligation under the supplied check meanings. Merely naming O does not force a caller to test those fields or to withhold on failure. An alternative complete answer may use other declared names or add a well-typed currentness policy, but must state the objects, verification rule and enforcement point rather than call the tuple an assurance proof.

Exercise 15.8. Chapter 14’s policy case asks what a service returns and what releases a recommendation: the same grounded *allow* can be retained diagnostically while unsupported premise evidence blocks release. It reuses semantic evaluation and contextual explanation for the new job of component responsibility. Chapter 15’s retraction case asks how receipt and time affect a later use: unchanged accepted conclusions at C_2 still require a current certificate and intent. It reuses those checks and Chapter 13’s version sensitivity for the new question of current dispatch and remaining goal. These are related system questions within Part III, not a claim that agentic reasoning abandons the earlier mathematical and public/practical methods.

Exercise 15.9. The certificate fields are proposal id, context version, policy version, paper, purpose, warning, scope and reviewer. For a valid exception, the proposal cites the current retracted paper; all four identifying fields match their current counterparts; both purpose fields say *discussRetraction*; both warnings are true; scope is discussion-only; and the reviewer is authorized by the current policy. These conditions are conjunctive and separate from intent membership. For all three mutations $J = 1$ and $Safe = 1$. (a) Wrong context gives $(Perm, U) = (0, 1)$ with its fresh intent, hence block. (b) Wrong paper also gives $(0, 1)$, hence block. (c) Correct k_1 with empty log gives $(1, 0)$, hence the citation’s audit block. Logging an invalid certificate establishes only that this exact invalid object was recorded. It does not change the field comparisons in *ValidEx* or grant permission. No citation warning branch applies.

Exercise 15.10. A complete constructed example fixes an institution’s norm: “Applications must be filed on paper.” An accessibility assistant has read the norm, knows that the applicant cannot use the required paper format, submits a digital alternative anyway, and publicly records: “I knowingly depart from the format rule to demonstrate its exclusionary effect and request an accessible filing route.” The

norm and alternative act establish the stipulated violation; the supplied knowledge establishes it is knowing; the explicit request supplies a reform purpose. This is a plausible miniature civil- disobedience interpretation. It neither proves the digital filing lawful nor endorses the violation, and it does not infer intent from file format alone. An adequate alternative supplies all four requested facts and keeps the institution's eventual response open.

Exercise 15.11. Define $Progress_3(Trace) = 1$ if at least one of the last three recorded steps records either acquisition of a newly eligible evidential source or an acknowledged request for missing information; otherwise it is 0. First run the original action monitor. If it blocks, preserve block regardless of progress. For an otherwise permitted handling step, record a progress warning when $Progress_3 = 0$, directing review of the remaining goal; do not convert the warning into authority for a blocked citation. The supplied three events contain neither qualifying event, so the condition is 0 and the additional progress warning applies. Append an acknowledged request as step 4: the new window is steps 2–4, containing that request, so the condition becomes 1. The acknowledgement still supplies no eligible source or evidential answer. Repeated acknowledged requests can keep the window test true indefinitely while every requested source fails to arrive; this is a counterexample to inferring either goal completion or general liveness. A complete alternative window convention must declare its bounds, observable qualifying events and response, evaluate both traces, and preserve the original blocking precedence.

Exercise 15.12. (a) B_0 gives vertices R_p, R_d , no defeats, accepted both and $\Gamma_0 = \{p, \neg q\}$. Receipt adds observation q , tree R_q and only defeat $R_q \rightarrow R_d$. The new accepted set is $\{R_p, R_q\}$ and $\Gamma_1 = \{p, q\}$, with default d retained. (b) For each stage, the record id is the current pointer, issuer is publisher-service and paper matches $m_0.paper = p_0$; the memory is one of the declared states. At the positive notice the literal is $retracted(p_0)$ and $q \in Obs$. Recomputed conclusions have exactly one status sign and agree with $s_0.retracted = 0$ or $s_1.retracted = 1$. Copying fixed recent/relevant flags and the assigned versions gives precisely C_0, C_1 . The accompanying receipt retains the checked route and source, not action authorization. (c) At C_1 , runtime grounded is $\{A_2, A_3\}$. Evidence with no exception has $(Safe, Perm, J, U) = (1, 0, 0, 1)$ and blocks; supplying the discussion exception does not make evidence permitted or justified. Discussion without the exception has $(1, 0, 1, 1)$ and blocks. Discussion with k_1 and exact current intent has $(1, 1, 1, 1)$ and passes, still subject to O1 for dispatch. (d) Accepting q identifies status, while purpose is the separately requested use, reviewer authority is a supplied policy fact, and source reliability is A1–A2. No one of these is inferred from the others. A notice whose paper field names another paper necessarily fails the paper equality and returns *handoff_error*, so it creates no authorized citation handoff. If its literal is also changed to that other paper, the literal equality fails too; changing only the paper field need not change the literal.

Exercise 15.13. (a) The observation set, revision arguments and defeats, accepted conclusions, runtime graph, grounded extension and action support are unchanged. The receipt identifier, document version, locator, provenance envelope, current pointer and context version change. Renewing authority and intent is an additional task, not a consequence of unchanged acceptance. (b) $J = Safe = 1$ in all rows. Old certificate and old intent give $(Perm, U) = (0, 0)$; old certificate with exact new intent gives $(0, 1)$; new certificate with only the old intent gives $(1, 0)$; new certificate and exact new intent give $(1, 1)$. The first three block and the last passes. (c) Replaying $(x_d, C_1, \Pi, k_1, [r_1])$ to the raw stateless monitor still passes. Its fields are mutually consistent and it does not know the caller's current pointer. O1 forbids dispatch at stage 2 because the associated receipt is s_1 while current is s_2 . This shows the role and limit of the trusted caller, not autonomous freshness detection by the checker. (d) Even a renewed, delivered discussion supplies status information rather than the original eligible-evidence answer. The only supplied candidate remains retracted, so the original goal is open.

Exercise 15.14: a complete nonprogramming design. (a) Receive the supplied m_0, s_0 and initialize B_0 . The receipt component checks the supplied record fields; on s_1 it admits q under A1–A2 and updates only *Obs*. The revision evaluator returns a graph, grounded argument set and conclusion projection. The partial adapter consumes that memory, current source record and fixed metadata and returns (C, P) or a handoff error. The proposal constructor consumes the requested purpose and current context, using the two fixed templates. The reviewer supplies the discussion exception. The caller computes runtime support, appends exact intent, obtains and records the monitor result, checks O1's receipt/template conditions, dispatches only after monitor pass and the O1 current-receipt/template checks, and then records the supplied outcome and remaining goal. Each arrow names a typed record and a conditional operation; it does not equate receipt with belief or monitor pass with execution.

(b) The revision objects are $F_{rev,0} = (\{R_p, R_d\}, \emptyset)$ and $F_{rev,1} = (\{R_p, R_d, R_q\}, \{R_q \rightarrow R_d\})$. The runtime objects are $F_{act,0} = (\{A_1\}, \emptyset)$ and $F_{act,1} = (\{A_1, A_2, A_3\}, \{A_2 \rightarrow A_1\})$. At stage 1 retain the pre-preference attacks $A_1 \leftrightarrow A_2$ and priority $r_2 \succ r_1$ as separate construction data. The relation sets just displayed are defeats for semantic evaluation. The two families' identifiers and conclusions differ.

(c) Grounded evaluation gives revision sets $\{R_p, R_d\}$ and $\{R_p, R_q\}$ with projections $\{p, \neg q\}$ and $\{p, q\}$; the runtime sets are $\{A_1\}$ and $\{A_2, A_3\}$. In each case the displayed set is reached from empty and is fixed. The adapter checks source agreement and returns C_0 then C_1 . The original evidence proposal is supported and permitted at C_0 , but fails both after receipt. Discussion at C_1 has accepted support and, with k_1 and current intent, passes all four monitor conditions. These are computations on the declared inputs, not claims that the notice was authenticated by the graph.

(d) A contextual explanation for the changed evidence result names

$$(F_{act,1}, \text{grounded}, A_1, \text{out}),$$

retains all three runtime arguments and the priority-filtered defeat, and supplies the checked witness $\{A_2, A_3\}$. Unattacked A_2 defeats the sole evidential support A_1 ; A_3 instead supports a marked discussion. The provenance receipt ties the retracted flag to s_1 and its locator. The explanation then distinguishes this support loss from the separate policy prohibition on evidential use and from the authorized discussion exception. Merely showing R_q -in would not answer the action question.

(e) Require $\text{Safe} \wedge \text{Perm} \wedge J \wedge U$ for every passed proposal and current receipt/template agreement for O1 citation dispatch. The monitor's branch argument establishes the first implication. The caller condition establishes that dispatched citations also meet its current-record and template checks. These are conditional guarantees under the reliable inputs, immutable versions and trusted log/caller assumptions. They establish neither global tool safety nor the truth of arbitrary source content.

(f) Remove k_1 but retain the discussion proposal and append a fresh intent recording no exception. Then $J = 1, U = 1, \text{Safe} = 1$, but $\text{Perm} = 0$, so block and do not dispatch. Alternatively, removing the intent alone leaves valid permission but $U = 0$ and also blocks. A complete failed variant identifies a concrete changed input and exact failed conjunct rather than just labelling it unsafe.

(g) Receive s_2 before dispatch. Keep Obs , both graphs and conclusions unchanged, update provenance and context to C_2 , obtain reviewer-supplied k_2 and append its exact current intent. Recheck the four conditions and O1; the current discussion template uses the new locator. A whole-old-bundle raw pass cannot bypass O1's current-record test. If renewal is unavailable, withhold citation and record that unmet condition instead of editing the reviewer field on the agent's own authority.

(h) Inference dominates the supplied system's formal development: a received observation defeats a default, and accepted runtime arguments support or oppose citation purposes. Explicit policy and authorization determine which supported response can be dispatched. This is not a numerical balancing model; no alternative/foil weights are supplied. Public dialogue could make a request for grounds answerable, but none is required to compute this finite trace. The successful supplied outcome delivers a current, authorized notice explaining why the paper is not used as evidence. The original scientific question still has no eligible evidential answer because its only candidate is retracted. Thus the system achieves a useful status-information subgoal and identifies a need for further evidence without pretending that the original goal is done.

Criteria for alternatives. A design based on another supplied book case must answer all eight parts with complete finite inputs. It must distinguish its argument identifiers from claims and actions, name its evaluation and output extraction, locate verification and enforcement, trace a real failed variant and a real update, and provide

a contextual explanation and honest remaining-goal assessment. Additional local policies are allowed if expressly stipulated and used consistently. An architecture picture alone, external factual assumptions with no evidence, or a claim that a passing local check assures the whole system is incomplete. No programming artifact is needed to meet these criteria.

Literature/research task 15.L1. *[prompt]* ReAct’s language steps update the trajectory’s context without themselves obtaining an external observation; environment actions can return observations. A transcript therefore records a generated sequence of proposed reasoning, action and feedback. It is not by itself a proof that a specified reasoning method has been preserved. In the supplied RAD manuscript, an exact alignment claim needs stated common input/output types, two defined routes and equality of their outputs over a specified domain; partial and approximate alignment are different relations. The chapter proves only its finite revision-family agreement. The composition discussion is outlook, not a theorem for this entire handoff. For dispatch here, the evidence must additionally include the current source-linked context and receipt, accepted action support, purpose-bound reviewer authorization, supplied successful safety check, exact intent and matching template. The caller enforces those particular conditions. Neither an articulate rationale nor route agreement establishes source truth, authentic identity, currentness independent of the caller or success of the original goal. A complete answer identifies the two sources’ distinct interface claims and then cites the local dispatch evidence without ranking empirical agent performance.

Literature/research task 15.L2 and research capstone. *[prompt]* One five-sentence route uses supplied RAD, the Handbook epistemic-planning chapter and ReAct: “The supplied RAD manuscript asks how a direct reasoning route relates to an explanatory one; epistemic planning asks how actions change information; ReAct supplies an interleaved language/action/observation interface. A typed route relation, an information-changing action model and a generated trajectory are consequently different objects. Exact alignment requires equality on a stated domain, whereas a transcript alone exhibits a trajectory rather than proving that relation. Our finite revision family provides a small alignment example, and our receipt-to-action caller adds separate currentness and authorization conditions. Neither that finite agreement nor a generated trajectory establishes general liveness or completes the eligible-evidence goal.” A successful alternative uses the named exact source versions, accurately distinguishes theorem, proposal and implementation, and identifies one concrete local transfer with its assumptions and limit.

For a research capstone, retain the eight-output solution contract above and add a source-evidence record: identify which domain facts and formal definitions come from which passages or data, distinguish those from pedagogical stipulations, and state which empirical claims remain untested. A complete defensible example may investigate whether a chosen institution actually supplies a source-status record and authorization role comparable to s_1 and reviewer 1, while keeping the

chapter's synthetic values as the worked finite instance. If the external evidence is absent, report the missing institutional premise rather than presenting a fictional authorization as fact. This is guidance for assessing an open investigation, not an assertion of such evidence for a particular institution.

Reader reference

These pages lead back to the book’s explanations. A summary here does not remove the hypotheses of a definition or identify objects that have different jobs. Use the chapter or section link when a calculation depends on the distinction.

Foundational definition spine

The opening supplies a concept when it first becomes useful; a later treatment may give it a more specific mathematical form. The following routes connect those encounters rather than assigning every term one universal definition.

Concepts	Preparation, formal treatment and return
Logic and proof	Truth and consequence , quantifiers , set orders and proof habits in Chapter 0; source-method choices in Chapter 1 ; output-family comparison in Chapter 7 .
Reasons and acts	Source-associated arguments in Chapter 2; content , warrants and acts in Chapter 3; inferential , public and practical questions in Chapter 4.
Structured arguments	Premises , rules and the ambient signature ; attacks ; defeats and preferences ; rationality of outputs .
Abstract semantics	The local stable test is used for representation in Chapter 1. Chapter 6 develops defense , extension semantics , grounded iteration and labellings and acceptance .
Representation and principles	Hard premises and extraction ; whole-family correspondence ; construction requirements ; principles of methods .
Dialogue	Early public responsibilities in Chapter 4 ; the records and transition rules of Chapter 9; trace checking and bounded continuation ; explanatory exchange .
Issues and practical choices	Comparative grounds in Chapter 4; premise roles , issue standards and two-coordinate comparisons in Chapter 10; permission then selection ; explanation and authorized revision .
Computation and explanation	Checking versus justifying ; representations and algorithms ; a query-specific reduction ; targets and contextual certifi- cates .
Systems and action	Computation , checks and release ; report admission ; adapters ; execution and remaining goals ; currentness and renewal .

Alphabetical glossary

Admissible set. A conflict-free set that defends each of its members. [Chapter 6](#), p. 54.

Agentic argumentation system. A system in which reasoning is connected to receiving information, revising represented inputs and acting. The book's finite case specifies each of these connections separately. [Chapter 15](#), p. 170.

Argument. A reason offered or reconstructed in an argumentative act; a structured derivation in the formal construction; or a node when internal structure is abstracted away. These uses are related but not identical. [Chapter 3](#); [Chapter 5](#); [Chapter 6](#).

Argumentation framework. An abstract framework is a pair (A, R) of arguments and directed attacks. Extra source, support or procedural information belongs to the specified enriched object. [Chapter 6](#), p. 53.

Attack. A directed challenge. Structured attacks identify a premise, conclusion or rule application as their target; the abstract relation retains the selected endpoints. An attack need not succeed as a defeat. [Chapter 5](#); [Chapter 6](#).

Attack assignment. A construction that assigns attacks to the arguments of an input. A condition on such assignments compares constructions across inputs, not only the extensions of one graph. [Chapter 8](#), p. 79.

Authorized revision. A permitted change to a specified versioned record by an identified authority, followed by the required reevaluation. A legal utterance alone is not a weight update. [Chapter 13](#), p. 150.

Balancing. Practical comparison of options through reasons and their specified force. The book's numerical model distinguishes justifying from requiring weight and tests an option against its foils. Issue proof standards are a different evaluative mechanism. [Chapter 4](#); [Chapter 10](#).

Burden of persuasion. The responsibility to establish an issue under the applicable proof standard. It differs from the task of making that issue live by producing material. [Chapter 10](#).

Burden of production. The responsibility to introduce sufficient material for an issue to be considered. [Chapter 10](#).

Certificate. A witness or record that a specified checker can inspect. Its guarantee depends on the checked claim: stable membership, grounded computation, contextual defense and release are different targets. [Chapter 0](#); [Chapter 12](#); [Chapter 13](#); [Chapter 14](#).

Claim and conclusion. A proposition at issue and, in an argument, the proposition supported by its premises and inference. The act of asserting it is distinct from its content. [Chapter 3](#); [Chapter 5](#).

Closure. The consequences of a set under a stated consequence operation or rule set. Closure of selected conclusions under strict rules is an output requirement, not a choice of which conclusions to accept. [Chapter 0](#); [Chapter 8](#).

Commitment. A participant's recorded public commitment in a specified protocol. Keep the local dialogue's active records and commitment ledger separate from its pending-response field. [Chapter 9](#).

Complete extension. An admissible set containing every argument it defends. [Chapter 6](#).

Conflict-free set. A set containing no pair x, y , possibly equal, with xRy . [Chapter 6](#).

Consequence relation. A relation specifying which formulas follow from given premises under a particular reasoning method. [Chapter 0](#); [Chapter 1](#).

Contrastive explanation. An explanation organized around why an observed result holds rather than a specified alternative. The target and retained context determine the relevant comparison. [Chapter 13](#).

Credulous acceptance. Membership in at least one extension of the stated semantics. [Chapter 6](#); [Chapter 12](#).

Currentness. Agreement between the context to which a checked record belongs and the current context of its proposed use. An unchanged belief need not make an old bundle current. [Chapter 15](#).

Defeasible rule. A rule whose application supports a conclusion that can be challenged in the stated argumentation construction. [Chapter 5](#).

Defeat. An attack that succeeds under the specified preference or priority policy. Where comparison is local, the target is the attacked subargument. [Chapter 5](#).

Defense. In an AF, a set defends an argument by attacking every attacker. The source-associated conditions of Chapter 2 add restrictions on who supplies the defenses; they are not the unqualified AF definition. [Chapter 2](#); [Chapter 6](#).

Dialogue. Interaction governed by a specified protocol. Moves, public records, commitments and pending replies cannot in general be read off from a static extension. [Chapter 4](#); [Chapter 9](#).

Dung semantics. Methods for selecting extensions or labellings from an attack framework using conflict, defense and acceptability. [Chapter 6](#).

Enrichment topic. A further reading or study route whose role is specified locally. Optionality does not make an ordinary task externally dependent. [Introduction](#); [reading routes](#).

Exception. In the Carneades treatment, a premise role naming a blocking circumstance. The holding condition depends on the statement's status and acceptability; an exception premise holding permits the argument's use rather than proving the exceptional circumstance. [Chapter 10](#).

Explanation target. The result to be explained together with the input, operation or semantics, query and result type that make the request determinate. Public and practical explanations retain their own context. [Chapter 13](#).

Extension. A set of arguments returned by an extension semantics. The family of such sets is another object; $\emptyset$ and $\{\emptyset\}$ must not be confused. [Chapter 6](#); [Chapter 11](#).

Factor. A represented feature used as a reason for or against a practical option, under a specified comparison. [Chapter 10](#).

First-reading route. A route through the connected argument before selected proof reconstructions, transfers or further reading. It does not change which material is available in the book. [Reading routes](#).

Grounded extension. The least complete extension. For the finite AFs treated here it is obtained by iterating the defense operator from the empty set. [Chapter 6](#).

Hard constraint. A requirement whose enforcement prevents a release or action when it fails, under the specified caller or policy. Logging a failed check is not itself enforcement. [Chapter 14](#); [Chapter 15](#).

Hard premises. Premises held fixed within a stated source-method comparison. Hardness is a role in that method, not a claim of unrevisable truth. [Chapter 1](#); [its representation](#).

Knowledge base. The represented formulas or other specified inputs to a reasoning method. A structured theory also has a language, contrary information, rule sets and other explicitly declared components. [Chapter 0](#); [Chapter 5](#).

Labelling. Assignment of in, out or undec under a stated semantic definition. Being outside an extension alone does not establish the out label. [Chapter 6](#).

Last link. Comparison through the selected set of last defeasible rules, LastDefRules, under the chapter's corrected lifting and stated firm, rule-only restrictions. There need not be one last rule. [Chapter 5](#).

Maximal consistent subset. A subset consistent with the stated hard premises that has no larger consistent superset within the given base. It need not have maximum cardinality. [Chapter 0](#); [Chapter 1](#).

Monitor. A checker of specified conditions on a proposed action and its current bundle. A pass, dispatch, supplied execution outcome and achievement of the goal remain different events or claims. [Chapter 15](#).

Nonmonotonic consequence. Consequence for which adding premises can withdraw a previous conclusion. Which additions do so depends on the source method. [Chapter 0](#); [Chapter 1](#).

Obligation. A responsibility specified by the relevant public or normative procedure. In the local dialogue, a pending response is a particular state component, not a general moral obligation. [Chapter 9](#); [Chapter 15](#).

Permission. In the practical model, survival of every required option–foil comparison. The permitted set does not itself choose an option. [Chapter 10](#); [Chapter 12](#).

Pipeline certificate. In the policy service, evidence attached to a particular computed result and its individual checks. Its precise grounded claim and the caller’s release decision have separate contracts. [Chapter 14](#).

Preferred extension. An inclusion-maximal admissible set. [Chapter 6](#).

Proof standard. The specified criterion for accepting an issue from its currently defensible pro and con arguments. The selected kernel uses SE, BA and DV, with their stated conditions. It is not the dinner’s two-coordinate practical comparison. [Chapter 10](#).

Rebut. An attack on a conclusion of the relevant defeasible subargument under the chapter’s precise contrary and target conditions. [Chapter 5](#).

Reasoning-alignment diagram. A diagram specifying a source route, an argumentative route and an extraction to compare. Equality of the outputs is a further claim requiring proof under stated assumptions. [Chapter 1](#); [Chapter 7](#).

Rejected-status explanation. An explanation of a specified failure of acceptance. Failure of membership in one extension, failure of credulous acceptance and absence of every semantic output differ. [Chapter 13](#).

Report admission. Selection of represented content from a received report under stated assumptions. It is distinct from authenticating the source and from the later revision of a belief state. [Chapter 14](#); [Study S14.2](#).

Selection. A further policy choosing among permitted options. Two supplied policies can select differently from the same permitted set. [Chapter 12](#); [Chapter 13](#).

Semantic support. Support with a specified effect on accepted sets or another evaluative output. Its role must be distinguished from argument construction, scoring, selection and explanatory display. [Chapter 11](#); [Chapter 14](#).

Skeptical acceptance. Here, the guarded query requires a nonempty extension family and membership in every extension. The guard distinguishes acceptance from empty-domain universal truth. [Chapter 6](#).

Stable extension. A conflict-free set attacking every argument outside it. A framework can have no stable extension. [Chapter 1](#); [Chapter 6](#).

Strict rule. A rule used without defeasible qualification in the specified structured-argument construction. Its role is relative to that construction's language and rule set. [Chapter 5](#).

Structured argument. A derivation retaining its premises, rules, subarguments and conclusion. Sparse displayed contrary information does not replace the full ambient signature. [Chapter 5](#).

Support. A positive relation whose interpretation is specified by its use: for example, constructing an argument, evaluating a bipolar framework, accumulating a score, comparing options or displaying an explanation. [Chapter 10](#); [Chapter 11](#).

System architecture. An arrangement of components and interfaces for a specified service purpose. The policy, public-response and practical services do not require a common sequence of all three perspectives. [Chapter 14](#).

Teaching convention. An explicitly stated pedagogical construction or restriction. Its worked result does not automatically transfer to a fuller source formalism. [Introduction](#).

Tool-use trace. A specified sequence of states, actions, observations and updates. Its records require interpretation relative to the action, information and authority assumptions. [Chapter 15](#).

Undercut. An attack on the applicability of a defeasible rule, using the naming and contrary conventions of the structured theory. [Chapter 5](#).

Weakest link. Comparison through the selected defeasible-rule sets under the corrected lifting and the chapter's stated restrictions; in the finite nonempty total-rank cases this compares minimum ranks. [Chapter 5](#).

Recurring examples and formal objects

Repair and representation. Chapter 1’s $K_0 = \{p, \neg p, q\}$ supplies two repairs and six base arguments. The separate valuation example changes hard premises for $K = \{p \rightarrow q\}$. Chapter 7 returns to source choice, hard-premise filtering and extraction to compare complete output families, including its Exercise 7.2.

Source-sensitive defense and aggregation. The Alice/Bob/Cayrol graph of Chapter 2 tests source conditions on a defending set. Its separate reported paths compare graph aggregation with aggregation of semantic answers. The grounded study supplies its own iteration. Neither graph is the later computational-reuse framework.

Reconstruction and requirements. The smoking exchange in Chapter 3 separates an act, its content, a warrant and analyst-added material. Changed graph queries expose what verification can leave unexplained. The supplied extraction study retains annotations and evaluation conventions for a supplied comparison of extraction accuracy and downstream query recall, using stipulated predictions rather than measured system outputs.

Structured rules and priorities. The Scot and professor cases connect construction and local defeat, semantic choice, priority-sensitive representation and rationality requirements. Sherlock’s added route and the bounded PDL witness in Chapter 8 test properties of constructions across inputs.

Linked cycles, F_{link} . Chapter 6’s linked two-cycle and three-cycle discriminate among complete, grounded, preferred and stable semantics. The connection to labellings is part of that example’s job.

Computational reuse, F_{reuse} . The different five-node graph in Exercise 6.2 is reprinted with its full edge set and derived outputs in Chapter 12. Its conditioned components and edge update lead to contextual certificates in Chapter 13. Similar node names do not identify the two graphs.

Custody and issue evaluation. Chapter 4 distinguishes an inferential conflict, public responsibilities and reasons for an arrangement. Chapter 10 reuses the fictional facts for an issue with premise roles, proof standards and burdens. Establishing that issue is not selecting or executing an arrangement.

Restaurant interaction. The public records of Chapter 9 return as a response/completion comparison, trace check and bounded continuation, explanatory exchange and public-response service. A checked offered plan is not an observed completed conversation.

Dinner and versioned weights. Chapter 10’s option–foil calculation supplies the pairwise totals, permitted set and two selectors. Later returns test a changed alternative domain, computation, grounds and authorized revision, and version-

[bound service results](#). [Study S10.4](#) asks what licenses a weight.

Policy service and report admission. [Chapter 14](#) stipulates extracted reasons and builds a grounded computation, diagnostic checks and a release caller with separate contracts. Its [report comparison](#) and [Study S14.2](#) concern admitted content, not authentication. Existing [application evidence](#) has a separate empirical role.

Citation currentness and remaining goals. [Chapter 15](#)'s finite research-assistant case connects receipt, revision, adapter, authority and action. The [complete-old-bundle counterexample](#) and [authorized discussion's informational outcome](#) ask different questions. The capstone retains all supplied inputs and a complete paper design in its collected answer.

Notation index

Symbols have the types declared where they are used. In particular, a structured argument, its conclusion, a selected set and an extracted family are different objects even when a display abbreviates their names.

Notation	Meaning and owner
K, P, Cn	Knowledge base, designated hard premises and the specified consequence/closure operation. Chapter 1 states the roles before varying an input.
$\text{Arg, Conc, Prem, Sub}$	Constructed arguments, conclusion, premises and subarguments; the full structured signature is in Chapter 5 .
$\text{DefRules, LastDefRules}$	Selected sets of defeasible rules and last defeasible rules, used by the corrected comparisons in Chapter 5 .
$F = (A, R), xRy$	An abstract framework and a directed attack from x to y ; Chapter 6 .
$E, S, \sigma(F)$	A candidate or selected set and a family of extensions under a semantics. The local definition determines the role of E or S ; Chapter 6 and Chapter 11 .
in, out, undec	Labelling values, with conditions fixed in Chapter 6 .
$\emptyset, \{\emptyset\}$	No members versus a family containing one empty set. Chapter 0 prepares the type distinction; Chapter 11 uses it for existence.
MCS, extraction	A source-method repair family and the separately specified map from argumentative answers. Keep whole families in Chapter 7 .
DOS, Com	Public dialogue records whose types and update rules are supplied in Chapter 9 . A pending reply is recorded separately.
R^+, Score	Support or scoring notation only under its local interpretation; Chapter 11 compares operations, and Study S14.1 fixes a particular score.
$J, W, \text{option/foil}$	Justifying and requiring totals in the numerical practical comparison of Chapter 10 ; these weights are not attacks or issue statuses.
Target, Cert_o	An explanation target with query and context in Chapter 13 ; the separate service certificate Cert_o in Chapter 14 .
Trace, Mon	Trace and monitor notation under the explicit local contracts of Chapter 15 ; no universal revision or action calculus is assumed.

References

- [1] Liuwen Yu, Leendert van der Torre, and Réka Markovich. “Thirteen Challenges in Formal and Computational Argumentation”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 847–976.
- [2] Pietro Baroni, Dov Gabbay, Massimiliano Giacomin, and Leendert van der Torre, eds. *Handbook of Formal Argumentation*. London: College Publications, 2018. ISBN: 978-1-84890-275-6.
- [3] Dov Gabbay, Massimiliano Giacomin, Guillermo R. Simari, and Matthias Thimm, eds. *Handbook of Formal Argumentation, Volume 2*. London: College Publications, 2021. ISBN: 978-1-84890-336-4.
- [4] Dov Gabbay, Gabriele Kern-Isberner, Guillermo R. Simari, and Matthias Thimm, eds. *Handbook of Formal Argumentation, Volume 3*. London: College Publications, 2024. ISBN: 978-1-84890-481-1.
- [5] Phan Minh Dung. “On the Acceptability of Arguments and Its Fundamental Role in Nonmonotonic Reasoning, Logic Programming and N-Person Games”. In: *Artificial Intelligence* 77.2 (1995), pp. 321–357. DOI: [10.1016/0004-3702\(94\)00041-X](https://doi.org/10.1016/0004-3702(94)00041-X).
- [6] Pietro Baroni, Martin Caminada, and Massimiliano Giacomin. “Abstract Argumentation Frameworks and Their Semantics”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 159–236.
- [7] Wolfgang Dvořák and Paul E. Dunne. “Computational Problems in Formal Argumentation and Their Complexity”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 631–688.
- [8] Herbert B. Enderton. *A Mathematical Introduction to Logic*. 2nd ed. Academic Press, 2001.
- [9] Heinz-Dieter Ebbinghaus, Jörg Flum, and Wolfgang Thomas. *Mathematical Logic*. 2nd ed. Springer, 1994.
- [10] David Makinson. *Bridges from Classical to Nonmonotonic Logic*. London: College Publications, 2005.
- [11] Tjitze Rienstra, Leendert van der Torre, and Liuwen Yu. “Reasoning Alignment for Agentic AI: Argumentation, Belief Revision, and Dialogue”. Supplied manuscript; cited page numbers refer to this version, distinct from the published IfCoLog article.
- [12] Sarit Kraus, Daniel Lehmann, and Menachem Magidor. “Nonmonotonic Reasoning, Preferential Models and Cumulative Logics”. In: *Artificial Intelligence* 44.1–2 (1990), pp. 167–207. DOI: [10.1016/0004-3702\(90\)90143-5](https://doi.org/10.1016/0004-3702(90)90143-5).
- [13] Philippe Besnard and Anthony Hunter. “A Review of Argumentation Based on Deductive Arguments”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 437–484.

- [14] Tjitze Rienstra, Leendert van der Torre, and Liuwen Yu. “Reasoning Alignment for Agentic AI: Argumentation, Belief Revision, and Dialogue”. In: *Journal of Applied Logics – IfCoLog Journal of Logics and their Applications* 12.6 (2025), pp. 1683–1712. URL: <https://hdl.handle.net/10993/66946>.
- [15] Liuwen Yu, Xu Li, and Leendert van der Torre. “Agent Defense in Abstract Argumentation: Semantics and Principle-Based Analysis”. Supplied 52-page proof, cited by internal pagination; distinct from the published 2026 article.
- [16] Dorothea Baumeister, Daniel Neugebauer, and Jörg Rothe. “Collective Acceptability in Abstract Argumentation”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 249–298.
- [17] Liuwen Yu, Xu Li, and Leendert van der Torre. “Agent Defense in Abstract Argumentation: Semantics and Principle-Based Analysis”. In: *Argument and Computation* 17.2 (2026), pp. 141–182. DOI: [10.1177/19462174261420549](https://doi.org/10.1177/19462174261420549).
- [18] Martin Caminada and Gabriella Pigozzi. “On Judgment Aggregation in Abstract Argumentation”. In: *Autonomous Agents and Multi-Agent Systems* 22.1 (2011), pp. 64–102. DOI: [10.1007/s10458-009-9116-7](https://doi.org/10.1007/s10458-009-9116-7).
- [19] Antonis Bikakis, Giorgos Flouris, João Leite, and Theodore Patkos. “Social Argumentation Systems”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 15–66.
- [20] Frans H. van Eemeren and Bart Verheij. “Argumentation Theory in Formal and Computational Perspective”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 3–74.
- [21] Fabrizio Macagno, Douglas Walton, and Chris Reed. “Argumentation Schemes”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 519–576.
- [22] Thomas F. Gordon. “Towards Requirements Analysis for Formal Argumentation”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 145–156.
- [23] Stephen E. Toulmin. *The Uses of Argument*. Cambridge University Press, 1958.
- [24] Douglas Walton, Chris Reed, and Fabrizio Macagno. *Argumentation Schemes*. Cambridge University Press, 2008.
- [25] Cha'im Perelman and Lucie Olbrechts-Tyteca. *The New Rhetoric: A Treatise on Argumentation*. University of Notre Dame Press, 1969.
- [26] Katarzyna Budzynska and Serena Villata. “Processing Natural Language Argumentation”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 577–628.

- [27] John Lawrence and Chris Reed. “Argument Mining: A Survey”. In: *Computational Linguistics* 45.4 (2019), pp. 765–818. DOI: [10.1162/coli_a_00364](https://doi.org/10.1162/coli_a_00364). URL: <https://aclanthology.org/J19-4006/>.
- [28] Marco Lippi and Paolo Torroni. “Argumentation Mining: State of the Art and Emerging Trends”. In: *ACM Transactions on Internet Technology* 16.2 (2016), 10:1–10:25. DOI: [10.1145/2850417](https://doi.org/10.1145/2850417).
- [29] Carlos I. Chesñevar, Jarred McGinnis, Sanjay Modgil, Iyad Rahwan, Chris Reed, Guillermo R. Simari, Matthew South, Gerard Vreeswijk, and Steven Willmott. “Towards an Argument Interchange Format”. In: *The Knowledge Engineering Review* 21.4 (2006), pp. 293–316. DOI: [10.1017/S0269888906001044](https://doi.org/10.1017/S0269888906001044).
- [30] Giorgos Flouris, John Lawrence, and Dimitra Zografistou. “Searching, Navigating, and Querying Arguments and Debates: Tools, Languages and Methodologies”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 67–100.
- [31] Henning Wachsmuth, Nona Naderi, Yufang Hou, Yonatan Bilu, Vinodkumar Prabhakaran, Tim Alberdingk Thijm, Graeme Hirst, and Benno Stein. “Computational Argumentation Quality Assessment in Natural Language”. In: *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers*. Association for Computational Linguistics, 2017, pp. 176–187. URL: <https://aclanthology.org/E17-1017/>.
- [32] Blanca Calvo Figueras, Jaione Bengoetxea, Maite Heredia, Ekaterina Sviridova, Elena Cabrio, Serena Villata, and Rodrigo Agerri. “Overview of the Critical Questions Generation Shared Task”. In: *Proceedings of the 12th Argument mining Workshop*. Association for Computational Linguistics, 2025, pp. 243–257. DOI: [10.18653/v1/2025.argmining-1.23](https://doi.org/10.18653/v1/2025.argmining-1.23). URL: <https://aclanthology.org/2025.argmining-1.23/>.
- [33] Antonio Rago, Kristijonas Čyras, Jack Mumford, and Oana Cocarascu. “Argumentation and Machine Learning”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 577–626.
- [34] Liuwen Yu and Leendert van der Torre. “The A-BDI Metamodel for Human-Level AI: Argumentation as Balancing, Dialogue and Inference”. In: *Logic and Argumentation*. Springer, 2025, pp. 361–379. DOI: [10.1007/978-981-96-7956-0_22](https://doi.org/10.1007/978-981-96-7956-0_22).
- [35] Henry Prakken. “Formal Systems for Persuasion Dialogue”. In: *The Knowledge Engineering Review* 21.2 (2006), pp. 163–188. DOI: [10.1017/S0269888906000865](https://doi.org/10.1017/S0269888906000865).
- [36] Douglas Walton and Erik C. W. Krabbe. *Commitment in Dialogue*. State University of New York Press, 1995.

- [37] Elizabeth Black, Nicolas Maudet, and Simon Parsons. “Argumentation-Based Dialogue”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 511–576.
- [38] Andreas Xydis, Federico Castagna, Elizabeth I. Sklar, and Simon Parsons. “Applications of Argumentation-Based Dialogues”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 225–304.
- [39] Katie Atkinson and Trevor Bench-Capon. “Value-Based Argumentation”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 299–354.
- [40] Srdjan Vesic and Bruno Yun. “Argumentation-Based Applications for Decision-Making”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 377–428.
- [41] Katie Atkinson and Trevor Bench-Capon. “Practical Reasoning as Presumptive Argumentation Using Action Based Alternating Transition Systems”. In: *Artificial Intelligence* 171.10–15 (2007), pp. 855–874. DOI: [10.1016/j.artint.2007.04.009](https://doi.org/10.1016/j.artint.2007.04.009).
- [42] Leila Amgoud and Henri Prade. “Using Arguments for Making and Explaining Decisions”. In: *Artificial Intelligence* 173.3–4 (2009), pp. 413–436. DOI: [10.1016/j.artint.2008.11.006](https://doi.org/10.1016/j.artint.2008.11.006).
- [43] Ralph L. Keeney. “Decision Analysis: An Overview”. In: *Operations Research* 30.5 (1982), pp. 803–838. DOI: [10.1287/opre.30.5.803](https://doi.org/10.1287/opre.30.5.803).
- [44] Henry Prakken and Giovanni Sartor. “A Formal Framework for Combining Legal Reasoning Methods”. In: *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law*. Locators refer to the ten-page author PDF. ACM, 2023. DOI: [10.1145/3594536.3595129](https://doi.org/10.1145/3594536.3595129). URL: <https://webpace.science.uu.nl/~prakk101/papers/everything23.pdf>.
- [45] Pere Pardo, Liuwen Yu, Chen Chen, and Leendert van der Torre. “Weakest Link, Prioritized Default Logic and Principles in Argumentation”. In: *Journal of Logic and Computation* 35.4 (2025), exaf007. DOI: [10.1093/logcom/exaf007](https://doi.org/10.1093/logcom/exaf007).
- [46] Sanjay Modgil and Henry Prakken. “A General Account of Argumentation with Preferences”. In: *Artificial Intelligence* 195 (2013), pp. 361–397. DOI: [10.1016/j.artint.2012.10.008](https://doi.org/10.1016/j.artint.2012.10.008).
- [47] Sanjay Modgil and Henry Prakken. “Abstract Rule-Based Argumentation”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 287–364.
- [48] Sanjay Modgil and Henry Prakken. “Corrigendum to A General Account of Argumentation with Preferences [Artif. Intell. 195 (2013) 361–397]”. In: *Artificial Intelligence* 263 (2018), pp. 107–110. DOI: [10.1016/j.artint.2018.05.001](https://doi.org/10.1016/j.artint.2018.05.001).

- [49] Kristijonas Čyras, Xiuyi Fan, Claudia Schulz, and Francesca Toni. “Assumption-Based Argumentation: Disputes, Explanations, Preferences”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 365–408.
- [50] Alejandro J. García and Guillermo R. Simari. “Argumentation Based on Logic Programming”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 409–436.
- [51] Souhila Kaci, Leendert van der Torre, Srdjan Vesic, and Serena Villata. “Preference in Abstract Argumentation”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 211–248.
- [52] Martin Caminada. “Argumentation Semantics as Formal Discussion”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 487–518.
- [53] Beishui Liao and Leendert van der Torre. “Attack-Defense Semantics of Argumentation”. In: *Computational Models of Argument*. Frontiers in Artificial Intelligence and Applications. IOS Press, 2024, pp. 133–144. DOI: [10.3233/FAIA240316](https://doi.org/10.3233/FAIA240316).
- [54] Anthony P. Young, Sanjay Modgil, and Odinaldo Rodrigues. “Prioritised Default Logic as Rational Argumentation”. In: *Proceedings of the 15th International Conference on Autonomous Agents and Multiagent Systems*. Ed. by J. Thangarajah, K. Tuyls, C. Jonker, and S. Marsella. International Foundation for Autonomous Agents and Multiagent Systems, 2016, pp. 626–634. URL: <https://www.ifaamas.org/Proceedings/aamas2016/pdfs/p626.pdf>.
- [55] Anthony P. Young, Sanjay Modgil, and Odinaldo Rodrigues. “On the Interaction between Logic and Preference in Structured Argumentation”. Author manuscript, 15 pages; cited page numbers refer to this version. 2017. URL: https://cdn.prod.website-files.com/674c48c6202b4f012a14375f/67d0880bb4ca0b03d957bedb_TAFA-17_paper_14.pdf.
- [56] Samy Sá, Wolfgang Dvořák, and Martin Caminada. “Syntactic and Semantic Connections between Logic Programming and Argumentation Systems”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 429–512.
- [57] Jesse Heyninck, Gabriele Kern-Isberner, Tjitze Rienstra, Kenneth Skiba, and Matthias Thimm. “The Semantical Structure of Conditionals, and Its Relation to Formal Argumentation”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 513–576.
- [58] Raymond Reiter. “A Logic for Default Reasoning”. In: *Artificial Intelligence* 13.1–2 (1980), pp. 81–132. DOI: [10.1016/0004-3702\(80\)90014-4](https://doi.org/10.1016/0004-3702(80)90014-4).

- [59] Gerhard Brewka. “Reasoning about Priorities in Default Logic”. In: *Proceedings of the Twelfth National Conference on Artificial Intelligence*. AAAI Press / MIT Press, 1994, pp. 940–945.
- [60] Michael Gelfond and Vladimir Lifschitz. “The Stable Model Semantics for Logic Programming”. In: *Proceedings of the Fifth International Conference and Symposium on Logic Programming*. MIT Press, 1988, pp. 1070–1080.
- [61] Martin Caminada, Samy Sá, João Alcântara, and Wolfgang Dvořák. “On the Equivalence between Logic Programming Semantics and Argumentation Semantics”. In: *International Journal of Approximate Reasoning* 58 (2015), pp. 87–111.
- [62] Andrei Bondarenko, Phan Minh Dung, Robert A. Kowalski, and Francesca Toni. “An Abstract, Argumentation-Theoretic Approach to Default Reasoning”. In: *Artificial Intelligence* 93.1–2 (1997), pp. 63–101. DOI: [10.1016/S0004-3702\(97\)00015-5](https://doi.org/10.1016/S0004-3702(97)00015-5).
- [63] Ringo Baumann and Leendert van der Torre. *Inference Operators for Argumentation Formalisms—The Case of Dung-style Frameworks*. NMR 2025, five-page short paper; cited page numbers refer to the paper PDF. 2025. URL: <https://ceur-ws.org/Vol-4071/short1.pdf>.
- [64] Phan Minh Dung. “An Axiomatic Analysis of Structured Argumentation with Priorities”. In: *Artificial Intelligence* 231 (2016), pp. 107–150. DOI: [10.1016/j.artint.2015.10.005](https://doi.org/10.1016/j.artint.2015.10.005).
- [65] Phan Minh Dung and Phan Minh Thang. “Fundamental Properties of Attack Relations in Structured Argumentation with Priorities”. In: *Artificial Intelligence* 255 (2018), pp. 1–42. DOI: [10.1016/j.artint.2017.11.002](https://doi.org/10.1016/j.artint.2017.11.002).
- [66] Martin Caminada. “Rationality Postulates: Applying Argumentation Theory for Non-Monotonic Reasoning”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 771–796.
- [67] Martin Caminada and Leila Amgoud. “On the Evaluation of Argumentation Formalisms”. In: *Artificial Intelligence* 171.5–6 (2007), pp. 286–310. DOI: [10.1016/j.artint.2007.02.003](https://doi.org/10.1016/j.artint.2007.02.003).
- [68] Nikos Gorogiannis and Anthony Hunter. “Instantiating Abstract Argumentation with Classical Logic Arguments: Postulates and Properties”. In: *Artificial Intelligence* 175.9–10 (2011), pp. 1479–1497.
- [69] Phan Minh Dung and Phan Minh Thang. “Closure and Consistency in Logic-Associated Argumentation”. In: *Journal of Artificial Intelligence Research* 49 (2014), pp. 79–109.
- [70] Jesse Heyninck and Christian Straßer. “Revisiting Unrestricted Rebut and Preferences in Structured Argumentation”. In: *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence*. 2017, pp. 1088–1092. URL: <https://www.ijcai.org/proceedings/2017/151>.

- [71] Y. Wu and M. Podlaszewski. “Implementing Crash-Resistance and Non-Interference in Logic-Based Argumentation”. In: *Journal of Logic and Computation* 25.2 (2015), pp. 303–333.
- [72] D. Grooters and Henry Prakken. “Two Aspects of Relevance in Structured Argumentation: Minimality and Paraconsistency”. In: *Journal of Artificial Intelligence Research* 56 (2016), pp. 197–245.
- [73] Ringo Baumann. “On the Nature of Argumentation Semantics: Existence and Uniqueness, Expressibility, and Replaceability”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 839–936.
- [74] Leendert van der Torre and Srdjan Vesic. “The Principle-Based Approach to Abstract Argumentation Semantics”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 797–838.
- [75] Peter McBurney and Simon Parsons. *Syntax and Semantics of the Fatio Argumentation Protocol*. Tech. rep. ULCS-04-002. Original report; textbook locators refer to positions in its ten-page PDF. Department of Computer Science, University of Liverpool, 2004. URL: <https://techreports.csc.liv.ac.uk/index.php?id=ULCS-04-002>.
- [76] Peter McBurney and Simon Parsons. “Locutions for Argumentation in Agent Interaction Protocols”. In: *Agent Communication*. Vol. 3396. Lecture Notes in Computer Science. Springer, 2005, pp. 209–225. DOI: [10.1007/978-3-540-32258-0_14](https://doi.org/10.1007/978-3-540-32258-0_14).
- [77] Foundation for Intelligent Physical Agents. *FIPA Communicative Act Library Specification*. 2002. URL: <http://www.fipa.org/specs/fipa00037/>.
- [78] Amit Chopra, Leendert van der Torre, Harko Verhagen, and Serena Villata, eds. *Handbook of Normative Multiagent Systems*. London: College Publications, 2018. ISBN: 978-1-84890-285-5.
- [79] Richard Booth, Dov Gabbay, Souhila Kaci, Tjitze Rienstra, and Leendert van der Torre. *Abduction and Dialogical Proof in Argumentation and Logic Programming*. Locators refer to arXiv version 1. 2014. arXiv: [1407.3896v1](https://arxiv.org/abs/1407.3896v1). URL: <https://arxiv.org/abs/1407.3896v1>.
- [80] Luca Pasetto and Christoph Benzmüller. “Implementing the Fatio Protocol for Multi-Agent Argumentation in LogiKEY”. In: *Automated Reasoning in Quantified Non-Classical Logics (ARQNL 2024)*. Locators refer to the ten-page author PDF. 2024. URL: <https://www.ruhr-uni-bochum.de/lodex/publication/pasetto-benzmuller-arqnl-2024/pasetto-benzm%C3%BCller-ARQNL-2024.pdf>.
- [81] Luca Pasetto, Thomas Skafte, Leendert van der Torre, and Liuwen Yu. “The Libra Dialogue Protocol for Comparative Deliberation”. COMMA 2026; cited version is the twelve-page author-supplied paper, with locators to its printed pages. 2026.

- [82] Thomas F. Gordon, Henry Prakken, and Douglas Walton. “The Carneades Model of Argument and Burden of Proof”. In: *Artificial Intelligence* 171.10–15 (2007), pp. 875–896. DOI: [10.1016/j.artint.2007.04.010](https://doi.org/10.1016/j.artint.2007.04.010).
- [83] Liuwen Yu, Chenyang Cai, Réka Markovich, and Leendert van der Torre. “Balancing for Agent Decision Making through Argumentation”. In: *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems*. Selected ten-page attributed author version; author order and PDF locators follow this copy. The Chapter 10 import is restricted to Definitions 3.1–3.4 on PDF page 3. International Foundation for Autonomous Agents and Multiagent Systems, 2026, pp. 2876–2885. DOI: [10.65109/VBZK6091](https://doi.org/10.65109/VBZK6091). URL: https://yuliuwen.github.io/assets/files/balancing_aamas2026.pdf.
- [84] Claudette Cayrol, Andrea Cohen, and Marie-Christine Lagasquie-Schiex. “Higher-Order Interactions (Bipolar or Not) in Abstract Argumentation: A State of the Art”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 15–130.
- [85] Trevor Bench-Capon, Katie Atkinson, Floris Bex, Henry Prakken, and Bart Verheij. “Computational Models of Legal Argument”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 101–224.
- [86] Bas van Gijzel and Henry Prakken. “Relating Carneades with Abstract Argumentation via the ASPIC+ Framework for Structured Argumentation”. In: *Argument and Computation* 3.1 (2012), pp. 21–47. DOI: [10.1080/19462166.2012.661766](https://doi.org/10.1080/19462166.2012.661766).
- [87] Claudette Cayrol and Marie-Christine Lagasquie-Schiex. “On the Acceptability of Arguments in Bipolar Argumentation Frameworks”. In: *Symbolic and Quantitative Approaches to Reasoning with Uncertainty*. Springer, 2005, pp. 378–389. DOI: [10.1007/11518655_33](https://doi.org/10.1007/11518655_33).
- [88] Antonis Bikakis, Andrea Cohen, Wolfgang Dvořák, Giorgos Flouris, and Simon Parsons. “Joint Attacks and Accrual in Argumentation Frameworks”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 131–210.
- [89] Stefano Bistarelli and Francesco Santini. “Weighted Argumentation”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 355–396.
- [90] Anthony Hunter, Sylwia Polberg, Nico Potyka, Tjitze Rienstra, and Matthias Thimm. “Probabilistic Argumentation: A Survey”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 397–444.

- [91] Elise Bonzon, Jérôme Delobelle, Sébastien Konieczny, and Nicolas Maudet. “A Comparative Study of Ranking-Based Semantics for Abstract Argumentation”. In: *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence*. 2016, pp. 914–920. URL: <https://cdn.aaai.org/ojs/10116/10116-13-13644-1-2-20201228.pdf>.
- [92] Mathieu Beirlaen, Jesse Heyninck, Pere Pardo, and Christian Straßer. “Argument Strength in Formal Argumentation”. In: *IfCoLog Journal of Logics and Their Applications* 5.3 (2018), pp. 629–676. URL: <https://collegepublications.co.uk/contents/ifcolog00023.pdf>.
- [93] Pietro Baroni, Antonio Rago, and Francesca Toni. “From Fine-Grained Properties to Broad Principles for Gradual Argumentation: A Principled Spectrum”. In: *International Journal of Approximate Reasoning* 105 (2019), pp. 252–286. DOI: [10.1016/j.ijar.2018.11.019](https://doi.org/10.1016/j.ijar.2018.11.019).
- [94] Henry Prakken. “An Abstract and Structured Account of Dialectical Argument Strength”. Sixty-page author manuscript; locators refer to this PDF. July 30, 2024. URL: <https://webpace.science.uu.nl/~prakk101/pubs/gradualS.pdf>.
- [95] Davide Liga, Réka Markovich, and Liuwen Yu. “Where Do Weights Come From? Weighting and Weighing by LLMs and Formal Argumentation in Discretionary Judicial Reasoning”. Author PDF for an AAAI 2026 workshop; locators refer to this version. 2026. URL: https://yuliuwen.github.io/assets/files/2026_AAAI%20bridge%20.pdf.
- [96] Liuwen Yu, Caren Al Anaissy, Srdjan Vesic, Xu Li, and Leendert van der Torre. “A Principle-Based Analysis of Bipolar Argumentation Semantics”. In: *Logics in Artificial Intelligence*. Springer, 2023, pp. 209–224. DOI: [10.1007/978-3-031-43619-2_15](https://doi.org/10.1007/978-3-031-43619-2_15).
- [97] Ringo Baumann, Sylvie Doutre, Jean-Guy Mailly, and Johannes P. Wallner. “Enforcement in Formal Argumentation”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 445–510.
- [98] Pietro Baroni and Massimiliano Giacomin. “On Principle-Based Evaluation of Extension-Based Argumentation Semantics”. In: *Artificial Intelligence* 171.10–15 (2007), pp. 675–700. DOI: [10.1016/j.artint.2007.04.004](https://doi.org/10.1016/j.artint.2007.04.004).
- [99] Pietro Baroni, Massimiliano Giacomin, and Beishui Liao. “Locality and Modularity in Abstract Argumentation”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 937–980.
- [100] Pietro Baroni, Antonio Rago, and Francesca Toni. “How Many Properties Do We Need for Gradual Argumentation?” In: *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*. 2018, pp. 1736–1743. DOI: [10.1609/aaai.v32i1.11544](https://doi.org/10.1609/aaai.v32i1.11544).

- [101] Ringo Baumann and Gerhard Brewka. “AGM Meets Abstract Argumentation: Expansion and Revision for Dung Frameworks”. In: *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence*. 2015, pp. 2734–2740.
- [102] Sylvie Doutre and Jean-Guy Mailly. “Constraints and Changes: A Survey of Abstract Argumentation Dynamics”. In: *Argument and Computation* 9.3 (2018), pp. 223–248. DOI: [10.3233/AAC-180425](https://doi.org/10.3233/AAC-180425).
- [103] Gianvincenzo Alfano, Francesco Parisi, Sergio Greco, Gerardo I. Simari, and Guillermo R. Simari. “On the Incremental Computation of Semantics in Dynamic Argumentation”. In: *Handbook of Formal Argumentation, Volume 2*. College Publications, 2021, pp. 663–718.
- [104] Federico Cerutti, Sarah A. Gaggl, Matthias Thimm, and Johannes P. Wallner. “Foundations of Implementations for Formal Argumentation”. In: *Handbook of Formal Argumentation*. College Publications, 2018, pp. 689–768.
- [105] Uwe Egly, Sarah A. Gaggl, and Stefan Woltran. “Answer-Set Programming Encodings for Argumentation Frameworks”. In: *Argument and Computation* 1.2 (2010), pp. 147–177. DOI: [10.1080/19462166.2010.486479](https://doi.org/10.1080/19462166.2010.486479).
- [106] International Competition on Computational Models of Argumentation. *ICCMA Competition Resources and Formats*. URL: <https://argumentationcompetition.org/>.
- [107] Piyawat Lertvittayakumjorn and Francesca Toni. “Argumentative explanations for pattern-based text classifiers”. In: *Argument & Computation* 14.2 (2023), pp. 163–234. DOI: [10.3233/AAC-220004](https://doi.org/10.3233/AAC-220004). URL: <https://journals.sagepub.com/doi/10.3233/AAC-220004>.
- [108] AnneMarie Borg and Floris Bex. “Minimality, Necessity and Sufficiency for Argumentation and Explanation”. In: *International Journal of Approximate Reasoning* 168 (2024), p. 109143. DOI: [10.1016/j.ijar.2024.109143](https://doi.org/10.1016/j.ijar.2024.109143).
- [109] Kristijonas Čyras, Antonio Rago, Emanuele Albini, Pietro Baroni, and Francesca Toni. “Argumentative XAI: A Survey”. In: *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence*. 2021, pp. 4392–4399. DOI: [10.24963/ijcai.2021/600](https://doi.org/10.24963/ijcai.2021/600).
- [110] Alexander Bochman, Federico Cerutti, and Tjitze Rienstra. “Causation and Argumentation”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 627–716.
- [111] Xu Li, Leon van der Torre, and Liuwen Yu. “Revealed Epistemic Trust”. In: *Proceedings of the 23rd International Conference on Principles of Knowledge Representation and Reasoning*. 2026, pp. 462–471. DOI: [10.24963/kr.2026/44](https://doi.org/10.24963/kr.2026/44). URL: <https://proceedings.kr.org/2026/44/>.

- [112] Louise Dupuis, Matteo Michelini, Dunja Šešelja, and Christian Straßer. “Argumentative Agent-Based Models”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 305–376.
- [113] Juan C. L. Teze, Lluís Godo, and Gerardo I. Simari. “Defeasible Argumentation-Based Epistemic Planning with Preferences”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 717–762.
- [114] Carlos Iván Chesnevar, Jürgen Dix, Beishui Liao, Jieting Luo, Carlo Proietti, and Antonio Yuste-Ginel. “Formal Argumentation and Modal Logic”. In: *Handbook of Formal Argumentation, Volume 3*. College Publications, 2024, pp. 763–846.
- [115] Christoph Benzmüller, Xavier Parent, and Leendert van der Torre. “Designing Normative Theories for Ethical and Legal Reasoning: LogiKEy Framework, Methodology, and Tool Support”. In: *Artificial Intelligence* 287 (2020), p. 103348. DOI: [10.1016/j.artint.2020.103348](https://doi.org/10.1016/j.artint.2020.103348).
- [116] Marija Slavkovik, Liuwen Yu, Leendert van der Torre, Réka Markovich, and Beishui Liao. “Disobedience in Normative Multi-Agent Systems”. In: *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems*. AAMAS 2026 (Paphos, Cyprus, May 25–29, 2026). International Foundation for Autonomous Agents and Multiagent Systems, 2026, pp. 2438–2446. DOI: [10.65109/LDPF7116](https://doi.org/10.65109/LDPF7116).
- [117] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. “ReAct: Synergizing Reasoning and Acting in Language Models”. In: *International Conference on Learning Representations*. Locators refer to the camera-ready arXiv version 3, 10 March 2023. 2023. arXiv: [2210.03629v3](https://arxiv.org/abs/2210.03629v3). URL: <https://arxiv.org/abs/2210.03629v3>.